$32,390,000
CITY OF CLEVELAND, OHIO
WATER POLLUTION CONTROL REVENUE BONDS
SERIES 2016 (Green Bonds)

Below please find a revision (the "Revision") with respect to the Official Statement dated March 31, 2016 for the captioned issuance. The Revision corrects the description of the regular record date as set forth in the First Supplemental Trust Agreement (as defined in the Official Statement). The Revision is intended to replace in its entirety the sentence noted below originally printed in the Official Statement.

1. The second sentence of the paragraph located under "**DESCRIPTION OF THE SERIES 2016 BONDS -- Payment of Principal and Interest**" shall be amended by deleting it in its entirety and replacing it with the following new sentence:

Interest on the Series 2016 Bonds will be payable by check or draft mailed or delivered on the Interest Payment Date to the registered holder (initially DTC or its nominee) at the address shown on the Register as of the close of business on the last day of the calendar month next preceding that Interest Payment Date; provided that, so long as the Series 2016 Bonds remain in book-entry form, DTC (or any successor depository company) may direct the Trustee to make such payment by wire transfer of immediately available funds on each Interest Payment Date.

Dated: April 12, 2016

NEW ISSUE-BOOK ENTRY ONLY

Ratings
Moody's: Aa3
Standard & Poor's: A+
(See "Ratings" herein)

In the opinion of Squire Patton Boggs (US) LLP, Bond Counsel, under existing law (i) assuming continuing compliance with certain covenants and the accuracy of certain representations, interest on the Series 2016 Bonds is excluded from gross income for federal income tax purposes and is not an item of tax preference for purposes of the federal alternative minimum tax imposed on individuals and corporations and (ii) interest on, and any profit made on the sale, exchange or other disposition of, the Series 2016 Bonds, are exempt from all Ohio state and local taxation, except the estate tax, the domestic insurance company tax, the dealers in intangibles tax, the tax levied on the basis of the total equity capital of financial institutions, and the net worth base of the corporate franchise tax. Interest on the Series 2016 Bonds may be subject to certain federal taxes imposed only on certain corporations, including the corporate alternative minimum tax on a portion of that interest. For a more complete discussion of tax aspects, see "Tax Matters."

$32,390,000
CITY OF CLEVELAND, OHIO
Water Pollution Control Revenue Bonds
Series 2016 (Green Bonds)

Dated: Date of Delivery **Due:** November 15, as shown on the inside cover page

The $32,390,000 City of Cleveland, Ohio, Water Pollution Control Revenue Bonds, Series 2016 (Green Bonds) (the "Series 2016 Bonds") are special obligations of the City of Cleveland, Ohio (the "City") issued under and secured by the provisions of the Trust Agreement between the City and Zions Bank, a Division of ZB, National Association, as trustee (the "Trustee"), to provide funds to (i) pay Capital Costs relating to the acquisition, construction and improvement of the System, (ii) fund the Debt Service Reserve Requirement for the Series 2016 Bonds, and (iii) pay the costs of issuance of the Series 2016 Bonds. See "Appendix B - Definitions and Summary of Certain Terms and Provisions of the Trust Agreement."

The Series 2016 Bonds will be secured under the Trust Agreement, on a parity with any Additional Bonds, by a pledge of and lien on the Net Revenues and the Special Funds (all as defined herein). See "Security and Sources of Payment of the Bonds."

Interest on the Series 2016 Bonds is payable each May 15 and November 15, beginning November 15, 2016. The Series 2016 Bonds will be initially issued as fully registered bonds under a book-entry system, and will be registered initially in the name of Cede & Co., as nominee for The Depository Trust Company, New York, New York ("DTC"). Purchases of book-entry interests in the Series 2016 Bonds will be made in denominations of $5,000 and any integral multiple thereof. DTC, or its nominee, will receive all payments with respect to the Series 2016 Bonds from the Trustee. DTC is required by its rules and procedures to remit such payments to participants in DTC for subsequent disbursement to the owners of book-entry interests. Holders of book-entry interests in the Series 2016 Bonds will not receive physical delivery of bond certificates. The Series 2016 Bonds in certificated form as such will not be transferable or exchangeable, except for transfer to a nominee of DTC or as otherwise described in this Official Statement. Principal of and redemption premium, if any, and interest on, the Series 2016 Bonds will be paid to Cede & Co. as the registered owner. See "Appendix G – Book-Entry Only System" hereto.

The Series 2016 Bonds are subject to optional and mandatory sinking fund redemption prior to maturity as described herein. See "Description of the Series 2016 Bonds – Redemption Provisions."

The Series 2016 Bonds will be special obligations of the City and will not represent or constitute a general obligation, bonded indebtedness or a pledge of the general credit or taxing power of the City, the State of Ohio or any political subdivision thereof. The holders of the Series 2016 Bonds will not have the right to have excises or taxes levied by the City, the State of Ohio or the taxing authority of any political subdivision thereof for the payment of debt service charges. See "Certain Investment Considerations and Risk factors – Limited Obligations."

The Series 2016 Bonds are offered when, as and if issued by the City and accepted by the Underwriters, subject to the opinions on certain legal matters relating to their issuance by Squire Patton Boggs (US) LLP, Bond Counsel, and certain other conditions. Certain legal matters will be passed upon for the City by Barbara A. Langhenry, its Director of Law. Certain legal matters will be passed upon for the Underwriters by their counsel, Calfee, Halter & Griswold LLP. It is expected that delivery of the Series 2016 Bonds in definitive form will be made to DTC on or about April 14, 2016.

Morgan Stanley

Barclays **Mesirow Financial, Inc.**

The date of this Official Statement is March 31, 2016, and the information speaks only as of that date.

$32,390,000
City of Cleveland, Ohio
Water Pollution Control Revenue Bonds
Series 2016 (Green Bonds)

MATURITY SCHEDULE

Dated: Date of Delivery Due: November 15, as shown below

$17,230,000 Serial Bonds

Maturity Date	Amount	Rate	Yield	Price	CUSIP No.[1]
11/15/2017	$550,000	4.000%	0.750%	105.114	186423AA7
11/15/2018	$570,000	3.000%	0.920%	105.303	186423AB5
11/15/2019	$590,000	4.000%	1.140%	110.020	186423AC3
11/15/2020	$610,000	4.000%	1.320%	111.887	186423AD1
11/15/2021	$635,000	4.000%	1.520%	113.232	186423AE9
11/15/2022	$660,000	5.000%	1.670%	120.685	186423AF6
11/15/2023	$695,000	5.000%	1.830%	122.356	186423AG4
11/15/2024	$730,000	5.000%	2.000%	123.558	186423AH2
11/15/2025	$765,000	5.000%	2.130%	124.764	186423AJ8
11/15/2026	$805,000	5.000%	2.270%	123.397[2]	186423AK5
11/15/2027	$845,000	5.000%	2.390%	122.240[2]	186423AL3
11/15/2028	$885,000	5.000%	2.510%	121.095[2]	186423AM1
11/15/2029	$930,000	5.000%	2.610%	120.151[2]	186423AN9
11/15/2030	$980,000	5.000%	2.660%	119.682[2]	186423AP4
11/15/2031	$1,025,000	5.000%	2.720%	119.123[2]	186423AQ2
11/15/2032	$1,080,000	5.000%	2.790%	118.474[2]	186423AR0
11/15/2033	$1,130,000	5.000%	2.860%	117.829[2]	186423AS8
11/15/2034	$1,190,000	5.000%	2.900%	117.462[2]	186423AT6
11/15/2035	$1,245,000	5.000%	2.980%	116.733[2]	186423AU3
11/15/2036	$1,310,000	5.000%	3.000%	116.552[2]	186423AV1

$7,595,000 5.000% Term Bonds due November 15, 2041 Yield 3.140% Price 115.291[2] CUSIP[1] No. 186423AW9

$7,565,000 5.000% Term Bonds due November 15, 2045 Yield 3.190% Price 114.845[2] CUSIP[1] No. 186423AX7

(1) Copyright American Bankers Association. CUSIP numbers have been assigned by an independent company not affiliated with the City and are included solely for the convenience of the holders of the Series 2016 Bonds. The City is not responsible for the selection or uses of these CUSIP numbers, and no representation is made as to their correctness on the Series 2016 Bonds or as indicated above. The CUSIP number for a specific maturity is subject to being changed after the issuance of the Series 2016 Bonds as a result of various subsequent actions including, but not limited to, a refunding in whole or in part of such maturity or as a result of the procurement of secondary market portfolio insurance or other similar enhancement by investors that is applicable to all or a portion of certain maturities of the Series 2016 Bonds.

(2) Priced to November 15, 2025 optional redemption date.

CITY OF CLEVELAND, OHIO
DIVISION OF WATER POLLUTION CONTROL

MAYOR
FRANK G. JACKSON

CITY COUNCIL PRESIDENT
KEVIN J. KELLEY

CITY COUNCIL UTILITIES COMMITTEE
TERRELL H. PRUITT, *Chair*

CLERK OF COUNCIL
PATRICIA J. BRITT

DEPARTMENT OF FINANCE
Director of Finance
SHARON A. DUMAS

Assistant Secretary, Sinking Fund Commission
ELIZABETH C. HRUBY

DEPARTMENT OF PUBLIC UTILITIES
Director of Public Utilities
ROBERT L. DAVIS

Commissioner of the Division of Water Pollution Control
RACHID ZOGHAIB

DEPARTMENT OF LAW
Director of Law
BARBARA A. LANGHENRY

TRUSTEE
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Cleveland, Ohio

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Cleveland, Ohio

CO-FINANCIAL ADVISERS
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Tuxedo Park, New York

PHOENIX CAPITAL PARTNERS, LLP
Philadelphia, Pennsylvania
Washington, DC

FEASIBILITY CONSULTANT
Arcadis U.S., Inc.
Cleveland, Ohio

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REGARDING THIS OFFICIAL STATEMENT

This Official Statement does not constitute an offering of any security other than the original offering of the Series 2016 Bonds identified on the cover hereof. No person has been authorized by the City or the Underwriters to give any information or to make any representation other than as contained in this Official Statement. Any other information or representation should not be relied upon as having been given or authorized by the City or the Underwriters. This Official Statement does not constitute an offer to sell or the solicitation of an offer to buy, and there shall not be any sale of, the Series 2016 Bonds by any person in any jurisdiction in which it is unlawful to make such offer, solicitation or sale.

The information and expressions of opinion in this Official Statement are subject to change without notice. Neither the delivery of this Official Statement nor any sale made hereunder shall create, under any circumstances, or give rise to any implication that there has been no change in the affairs of the City since its date.

Any statements made in this Official Statement which involve opinions or estimates, whether expressly stated to be such, are made as such and are not representations of fact or certainty, and no representation is made that any of those statements have been or will be realized. Information in this Official Statement that has been derived by the City from its officials and other sources is believed to be accurate and reliable. Information other than that obtained from official records of the City has not been independently confirmed or verified by the City, and its accuracy is not guaranteed.

This Official Statement contains statements which, to the extent they are not recitations of historical fact, constitute "forward-looking statements." The words "estimate," "project," "anticipate," "expect," "intend," "believe," and similar expressions are intended to identify forward-looking statements. A number of factors affecting the City's financial results could cause actual results to differ materially from those stated in the forward-looking statements. Readers are cautioned not to place undue reliance on these forward-looking statements, which speak only as of the date hereof.

Zions Bank, a Division of ZB, National Association, by acceptance of its duties as Trustee under the Trust Agreement, has not reviewed this Official Statement and has made no representations as to the information contained herein, including but not limited to, any representations as to the financial feasibility or related activities.

THE COVER PAGE CONTAINS CERTAIN INFORMATION FOR QUICK REFERENCE ONLY. IT IS NOT A SUMMARY OF THIS ISSUE. INVESTORS MUST READ THE ENTIRE OFFICIAL STATEMENT TO OBTAIN INFORMATION ESSENTIAL TO THE MAKING OF AN INFORMED INVESTMENT DECISION.

THE SERIES 2016 BONDS HAVE NOT BEEN REGISTERED UNDER THE SECURITIES ACT OF 1933, AND THE TRUST AGREEMENT HAS NOT BEEN QUALIFIED UNDER THE TRUST INDENTURE ACT OF 1939, IN RELIANCE UPON EXEMPTIONS CONTAINED IN SUCH ACTS. THE REGISTRATION OR QUALIFICATION OF THE SERIES 2016 BONDS IN ACCORDANCE WITH APPLICABLE PROVISIONS OF

SECURITIES LAWS OF THE STATES IN WHICH THE SERIES 2016 BONDS HAVE BEEN REGISTERED OR QUALIFIED AND THE EXEMPTION FROM REGISTRATION OR QUALIFICATION IN OTHER STATES CANNOT BE REGARDED AS A RECOMMENDATION THEREOF. NEITHER THE U.S. SECURITIES AND EXCHANGE COMMISSION NOR ANY OTHER FEDERAL, STATE OR OTHER GOVERNMENTAL ENTITY OR AGENCY WILL HAVE PASSED UPON THE ACCURACY OR ADEQUACY OF THIS OFFICIAL STATEMENT OR APPROVED THE SERIES 2016 BONDS FOR SALE. ANY REPRESENTATION TO THE CONTRARY MAY BE A CRIMINAL OFFENSE. THE UNDERWRITERS HAVE PROVIDED THE FOLLOWING SENTENCE FOR INCLUSION IN THIS OFFICIAL STATEMENT. THE UNDERWRITERS HAVE REVIEWED THE INFORMATION IN THIS OFFICIAL STATEMENT IN ACCORDANCE WITH, AND AS PART OF, THEIR RESPONSIBILITIES TO INVESTORS UNDER THE FEDERAL SECURITIES LAWS AS APPLIED TO THE FACTS AND CIRCUMSTANCES OF THIS TRANSACTION, BUT THE UNDERWRITERS DO NOT GUARANTEE THE ACCURACY OR COMPLETENESS OF SUCH INFORMATION.

In connection with this offering of the Series 2016 Bonds, the Underwriters may overallot or effect transactions that stabilize or maintain the market prices of the Series 2016 Bonds at levels above those which might otherwise prevail in the open market. Such stabilizing, if commenced, may be discontinued at any time without prior notice. The prices and other terms of the offering and sale of the Series 2016 Bonds may be changed from time to time by the Underwriters after the Series 2016 Bonds are released for sale, and the Series 2016 Bonds may be offered and sold at prices other than the initial offering prices, including sales to dealers, without prior notice.

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OFFICIAL STATEMENT

Relating to

$32,390,000
CITY OF CLEVELAND, OHIO
WATER POLLUTION CONTROL REVENUE BONDS
SERIES 2016 (GREEN BONDS)

INTRODUCTION

The City of Cleveland, Ohio (the "City") is furnishing this Official Statement, including the cover page and appendices, to provide certain information in connection with the issuance and sale by the City of its $32,390,000 City of Cleveland, Ohio, Water Pollution Control Revenue Bonds, Series 2016 (Green Bonds) (the "Series 2016 Bonds"). The Series 2016 Bonds and any series of Additional Bonds hereafter issued under the Trust Agreement are referred to collectively as the "Bonds." All capitalized terms used but not otherwise defined in this Official Statement shall have the meanings ascribed to them in the Trust Agreement. See "APPENDIX B – DEFINITIONS AND SUMMARY OF CERTAIN TERMS AND PROVISIONS OF THE TRUST AGREEMENT."

The Series 2016 Bonds are being issued (i) to pay Capital Costs relating to the acquisition, construction and improvement of the System, (ii) fund the Debt Service Reserve Requirement for the Series 2016 Bonds, and (iii) to pay the costs of issuance of the Series 2016 Bonds, as described under "Purpose of Issue" herein.

This introduction is not a summary of this Official Statement. It is only a brief description of and guide to, and is qualified by, more complete and detailed information contained in the entire Official Statement, including the cover page and appendices hereto, and the documents summarized or described herein. A full review should be made of the entire Official Statement. The offering of the Series 2016 Bonds to potential investors is made only by means of the entire Official Statement.

Issuer

The City is a municipal corporation duly organized and validly existing under its Charter and the Constitution and laws of the State of Ohio (the "State"). The Division of Water Pollution Control (the "Division") along with the Division of Water, Cleveland Public Power, the Division of Fiscal Control, and the Office of Utilities General Administration make up the City's Department of Public Utilities. The Division is responsible for cleaning and maintaining the network of sewers in the City. The network carries sanitary sewage, combined stormwater and sewage, and industrial waste from their points of origin to sewage processing facilities for treatment and disposal. See "THE DIVISION OF WATER POLLUTION CONTROL."

Security and Sources of Payment for the Series 2016 Bonds

The Series 2016 Bonds will be special obligations of the City and will not represent or constitute a general obligation, bonded indebtedness or a pledge of the general credit or taxing power of the City, the State of Ohio or any political subdivision thereof. The holders of the

Series 2016 Bonds will not have the right to have excises or taxes levied by the City, the State of Ohio or the taxing authority of any political subdivision thereof for the payment of debt service charges. See "CERTAIN INVESTMENT CONSIDERATIONS AND RISK FACTORS – Limited Obligations."

The Series 2016 Bonds are issued and secured under the Master Trust Agreement, dated as of April 14, 2016 (the "Master Trust Agreement"), by and between the City and Zions Bank, a Division of ZB, National Association, Cleveland, Ohio, as trustee (the "Trustee"), as supplemented by the First Supplemental Trust Agreement dated as of April 14, 2016, by and between the City and the Trustee (the "First Supplemental Trust Agreement" and together with the Master Trust Agreement and as the Master Trust Agreement may be further amended and supplemented from time to time, the "Trust Agreement").

Upon issuance and delivery, the Series 2016 Bonds will be secured under the Trust Agreement on parity with any Additional Bonds subsequently issued. After the issuance of the Series 2016 Bonds, the aggregate outstanding principal amount of Bonds secured by the Trust Agreement will be $32,390,000. See "OUTSTANDING BONDS AND OTHER OBLIGATIONS – Outstanding Bonds" and "CERTAIN INVESTMENT CONSIDERATIONS AND RISK FACTORS – Additional Bonds."

Purpose of Issue

The City is issuing the Series 2016 Bonds to (a) pay Capital Costs relating to the acquisition, construction and improvement of the System, (b) fund the Debt Service Reserve Requirement for the Series 2016 Bonds, and (c) pay the costs of issuance of the Series 2016 Bonds. The "System" consists of the wastewater collection and disposal system owned and operated by the City, including all Waste Water Facilities thereof, as extant or as that system may be hereafter modified, improved, supplemented or extended. "Waste Water Facilities" consist of facilities for the purpose of treating, neutralizing, disposing of, stabilizing, cooling, segregating, or holding wastewater. "Capital Costs" means the costs of acquisition and construction of Waste Water Facilities of the System and all other costs and expense as may be necessary or incident thereto. See "APPENDIX B – DEFINITIONS AND SUMMARY OF CERTAIN TERMS AND PROVISIONS OF THE TRUST AGREEMENT."

Green Bond Designation & Green Projects

The mission of the Division is to provide superior leadership, excellent customer service and innovative stormwater and wastewater infrastructure management for a safe and sustainable environment. The Division is currently undertaking a number of capital improvement projects relating to the acquisition, construction and operation of Waste Water Facilities designed to fulfill its mission. Such facilities include wastewater collection facilities which meet the discharge standards in the National Pollution Discharge Elimination System ("NPDES") permit and wastewater collection and pumping facilities which reduce sewage spills. Accordingly, the City has designated the capital improvements to be financed with the proceeds of the Series 2016 Bonds as "Green Projects" based on the environmental benefits of these capital improvements and has designated the Series 2016 Bonds, the proceeds of which will be used to pay Capital Costs of such capital improvements, as "Green Bonds."

The purpose of labeling the Series 2016 Bonds as Green Bonds is to allow investors to invest in environmentally beneficial projects. The terms "Green Project" and "Green Bond" are neither defined in nor related to provisions in the Trust Agreement. Owners of the Series 2016 Bonds do not have any security other than as provided in the Trust Agreement, and owners of the Green Bonds do not assume any specific project risk related to any of the projects funded thereby. The Green Bonds, as with all other Bonds, are revenue bonds secured by a pledge of and lien on the Net Revenues of the System. See "SECURITY AND SOURCES OF PAYMENT OF THE BONDS."

The Green Projects to be funded with the proceeds of the Series 2016 Bonds fall into the following two categories:

- *Stormwater Projects*: Stream channel restoration, removal of shoaling and woody growth, and other restoration, rehabilitation and improvements to the Euclid Creek Flood Control project between Lake Eerie and Villaview Road, located on the east side of the City; constructing and repairing catch basins and manholes throughout the City.

- *Sewer Projects*: Constructing and installing replacement sewers and repairing and rehabilitating existing sewers throughout the City as needed; rehabilitating and relining sewers at various locations in the City; rehabilitating and repairing sewer service connections throughout the City; replacing structurally deteriorated sewer lines at various locations throughout the City to prevent flooding to properties, basement flooding, street flooding, and illicit discharges; acquiring equipment and professional services; investing in information technology projects.

For additional information concerning the Green Projects, see "THE DIVISION OF WATER POLLUTION CONTROL — Capital Improvement Programs" and "APPENDIX E - Feasibility Report" and Appendix B thereto.

A sub-account will be created within the Construction Fund for each Series of Bonds issued under the Trust Agreement, including a sub-account for the proceeds of the Series 2016 Bonds to facilitate tracking the spending of the proceeds. The net proceeds of the Series 2016 Bonds will be spent only on the above-described Green Projects.

The City assumes no obligation to ensure that these projects designated as Green Projects comply with any legal or other standards or principles that relate to Green Projects. The City will provide annual updates regarding the Green Projects and use of the proceeds of the Green Bonds in the form of Appendix F on its website: http://www.clevelandwpc.com. See "APPENDIX F – FORM OF UPDATES REGARDING GREEN PROJECTS AND GREEN BONDS." The City will provide such updates until the City completes these projects or until all proceeds of the Series 2016 Bonds have been spent, at which time no further updates will be provided. For the avoidance of doubt, such updates related to the Green Projects are not part of the Continuing Disclosure Annual Filing (see "APPENDIX D – FORM OF CONTINUING DISCLOSURE AGREEMENT").

Description of the Series 2016 Bonds

Denominations: The Series 2016 Bonds will be issued in fully registered form under a book-entry only system. Book-entry interests in the Series 2016 Bonds will be originally available for purchase in principal amounts of $5,000 and integral multiples thereof.

Redemption: The Series 2016 Bonds maturing on or after November 15, 2026 are subject to prior redemption, by and at the option of the City, in whole or in part (in whole multiples of $5,000), on any date on or after November 15, 2025, in any order of maturity, at a redemption price of 100% of the principal amount thereof, plus accrued interest to the redemption date. Certain of the Series 2016 Bonds are also subject to mandatory sinking fund redemption prior to maturity. See "DESCRIPTION OF THE SERIES 2016 BONDS – Redemption Provisions."

Book-Entry System: The Series 2016 Bonds will be issued under a book-entry only system, registered in the name of Cede & Co., as bond owner and nominee for The Depository Trust Company ("DTC"), New York, New York, to which payments of principal of, and interest and any premium on ("Debt Service"), the Series 2016 Bonds will be made. Purchases and transfers of the Series 2016 Bonds may be made in book-entry form only; purchasers will not receive bond certificates. See "APPENDIX G – BOOK-ENTRY ONLY SYSTEM."

The City and the Trustee will recognize and treat DTC (or any successor securities depository) or its nominee as the owner of the Series 2016 Bonds for all purposes, including payment of Debt Service, notices and enforcement of remedies. Crediting of Debt Service payments and transmittal of notices and other communications by DTC to Direct Participants and Indirect Participants, and by Direct Participants and Indirect Participants to the beneficial owners of the Series 2016 Bonds, are the responsibilities of those persons and will be handled by arrangements among them. The City and the Trustee have no responsibility or liability for any aspect of the records or notices to, or payments made on account of, book-entry ownership, or for maintaining, supervising or reviewing any records relating to that ownership. See "APPENDIX G – BOOK-ENTRY ONLY SYSTEM."

Tax Matters

In the opinion of Squire Patton Boggs (US) LLP, Bond Counsel, under existing law (i) assuming continuing compliance with certain covenants and the accuracy of certain representations, interest on the Series 2016 Bonds is excluded from gross income for federal income tax purposes and is not an item of tax preference for purposes of the federal alternative minimum tax imposed on individuals and corporations and (ii) interest on, and any profit made on the sale, exchange or other disposition of, the Series 2016 Bonds, are exempt from all Ohio state and local taxation, except the estate tax, the domestic insurance company tax, the dealers in intangibles tax, the tax levied on the basis of the total equity capital of financial institutions, and the net worth base of the corporate franchise tax. Interest on the Series 2016 Bonds may be subject to certain federal taxes imposed only on certain corporations, including the corporate alternative minimum tax on a portion of that interest. For a more complete discussion of the tax aspects, see "TAX MATTERS," "CERTAIN INVESTMENT CONSIDERATIONS AND RISK FACTORS – Tax Considerations," and "APPENDIX C - FORM OF BOND COUNSEL OPINION."

Authorization

The Series 2016 Bonds will be issued under authority of Article XVIII of the Ohio Constitution, the Charter of the City, Ordinance No. 29-16 passed by the Council of the City on January 25, 2016 (the "General Bond Ordinance"), Ordinance No. 28-16 passed by the Council of the City on January 25, 2016 (the "Series 2016 Bond Ordinance") and a Certificate of Award delivered pursuant thereto (the "Certificate of Award," and together with the General Bond Ordinance and the Series 2016 Bond Ordinance, the "Series 2016 Bond Legislation"). The Series 2016 Bonds will be secured by the Trust Agreement.

Concerning This Official Statement

All financial and other information presented in this Official Statement relating to the City has been provided by the City from its records, except for information expressly attributed to other sources. The presentation of information, including tables of receipts from sewer revenues and other sources, is intended to show recent historic information and is not intended to indicate current results or future or continuing trends. No representation is made that past experience, as shown by that financial and other information, will necessarily continue or be repeated in the future. See "CERTAIN INVESTMENT CONSIDERATIONS AND RISK FACTORS – Factors That May Cause Insufficiency of Expected Revenues" and "CERTAIN INVESTMENT CONSIDERATIONS AND RISK FACTORS – Historical Information."

This Official Statement should be considered in its entirety, and no one subject discussed should be considered less important than any other by reason of its location in the text. Reference should be made to laws, reports or documents referred to in this Official Statement for more complete information regarding their contents. In particular, the description of the Series 2016 Bonds and the Trust Agreement herein are qualified by reference to such documents in their entirety.

References to provisions of Ohio law, whether codified in the Ohio Revised Code (the "Revised Code") or un-codified, or of the Ohio Constitution, or the Charter or Codified Ordinances of the City are references to current provisions. Those provisions may be amended, repealed or supplemented from time to time. See "CERTAIN INVESTMENT CONSIDERATIONS AND RISK FACTORS – Changes in Law."

Availability of Documents; Inquiries; Continuing Disclosure

Prior to issuance and delivery of the Series 2016 Bonds, copies of the Trust Agreement may be obtained at the office of the City's Department of Finance, 601 Lakeside Avenue, Cleveland, Ohio 44114; thereafter, copies will be available for inspection at the designated corporate trust office of the Trustee.

Pursuant to Rule 15c2-12(b)(5) promulgated by the U.S. Securities and Exchange Commission under the Securities Exchange Act of 1934, as amended (the "Rule"), the City has agreed, for the benefit of the holders and beneficial owners of the Series 2016 Bonds, to furnish annually certain financial information and operating data and to give notice of certain material events. See "CONTINUING DISCLOSURE AGREEMENT."

SECURITY AND SOURCES OF PAYMENT OF THE BONDS

Sources of Payment

Under the terms of the Trust Agreement, the Bonds are special obligations payable solely from and secured by a pledge of and lien on the Net Revenues and the Special Funds, subject to bankruptcy laws and other laws affecting creditors' rights and to the exercise of judicial discretion. See "CERTAIN INVESTMENT CONSIDERATIONS AND RISK FACTORS – Limited Obligations" and "CERTAIN INVESTMENT CONSIDERATIONS AND RISK FACTORS – Bankruptcy." The Net Revenues generally consist of the Revenues derived from the ownership, operation, use and services of the System less the Operating Expenses thereof. The Special Funds consist of the Construction Fund, the Debt Service Fund, the Debt Service Reserve Fund (to the extent that a series of Bonds is secured by the Debt Service Reserve Fund), the Subordinated Debt Service Fund, the Rate Stabilization Fund, the Contingency Fund, and the Balance Subfund. See "APPENDIX B – DEFINITIONS AND SUMMARY OF CERTAIN TERMS AND PROVISIONS OF THE TRUST AGREEMENT – Pledge of and Lien on Net Revenues and Special Funds; Funds and Subfunds." Any pledge of or lien on any revenues, money or other intangible property not in the custody of the Trustee will be valid and enforceable only to the extent permitted by law.

The Series 2016 Bonds will be special obligations of the City and will not represent or constitute a general obligation, bonded indebtedness or a pledge of the general credit or taxing power of the City, the State of Ohio or any political subdivision thereof. The holders of the Series 2016 Bonds will not have the right to have excises or taxes levied by the City, the State of Ohio or the taxing authority of any political subdivision thereof for the payment of debt service charges. See "CERTAIN INVESTMENT CONSIDERATIONS AND RISK FACTORS – Limited Obligations."

Rate Covenant

The City covenants in the Trust Agreement, subject to all applicable requirements and restrictions imposed by law, that at all times it will prescribe, charge and collect such rates, rentals and other charges for the use of the services of the System and shall restrict the Operating Expenses of the System in such a way that Net Revenues Available for Debt Service (adjusted for purposes of clause (i) as described in the following sentence) in each Fiscal Year shall be not less than the greater of (i) 120% of Debt Service on all Bonds then Outstanding payable during that Fiscal Year or (ii) 100% of the sum of: (A) Debt Service on all Bonds then Outstanding payable during that Fiscal Year, (B) all required deposits to the Debt Service Reserve Fund during that Fiscal Year, (C) all required deposits to the Subordinated Debt Service Fund and to any reserve fund securing Subordinated Indebtedness during that Fiscal Year, (D) all required deposits to the Contingency Fund during that Fiscal Year, and (E) debt service payable on any other obligations payable from the Balance Subfund and deposits to any reserve funds securing such other obligations during that Fiscal Year. For purposes of clause (i) of the preceding sentence, but not for the purpose of clause (ii) of the preceding sentence, "Net Revenues Available for Debt Service" for any Fiscal Year shall include any amounts transferred from the Rate Stabilization Fund to the Revenue Fund during such Fiscal Year, but shall be reduced by any amounts transferred from the Revenue Fund to the Rate Stabilization Fund during that period. For purposes of determining the City's compliance with the rate covenant set forth

herein, the amount of any Direct Payments received by the City or the Trustee in any Fiscal Year shall be credited against (i) Debt Service in such Fiscal Year if such Direct Payment related to Bonds or (ii) debt service on Subordinated Indebtedness in such Fiscal Year if such Direct Payment related to Subordinated Indebtedness. The City currently is not receiving any Direct Payment.

If in any Fiscal Year the Net Revenues Available for Debt Service shall be less than the amount required under the Trust Agreement, the City shall, within thirty (30) days following receipt by the City of its audited annual financial statement and the written request of the Trustee, employ a Consulting Engineer to make recommendations within forty-five (45) days as to a revision of the rates, fees and charges or Operating Expenses or methods of operation of the System, if any, that will result in producing the amount so required in the next succeeding Fiscal Year. The City covenants and agrees, promptly upon its receipt of such recommendations, to revise the rates, fees, and charges or Operating Expenses or methods of operation of the System and to take such other action as shall be in conformity with such recommendations to the extent the City feasibly may do so and to notify the Trustee of the actions it has taken.

Upon the taking by the City of such actions, its failure to comply with the foregoing requirements shall not constitute an Event of Default. See "CERTAIN INVESTMENT CONSIDERATIONS AND RISK FACTORS – Changes in Market Conditions; Conditions of the City; Ability to Charge Rates."

Debt Service Reserve Fund

The Trust Agreement establishes a Debt Service Reserve Fund and permits the City to establish a Debt Service Reserve Requirement for each Series of Bonds. The Trust Agreement requires that if on any Interest Payment Date, the amount in the subaccount in the Interest Subfund or the Principal Subfund for any Series of Bonds shall be less than the amount required to be therein on deposit, the Trustee shall, after deposits are made, if any, into the Subordinated Debt Service Fund, the Contingency Fund, the Rate Stabilization Fund, and the Balance Subfund, apply the amounts in the applicable Subfund, if any, in the Debt Service Reserve Fund to the extent necessary to make good any such deficiency (or the entire amount in the applicable Subfund if less than sufficient). The City may, so long as it will not result in a reduction of the rating assigned to any of the Bonds, substitute for funds on deposit in the Debt Service Reserve Fund a letter of credit, surety bond or other credit enhancement facility issued by a Qualified Financial Institution. The City has established a Debt Service Reserve Requirement for the Series 2016 Bonds. As a result, deposits will be made to the Debt Service Reserve Fund for the Series 2016 Bonds.

Additional Bonds

The City may issue Additional Bonds (i.e., additional Series of Bonds payable from Net Revenues and secured by the Trust Agreement on parity with the Series 2016 Bonds and any other Series of Bonds hereafter issued) for the purposes of (a) paying Capital Costs or (b) refunding, including advance refunding, for any lawful purpose, any Outstanding Bonds or Subordinated Indebtedness where such refunding will not adversely affect the exclusion of the interest on any Tax-Exempt Bonds from gross income for purposes of federal income taxation.

Additional Bonds shall be equally and ratably payable from Net Revenues and shall be secured by a pledge of and lien on the Net Revenues and the Special Funds on a parity with any other then Outstanding Bonds (subject to the provisions of the Trust Agreement providing for the application thereof).

For any Series of Additional Bonds that the City issues for the payment of Capital Costs or for the purpose of refunding Subordinated Indebtedness or other obligations other than Outstanding Bonds, the City must furnish the Trustee with either (A) a Certificate of the Director of Finance certifying that the Net Revenues Available for Debt Service for the preceding Fiscal Year or any consecutive twelve (12) calendar months during the eighteen (18)-month period immediately preceding the date of original issuance of the proposed Series of Bonds (in each case, the "Test Period") are in an amount not less than the greater of (1) 120% of maximum annual Debt Service on all Bonds then Outstanding and on the proposed Series of Additional Bonds or (2) 100% of the sum of: (A) maximum annual Debt Service on any Bonds then Outstanding and on the proposed Series of Bonds, (B) all required deposits to the Debt Service Reserve Fund, (C) all required deposits to the Subordinated Debt Service Fund and to any reserve fund securing Subordinated Indebtedness, (D) all required deposits to the Contingency Fund, and (E) debt service payable on any other obligations payable from the Balance Subfund and deposits to any reserve funds securing such other obligations, during that Test Period (the amounts described in clauses (A) through (E) being referred to collectively as the "Test Amount"), or (B) a Certificate of the Director of Finance, which may be, but is not required to be, accompanied by or based upon a Certificate of a Consulting Engineer, certifying that the Adjusted Net Revenues for the Test Period are in an amount not less than the greater of (1) 120% of maximum annual Debt Service on any Bonds then Outstanding and on the proposed Series of Bonds or (2) 100% of the Test Amount during that Test Period.

No certification shall be required for any Series of Bonds being issued for the purpose of refunding Outstanding Bonds, provided that the issuance of the refunding Bonds will not result in any increase in Debt Service in any Fiscal Year; otherwise, the same certification shall be required as for a Series of Bonds for the payment of Capital Costs.

For purposes of determining the City's compliance with the Additional Bonds test set forth herein, the amount of any Direct Payment received or expected to be received by the City or the Trustee in any Fiscal Year shall be credited against (i) Debt Service in such Fiscal Year if such Direct Payment relates to Bonds or (ii) debt service on Subordinated Indebtedness in such Fiscal Year if such Direct Payment relates to Subordinated Indebtedness. See "CERTAIN INVESTMENT CONSIDERATIONS AND RISK FACTORS – Additional Bonds." The City currently is not receiving any Direct Payment.

Subordinated Indebtedness

The City may, at any time and from time to time, issue or incur Subordinated Indebtedness for the purpose of financing any Capital Costs, provided that such Subordinated Indebtedness is secured solely by, and is payable solely from, moneys on deposit in the Subordinated Debt Service Fund, or the Balance Subfund, or both. Any Supplemental Trust Agreement providing for the issuance of Subordinated Indebtedness shall provide that the trustee for such Subordinated Indebtedness shall have no right, power or authority to cause acceleration

of such Subordinated Indebtedness unless and until the Trustee has exercised its power to accelerate the Bonds. See "CERTAIN INVESTMENT CONSIDERATIONS AND RISK FACTORS – Subordinated Indebtedness."

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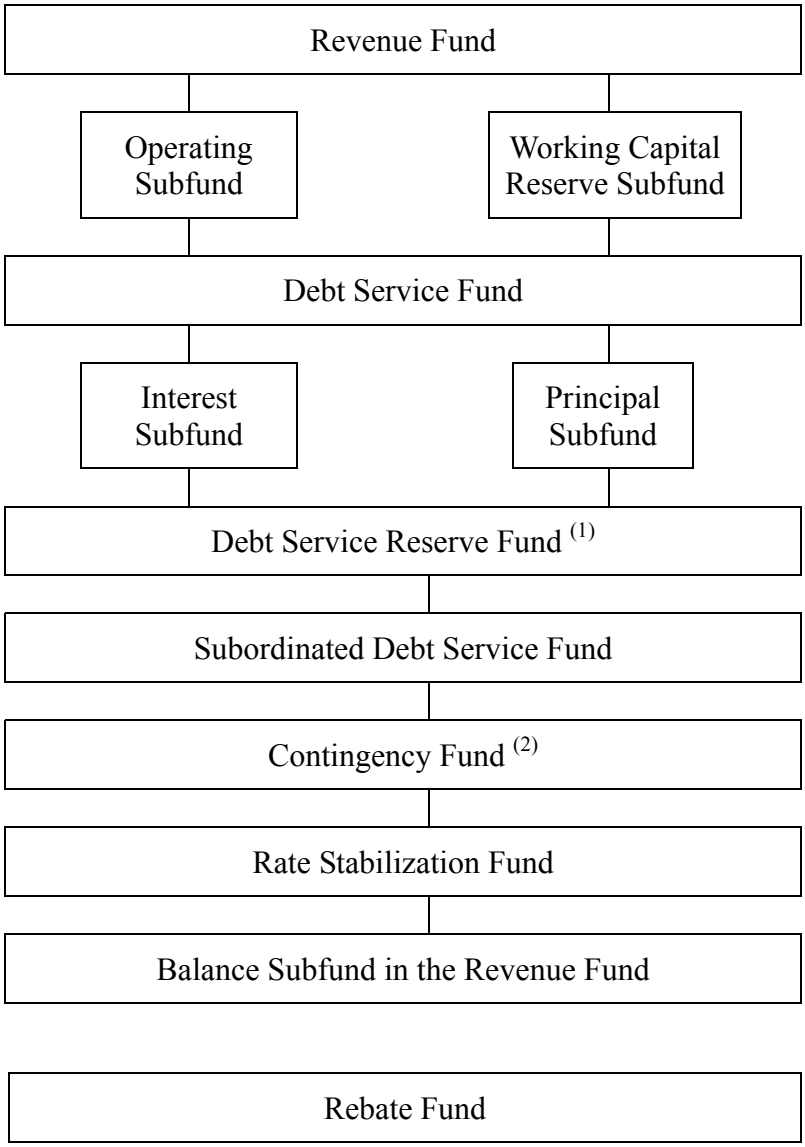
APPLICATION OF REVENUES

The Trust Agreement establishes the following application of Revenues:



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(1) Debt Service Reserve Requirement: as of any date of calculation with respect to any Series of Bonds, as specified in the Supplemental Trust Agreement applicable thereto, but subject to the requirements of the Master Trust Agreement.

(2) Contingency Fund Requirement: $1,000,000 or such higher amount as the Director of Public Utilities may from time to time specify (in writing delivered to the Trustee). Subsequent to any such specification, the Director of Public Utilities may reduce the Contingency Fund Requirement from the specified amount, but not below $1,000,000. Upon the issuance of the Series 2016 Bonds, the City will have a balance of $1,000,000 in the Contingency Fund funded from cash on hand.

Revenue Fund. So long as any of the Bonds remain outstanding, all Revenues of the System are required to be deposited in the Revenue Fund, which, pursuant to the Trust Agreement, will be held by the City as a trust fund separate and distinct from all other funds and moneys of the City (except for purposes of investment). In each month, the amounts in the Revenue Fund shall be deposited in the Operating Subfund and first applied to the payment of Operating Expenses. Thereafter, the balance in the Revenue Fund up to the Working Capital Reserve shall be deposited in the Working Capital Reserve Subfund and retained as a reserve for the payment of Operating Expenses and used for that purpose if the Operating Subfund shall have been fully expended. Thereafter, the balance in the Revenue Fund shall be paid at the times and in the amounts as follows.

Debt Service Fund. To the Debt Service Fund:

(i) To each subaccount in the Interest Subfund in the Debt Service Fund, monthly and on or before the last day of each month (or such other date as may be provided in any Supplemental Trust Agreement), after giving effect to amounts therein on deposit, an amount such that, if the same amount were paid in each of the months preceding the next Interest Payment Date, the aggregate of the amounts so paid would be sufficient to pay the interest due and payable on that next Interest Payment Date on all Outstanding Bonds of the Series to which such subaccount relates; and

(ii) To each applicable subaccount in the Principal Subfund in the Debt Service Fund, monthly and on or before the last day of each of the twelve months that precede a month in which a Serial Maturity Date occurs or a Sinking Fund Installment is due (or such other date as may be provided in any Supplemental Trust Agreement), after giving effect to amounts therein on deposit, an amount such that, if the same amount were so paid in each of the months preceding the next Serial Maturity Date or the next Sinking Fund Installment, the aggregate of the amounts so paid would be sufficient to pay the principal due on that next Serial Maturity Date or that next Sinking Fund Installment on all Outstanding Bonds of the Series to which such subaccount relates.

Debt Service Reserve Fund. To the subaccount established in the Debt Service Reserve Fund, monthly and on or before the last day of each month (or such other date as may be provided in any Supplemental Trust Agreement), commencing on the date set forth in the applicable Supplemental Bond Ordinance with respect to any Series of Bonds for which the Debt Service Reserve Requirement is to be deposited in the Debt Service Reserve Fund, after giving effect to any amounts already on deposit therein and pledged to such Series of Additional Bonds, an amount such that, if that same amount were deposited in equal monthly installments, the aggregate of the amounts so deposited in such subaccount would be equal to the Debt Service Reserve Requirement with respect to such Series of Additional Bonds on the fifth (5th) anniversary of such commencement date.

Thereafter, if the amount on deposit in such subaccount at any time is less than the Debt Service Reserve Requirement with respect to such Series of Additional Bonds, the City shall pay to the Trustee for deposit in such subaccount, on or before the last day of each month following the date of such deficiency, an amount equal to one-twelfth of such deficiency. The frequency and amount of any deposits required shall be subject to adjustment pursuant to the Trust

Agreement so that the deposits required into the Debt Service Reserve Fund for any Series of Bonds shall be no less frequent than the deposits required into the Debt Service Fund for the principal of and interest on such Bonds.

Subordinated Debt Service Fund. To any custodians of the Subordinated Debt Service Fund for deposit therein, such amounts as are required to be deposited therein on such dates as are specified in the Supplemental Trust Agreement or other contracts between the City and the holder or holders of the Subordinated Indebtedness. If any Supplemental Trust Agreement or other contract between the City and the holder or holders of Subordinated Indebtedness requires the establishment and funding a reserve account related to such Subordinated Indebtedness, then any required deposits to such reserve account may be made after any deposits otherwise required into the Subordinated Debt Service Fund but before any deposits required into the Contingency Fund and the Rate Stabilization Fund.

Contingency Fund. To the Contingency Fund, monthly on or before the last day of each month on which the balance in the Contingency Fund is less than the Contingency Fund Requirement, not less than one-twelfth (1/12) of the amount by which the Contingency Fund Requirement exceeds the balance. "Contingency Fund Requirement" means $1,000,000 or such higher amount as the Director of Public Utilities may from time to time specify (in writing delivered to the Trustee) as the Contingency Fund Requirement. Subsequent to any such specification, the Director of Public Utilities may reduce the Contingency Fund Requirement from the specified amount, but not below $1,000,000. Upon the issuance of the Series 2016 Bonds, the City will have a balance of $1,000,000 in the Contingency Fund funded from cash on hand.

Rate Stabilization Fund. To the Rate Stabilization Fund, on any date on which deposits are required into the Debt Service Fund, the Debt Service Reserve Fund, the Subordinated Debt Service Fund, and the Contingency Fund, and after all such deposits are made, any amounts that the City elects to make to the Rate Stabilization Fund.

Balance Subfund. To the credit of the Balance Subfund, on the first day of each Fiscal Year or on the first practicable date thereafter, the City shall deposit any balance remaining in the Operating Subfund and the Working Capital Reserve Subfund in excess of the Working Capital Reserve at the end of the preceding Fiscal Year; provided that the City may at any time provide for semiannual or quarterly deposits of such Net Revenues to the Balance Subfund.

Direct Payments. Upon receipt of any Direct Payment, the City or the Trustee shall cause it to be deposited (i) in the appropriate subaccount in the Debt Service Fund if such Direct Payment relates to Bonds, and (ii) in the appropriate account in the Subordinated Debt Service Fund if such Direct Payment relates to Subordinated Indebtedness, and shall cause it to be applied solely to the purposes to which the Trust Agreement permits funds in such subaccount and account to be applied. The City currently is not receiving any Direct Payment.

Rebate Fund. The Trust Agreement establishes the Rebate Fund as a separate deposit account (except when invested as therein provided) in the custody of the Trustee and requires separate accounts to be created in the Rebate Fund for each Series of Bonds that are Tax-Exempt Bonds, commencing with the Series 2016 Bonds. The Trust Agreement requires that, at least five

days prior to the end of the fifth (5th) Bond Year and each fifth (5th) Bond Year thereafter for each Series of Bonds, including the Series 2016 Bonds, and within five (5) days after the date on which all outstanding Bonds of such Series are paid or deemed paid in full, the City shall calculate, or shall furnish information to and engage an independent certified public accounting firm to calculate, the Rebate Amount as of the end of that Bond Year or the date of such final payment. Within thirty (30) days after the end of the fifth (5th) Bond Year and every fifth (5th) Bond Year thereafter, the Trustee, acting on behalf of the City and at the written direction of the City, shall pay to the United States in accordance with Section 148(f) of the Code from the moneys then on deposit in the applicable account in the Rebate Fund an amount equal to 90% (or such greater percentage not in excess of 100% as the City may direct the Trustee in writing to pay) of the Rebate Amount determined from the date of the original delivery of the Bonds of that Series (less the portion of the Rebate Amount, if any, previously paid to the United States). Within sixty (60) days after the payment in full of all outstanding Bonds of such Series and upon written direction of the City, the Trustee shall pay to the United States in accordance with Section 148(f) of the Code from the moneys then on deposit in the applicable account in the Rebate Fund an amount equal to 100% of the Rebate Amount determined from the date of the original delivery of the Bonds of that Series to the date of such payment (but less that portion of the Rebate Amount, if any, previously paid to the United States) and any moneys remaining in the applicable account in the Rebate Fund following such payment shall be paid to the City. Notwithstanding anything to the contrary, the moneys needed for any payment required as set forth herein may be derived from balances in any of the Special Funds other than the Debt Service Fund and the Debt Service Reserve Fund, and the City agrees to apply those balances to that purpose in the following order: the Balance Subfund, the Rate Stabilization Fund, the Contingency Fund, the Subordinated Indebtedness Fund and any related reserve fund or account, and the Construction Fund.

If all the gross proceeds of the Bonds of any Series, within the meaning of Section 148(f) of the Code, are expended for the governmental purpose for which that Series of Bonds was issued within six (6) months of the date of issuance of that Series and it is not expected that any other gross proceeds of that Series of Bonds will arise in the future, then the requirements set forth herein apply only to gross proceeds of that Series of Bonds, if any, that actually do subsequently arise.

Uses of Funds. No payment needs to be made into a Fund if the moneys in such Fund are equal to the required Fund balance, if any. Amounts in the various Funds discussed above are required and permitted to be used as follows:

(i) The Debt Service Fund shall be used only for the payment of Debt Service Charges on Bonds and, to the extent provided in the Trust Agreement, for the purchase for cancellation or redemption of Bonds.

(ii) The Debt Service Reserve Fund shall be used only to make up any deficiency in the Debt Service Fund with respect to Debt Service Charges on Bonds that are secured thereby, subject to the requirement that amounts in the Balance Subfund, the Rate Stabilization Fund, the Contingency Fund, and the Subordinated Debt Service Fund be applied to make up such deficiency prior to drawing from the Debt Service Reserve Fund.

(iii) The Subordinated Debt Service Fund shall be used to make up any deficiency in the Subfunds in the Debt Service Fund or any deficiency in the Debt Service Reserve Fund, to the extent necessary, subject to the requirement that amounts in the Balance Subfund, the Rate Stabilization Fund, and the Contingency Fund be applied to make up such deficiency prior to drawing from the Subordinated Debt Service Fund. The remaining amounts on deposit in the Subordinated Debt Service Fund shall be applied to the payment of the principal of and interest and any premium on Subordinated Indebtedness as and when due or to the required funding of any special reserve account established for any Subordinated Indebtedness.

(iv) The Contingency Fund shall be used to make up any deficiency in the Subfunds in the Debt Service Fund or any deficiency in the Debt Service Reserve Fund, to the extent necessary, after first drawing on the Balance Subfund and the Rate Stabilization Fund, and otherwise may be used by the City for the payment of Capital Costs or for the purpose of redeeming Bonds.

(v) The Rate Stabilization Fund shall be used to make up any deficiency in the Subfunds in the Debt Service Fund or any deficiency in the Debt Service Reserve Fund, after first drawing on the Balance Subfund. In addition, the City may from time to time transfer amounts on deposit in the Rate Stabilization Fund to the Revenue Fund and cause them to be applied as Revenues.

(vi) The Balance Subfund shall be used to make up any deficiency in the Subfunds in the Debt Service Fund or any deficiency in the Debt Service Reserve Fund, and may be used for any lawful System purpose, including, but not limited to, transfers to the Subordinated Debt Service Fund for payment of the principal of and interest on any Subordinated Indebtedness and the funding or replenishment of any special reserve account established for any Subordinated Indebtedness, but amounts on deposit in the Balance Subfund may not be transferred or credited to the Rate Stabilization Fund.

(vii) The Rebate Fund is not pledged to the payment of Bond Service Charges, is not one of or part of any of the Special Funds, and is free and clear of any lien thereunder. At the written request of the City, any moneys held as part of the Rebate Fund shall be invested or reinvested by the Trustee in Eligible Investments.

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SOURCES AND USES OF FUNDS

The table below shows the estimated sources and uses of funds from the Series 2016 Bonds and other funds of the City.

Sources of Funds:	
Bond Proceeds:	
Principal Amount	$32,390,000.00
Original Issue Premium	5,384,899.90
Other Sources of Funds:	
Cash on Hand	1,000,000.00
Total Sources	**$38,774,899.90**
Uses of Funds:	
Deposit to Project Fund:	$35,000,000.00
Deposit to Debt Service Reserve Fund:	2,136,000.00
Deposit to Contingency Fund:	1,000,000.00
Costs of Issuance: [1]	381,607.04
Underwriting Discount:	257,292.86
Total Uses	**$38,774,899.90**

(1) Costs of issuance of the Series 2016 Bonds include costs of legal, printing, advisory and rating agency fees, financial and other miscellaneous fees and expenses.

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DESCRIPTION OF THE SERIES 2016 BONDS

Authorization

The Series 2016 Bonds are being issued pursuant to the Constitution and laws of Ohio, the Charter of the City (the "Charter"), the Series 2016 Bond Legislation, and the Trust Agreement.

General Description

The Series 2016 Bonds will be issued under the Trust Agreement in the aggregate principal amount set forth on the cover page hereof. The Series 2016 Bonds will be dated their respective dates of delivery, will bear interest from that date as described herein, payable semi-annually on May 15 and November 15 of each year (each, an "Interest Payment Date"), commencing November 15, 2016 and will mature November 15 in the years and in the principal amounts set forth on the inside front cover page of this Official Statement. Certain of the Series 2016 Bonds are subject to redemption prior to maturity. See "REDEMPTION PROVISIONS."

The Series 2016 Bonds will be issued initially as fully registered bonds without coupons and in authorized denominations of $5,000 or any integral multiple thereof. The authorized denominations of the Series 2016 Bonds are hereinafter called "Authorized Denominations."

The Series 2016 Bonds will be registered in the name of Cede & Co., as nominee of DTC, and held in the custody of DTC pursuant to DTC's book entry only system. Purchasers of beneficial interests in the Series 2016 Bonds will be made in book-entry form, without certificates.

If at any time the book entry system is discontinued for the Series 2016 Bonds, the City will take the actions necessary to provide for the issuance of Bond certificates to the Holders of the Series 2016 Bonds. See "APPENDIX G – BOOK-ENTRY ONLY SYSTEM."

So long as the Series 2016 Bonds are held by DTC or its nominee, Cede & Co., in book-entry only form, the Trustee will recognize and treat DTC or its nominee, Cede & Co., as the Holder of the Series 2016 Bonds for all purposes under the Trust Agreement, provided that the Trustee will recognize Beneficial Owners for purposes of the purchase of Book Entry Interests. See "Appendix G – Book-Entry Only System." Each Beneficial Owner is responsible for observing the procedures of DTC, the Direct Participant, any Indirect Participant and the Trustee, as set forth in the Trust Agreement, in order to permit the timely observance of the tender process with respect to Book Entry Interests.

Payment of Principal and Interest

The principal of and premium, if any, on the Series 2016 Bonds will be payable to the registered owner (initially DTC or its nominee) upon their presentation and surrender at the designated office of the Paying Agent. The Series 2016 Bonds will bear interest on their unpaid principal amounts payable on each Interest Payment Date. Interest on the Series 2016 Bonds will be payable by check or draft mailed or delivered on the Interest Payment Date to the registered holder (initially DTC or its nominee) at the address shown on the Register as of the

close of business on the 15th day of the calendar month next preceding such Interest Payment Date; provided that, so long as the Series 2016 Bonds remain in book-entry form, DTC (or any successor depository company) may direct the Trustee to make such payment by wire transfer of immediately available funds on each Interest Payment Date.

Transfer and Exchange of the Series 2016 Bonds; Persons Treated as Holders

The person in whose name any Series 2016 Bond is registered will be deemed and regarded as the absolute owner thereof for all purposes, and payment of or on account of the principal or purchase price (which is an amount equal to the principal amount thereof plus, if such purchase date is not an Interest Payment Date therefor accrued and unpaid interest thereon to such purchase date) of and interest on the Series 2016 Bonds will be made only to or upon the order of the registered owner thereof or his or her legal representative. All such payments will be valid and effective to satisfy and discharge the liability upon that Series 2016 Bond to the extent of the sum or sums so paid.

So long as the Series 2016 Bonds are held in book-entry form, transfers of the Series 2016 Bonds by Beneficial Owners may be made only as described in "APPENDIX G – BOOK-ENTRY ONLY SYSTEM." At any other time, any Series 2016 Bonds may be transferred or exchanged only upon the books kept for the registration and transfer of Series 2016 Bonds as provided in the Trust Agreement.

Redemption Provisions

Optional Redemption

The Series 2016 Bonds maturing on or after November 15, 2026 are subject to redemption in whole or in part (in whole multiples of $5,000) on any date on or after November 15, 2025 in any order of maturity as directed by the City at a redemption price of 100% of the principal amount redeemed plus interest accrued to the redemption date.

Mandatory Sinking Fund Redemption – 2041 Term Bonds

The Series 2016 Bonds maturing on November 15, 2041 in the amount of $7,595,000 and bearing interest at 5.000% per year (the "2041 Term Bonds") are subject to mandatory sinking fund redemption prior to maturity at a redemption price equal to 100% of the principal amount thereof to be redeemed, plus interest, if any, accrued thereon to the date fixed for redemption, on each November 15 of the years and in the principal amounts set forth below:

Year	Principal Amount
2037	$1,375,000
2038	$1,445,000
2039	$1,515,000
2040	$1,590,000

If not previously redeemed, there will remain $1,670,000 principal amount of Series 2016 Bonds to be retired at stated maturity on November 15, 2041. 2041 Term Bonds redeemed by

optional redemption or purchased for cancellation may be credited against the mandatory sinking fund redemption requirements.

Mandatory Sinking Fund Redemption – 2045 Term Bonds

The Series 2016 Bonds maturing on November 15, 2045 in the amount of $7,565,000 and bearing interest at 5.000% per year (the "2045 Term Bonds") are subject to mandatory sinking fund redemption prior to maturity at a redemption price equal to 100% of the principal amount thereof to be redeemed, plus interest, if any, accrued thereon to the date fixed for redemption, on each November 15 of the years and in the principal amounts set forth below:

Year	Principal Amount
2042	$1,755,000
2043	$1,845,000
2044	$1,935,000

If not previously redeemed, there will remain $2,030,000 principal amount of Series 2016 Bonds to be retired at stated maturity on November 15, 2045. 2045 Term Bonds redeemed by optional redemption or purchased for cancellation may be credited against the mandatory sinking fund redemption requirements.

Notice of Redemption

If less than all of the Series 2016 Bonds that are stated to mature on different dates are called for redemption at one time, the maturities of the Series 2016 Bonds that are to be called will be determined by the City in its sole discretion. If less than all of the Series 2016 Bonds of a single maturity are to be redeemed, the selection of the Series 2016 Bonds to be redeemed, or portions thereof in amounts of $5,000 or any integral multiple thereof, so long as the Series 2016 Bonds remain in book-entry form, will be made by the Depository (or any successor Depository) in accordance with the Depository's procedures and otherwise will be made as specified by and selected at the sole discretion of the City. In the case of a partial redemption of the Series 2016 Bonds by lot when the Series 2016 Bonds of Authorized Denominations greater than $5,000 are then outstanding, each $5,000 unit of principal thereof shall be treated as though it were a separate Series 2016 Bond of the denomination of $5,000.

The Trustee shall cause notice of any redemption of Series 2016 Bonds to be (a) mailed to the Holders of all Series 2016 Bonds to be redeemed at the addresses appearing in the Register and (b) transmitted by Electronic Means to each Depository and to the Municipal Securities Rulemaking Board, by means of its Electronic Municipal Market Access ("EMMA") system. Each such notice shall be sent not more than 45 nor fewer than 20 calendar days prior to the redemption date. Selection of book-entry interests in the Series 2016 Bonds called, and notice of the call to the owners of those interests called, is the responsibility of the Depository pursuant to its Rules and Procedures and of its Participants and Indirect Participants. Any failure of the Depository to advise any Participant, or of any Participant or any Indirect Participant to notify the book-entry interest owners, of any such notice and its content or effect will not affect the validity of any proceedings for the redemption of any Series 2016 Bonds.

Each notice of redemption will (a) identify the Series 2016 Bonds to be redeemed (specifying the CUSIP numbers, if any, assigned to the Series 2016 Bonds), (b) specify the redemption date and the redemption price, (c) set forth the name, address and telephone number of the person from whom information pertaining to the redemption may be obtained, and (d) state that on the redemption date the Series 2016 Bonds called for redemption will be payable at the Designated Office of the Trustee, that from that date interest will cease to accrue, and that no representation is made as to the accuracy or correctness of the CUSIP numbers printed therein or on the Series 2016 Bonds. No defect affecting any Series 2016 Bond, whether in the notice of redemption or the delivery thereof (including any failure to mail such notice), shall affect the validity of the redemption proceedings for any other Series 2016 Bonds.

If at the time of mailing of notice of redemption of Series 2016 Bonds as described above, there shall not have been deposited with the Trustee moneys or federal securities sufficient to redeem all Series 2016 Bonds called for redemption, the notice of redemption is to state that the redemption is conditional upon the deposit of moneys or federal securities with the Trustee not later than the redemption date in an amount sufficient to pay the redemption price of all Series 2016 Bonds or portions of Series 2016 Bonds that are to be redeemed on the applicable redemption date, and the notice will be of no effect and the Series 2016 Bonds will not be redeemed unless such moneys or federal securities are so deposited.

Effect of Redemption

If moneys sufficient to pay the redemption price of Series 2016 Bonds to be redeemed have been deposited with the Trustee on or prior to the redemption date, then such Series 2016 Bonds (or the portion thereof to be redeemed) will become due and payable on the redemption date, and interest on such Series 2016 Bonds will cease to accrue from and after the date of redemption.

Purchase in Lieu of Redemption

By their acceptance of the Series 2016 Bonds, the Holders irrevocably grant to the City the option to purchase any Series 2016 Bond that is redeemable by optional redemption on any date on which the Series 2016 Bond is so redeemable at a purchase price no less than the redemption price to be paid to Holders upon optional redemption. The City may exercise such option by written request delivered to the Trustee within the time period specified in the Trust Agreement, as described above. In the case of the purchase of less than all of the Series 2016 Bonds, the particular Series 2016 Bonds to be purchased will be selected in accordance with the provisions of the Trust Agreement as though such purchase were a redemption or in such other manner as the City directs.

Revision of Book-Entry System; Replacement Bonds

The Trust Agreement provides for issuance of fully registered Series 2016 Bonds ("Replacement Bonds") directly to assignees of DTC or its nominee only in the event that the use of the book-entry system is discontinued for the Series 2016 Bonds. If the discontinuance of a securities depository book-entry system for the Series 2016 Bonds is not the result of City action

or inaction, the delivery of Replacement Bonds will be at the expense (including printing costs) of any persons requesting such Replacement Bonds.

In the event that the book-entry system is discontinued, Series 2016 Bonds will be transferable or exchangeable for fully registered Series 2016 Bonds of any denomination or denominations authorized by the Trust Agreement in the aggregate principal amount equal to the unmatured and unredeemed principal amount of, and bearing interest at the same rate and maturing on the same date as, the Series 2016 Bonds being transferred or exchanged. Replacement Bonds will be transferable or exchangeable at the Designated Office of the Registrar or any Authentication Agent, without charge (except any tax, fee, or other governmental charge required to be paid with respect to the exchange or transfer). See "APPENDIX G – BOOK-ENTRY ONLY SYSTEM."

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OUTSTANDING BONDS AND OTHER OBLIGATIONS

Outstanding Bonds

The Series 2016 Bonds are the first Series of Bonds issued under the Trust Agreement, and as a result, as of the time of the issuance of the Series 2016 Bonds, the Series 2016 Bonds will be the only Outstanding Bonds until the issuance of any Additional Bonds in the future.

Other Obligations

As of January 1, 2016, the Division had total debt outstanding of $753,120 (principal only) associated with four Ohio Water Development Authority ("OWDA") construction loans and two Ohio Public Works Commission ("OPWC") construction loans. These loans are payable from the Subordinated Debt Service Fund. Future OWDA or OPWC loans are not contemplated at this time.

The table below provides a summary of debt service on the existing outstanding OWDA and OPWC loans.

Period Ending December 31	Existing OPWC Loans	Existing OWDA Loans	Total
2016	$24,060	$467,930	$491,990
2017	$24,060	$155,135	$179,195
2018	$24,060	--	$24,060
2019	$24,060	--	$24,060
2020	$24,060	--	$24,060
2021	$11,201	--	$11,201
2022	$11,201	--	$11,201
Total	**$142,702**	**$623,065**	**$765,767**

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DEBT SERVICE SCHEDULE

The following schedule sets forth the annual Debt Service on the Series 2016 Bonds, which constitute all the Outstanding Bonds as of the time of issuance of the Series 2016 Bonds and before the issuance of any Additional Bonds in the future.

Calendar Year	Principal on the Series 2016 Bonds	Interest on the Series 2016 Bonds	Total Debt Service on the Series 2016 Bonds[1]
2016	--	$928,546.53	$928,546.53
2017	$550,000	$1,584,250.00	$2,134,250.00
2018	$570,000	$1,562,250.00	$2,132,250.00
2019	$590,000	$1,545,150.00	$2,135,150.00
2020	$610,000	$1,521,550.00	$2,131,550.00
2021	$635,000	$1,497,150.00	$2,132,150.00
2022	$660,000	$1,471,750.00	$2,131,750.00
2023	$695,000	$1,438,750.00	$2,133,750.00
2024	$730,000	$1,404,000.00	$2,134,000.00
2025	$765,000	$1,367,500.00	$2,132,500.00
2026	$805,000	$1,329,250.00	$2,134,250.00
2027	$845,000	$1,289,000.00	$2,134,000.00
2028	$885,000	$1,246,750.00	$2,131,750.00
2029	$930,000	$1,202,500.00	$2,132,500.00
2030	$980,000	$1,156,000.00	$2,136,000.00
2031	$1,025,000	$1,107,000.00	$2,132,000.00
2032	$1,080,000	$1,055,750.00	$2,135,750.00
2033	$1,130,000	$1,001,750.00	$2,131,750.00
2034	$1,190,000	$945,250.00	$2,135,250.00
2035	$1,245,000	$885,750.00	$2,130,750.00
2036	$1,310,000	$823,500.00	$2,133,500.00
2037	$1,375,000	$758,000.00	$2,133,000.00
2038	$1,445,000	$689,250.00	$2,134,250.00
2039	$1,515,000	$617,000.00	$2,132,000.00
2040	$1,590,000	$541,250.00	$2,131,250.00
2041	$1,670,000	$461,750.00	$2,131,750.00
2042	$1,755,000	$378,250.00	$2,133,250.00
2043	$1,845,000	$290,500.00	$2,135,500.00
2044	$1,935,000	$198,250.00	$2,133,250.00
2045	$2,030,000	$101,500.00	$2,131,500.00

[1] Does not include OWDA/OPWC loans, which are payable from the Subordinated Debt Service Fund.

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THE CITY OF CLEVELAND

Introduction

The City is a municipal corporation and political subdivision of the State of Ohio (the "State"). It is located on the southern shore of Lake Erie and is the county seat of Cuyahoga County in northeastern Ohio. The City was incorporated as a village in 1814 and became a city in 1836. For a more extensive discussion of the City, see "ECONOMIC AND DEMOGRAPHIC INFORMATION CONCERNING THE CITY OF CLEVELAND" in APPENDIX A.

City Government

The City operates under and is governed by the Charter, which was first adopted by the voters in 1913 and has been and may be further amended by the voters from time to time. The City is also subject to certain general State laws that are applicable to all cities in the State. In addition, under Article XVIII, Section 3 of the Ohio Constitution, the City may exercise all powers of local self-government and may exercise police powers to the extent not in conflict with applicable general State laws. The Charter provides for a mayor-council form of government.

Legislative authority is currently vested in a 17-member City Council ("Council"). The terms of Council members and the Mayor are four years. All Council members are elected from wards. The current terms of the Mayor and Council members expire on January 1, 2018. Council fixes compensation of City officials and employees and enacts ordinances and resolutions relating to City services, tax levies, appropriating and borrowing money, licensing and regulating businesses and trades, and other municipal functions. The presiding officer is the President of Council, who is elected by the Council members. Kevin J. Kelley was elected as President of Council for the term beginning January 6, 2014. The Clerk of Council is appointed by Council.

Pursuant to the City Charter, City Council is required to re-divide the City into wards based on the City's population. Currently, there are 17 wards in the City.

The City's chief executive and administrative officer is the Mayor who is elected by the voters for a four-year term. Frank G. Jackson was re-elected to his third term as Mayor of the City on November 5, 2013 and began his current term on January 6, 2014. Prior to assuming office as Mayor, Mr. Jackson served as President of Council from January 2002 through December 2005 and as Ward 5 Council Member since 1989. The Mayor may veto any legislation passed by Council. A veto may be overridden by a two-thirds vote of all members of Council.

The Charter establishes certain administrative departments, and Council may establish divisions thereof or additional departments. The Mayor appoints all of the directors of the City's 14 departments.

Financial Matters

The responsibilities for the City's major financial functions are vested in the Director of Finance. The Director of Finance is responsible for preparing and implementing the City's current operating budget and capital improvement plan, collecting the City's revenues, and procuring the City's goods and services and making payments therefor. The Director of Finance is also responsible for maintaining an effective system of internal accounting control which includes the maintenance of a centralized accounting system and the supervision of the City's internal audit staff. Preparation and issuance of the City's internal and external financial reports are supervised by the Director of Finance.

The Director of Finance for the City is Sharon A. Dumas. Ms. Dumas served as Assistant Director of Finance for Budget and Capital from October 13, 2003 until her appointment as Director of Finance on March 29, 2006. Prior to joining the City of Cleveland Finance Department, Ms. Dumas worked as Assistant Director of Community Development for the City. Ms. Dumas also served as Finance Director for the City of East Cleveland, Ohio from 1988 to 1994. Ms. Dumas holds a Masters of Accounting and Financial Information Systems from Cleveland State University and has over 30 years of experience in private and public sector accounting.

James T. Hartley was named an Assistant Finance Director in April 2007 and, in addition, has served as acting City Treasurer and now City Treasurer since April 2008. Prior to accepting his current positions, Mr. Hartley was the Chief Investment Officer for the Ohio Treasurer of State from 1999 until 2007. As a member of the Treasurer's senior staff, he was responsible for overseeing and directing the State's investment program, including the State Operating Fund, STAR Ohio Local Government Investment Pool, and the State Tobacco Settlement Funds.

James E. Gentile, CPA, returned to the City as Controller in February 2002. Prior to accepting the position, Mr. Gentile was Deputy Auditor for the Auditor of State's Office since 1995 where he planned and supervised audits of cities, school districts and other local government agencies. From 1991 through 1995, he was employed by the City as an accountant and, in his final year, as Acting City Controller.

Elizabeth C. Hruby has served as the City's debt manager since 1996. She is the Assistant Secretary to the Sinking Fund Commission. She has been employed by the City since 1982 when she began as a Budget Analyst in the Office of Budget and Management. She was promoted to Operating Budget Manager in 1987 and was responsible for the development and monitoring of the City's annual operating budget. From January 7, 2002 until April 1, 2002, Ms. Hruby served as interim Director of Finance for the City.

THE DIVISION OF WATER POLLUTION CONTROL

Overview and History

The Division is responsible for managing the sanitary sewage and stormwater collection system and for overseeing matters relating to the elimination, control and regulation of water pollution within City limits. The City derives its authority to own and operate a water pollution control utility from the municipal utility provisions in Article XVIII, Sections 4, 5, 6 and 12 of the Ohio Constitution and Section I of the Charter. The City is empowered by such provisions to establish rates and charges for the services provided by the Division, acquire property, and construct facilities to provide sewage and stormwater collection services throughout its Service Area and perform other necessary functions to operate and maintain the Division.

The Division was created for the purpose of supplying sewer services to customers within the Cleveland metropolitan area. Embarking with a rudimentary system in the late 1800's, the Division developed as the City itself expanded. Until the early 1970's, the City operated the entire system and managed all aspects of sewage treatment and disposal. In 1972, a court order (the "1972 Court Order") created the Northeast Ohio Regional Sewer District ("NEORSD") and transferred the operation of all wastewater treatment plants and interceptors to the NEORSD in December 1973. The City retained responsibility for the sewer collector system in the City. The Division services a significant portion of the entire Cleveland metropolitan area by managing the sanitary sewage and storm water drainage collection system. The sewer collection system transfers sanitary and storm sewage from its point of origin to an interceptor sewer or treatment plant for processing. The System is comprised of approximately 1,445 miles of sewer lines with attendant catch basins and includes 12 pump/lift stations (not including the 3 pump stations that will be transferred to NEORSD by May 15, 2016). The Division is also responsible for the cleaning of 43,500 catch basins.

Service Area

The Service Area of the Division includes all residential, commercial, governmental, and non-profit customers located within City limits. The Service Area also includes customers that are physically located outside City limits but whose water meters are located within City limits.

Sewer Collection

The Division has responsibility over the sewer collection system within the City. The Division is responsible for cleaning and maintaining the network of sewers within the City which carries sanitary sewage and industrial waste (which may require pre-treatment) from their points of origin to sewage processing facilities for treatment and disposal by NEORSD. The Division maintains, cleans, repairs and improves sewers and their appendages. Protecting the health and environment of City residents and visitors is vitally important to the Division.

The System includes three types of sewers. The most prevalent type is sewer pipes that collect and transfer combined sanitary and stormwater flows. Approximately ninety-two percent (92%) of the System is served by combined sewers. The System also includes separate sanitary and storm sewers, as well as a third category of segmented block pipe wherein separate sanitary and storm sewers are encased within the same segmented block pipe.

Stormwater Management

In addition to operating and maintaining a sewer collection system, the Division is the stormwater manager for the City. The Commissioner of Water Pollution Control serves as the City's stormwater manager. A steering committee consisting of representatives from the Division, the Cuyahoga Soil and Water Conservation District, the Department of Public Works, the Office of Sustainability, the Department of Law, the Department of Building and Housing, the Cleveland City Planning Commission, the Division of Waste Collection and Disposal, the Office of Capital Projects, and the City Council support the management of the stormwater system and compliance with the City's Small Municipal Separate Storm Sewer System ("MS4") permit.

Non-point stormwater runoff is rainwater and snowmelt that flows over solid, impervious surfaces such as streets, sidewalks and parking lots. Hard, compact soil also can prevent stormwater from infiltrating the ground, which can lead to street flooding and erosion. Stormwater runoff can carry debris, chemicals, sediment or other pollutants on the ground into sewer catch basins and can flow into local waterways untreated. This can cause contamination of drinking water and compromise recreation areas along Lake Erie, which is Northeast Ohio's main source of drinking water.

The Division's responsibilities include overall management and reporting on stormwater compliance including compliance with the Six Minimum Control Measures specified in the MS4 permit. Of these six measures, the Division is responsible for the first three: Public Education and Outreach, Public Involvement and Participation, and Illicit Connection Detection and Elimination. The City's Department of Building and Housing is responsible for the fourth and fifth minimum control measures: Construction Site Stormwater Runoff Control, and Post Construction Stormwater Runoff Control. The City's Department of Public Works and other affected departments are responsible for the sixth minimum control measure: Pollution Prevention and Good Housekeeping.

The City complies with Environmental Protection Agency ("EPA") requirements to maintain its stormwater permit. The City's Public Involvement and Public Education ("PIPE") program focuses on educating the entire community, but pays particular attention to its separate sewer areas, which have separate sewer pipes for stormwater and wastewater. In those areas, also called MS4 areas, runoff is discharged untreated into local waterways. Those discharges can cause water pollution and flooding.

Public education is an important component of the Division's authority as the City's stormwater manager. Annually, the Division participates in numerous community events to inform residents and others about how their personal actions related to stormwater runoff can impact water quality and impede the free-flowing capacity of the System.

The Division promotes the use of low-impact best management practices ("BMP") to help prevent flooding, protect stream channels and improve water quality. Examples of BMPs suitable for residential and commercial properties are green roofs covered with grass and plants to absorb rain water, downspouts connected to rain barrels to collect rain water for re-use, rain

gardens with native plants that absorb large amounts of water, and pervious pavements that absorb and filter rain water.

Organizational Structure

The Department of Public Utilities is composed of five divisions, including the Division, the Division of Water, Cleveland Public Power, the Division of Fiscal Control, and the Office of Utilities General Administration. Each division is revenue generating except for the Division of Fiscal Control, which provides general administration, accounting, cash management and other financial services to the other four divisions.

The Division is administered and controlled by the Commissioner of Water Pollution Control, who manages all of the personnel, procurement, budget and overall operations of the Division, including the System. The Commissioner of Water Pollution Control reports to the Director of Public Utilities who reports to the Mayor and City Administration. The Deputy Commissioner of Water Pollution Control assists the Commissioner of Water Pollution Control in managing the Division.

Key Management Personnel

Director of Public Utilities – Robert L. Davis, as Director of the City of Cleveland Department of Public Utilities, is responsible for management and oversight of more than 1700 employees, a budget of approximately $570 million, and all operations within the Division, the Division of Water, Cleveland Public Power, the Division of Fiscal Control, the Office of Utilities General Administration, as well as Cleveland's public access channel, TV20. Before taking the position of Director in May 2015, Mr. Davis served the City of Warren, Ohio as the Director of Utility Services. In this role, Mr. Davis was responsible for the operation and maintenance of Water, Sewer, Sanitation, and Water Pollution Control. Additionally, Mr. Davis oversaw Distribution Collections systems and Customer Service and Billing for each of these divisions. Mr. Davis has a strong foundation of both administrative and hands on experience having risen through the ranks throughout his 28-year career as a utility professional. Mr. Davis holds a Class IV Water License and has been a member of the American Water Works Association ("AWWA") for more than 25 years. Mr. Davis is currently the Past Chair of the Ohio Section of AWWA in addition to having served 5 years on the Ohio Governing Board. Mr. Davis has also served as Chair of the Membership Committee for the Ohio Section and works very closely with the Young Professionals, Diversity, Strategic Planning, and Finance and Audit committees. In 2015, Mr. Davis was inducted into Warren High School's Distinguished Alumni Hall of Fame.

Chief Financial Officer of Public Utilities - Frank M. Badalamenti has been serving as the Chief Financial Officer for the Department of Public Utilities since May 2015. Prior to that, Mr. Badalamenti was an Assistant Finance Director for the City from July 2014 to April 2015. Previously Mr. Badalamenti was Chief Financial Officer for the Department of Public Utilities from May 2012 to June 2014. Mr. Badalamenti was the Manager of Internal Audit for the City from October 1999 to April 2012. Prior to working for the City, Mr. Badalamenti served in the United States Air Force for 20 years in various financial operational capacities.

Commissioner of Water Pollution Control – Rachid Zoghaib was named Commissioner of Water Pollution Control in April 2011. Prior to that, Mr. Zoghaib served as the Deputy Commissioner of Water Pollution Control from 2004 to 2011 and served as the Director of Engineering of Water Pollution Control from 1994 to 2004. Mr. Zoghaib has more than 25 years of wastewater, environmental and public management experience. As Commissioner, Mr. Zoghaib's responsibilities include managing and directing all administrative, operational and financial activities and programs of the Division. Mr. Zoghaib earned a Bachelor of Science Degree in Civil Engineering and a Master of Science Degree in Civil Engineering from the Youngstown State University and a Leadership Certificate from Cleveland Management Academy at Cleveland State University. Mr. Zoghaib is also a Licensed Professional Engineer in the State of Ohio.

Deputy Commissioner of Water Pollution Control – Ramona Lowery is responsible for managing the sanitary sewage and stormwater collection system in the City. Ms. Lowery also oversees the implementation of pollution prevention programs through Public Involvement and Public Education ("PIPE"). Ms. Lowery provides executive supervision in planning and managing the Division. Ms. Lowery also assists the Commissioner of Water Pollution Control in managing daily operations and coordinating reliable sewer maintenance operations. Ms. Lowery has 20 years of experience working in the wastewater field. Prior to her employment with the Division, Ms. Lowery worked for NEORSD for 7 years as Manager of Operations and Maintenance. Ms. Lowery has a Master of Business Administration and Master of Public Administration from the University of Phoenix, and a Bachelor of Science in Civil Engineering from Cleveland State University. Ms. Lowery holds a Class II Wastewater Collection System License and a Class II Water Distribution System License. Ms. Lowery is also affiliated with several professional organizations such as the Water Environment Federation, the Cleveland Engineering Society, and the National Society of Black Engineers.

Employees and Employee Relations

The Division's personnel budget for Fiscal Year 2016 includes 162 full-time and 4 part-time positions. Of this amount, 125 full-time and all 4 part-time positions are currently filled. The City recently reorganized and moved 16 financial, public affairs, geographic information system, information technology, and human resource positions that were within the Division to other divisions within the Department of Public Utilities to consolidate responsibilities and enhance efficiency and coordination. Although these positions were moved out of the Division personnel budget, these staff continue to support the Division as needed. In addition, the staffing level for the Division's personnel budget for Fiscal Year 2016 remains the same as the Division's personnel budget for Fiscal Year 2015 at 166. The Division is in the process of increasing operation, maintenance and engineering staff to meet future needs.

Each of the full-time and part-time employees of the Division is an employee of the City. Approximately eighty-nine percent (89%) of the Division's employees belong to one of several collective bargaining units represented within the City as listed below. Currently fourteen employees, primarily upper-level professionals and administrators, do not belong to unions. The contracts with all the collective bargaining units listed below are effective from April 1, 2013 and through March 31, 2016. The City is in the process of negotiating new three-year contracts to

replace the expiring contracts. At this time, the Division does not anticipate that the new contracts will have any material impact on the Division's operations or financial projections.

- Local 100 – Ohio American Federation of State, County and Municipal Employees ("AFSCME"), Ohio Council 8, AFL-CIO
- Local 10 – International Union of Operating Engineers
- Local 439 – Machinist District 54 Union
- Cleveland Building & Construction Trades Council
- Local 1099 – Municipal Foreman and Laborer's Union
- SEIU – Service Employees International Union
- SEME – Federal Credit Union (formerly known as "SEME")
- CEO2 – Construction Equipment Operator

There have been no significant labor disputes or work stoppages in the City within the last 25 years.

Interactions with NEORSD; Key Permits

Prior to 1973, the City was responsible for both wastewater collections and wastewater treatment. Beginning in December 1973, the ownership and operation of all of the wastewater treatment plants and interceptors were transferred to NEORSD by the 1972 Court Order.

The Division's discharge of flows to the NEORSD interceptor system for transfer and treatment at one of NEORSD's wastewater treatment plants is governed by the City's Community Discharge Permit for sanitary sewers and the City's Community Combined Discharge Permit (collectively, the "Discharge Permits") for combined stormwater and sewers. Neither Discharge Permit has an expiration date; however, the Discharge Permits may be modified from time to time as necessary to reflect the changing needs and circumstances of NEORSD and the Division. The Division is required to submit a Compliance Report Checklist to NEORSD on an annual basis to demonstrate the Division's compliance with the Discharge Permits and NEORSD's BMP guidelines.

The City consistently maintains compliance with the Discharge Permits' requirements which require implementation of a number of BMPs. Under its Community Discharge Permit for sanitary sewers, which represents less than 8% of the collection by the System, a peak flow limitation of 14.284 million gallons per day is imposed at the Grayton Road connector. Under its Community Combined Discharge Permit for combined sewers, no peak flow limits or restrictions are currently imposed. However, NEORSD has the right to impose future limitations if required. No limitations, however, are anticipated to be imposed in the foreseeable future.

Discharge of separated storm water for the Division is regulated by the City's MS4 permit. A new MS4 permit was issued on September 11, 2014. Only a small portion of the System consists of separated stormwater sewers. The majority of the System is combined storm and sewer service that is regulated under the City's Discharge Permits.

Customer Base

The Division provides services to approximately 126,000 active metered water connections that represent essentially 100% of the population of the City including all residential, commercial, industrial, governmental, and non-profit customers. The table below illustrates the current number of active water metered accounts broken down by meter size (and homestead rate accounts whose charges are not dependent upon meter size). The largest percentage of Division customers (96%) are residential (who typically are served by 5/8 inch water meters), while the remaining 4% of the customers are primarily commercial, governmental, and industrial accounts.

Number of Accounts by Water Meter Size (Including Homestead Rate Accounts)

Water Meter Size (Inch) (or Homestead Rate)	Number of Accounts
1/2	1
5/8	107,807
Homestead	10,993
3/4	420
1	2,266
1 ½	1,181
2	1,717
3	716
4	419
6	218
8	66
12	7
Total	**125,811**

Historical Customer Accounts and Consumption

The table below provides the historic trend in the number of customers and the volume of consumption over the past four (4) years. The number of customers and the volume of consumption varied from year to year, but overall consumption has seen a declining trend over those years. Overall, the Division has seen an aggregate decline of 2% over the past four (4) years. The rate of decline, however, is expected to be less over time as the City has diversified its economic base and as economic conditions in and around the City continue to improve. However, a continued decline in billed consumption per customer account is expected as customers continue to upgrade to water-efficient fixtures and increase water conservation efforts.

Description	2012	2013	2014	2015
Water Meter Connections	125,577	126,555	125,768	125,811
Consumption (MCF)	1,826,888	1,822,980	1,823,983	1,790,291
Annual Consumption Change	---	(0.21%)	0.06%	(1.85%)

The Division experienced a slight increase in consumption in 2014. This increase was due in part to the Division of Water implementing an Automated Meter Reading ("AMR") system which served to reduce the number of estimated bills and improve meter reading accuracy. This is anticipated to be a one-time increase in response to the improved AMR system. Over the long-term, however, consumption per account is anticipated to decrease in response to

water conservation efforts. This reduction is offset to some extent by the Division's minimum bill charge.

Projected Customer Accounts and Consumption

The Division is conservatively projecting continued decreases in consumption in Fiscal Years 2016 through 2020. A 2.2% annual decrease in consumption is being projected to account for potential decreases in customer usage and other unknown circumstances. This projection is consistent with the Division of Water's projections for water consumption. The total number of customers is projected to remain static. The number of customers is based on buildings with metered water service that are connected to the System. As long as a building is connected to the System, the customers in the building will be billed a fixed charge regardless of the amount of water consumed. The Division does not anticipate any net disconnection of buildings or establishments from the System in the foreseeable future.

A summary of the projected fixed charge customer accounts and billable consumption forecast for the years 2016 through 2020 is shown in the table below.

Description	2015[1]	2016	2017	2018	2019	2020
Fixed Charge Customer Accounts	125,811	125,811	125,811	125,811	125,811	125,811
Annual Change in Customer Accounts	(0%)	(0%)	(0%)	(0%)	(0%)	(0%)
Consumption Forecast (MCF)	1,790,291	1,750,905	1,712,385	1,674,712	1,637,869	1,601,835
Annual Change in Consumption	(1.8%)	(2.2%)	(2.2%)	(2.2%)	(2.2%)	(2.2%)

[1] Actual results; provided to show forecasted changes in subsequent years.

Largest Customers of the Division

The following table sets forth the ten (10) largest customers for 2015 by billed consumption, which accounted for 12.6% of the billable consumption.

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City of Cleveland - Division of Water Pollution Control
Largest Customers of the Division
2015

	Consumer	Total Accounts	Billable Consumption (MCF)	% of Total Consumption
1	Cuyahoga Metropolitan Housing Authority	700	72,396.20	4.0%
2	Cleveland Clinic	37	43,453.40	2.4%
3	Case Western Reserve University	62	22,631.80	1.3%
4	Cuyahoga County	22	17,009.60	1.0%
5	University Hospitals Health System	15	16,543.10	0.9%
6	MetroHealth Medical Centers	11	15,465.10	0.9%
7	Cleveland Thermal	5	12,292.30	0.7%
8	Cleveland Board of Education	80	9,357.80	0.5%
9	Cleveland State University	27	9,197.80	0.5%
10	National Aeronautics and Space Administration[1]	3	7,429.00	0.4%
	Total			**12.6%**

[1] Physically located outside City limits with its water meter being located within City limits.

Billing and Collection Procedures

The Division's customers are billed quarterly based on their meter size (except the homestead rate accounts whose charges are not dependent upon meter size) and metered water consumption (with a minimum bill charge). Bills are issued by the City's Division of Water on behalf of the Division. The Division's charges are found on the same bill for water and waste collection. The Division of Water charges the Division its allocable share for provision of the billing services based on the Division of Water's cost and number of bills issued.

The Division typically anticipates a billing collection rate of 90%. This rate is consistent with the Division's current and historical billing collection rates.

The Division of Water, as the billing agent for the Division, anticipates changing from quarterly billing to monthly billing in the first quarter of Fiscal Year 2017 in response to customer requests. While the overall charges from the Division of Water are anticipated to increase due to increased costs associated with additional printing and postage costs, the charges per bill will decrease due to the increase in billing frequency. Furthermore, because no increase in staffing is anticipated to be required and the increased postage and printing costs will be shared among the Division, the Division of Water, and the Division of Waste Collection and Disposal, the net increase in the Division's share of the cost of billing services is anticipated to be minimal over the current quarterly billing costs. This increase in billing costs is projected to be incurred in Fiscal Year 2017.

Sewer Rates

The Trust Agreement provides that the City must charge sewer rates and so restrict Operating Expenses as will result in Net Revenues Available for Debt Service in each Fiscal Year being not less than 120% of Debt Service on all Bonds then Outstanding payable during that Fiscal Year. See "SECURITY AND SOURCES OF PAYMENT OF THE BONDS – Rate Covenant" and "APPENDIX B – DEFINITIONS AND SUMMARY OF CERTAIN TERMS AND PROVISIONS OF THE TRUST AGREEMENT."

The Division is responsible for establishing sewer rates and other fees and charges for its customers which must be approved by City Council. The rates must be at sufficient levels to pay for operating and maintenance costs of the System and debt service on outstanding debt and to provide for appropriate contingencies and reserves.

Sewer rates provide funding for the Division to conduct daily maintenance and repairs to the System, including its pump stations. The table below presents the new sewer rates that went into effect on April 1, 2015. These rates included both fixed charges and consumption charges. Effective April 1, 2015, City Council approved a new fixed fee charge based upon meter size in accordance with Ordinance No. 1030-14. The fixed fee will raise about $39 million over five years to address sewer issues more proactively and better serve City residents and business owners. The last time the City had a sewer rate increase was in 2006. Under the new sewer fixed charges, City residential properties are assessed $12 on their current quarterly payment, and that fee will increase by $3.00 per quarter in 2017 and in 2019. Properties with larger meters will be assessed at a higher rate. In addition, the fixed charges are not tied to water consumption, resulting in more predictability of charges to be collected. Sewer consumption charges are based on water consumption at each property location. See "REVENUES, EXPENSES AND DEBT SERVICE - General."

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Cleveland Fixed Quarterly Sewer Rates
2015 Through 2019

Fixed Charge (meter size) (or homestead rate)	2015	2016	2017	2018	2019
1/2 to 1 inch	$12.00	$12.00	$15.00	$15.00	$18.00
1/2 to 2 inch	$30.00	$30.00	$40.00	$40.00	$55.00
3 to 4 inch	$99.00	$99.00	$115.00	$115.00	$140.00
6 inch	$198.00	$198.00	$225.00	$225.00	$260.00
8 inch	$300.00	$300.00	$325.00	$325.00	$370.00
12 inch	$450.00	$450.00	$500.00	$500.00	$550.00
Homestead	$7.20	$7.20	$9.00	$9.00	$10.80

Cleveland Fixed Quarterly Consumption Charges
2015 Through 2019

Consumption Charge	2015	2016	2017	2018	2019
Regular (per MCF)	$12.53	$12.53	$12.53	$12.53	$12.53
Homestead (per MCF)	7.43	7.43	7.43	7.43	7.43

The Division has a special homestead rate for homesteads that are single family residential property owned and occupied by a person sixty-five (65) years of age or older, or homesteads that are single family residential property owned and occupied by a person permanently and totally disabled, whose total annual income does not exceed the applicable limit, which is $32,500 for 2016.

The Division has also instituted an affordability program to provide assistance to eligible low-income customers. The program offers a reduction of 40%, regardless of the meter size, from each quarterly sewer bill for residential customers who have been qualified to receive assistance through the federally funded Home Energy Assistance Program ("HEAP"). It is anticipated that the resulting reduction in sewer services sales revenue for the Division because of the homestead rate will be insignificant.

Maintenance Practices

The Division maintains a sewer system preventative maintenance program that includes implementation of BMPs, including a comprehensive Capacity Management, Operation, and Maintenance ("CMOM") program. Historically, the maintenance program focused first on addressing any complaints and known problem areas and second on conducting preventative maintenance. While these priorities remain, the Division is increasing both its preventative maintenance efforts and resources and rate of replacement of sewers which are past their useful life. The Division has developed a comprehensive CMOM program to increase preventative maintenance investment and reduce the potential of sanitary sewer overflows ("SSOs") in a cost effective manner. The program includes the following components: Management Plan, Operation and Maintenance Plan, Capacity Plan, Overflow Response Plan, Condition Assessment Plan, Communications Plan, and Audit Plan. Key elements of the program include:

- Using a computerized maintenance and management system to track all activities including inspections, cleaning, and rehabilitation and replacement projects.
- Continually updating and maintaining Division sewer maps in Graphic Information Systems.
- Conducting condition assessments to budget and plan for upcoming rehabilitation and replacement programs.
- Maintaining adequate inventory of equipment and spare parts to allow immediate responses to emergencies and prevent SSOs.
- Developing Standard Operating Procedures.
- Establishing performance measures and goals.
- Expanding safety training for employees.
- Establishing operation and maintenance programs for cleaning and inspections.

Capital Improvement Programs

The Division historically employed a pay-as-you-go capital improvement plan, supplemented with grants and low cost loans from the OPWC and OWDA for eligible projects. The Division's past plans focused primarily on projects to address street and basement flooding, to acquire equipment, and to implement information technology systems. For more information on the Division's past capital improvement projects, see "APPENDIX E - Feasibility Report - 6.1 Past Capital Improvements."

The Division plans to use approximately $35 million of the proceeds of the Series 2016 Bonds to fund the majority of its capital improvement program for the next three years (2016-2018), supplemented with cash funding of vehicles, equipment, and technology projects. For more information on the projects to be funded with proceeds of the Series 2016 Bonds, see "APPENDIX E - Feasibility Report - 6.2 Projects to be Funded with Series 2016 Bonds" and Appendix B thereto.

In addition to the projects anticipated to be funded with proceeds of the Series 2016 Bonds, the Division anticipates funding equipment purchases and technology investments with cash on hand and issuing additional Bonds as needed to fund future capital improvement projects. The capital improvement program over the five-year period from 2016 through 2020 includes project commitments of approximately $64 million. The principal focus of this five-year capital improvement program is sewer repair and replacement to rehabilitate and repair aging sewer pipes. For more information on the Division's future capital improvement projects, see "APPENDIX E - Feasibility Report - 6.3 Future Capital Improvements" and Appendix B thereto.

The following table summarizes the selected sources for the capital required to finance the current and future capital improvement program for the years 2016-2020.

City of Cleveland - Division of Water Pollution Control
Projected Funding Sources for Capital Improvement Program
2016-2020

	2016	2017	2018	2019	2020	Total	% of Total
Cash/Reserves	$2,940,080	$2,071,600	$2,215,800	$2,382,500	$3,250,000	$12,859,980	20%
Proceeds of the Series 2016 Bonds	$16,250,000	$9,400,000	$9,350,000	$0	$0	$35,000,000	55%
Future Bonds	$0	$0	$0	$8,042,000	$8,091,000	$16,133,000	25%
Total	$19,190,080	$11,471,600	$11,565,800	$10,424,500	$11,341,000	$63,992,980	100%

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REVENUES, EXPENSES AND DEBT SERVICE

General

The Revenues of the System include revenues derived from user consumption charges, fixed meter charges, and other miscellaneous revenue.

User Consumption Charges: Historically, the Division relied solely on user charges that were based on each customer's billable metered water consumption. The Division still relies on such consumption charges, but in Fiscal Year 2015 the Division also implemented a fixed charge component based on meter size. For the consumption charge component, customers are charged the same rate per thousand cubic feet ("MCF") of consumption as metered by the City's Division of Water. The only exception is that certain customers pay a discounted homestead rate if they are eligible and have been approved based on established economic and social criteria. The homestead rate provides customers with a discounted rate which is approximately 40% lower than the regular rate to maintain affordability of such rates. The consumption charge also includes a minimum bill charge based on a minimum consumption of one MCF per quarter. When the Division moves to monthly billing, the minimum charge per bill will be prorated to adjust for monthly as opposed to quarterly billing.

Fixed Meter Charges: Beginning in Fiscal Year 2015, for customers with a water meter, the Division instituted a new fixed charge component that varies based on the size of the customer's incoming water meter service. The fixed charge is in addition to the consumption charge and was instituted to address the infrastructure of the System. The fixed charge is assessed even if the customer had zero water consumption. Similar to the consumption charge, a 40% discounted homestead fixed charge rate is available for qualifying customers to maintain affordability of the rates. When the Division moves to monthly billing, which is anticipated to be in 2017, the Fixed Meter Charge per bill will be prorated to adjust for monthly as opposed to quarterly billing.

Other Revenue: Other revenue includes interest income, licenses and permit fee revenue, reimbursements/credits from the City, and other miscellaneous revenue. Other revenue represents a very small percentage of the Division's total revenues.

Historical Revenues and Expenses

The Division's revenues, expenses and debt service for fiscal years 2011 through 2015 are set forth in the following table. The table was prepared using information contained in the audited financial statements of the Division for the years ended December 31, 2014, 2013, 2012, 2011 and unaudited figures for 2015.

City of Cleveland - Division of Water Pollution Control
SUMMARY OF HISTORICAL REVENUES, EXPENSES AND DEBT SERVICE [1]
Year Ended December 31

	2011	2012	2013	2014	2015 - est.[2]
Operating Revenues					
Charges for Services	$22,199,000	$22,876,000	$22,549,000	$22,547,000	$28,744,000
Total Operating Revenues	$22,199,000	$22,876,000	$22,549,000	$22,547,000	$28,744,000
Operating Expenses					
Operations	$9,315,000	$10,194,000	$9,232,000	$10,670,000	$10,818,000
Maintenance	$8,835,000	$9,075,000	$8,901,000	$8,459,000	$8,170,000
Total Operating Expenses	$18,150,000	$19,269,000	$17,323,000	$19,129,000	$ 18,988,000
Operating Income (Loss) Before Depreciation	$4,049,000	$3,607,000	$5,226,000	$3,418,000	$9,756,000
Depreciation	$4,734,000	$5,108,000	$5,221,000	$5,579,000	$5,700,000
Operating Income (Loss) After Depreciation	($685,000)	($1,501,000)	$5,000	($2,161,000)	$4,056,000
Non-Operating Revenue (Expense)					
Investment Income	$53,000	$59,000	$58,000	$54,000	$51,000
Interest Expense	($121,000)	($102,000)	($82,000)	($61,000)	($40,000)
Other	$53,000	$21,000	$99,000	$86,000	$80,000
Total Non-Operating Revenue (Expense), Net	($15,000)	($22,000)	$75,000	$79,000	$91,000
Income (Loss) Before Other Contributions	($700,000)	($1,523,000)	$80,000	($2,082,000)	$4,147,000

(1) Information derived from audited financial statements of the Division (other than the 2015 information).
(2) Estimated figures.

The Summary of Historical Revenues, Expenses and Debt Service reflects the results of past operations and is not necessarily indicative of results of operations for any future period. Future operations will necessarily be affected by various factors and developments, including possible changes in sewer rates, the size of the Service Area, business activity, labor contracts, electricity costs, changes in population, weather, future plant construction programs and other matters, the nature and effect of which cannot now be determined. See "CERTAIN INVESTMENT CONSIDERATIONS AND RISK FACTORS – Historical Information."

Management Discussion of Results of Operations

The comparative statement of revenues and expenses for Fiscal Years 2011-2014 set forth below has been derived from the financial statements of the Division. The comparative statement of revenues and expenses for Fiscal Year 2015 set forth below is based on unaudited figures from the Division. The Division is considered an enterprise fund because the operations of the Division are similar to a private sector business enterprise. Accordingly, in accounting for the

activities of the Division, the economic resources measurement focus and the accrual basis of accounting are used.

Fiscal Year 2011

Operating revenues in 2011 ($22,199,000) decreased by $2,911,000 (11.6%) from 2010 ($25,110,000), mainly due to a decrease in consumption.

Total operating expenses in 2011 decreased by $1,042,000 (4.4%) to $22,884,000 compared to $23,926,000 in 2010. The components of the decrease in operating expenses in 2011 were a decrease in operation costs of $683,000, a decrease in maintenance costs of $92,000, and a decrease in depreciation expense of $267,000.

The Division incurred an operating loss after depreciation of $685,000 in 2011; the Division used cash on hand to fund the operating loss.

Fiscal Year 2012

Operating revenues in 2012 ($22,876,000) increased by $677,000 (3%) from 2011 ($22,199,000), mainly due to an increase in consumption.

Total operating expenses in 2012 increased by $1,493,000 (6.5%) to $24,377,000 compared to $22,884,000 in 2011. There was an increase in operation costs of $879,000. The major component was accrued bad debt expense, which increased by $1,061,000.

The Division incurred an operating loss after depreciation of $1,501,000 in 2012; the Division used cash on hand to fund the operating loss.

Fiscal Year 2013

Operating revenues in 2013 ($22,549,000) decreased by $327,000 (1.4%) compared to 2012 ($22,876,000), mainly due to a decrease in consumption.

Total operating expenses in 2013 ($22,544,000) decreased by $1,833,000 (7.5%) compared to $24,377,000 in 2012. There was a decrease in operation costs of $962,000. The major component of the decrease was accrued bad debt expense, which decreased by $1,945,000. In addition, maintenance costs decreased by $984,000. The major components were transfers of wages and benefits to capital and maintenance utility systems, which decreased by $374,000 and $309,000, respectively. The transfer of wages and benefits to capital was calculated using the Safe Harbor Rate Method, which decreased overhead costs and increased capital project costs.

Fiscal Year 2014

Operating revenues in 2014 ($22,547,000) remained approximately the same as in 2013 ($22,549,000), as the 2014 consumption was stable in comparison to 2013.

Total operating expenses in 2014 increased by $2,164,000 (9.6%) to $24,708,000 compared to $22,544,000 in 2013. The major components of the $2,164,000 increase in

operating expenses in 2014 were increases in bad debt expense of $1,744,000 and depreciation expense of $358,000, offset by a decrease in charges in the Division of Water of $462,000.

The Division incurred an operating loss after depreciation of $2,161,000 in 2014; the Division used cash on hand to fund the operating loss.

Fiscal Year 2015

Operating revenues in 2015 ($28,744,000) increased by $6,197,000 (27.5%) compared to 2014 ($22,547,000), mainly due to the implementation of the fixed fee as part of the quarterly charged which was in effect for three quarters in 2015.

Total operating expenses in 2015 ($24,688,000) remained approximately the same as in 2014 ($24,708,000).

Projected Cash Flow 2016-2020

The following table, prepared by the Feasibility Consultant, presents a statement of the estimated cash flow for the Division for Fiscal Years 2016 through 2020.

See "APPENDIX E – FEASIBILITY REPORT" herein for the full text of the Feasibility Report. The Feasibility Report should be read in its entirety, including but not limited to the underlying assumptions and the basis and support therefor. As stated in the Feasibility Report, actual results may differ materially from those projected, as influenced by the conditions, events and circumstances that actually occur that are unknown at this time and/or which are beyond the control of the Feasibility Consultant or the Division.

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Cash Flow Projection (FY 2016 through FY2020[1])

Line No.	Description	Fiscal Year Ending December 31, FY2016	FY2017	FY2018	FY2019	FY2020
	Customer Information					
1	Number of Customer Meter Accounts	125,811	125,811	125,811	125,811	125,811
2	Regular Consumption User Charge ($/MCF)	12.53	12.53	12.53	12.53	13.09
3	**Annual Rate Increase**	0.0%	1.6%	0.0%	2.0%	4.5%
	Revenues					
4	Consumption User Charge Revenue	$20,374,361	$19,926,125	$19,487,750	$19,059,020	$19,478,509
5	Fixed Charge Revenue	6,013,969	7,476,678	7,476,678	9,039,355	9,446,126
6	Miscellaneous Revenue	865,000	865,000	865,000	865,000	865,000
7	**Total Revenues**	$27,253,330	$28,267,803	$27,829,428	$28,963,375	$29,789,635
	Operating Expenses					
8	Personnel Expenses	$11,146,302	$11,476,837	$11,819,363	$12,174,390	$12,542,448
9	Other Training and Professional Dues	25,500	26,265	27,053	27,865	28,700
10	Utilities	389,513	401,198	413,234	425,631	438,400
11	Contractual Services	1,679,910	1,730,307	1,782,217	1,835,683	1,890,754
12	Materials and Supplies	447,000	460,410	474,222	488,449	503,102
13	Maintenance	1,904,000	1,961,120	2,019,954	2,080,552	2,142,969
14	Claims	983,780	1,013,293	1,043,692	1,075,003	1,107,253
15	Interdepartmental Charges	5,125,544	5,279,310	5,437,690	5,600,820	5,768,845
16	**Total Operating Expenses**	$21,701,549	$22,348,741	$23,017,425	$23,708,393	$24,422,472
17	**Net Revenues Available for Debt Service**	$5,551,781	$5,919,061	$4,812,003	$5,254,982	$5,367,163
	Debt Service					
18	Existing Debt Service	491,990	179,195	24,060	24,060	24,060
19	Series 2016 Revenue Bond	928,547	2,134,250	2,132,250	2,135,150	2,131,550
20	Future Revenue Bond	-	-	-	567,936	1,139,332
21	**Total Debt Service**	$1,420,536	$2,313,445	$2,156,310	$2,727,145	$3,294,942
	Debt Service Coverage					
22	DSC (Total Revenue Bonds)	598%	277%	226%	194%	164%
23	DSC (All-Debt)	391%	256%	223%	193%	163%

Line No.	Description	Fiscal Year Ending December 31, FY2016	FY2017	FY2018	FY2019	FY2020
	Total Capital Plan	$19,190,080	$11,471,600	$11,565,800	$10,424,500	$11,341,000
	Capital Project Funding					
24	Cash from Operating Fund	2,940,080	2,071,600	2,215,800	2,382,500	3,250,000
25	Series 2016 Revenue Bond	16,250,000	9,400,000	9,350,000	-	-
26	New Debt	-	-	-	8,042,000	8,091,000
27	**Total Capital Project Funding**	$19,190,080	$11,471,600	$11,565,800	$10,424,500	$11,341,000
28	**Minor Capital Expense**	$200,000	$206,000	$212,180	$218,545	$225,102
29	**Total Revenue Requirements**	$23,322,085	$24,868,186	$25,385,914	$26,654,084	$27,942,515
30	**Revenues Over (Under) Expenses**	$3,931,245	$3,399,617	$2,443,514	$2,309,291	$1,847,120
	WPC Cash Fund Balance					
31	**Beginning Balance**	$16,600,000	$17,591,165	$18,919,182	$19,146,896	$19,073,687
32	**Revenues Over (Under) Expenses**	3,931,245	3,399,617	2,443,514	2,309,291	1,847,120
33	Transfers to Capital Projects Fund	(2,940,080)	(2,071,600)	(2,215,800)	(2,382,500)	(3,250,000)
34	**Ending Balance**	$17,591,165	$18,919,182	$19,146,896	$19,073,687	$17,670,807
35	Minimum Operating Reserve (6 Months)	$10,850,775	$11,174,371	$11,508,712	$ 11,854,196	$12,211,236

(1) The revenue and coverage projections for FY 2020 reflect a targeted rate increase that has not yet been proposed to or approved by City Council.

FINANCIAL ADVISERS

The City has retained Government Capital Management, LLC and Phoenix Capital Partners, LLP as co-financial advisers (the "Financial Advisers") to the City, in connection with the issuance of the Series 2016 Bonds. The Financial Advisers are not obligated to undertake, and have not undertaken to make, an independent verification or to assume the responsibility for the accuracy, completeness or fairness of the information contained in this Official Statement. Government Capital Management, LLC and Phoenix Capital Partners, LLP are independent financial advisory firms and are not engaged in the business of underwriting, trading or distributing municipal or other public securities.

FEASIBILITY REPORT

Arcadis U.S., Inc., as feasibility consultant (the "Feasibility Consultant") to the City and the Division, has reviewed those plans and operations, as well as the historical and current financial records of the System, as were made available to Arcadis, and analyzed the projections of net revenue sufficiency for service of debt in accordance with the terms of the Trust Agreement. The Feasibility Report is attached hereto as Appendix E. The opinions rendered in the Feasibility Report are stated only as of the date therein, and are based upon the information provide to Arcadis, and no other opinion shall be implied or inferred as a result of anything contained in or omitted from the Feasibility Report. See "APPENDIX E – FEASIBILITY REPORT."

LITIGATION

No litigation or administrative action or proceeding is pending or, to the knowledge of the appropriate officials of the City, threatened, restraining or enjoining, or seeking to restrain or enjoin, the issuance, sale or delivery of the Series 2016 Bonds or the pledge of the Net Revenues to pay debt service on the Series 2016 Bonds, or in any way contesting or affecting the validity or enforceability of the Series 2016 Bond Legislation, the Master Trust Agreement, the First Supplemental Trust Agreement, or the Series 2016 Bonds, or with respect to the fixing and collecting of rates and charges for the Division or challenging the Division's right to own or operate the System. The City will deliver a certificate to that effect to the Underwriters at the time of original delivery of the Series 2016 Bonds to the Underwriters.

The City does not now maintain nor has it previously maintained liability or property insurance with respect to the Division. The City has no practice of setting aside or accumulating funds for the purpose of insurance, but has been able to satisfy Division liabilities in a timely manner by absorbing the costs of any such liabilities into the operating expenses of the Division.

The Division is a party to various legal proceedings seeking damages or injunctive or other relief generally incidental to its operations. These proceedings are not directly related to the Series 2016 Bonds or the security for the Series 2016 Bonds. If decided adversely to the Division, however, money judgments rendered in such proceedings for damages, if not otherwise satisfied by the Division, could have an indirect effect on the Division since judgment creditors of the Division may seek a levy of execution against properties of the Division. The ultimate disposition of these proceedings is not now determinable. It is the opinion of the City's Director of Law, based upon her present understanding and knowledge of these proceedings, that the

disposition of these proceedings, individually or in the aggregate, will not result in liabilities in an amount which, in the opinion of the City's Director of Finance, would have a material adverse effect on the Series 2016 Bonds, the security for the Series 2016 Bonds or the operations of the Division.

LEGAL MATTERS

Certain legal matters incident to the issuance of the Series 2016 Bonds and with regard to the tax-exempt status of the interest on the Series 2016 Bonds (see "TAX MATTERS") are subject to the opinion of Squire Patton Boggs (US) LLP, Bond Counsel to the City. The signed legal opinion of Bond Counsel, substantially in the forms attached hereto as APPENDIX C, dated and premised on law in effect on the dates of issuance of the Series 2016 Bonds, will be delivered on the dates of issuance of the Series 2016 Bonds. The text of the opinions to be delivered may vary from the text as set forth in APPENDIX C if necessary to reflect facts and law on the dates of delivery. Each opinion will speak only as of its date, and subsequent distribution of such opinion by recirculation of this Official Statement or otherwise shall create no implication that Bond Counsel has reviewed or expresses any opinion concerning any of the matters referred to in that opinion subsequent to its date.

The opinion of Bond Counsel and any other legal opinions and letters of counsel to be delivered concurrently with the delivery of the Series 2016 Bonds express the professional judgment of the attorneys rendering the opinions or advice regarding the legal issues and other matters expressly addressed therein. By rendering a legal opinion or advice, the giver of such opinion or advice does not become an insurer or guarantor of the result indicated by that opinion, or the transaction on which the opinion or advice is rendered, or of the future performance of parties to the transaction. Nor does the rendering of an opinion guarantee the outcome of any legal dispute that may arise out of the transaction.

Certain legal matters will be passed upon for the Underwriters by their counsel, Calfee, Halter & Griswold LLP.

Certain legal matters will be passed upon for the City by Barbara A. Langhenry, its Director of Law.

The City has also retained the legal services of both Squire Patton Boggs (US) LLP and Calfee, Halter & Griswold LLP from time to time as special counsel in connection with matters that do not relate to City financings. Squire Patton Boggs (US) LLP and Calfee, Halter & Griswold LLP also serve and have served as bond counsel for one or more of the political subdivisions that the City territorially overlaps. Squire Patton Boggs (US) LLP has served as counsel to one or more of the Underwriters in connection with matters that do not relate to the City.

TAX MATTERS

General

In the opinion of Squire Patton Boggs (US) LLP, Bond Counsel, under existing law: (i) interest on the Series 2016 Bonds is excluded from gross income for federal income tax purposes under Section 103 of the Internal Revenue Code of 1986, as amended (the "Code"), and is not an item of tax preference for purposes of the federal alternative minimum tax imposed on individuals and corporations; and (ii) interest on, and any profit made on the sale, exchange or other disposition of, the Series 2016 Bonds are exempt from all Ohio state and local taxation, except the estate tax, the domestic insurance company tax, the dealers in intangibles tax, the tax levied on the basis of the total equity capital of financial institutions, and the net worth base of the corporate franchise tax. Bond Counsel expresses no opinion as to any other tax consequences regarding the Series 2016 Bonds.

The opinion on tax matters will be based on and will assume the accuracy of certain representations and certifications, and continuing compliance with certain covenants, of the City contained in the transcript of proceedings and that are intended to evidence and assure the foregoing, including that the Series 2016 Bonds are and will remain obligations the interest on which is excluded from gross income for federal income tax purposes. Bond Counsel will not independently verify the accuracy of the City's certifications and representations or the continuing compliance with the City's covenants.

The opinion of Bond Counsel is based on current legal authority and covers certain matters not directly addressed by such authority. It represents Bond Counsel's legal judgment as to exclusion of interest on the Series 2016 Bonds from gross income for federal income tax purposes but is not a guaranty of that conclusion. The opinion is not binding on the Internal Revenue Service ("IRS") or any court. Bond Counsel expresses no opinion about (i) the effect of future changes in the Code and the applicable regulations under the Code or (ii) the interpretation and the enforcement of the Code or those regulations by the IRS.

The Code prescribes a number of qualifications and conditions for the interest on state and local government obligations to be and to remain excluded from gross income for federal income tax purposes, some of which require future or continued compliance after issuance of the obligations. Noncompliance with these requirements by the City may cause loss of such status and result in the interest on the Series 2016 Bonds being included in gross income for federal income tax purposes retroactively to the date of issuance of the Series 2016 Bonds. The City has covenanted to take the actions required of it for the interest on the Series 2016 Bonds to be and to remain excluded from gross income for federal income tax purposes, and not to take any actions that would adversely affect that exclusion. After the date of issuance of the Series 2016 Bonds, Bond Counsel will not undertake to determine (or to so inform any person) whether any actions taken or not taken, or any events occurring or not occurring, or any other matters coming to Bond Counsel's attention, may adversely affect the exclusion from gross income for federal income tax purposes of interest on the Series 2016 Bonds or the market value of the Series 2016 Bonds.

A portion of the interest on the Series 2016 Bonds earned by certain corporations may be subject to a federal corporate alternative minimum tax. In addition, interest on the Series 2016 Bonds may be subject to a federal branch profits tax imposed on certain foreign corporations doing business in the United States and to a federal tax imposed on excess net passive income of certain S corporations. Under the Code, the exclusion of interest from gross income for federal income tax purposes may have certain adverse federal income tax consequences on items of income, deduction or credit for certain taxpayers, including financial institutions, certain insurance companies, recipients of Social Security and Railroad Retirement benefits, those that are deemed to incur or continue indebtedness to acquire or carry tax-exempt obligations, and individuals otherwise eligible for the earned income tax credit. The applicability and extent of these and other tax consequences will depend upon the particular tax status or other tax items of the owner of the Series 2016 Bonds. Bond Counsel will express no opinion regarding those consequences.

Payments of interest on tax-exempt obligations, including the Series 2016 Bonds, are generally subject to IRS Form 1099-INT information reporting requirements. If a Series 2016 Bond owner is subject to backup withholding under those requirements, then payments of interest will also be subject to backup withholding. Those requirements do not affect the exclusion of such interest from gross income for federal income tax purposes.

Bond Counsel's engagement with respect to the Series 2016 Bonds ends with the issuance of the Series 2016 Bonds, and, unless separately engaged, Bond Counsel is not obligated to defend the City or the owners of the Series 2016 Bonds regarding the tax status of interest thereon in the event of an audit examination by the IRS. The IRS has a program to audit tax-exempt obligations to determine whether the interest thereon is includible in gross income for federal income tax purposes. If the IRS does audit the Series 2016 Bonds, under current IRS procedures, the IRS will treat the City as the taxpayer and the beneficial owners of the Series 2016 Bonds will have only limited rights, if any, to obtain and participate in judicial review of such audit. Any action of the IRS, including but not limited to selection of the Series 2016 Bonds for audit, or the course or result of such audit, or an audit of other obligations presenting similar tax issues, may affect the market value of the Series 2016 Bonds.

Prospective purchasers of the Series 2016 Bonds upon their original issuance at prices other than the respective prices indicated on the inside cover of this Official Statement, and prospective purchasers of the Series 2016 Bonds at other than their original issuance, should consult their own tax advisers regarding other tax considerations such as the consequences of market discount, as to all of which Bond Counsel expresses no opinion.

Risk of Future Legislative Changes and/or Court Decisions

Legislation affecting tax-exempt obligations is regularly considered by the United States Congress and may also be considered by the State legislature. Court proceedings may also be filed, the outcome of which could modify the tax treatment of obligations such as the Series 2016 Bonds. There can be no assurance that legislation enacted or proposed, or actions by a court, after the date of issuance of the Series 2016 Bonds will not have an adverse effect on the tax status of interest or other income on the Series 2016 Bonds or the market value or marketability of the Series 2016 Bonds. These adverse effects could result, for example, from changes to

federal or state income tax rates, changes in the structure of federal or state income taxes (including replacement with another type of tax), or repeal (or reduction in the benefit) of the exclusion of interest on the Series 2016 Bonds from gross income for federal or state income tax purposes for all or certain taxpayers.

For example, recent presidential and legislative proposals would eliminate, reduce or otherwise alter the tax benefits currently provided to certain owners of state and local government bonds, including proposals that would result in additional federal income tax on taxpayers that own tax-exempt obligations if their incomes exceed certain thresholds. Investors in the Series 2016 Bonds should be aware that any such future legislative actions (including federal income tax reform) may retroactively change the treatment of all or a portion of the interest on the Series 2016 Bonds for federal income tax purposes for all or certain taxpayers. In such event, the market value of the Series 2016 Bonds may be adversely affected and the ability of holders to sell their Series 2016 Bonds in the secondary market may be reduced. The Series 2016 Bonds are not subject to special mandatory redemption, and the interest rates on the Series 2016 Bonds are not subject to adjustment in the event of any such change.

Investors should consult their own financial and tax advisers to analyze the importance of these risks.

Original Issue Premium

All of the Series 2016 Bonds ("Premium Bonds") were offered and sold to the public at a price in excess of their stated redemption price at maturity (the principal amount). That excess constitutes bond premium. For federal income tax purposes, bond premium is amortized over the period to maturity of a Premium Bond, based on the yield to maturity of that Premium Bond (or, in the case of a Premium Bond callable prior to its stated maturity, the amortization period and yield may be required to be determined on the basis of an earlier call date that results in the lowest yield on that Premium Bond), compounded semiannually. No portion of that bond premium is deductible by the owner of a Premium Bond. For purposes of determining the owner's gain or loss on the sale, redemption (including redemption at maturity) or other disposition of a Premium Bond, the owner's tax basis in the Premium Bond is reduced by the amount of bond premium that is amortized during the period of ownership. As a result, an owner may realize taxable gain for federal income tax purposes from the sale or other disposition of a Premium Bond for an amount equal to or less than the amount paid by the owner for that Premium Bond. A purchaser of a Premium Bond in the initial public offering at the price for that Premium Bond stated on the cover of this Official Statement who holds that Premium Bond to maturity (or, in the case of a callable Premium Bond, to its earlier call date that results in the lowest yield on that Premium Bond) will realize no gain or loss upon the retirement of that Premium Bond.

Owners of Premium Bonds should consult their own tax advisers as to the determination for federal income tax purposes of the amount of bond premium properly accruable or amortizable in any period with respect to the Premium Bonds and as to other federal tax consequences and the treatment of bond premium for purposes of state and local taxes on, or based on, income.

FINANCIAL MATTERS

The financial statements of the City of Cleveland, Department of Public Utilities - Division of Water Pollution Control, for the year ended December 31, 2014 and December 31, 2013, have been audited by Clark, Schaefer, Hackett & Co. and are currently located at www.auditor.state.oh.us. The City continues to maintain an internal audit function and an active external audit committee.

CERTAIN INVESTMENT CONSIDERATIONS AND RISK FACTORS

The Series 2016 Bonds are subject to certain investment risks, which are described throughout this Official Statement. Accordingly, each prospective purchaser should make an independent evaluation of all of the information presented in this Official Statement in order to make an informed investment decision. This section of the Official Statement does not include all risks to which payment of the Series 2016 Bonds is subject, but is merely an attempt to summarize certain of those risks. Each potential purchaser of the Series 2016 Bonds should read this Official Statement in its entirety.

Limited Obligations

The Series 2016 Bonds will be special obligations of the City and will not represent or constitute a general obligation, bonded indebtedness or a pledge of the general credit or taxing power of the City, the State of Ohio or any political subdivision thereof. The holders of the Series 2016 Bonds will not have the right to have excises or taxes levied by the City, the State of Ohio or the taxing authority of any political subdivision thereof for the payment of debt service charges. The Series 2016 Bonds are payable solely from the Net Revenues derived by the City from charges for use of the System. Although the City has covenanted to charge such rates as shall at all times produce Net Revenues necessary to meet the requirements of the funds and accounts created by the Trust Agreement and the Series 2016 Bond Legislation, no representation or assurance can be made or given that Net Revenues, as presently collected, projected, or otherwise, will be realized by the City in amounts sufficient to pay maturing principal and interest on the Series 2016 Bonds. The holders of the Series 2016 Bonds will not have the right to have excises or taxes levied by the City, the State of Ohio or the taxing authority of any political subdivision thereof for the payment of debt service charges.

Additional Bonds

Under the Trust Agreement, the City is permitted to incur other debt payable on a parity with the Series 2016 Bonds. See "SECURITY AND SOURCES OF PAYMENT OF THE BONDS - Additional Bonds". Debt service on all parity Bonds of the City will be payable from Net Revenues on a pro-rata basis. Accordingly, to the extent that future obligations are issued on a parity with the Series 2016 Bonds, the security for the Series 2016 Bonds may be diluted. See "SECURITY AND SOURCES OF PAYMENT OF THE BONDS - Additional Bonds."

Subordinated Indebtedness

Under the Trust Agreement, the City may also issue Subordinated Indebtedness that are secured on a basis subordinate to the Series 2016 Bonds and any Additional Bonds payable from

subordinate pledged Revenues. See "SECURITY AND SOURCES OF PAYMENT OF THE BONDS - Subordinated Indebtedness."

Changes in Market Conditions; Conditions of the City; Ability to Charge Rates

The Series 2016 Bonds, like all obligations of state and local government, are subject to changes in value due to changes in the condition of the tax-exempt bond market and/or changes in the financial condition of the City. It is possible under certain market conditions, or if the financial condition of the City should change, that the market price of the Series 2016 Bonds could be adversely affected. The generation of Net Revenues is dependent upon several factors outside the City's control, such as the economy, collections of sewer and stormwater service charges and fees, continued growth (or lack thereof) and changes in law. It is impossible to predict whether current economic conditions will continue or worsen or to predict how future conditions will affect the operation of the Division or the City's finances in general.

Although the operations of the Division and the rates established for the Division by the City are not currently subject to direct regulation by the State of Ohio or the United States, such operations and rates can be adversely affected at any time by laws enacted by the Ohio General Assembly or Congress or by rules, regulations, orders, or determinations by State or federal agencies which have the effect of increasing the expenses of the Division and/or impairing or limiting the City's ability to increase rates.

The City covenants in the Trust Agreement that if in any Fiscal Year the Net Revenues Available for Debt Service shall be less than the amount required under the Trust Agreement, it will employ a Consulting Engineer to make recommendations as to a revision of the rates, fees and charges, or Operating Expenses, or methods of operations of the System, if any, that will result in producing the amount so required in the next succeeding Fiscal Year. The City covenants and agrees, promptly upon its receipt of such recommendations, to revise the rates, fees, and charges or Operating Expenses or methods of operation of the System and to take such other action as shall be in conformity with such recommendations to the extent the City feasibly may do so and to notify the Trustee of the actions it has taken. Upon the taking by the City of such actions, its failure to comply with the foregoing requirements shall not constitute an Event of Default. See "SECURITY AND SOURCES OF PAYMENT OF THE BONDS – Rate Covenant."

Factors That May Cause Insufficiency of Expected Revenues

General. Payment of the Series 2016 Bonds is dependent upon the generation of sufficient Net Revenues of the System. If the System becomes inoperable due to damage, destruction, environmental restriction or for any other reason; if the Division has inadequate storage or distribution facilities; if the Division is unable to increase rates and charges for any reason; or if the Division incurs unanticipated expenses or reduced Net Revenues for any other reason, the Division may be unable to generate adequate Net Revenues from the System to pay debt service on the Series 2016 Bonds (subject to the requirement that the City must subsequently raise rates if necessary to satisfy the Rate Covenant).

Environmental Requirements. The System is subject to numerous federal and State regulatory requirements, including but not limited to the requirement for the City's stormwater

system to be operated and maintained pursuant to its National Pollution Discharge Elimination System ("NPDES") permit (NPDES No. OHQ000003) for Small Municipal Separate Storm Sewer System ("MS4") permit, the City's Community Discharge Permit for sanitary sewers, and the City's Community Combined Discharge Permit for combined stormwater and sewers. Those regulations are subject to change at any time. In addition, failure to comply with regulatory changes, or the inability to comply with them in a timely manner could cause portions of the System to become unavailable. Any disruption of service could negatively impact Net Revenues.

In operating the Division, the City also may be subject to various environmental regulations which could subject the City to increased operating costs or capital expenditures. Such risks include the use of hazardous materials in the sewage and stormwater treatment process and the disposal of such materials or other factors. Such increased costs could reduce the amount of Net Revenues available to pay debt service on the Series 2016 Bonds.

Historical Information

The amount of Revenues available for the payment of Operating Expenses and the amount of Net Revenues available for the payment of debt service on the Bonds will be affected by the future levels of utilization of the System and the rates and charges that the City can impose in connection with the use thereof. All financial and other information presented herein has been provided by the City from its records, except for information expressly attributed to other sources. The presentation of information, including tables of receipts from rates, charges, taxes, and other sources, is intended to show recent historical information, and is not intended to indicate future or continuing trends in the financial position or other affairs of the City or the Division except as otherwise especially indicated herein. No representation is made that past experience, as might be shown by such financial and other information, will necessarily continue or be repeated.

Secondary Market

There is no guarantee that a secondary market for the Series 2016 Bonds will be maintained by the Underwriters or others. Thus, owners of the Series 2016 Bonds should be prepared to hold their Series 2016 Bonds to maturity.

Changes in Law

Various federal and Ohio laws and constitutional provisions, statutes, rules and regulations apply to the collection and expenditure of the Net Revenues, the obligations created by the issuance of the Series 2016 Bonds, and the operation of the Division and the City. There is no assurance that there will not be any change in, interpretation of, or addition to the applicable laws, provisions, and regulations which would have a material effect, directly or indirectly, on the affairs of the City or the Division, and the collection and expenditure of the Net Revenues.

Bankruptcy

The enforceability of the rights and remedies of owners of the Series 2016 Bonds may be limited to the extent that the obligations incurred by the City in issuing the Series 2016 Bonds are subject to the Federal Bankruptcy Code and applicable bankruptcy, insolvency,

reorganization, moratorium or similar laws relating to or affecting the enforcement of creditors' rights generally, now or hereafter in effect; to usual equity principles which may limit the specific enforcement under State law of certain remedies; to the exercise by the United States of America of the powers delegated to it by the United States Constitution; and to the reasonable and necessary exercise, in certain exceptional situations, of the police power inherent in the sovereignty of the State and its governmental bodies. Bankruptcy proceedings or the exercise of powers by the federal or State government, if initiated, could subject the owners of the Series 2016 Bonds to judicial discretion and interpretation of their rights in bankruptcy or otherwise, and may entail risks of delay, limitation or modification of their rights.

Chapter 9 of the Federal Bankruptcy Code contains provisions relating to the adjustment of debts of a State's political subdivisions, public agencies and instrumentalities ("eligible entity"), such as the City. Under the Federal Bankruptcy Code and in certain circumstances described therein, an eligible entity may be authorized to initiate Chapter 9 proceedings without prior notice to or consent of its creditors, which proceedings may result in material and adverse modification or alteration of the rights of its secured and unsecured creditors, including holders of its bonds and notes.

The Federal Bankruptcy Code protects holders of municipal revenue bonds by providing that special revenues acquired by the eligible entity after the commencement of bankruptcy proceedings remain subject to any lien resulting from any security agreement entered into by the eligible entity before commencement of the proceedings.

Section 133.36 of the Ohio Revised Code permits a political subdivision, such as the City, for the purpose of enabling such subdivision to take advantage of the provisions of the Federal Bankruptcy Code, and for that purpose only, and upon approval of the State Tax Commissioner, to file a petition stating that the subdivision is insolvent or unable to meet its debts as they mature, and that it desires to effect a plan for the composition or readjustment of its debts, and to take such further proceedings as are set forth in the Federal Bankruptcy Code as they relate to such subdivision. The taxing authority of such subdivision may, upon like approval of the State Tax Commissioner, refund its outstanding securities, whether matured or un-matured, and exchange refunding bonds for the securities being refunded. In its order approving such refunding, the State Tax Commissioner shall fix the maturities of the bonds to be issued, which shall not exceed thirty years. No taxing subdivision is permitted, in availing itself of the provisions of the Federal Bankruptcy Code, to scale down, cut down or reduce the principal sum of its securities except that interest thereon may be reduced in whole or in part. To the knowledge of the City, to date no municipality in Ohio has initiated a proceeding under Chapter 9 of the Federal Bankruptcy Code. As a result, the City has no basis to know whether such provision under Section 133.36 of the Ohio Revised Code would or would not be enforced by a federal bankruptcy court.

Local Fiscal Emergency Legislation

The Director of Finance has reviewed applicable portions of Chapter 118 of the Revised Code of Ohio (hereinafter in this section of this Official Statement, the "Act"). The Director of Finance, based upon her understanding of the Act and the City's finances, is of the opinion that, with

respect to the City, a fiscal emergency condition does not currently exist. However, no assurances can be given that fiscal emergency conditions will not exist in the future.

The Act provides methods for dealing with fiscal emergencies of municipal corporations, counties and townships in Ohio. The Act applies only to those municipal corporations, counties and townships that are determined to have circumstances that constitute the existence of a fiscal watch or a fiscal emergency condition and therefore a fiscal watch or a fiscal emergency pursuant to the Act.

The Act sets forth a series of conditions that constitute grounds for a fiscal watch. If a fiscal watch is determined to exist, the municipality, county or township is provided technical and support services by the State Auditor's Office to restore financial stability. If the fiscal watch conditions are not remedied, the municipality, county or township will remain under fiscal watch or be reclassified to a fiscal emergency.

The Act also sets forth a series of circumstances that are defined "fiscal emergency conditions." If a fiscal emergency condition is determined to exist, the municipality, county or township is subjected to state oversight through a seven-member Financial Planning and Supervision Commission (the "Commission"). The Commission is assisted by certified public accountants designated as the Financial Supervisor to be engaged by the Commission. The Auditor of State may also be required to assist the Commission.

A municipal corporation, county or township subject to the Act because of the existence of a fiscal emergency must develop and submit a detailed financial plan for the approval or rejection of the Commission. Among other matters, the financial plan must show the actions to be taken by such a municipal corporation, county or township to eliminate existing fiscal emergency conditions, avoid future fiscal emergency conditions, and to restore such a municipality's, county's or township's ability to market long-term debt obligations under State laws generally applicable to Ohio political subdivisions.

The Commission must approve the amount and purpose of any issue of debt obligations. The Commission, or when authorized by the Commission, the Financial Supervisor, among other powers, shall require the municipal corporation, county or township to establish monthly levels of expenditures and encumbrances consistent with the financial plan and shall monitor such monthly levels and require justification to substantiate any departure from an approved level. The Commission must disapprove the issuance of debt obligations if the issuance would impede the purposes of the financial plan or be inconsistent with the financial plan or the Act; debt limits would be exceeded; the ability of overlapping subdivisions to issue un-voted faith and credit debt obligations would be impaired; and their issuance would be likely to lead to the reallocation of minimum levies of other political subdivisions. Expenditures may not be made contrary to an approved financial plan; expenditures may not be made contrary to a proposed financial plan after it is submitted to the Commission and before it is approved or disapproved; and if it is disapproved, no expenditures may be made which are inconsistent with the reasons given for disapproval.

The Act provides, among other requirements and provisions, that a municipality, county or township subject to such Act (i) must develop an effective financial accounting and reporting

system; (ii) must ensure that budgets, appropriations and expenditures are consistent with the purposes of the financial plan; (iii) may issue Local Government Fund Bonds, payable solely from such a municipal corporation's, county's or township's share of the local government fund pursuant to restrictions imposed by such Act; and (iv) may issue current revenue notes and advanced tax payment notes pursuant to the authorization and subject to the restrictions of such Act.

Unlike the Federal Bankruptcy Code which allows debtors to seek an adjustment or reduction of their debts without the consent of their creditors, the Act does not allow for such adjustment or reductions without the consent of the holders of the debt obligations.

Tax Considerations

With regard to tax considerations with respect to the Series 2016 Bonds, see "TAX MATTERS" herein.

Ratings

With regard to the risk involved in a lowering of the City's bond rating, see "RATINGS" herein.

UNDERWRITING

Morgan Stanley, as the Representative of the Underwriters, has agreed, subject to certain conditions, to purchase the Series 2016 Bonds from the City at an aggregate purchase price of $37,517,607.04 (the principal amount of the Series 2016 Bonds ($32,390,000), plus $5,384,899.90 of original issue premium, less $257,292.86 Underwriter's discount). The Bond Purchase Agreement between the Representative and the City, with respect to the Series 2016 Bonds, provides that the Underwriters will purchase all of the applicable Series 2016 Bonds if any are purchased and that the purchase is subject to certain conditions.

The Underwriters are purchasing the Series 2016 Bonds for purposes of resale. The Underwriters have furnished the information in this Official Statement pertaining to the public offering prices of the Series 2016 Bonds and has participated in the preparation of portions of this Official Statement. The public offering prices of the Series 2016 Bonds may be changed from time to time by the Underwriters, and the Underwriters may offer and sell the Series 2016 Bonds to certain dealers (including dealers depositing into investment trusts) and others at prices lower than the public offering prices shown on the inside cover hereof.

The Underwriters and their respective affiliates are full-service financial institutions engaged in various activities that may include securities trading, commercial and investment banking, municipal advisory, brokerage, and asset management. In the ordinary course of business, the Underwriters and their respective affiliates may actively trade debt and, if applicable, equity securities (or related derivative securities) and provide financial instruments (which may include bank loans, credit support or interest rate swaps). The Underwriters and their respective affiliates may engage in transactions for their own accounts involving the securities and instruments made the subject of this Official Statement or other offering of the City. The Underwriters and their respective affiliates may also communicate independent investment

recommendations, market color or trading ideas and publish independent research views in respect of this securities offering or other offerings of the City. The Underwriters do not make a market in credit default swaps with respect to municipal securities at this time but may do so in the future.

Morgan Stanley, parent company of Morgan Stanley & Co. LLC, an Underwriter of the Series 2016 Bonds, has entered into a retail distribution arrangement with its affiliate Morgan Stanley Smith Barney LLC. As part of the distribution arrangement, Morgan Stanley & Co. LLC may distribute municipal securities to retail investors through the financial advisor network of Morgan Stanley Smith Barney LLC. As part of this arrangement, Morgan Stanley & Co. LLC may compensate Morgan Stanley Smith Barney LLC for its selling efforts with respect to the Series 2016 Bonds.

RATINGS

Moody's Investors Service, Inc. ("Moody's") and Standard & Poor's Rating Services, a division of The McGraw-Hill Companies ("S&P"), have assigned long-term ratings of "Aa3" and "A+," respectively, to the Series 2016 Bonds. Such ratings reflect only the views of such organizations and any desired explanation of the significance of such ratings should be obtained from the rating agency furnishing the same, at the following addresses: Moody's Investors Service, 7 World Trade Center at 250 Greenwich Street, New York, New York 10007; Standard & Poor's Ratings Services, 25 Broadway, New York, New York 10004.

No application for a rating has been made to any other rating service for the purpose of obtaining an additional rating on the Series 2016 Bonds. A rating reflects only the views of the rating agency, and any explanations of the meaning or significance of those ratings may only be obtained from the respective rating agencies. There is no assurance that those ratings when assigned will continue for any given period of time or that they will not be revised downward or withdrawn entirely by either or both of the rating agencies, if, in their judgment, circumstances so warrant. Any downward revision or withdrawal of those ratings may have an adverse effect on the market price of the Series 2016 Bonds. The City and the Underwriters have undertaken no responsibility either to bring to the attention of the holders of the Series 2016 Bonds any proposed change in or withdrawal of the ratings or to oppose any revision or withdrawal. A securities rating is not a recommendation to buy, sell or hold securities and may be subject to revision or withdrawal at any time.

CONTINUING DISCLOSURE AGREEMENT

The City has agreed, for the benefit of the holders and beneficial owners from time to time of the Series 2016 Bonds, in accordance with SEC Rule 15c2-12 (the "Rule"), to provide or cause to be provided to the Municipal Securities Rulemaking Board such annual financial information and operating data, audited financial statements and notices of the occurrence of certain events in such manner as may be required for purposes of paragraph (b)(5)(i) of the Rule (the "Continuing Disclosure Agreement"). See "APPENDIX D – FORM OF CONTINUING DISCLOSURE AGREEMENT." The performance by the City of the Continuing Disclosure Agreement will be subject to the annual appropriation by the City of any funds that may be necessary to perform it. The Continuing Disclosure Agreement will remain in effect only for

such period that the Series 2016 Bonds are outstanding in accordance with their terms and the City remains an Obligated Person with respect to the Series 2016 Bonds within the meaning of the Rule.

The City believes that it has complied in all material respects with its previous continuing disclosure undertakings under the Rule during the past five years, including the most recent continuing disclosure filing posted to EMMA on March 1, 2016, but the City notes the following instances of noncompliance, including: (i) it did not file or did not file on a timely basis event notices relating to certain changes to ratings assigned to the insurers of insured bonds or to the underlying ratings, (ii) it filed certain annual reports on June 30 of each year, which at times was one or two days later than the 180th day following the end of the fiscal year of the City, (iii) it filed certain annual reports on July 30 or July 31 of each year, which at times was one or two days later than the 210th day following the end of the fiscal year of the City, (iv) in 2012 it mistakenly uploaded the incorrect file for its parking revenue bonds 2011 annual information filing but corrected that filing as soon as the mistake was brought to its attention, (v) it did not file a statement every year that Continental/United Airlines' annual reports were available on the SEC's EDGAR website, (vi) certain annual reports filed by the City were not properly matched with all required CUSIP numbers, (vii) it amended one page on its 2013 Airport Annual Filing to add a column of information regarding enplaned passengers that was mistakenly excluded on the original version, (viii) it mistakenly omitted the required "CPP Net Gain of CEI Meters" table from its CPP Annual Filings for Fiscal Years 2010-2013, and (ix) it mistakenly omitted certain required information from the "Debt Service Requirements and Pledged Taxes" table in filings related to the Income Tax Receipts Obligations and the "Total Restricted and Unrestricted Income Tax Receipts" table in filings related to the Series 2008 Subordinate Lien Unrestricted Income Tax Bonds for Fiscal Years 2010-2013, which omissions were corrected in the City's 2014 Annual Filings for those bonds. All necessary notices and omitted information have been filed prior to the date of this Official Statement. The foregoing description of instances of non-compliance by the City with continuing disclosure undertakings should not be construed as an acknowledgment that any such instance was material. The City has reviewed the current requirements of the Rule and adopted procedures to ensure full compliance with the Rule.

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CONCLUDING STATEMENT

This Official Statement has been duly prepared and delivered by the City and executed for and on behalf of the City by its Director of Finance and Director of Public Utilities.

To the extent that any statements made in this Official Statement involve matters of opinion, forecasts or estimates, whether or not expressly stated to be such, they are made as such and not as representations of fact or certainty, and no representation is made that any of such statements have been or will be realized. Information in this Official Statement has been derived by the City from official and other sources and is believed by the City to be accurate and reliable. Information other than that obtained from official records of the City has not been independently confirmed or verified by the City and its accuracy is not guaranteed.

The summaries and descriptions of provisions of the Trust Agreement and all references to other materials not purporting to be quoted in full are qualified in their entirety by reference to the complete provisions of the documents and other materials summarized or described.

The agreement of the City with the owners of the Series 2016 Bonds is fully set forth in the Trust Agreement. Neither this Official Statement nor any statement that may have been or that may be made orally or in writing is to be construed as, or as part of, a contract with the original purchasers or subsequent holders of the Series 2016 Bonds.

CITY OF CLEVELAND, OHIO

BY: /s/ Sharon A. Dumas
Director of Finance

BY: /s/ Robert L. Davis
Director of Public Utilities

APPENDIX A

ECONOMIC AND DEMOGRAPHIC INFORMATION CONCERNING THE CITY OF CLEVELAND

In the 2010 Census classifications, the City was in the Cleveland-Elyria-Mentor Metropolitan Statistical Area ("MSA"), which consists of Cuyahoga (the "County"), Geauga, Lake, Lorain and Medina counties. In 2010, the MSA had a population of 2,077,240, ranking it the 28th largest metropolitan area (out of 366) in the United States. The City was also in the Cleveland-Lorain-Elyria Primary Metropolitan Statistical Area ("PMSA"), which consisted of the counties of Ashtabula, Cuyahoga, Geauga, Lake, Lorain and Medina and the Cleveland-Akron Consolidated Metropolitan Statistical Area ("CMSA"), which consisted of eight northeast Ohio counties until June of 2003 when the U.S. Census Bureau ceased using the PMSA and CMSA distinctions.

Population

Set forth below are population statistics from the U.S. Bureau of the Census for the City, the County, the PMSA, the CMSA and the State for each decade from 1970 to 2000 and for the City, the County, the PMSA and the State for 2010 and estimated for 2014.

Year	City	County	PMSA	CMSA	State
1970	750,973	1,721,330	2,418,809	2,999,811	10,652,017
1980	573,822	1,498,400	2,277,949	2,834,412	10,797,630
1990	505,616	1,412,140	2,202,069	2,859,644	10,847,115
2000	478,403	1,393,978	2,250,871	2,945,831	11,353,140
2010	396,815	1,280,122	2,077,240[a]	N/A	11,536,504
2014-est.	392,114	1,267,513	2,063,598[a]	N/A	11,594,163
2015-est.	---[b]	---[b]	---[b]	N/A	11,613,423

[a] Indicates population for the MSA.
[b] Data not available.

Source: U.S. Bureau of the Census.

Employment

The following table compares estimated employment and unemployment statistics (annual averages) for the City, the County and the MSA, including comparisons with unemployment rates for the State and the United States.

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Employment Statistics[a]

Year	Employed City	Employed County	Employed MSA	Unemployed City	Unemployed County	Unemployed MSA	Unemployment Rate City	Unemployment Rate County	Unemployment Rate MSA	Unemployment Rate Ohio	Unemployment Rate U.S.
2006	173,200	619,300	1,033,500	12,000	33,100	53,100	6.5%	5.1%	4.9%	5.4%	4.6%
2007	172,400	616,500	1,035,100	13,400	37,200	59,700	7.2	5.7	5.5	5.6	4.6
2008	168,900	604,200	1,020,200	14,300	40,000	64,900	7.8	6.2	6.0	6.4	5.8
2009	159,800	571,100	972,500	18,900	54,500	91,200	10.6	8.7	8.6	10.3	9.3
2010	148,800	577,900	964,100	17,900	53,800	87,900	10.8	8.5	8.4	10.3	9.6
2011	148,200	577,000	965,800	16,000	47,600	76,800	9.7	7.6	7.4	8.8	8.9
2012	148,700	579,100	970,900	14,200	42,400	68,400	8.7	6.8	6.6	7.4	8.1
2013	148,200	577,500	970,600	14,700	44,300	72,000	9.0	7.1	6.9	7.5	7.4
2014	150,000	584,400	982,000	13,100	39,900	64,500	8.0	6.4	6.2	5.7	6.2
2015[b]	151,100	588,900	989,600	11,200	33,800	54,900	6.9	5.4	5.3	4.9	5.3
2016[c]	147,000	572,300	964,300	9,800	29,500	49,200	6.2	4.9	4.9	6.0	6.1

[a] Rounded to the nearest hundred.
[b] As of December 2015.
[c] As of January 2016.
Source: Ohio Department of Job & Family Services, Labor Market Information Division. City and County estimates are NOT seasonally adjusted.

The following table indicates the distribution of employee classifications in the MSA for the years 2009 through 2015:

	Distribution of Employees by Sector (Amounts in 000's)						
	2009	**2010**	**2011**	**2012**	**2013**	**2014**	**2015**
Goods Producing Industries							
Mining, Logging, Construction	33.2	31.6	32.8	33.4	34.1	35.2	35.8
Primary Metal	8.2	8.1	8.2	8.7	8.8	8.7	8.3
Fabricated Metal	25.9	25.7	27.2	27.7	27.5	27.9	27.3
Transportation Equipment	10.9	11.2	11.8	12.2	12.6	12.5	12.8
Other	73.5	71.5	72.9	74.4	74.9	74.9	76.4
Total Goods Producing Industries	**151.7**	**148.1**	**152.9**	**156.4**	**157.9**	**159.2**	**160.6**
Service Producing Industries							
Transportation & Public Utilities	30.6	29.3	29.2	29.8	30.3	30.2	30.1
Wholesale Trade	48.3	46.6	47.9	49.6	49.4	50.2	51.4
Retail Trade	101.9	100.2	100.2	101.2	101.8	101.3	104.5
Finance, Insurance & Real Estate	65.8	64.6	63.4	62.9	62.8	64.9	68.5
Health Services	153.1	156.0	157.3	161.6	164.0	163.0	169.4
Other Services	308.9	307.6	315.0	321.7	327.3	334.2	341.9
Federal Government	19.0	20.1	19.0	18.5	18.3	18.3	18.7
State Government	7.5	7.5	7.2	7.0	6.9	7.3	7.7
Local Government	114.1	111.1	109.1	107.9	107.6	108.2	110.7
Total Service Producing Industries	**849.2**	**843.0**	**848.3**	**860.2**	**868.4**	**877.6**	**902.9**
Total	**1,000.9**	**991.1**	**1,001.2**	**1,016.6**	**1,026.3**	**1,036.7**	**1,063.5**
Goods Producing Percentage	**15.2%**	**14.9%**	**15.3%**	**15.4%**	**15.4%**	**15.4%**	**15.1%**
Service Producing Percentage	**84.8%**	**85.1%**	**84.7%**	**84.6%**	**84.6%**	**84.6%**	**84.9%**

Source: Ohio Department of Job & Family Services, Labor Market Information Division.

The following table indicates the per capita income for the County, the MSA, the State, and the United States for the years 2004 through 2014.

Per Capita Income

Year	County	MSA	Ohio	U.S.
2004	$35,969	$34,853	$31,776	$34,300
2005	37,235	36,161	32,758	35,888
2006	39,621	38,350	34,422	38,127
2007	41,012	39,751	36,199	39,804
2008	42,432	40,827	36,399	40,873
2009	40,369	39,192	35,527	39,379
2010	41,325	40,114	36,199	40,144
2011	44,216	42,964	38,631	42,332
2012	46,395	44,937	40,230	44,200
2013	47,294	45,747	41,049	44,765
2014	48,521	46,960	42,236	46,049

Source: U.S. Bureau of Economic Analysis.

The income per household in the City and the County is estimated to be distributed as set forth in the following table:

	City		County	
Income and Benefits(a)	# Households	% Households	# Households	% Households
Less than $10,000	33,753	20.3%	58,103	10.9%
$10,000 to $14,999	17,676	10.6%	35,347	6.6%
$15,000 to $24,999	28,946	17.4%	67,356	12.6%
$25,000 to $34,999	20,934	12.6%	59,232	11.1%
$35,000 to $49,000	22,364	13.4%	73,234	13.7%
$50,000 to $74,999	21,377	12.8%	89,222	16.7%
$75,000 to $99,999	10,220	6.1%	56,239	10.5%
$100,000 to $149,999	7,806	4.7%	56,481	10.6%
$150,000 to $199,999	1,792	1.1%	19,176	3.6%
$200,000 or more	1,782	1.1%	20,331	3.8%

(a) In 2014 inflation-adjusted dollars.
Source: U.S. Census Bureau Selected Economic Characteristics in the United States 2010-2014.

The U.S. Census Bureau also estimates that 35.4% of the people in the City and 18.8% of the people in the County have incomes that fall below the poverty level.

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Corporate Headquarters

Listed below are 12 corporations (representing 12 different industries) among the Fortune 1000 largest corporations of 2015 (ranked by worldwide revenues) that have headquarters in the County.

Corporations Headquartered in County
Among Fortune's Top 1000
Within the 1,000 Largest U.S. Corporations Ranked by Revenues[(a)]

Rank	Company	Major Products
153	Progressive	Insurance
230	Parker Hannifin Corp	Hydraulic Components
266	The Sherwin Williams Company	Paints & Chemicals
365	TravelCenters of America	National Travel Center Chain
547	Aleris International	Metals
560	Cliffs Natural Resources	Mining, Crude Oil Production
592	KeyCorp	Financial Services
792	Lincoln Electric Holdings	Industrial Equipment
805	Hyster-Yale Materials Handling	Industrial Machinery
879	Medical Mutual	Health Care Insurance
880	Applied Industrial Technologies Inc.	Industrial Components
904	TransDigm Group, Inc.	Aircraft Components

(a) Source: 2015 Fortune Directory of the Largest U.S. Corporations.

Home Values, Housing Units and Home Sales

The 2013 estimated median value of owner-occupied homes in the City, the County and the MSA were $76,700, $125,700, and $141,700, respectively, compared with $130,800 in the State and $176,700 in the United States. The number of housing units within the City for the nine-year period from 2005 to 2013 decreased by 3.73%, from 214,821 to 206,803, compared with a decrease of 0.49% for the County, from 620,564 to 617,545. (All figures in this paragraph are derived from the U.S. Bureau of the Census.) In recent years, as part of the City's community development initiatives, the City has provided financial assistance to nonprofit and for-profit developers to stimulate new housing construction in the City and made additional efforts to address increased foreclosures. See **"Housing And Neighborhood Development"** below.

Listed below are sale price summary statistics for the City and the County, respectively.

Housing Sales Statistics 2010-2014

	City		County	
Year	**Number of Sales**	**Average Sales Price**	**Number of Sales**	**Average Sales Price**
2010	2,005	$60,398	9,265	$142,804
2011	2,307	54,638	10,132	128,300
2012	2,432	57,842	11,915	133,129
2013	2,809	59,737	13,674	139,950
2014	3,761	54,548	16,021	129,634

Source: Cuyahoga County, Ohio

Building Permits

The following table shows information concerning the filing with the County of building permits for construction and demolition and the net assessed valuation (not the actual construction or demolition cost) of those building permits as determined by the County, for the City by class:

	2010		2011		2012		2013		2014	
	# of Permits	Assessed Value[(a)]	# of Permits	Assessed Value[(a)]	# of Permits	Assessed Value[(a)]	# of Permits	Assessed Value[(a)]	# of Permits	Assessed Value[(a)]
Commercial	880	$74,221	781	$192,044	698	$63,958	786	$157,917	758	$153,627
Industrial	103	5,206	84	3,428	106	4,470	154	8,858	105	4,403
Exempt	143	10,868	150	6,300	393	0	381	713	318	427
Public Utility	1	0	1	0	2	0	0	0	0	0
Residential	3,417	10,183	3,253	7,394	3,616	6,838	4,330	13,622	4,907	7,853
Total:	4,544	$100,478	4,269	$209,166	4,815	$75,266	5,651	$181,110	6,088	$166,310

[(a)] In thousands.

Source: Cuyahoga County, Ohio

Utilities

The MSA is well served with adequate and reliable water and energy resources. The principal source of water in the MSA is Lake Erie, the twelfth largest lake in the world. The principal provider of potable water in the County is the City's Division of Water. A large amount of fresh water is available to the area for its foreseeable needs. The two principal providers of electric energy in the MSA are the City's Cleveland Public Power and the Cleveland Electric Illuminating Company, a wholly-owned electric utility operating as a subsidiary of FirstEnergy Corp. Sewer services in the MSA are provided by the Northeast Ohio Regional Sewer District.

Transportation

The City is a major regional center for economic and commercial activity and is served by diversified transportation facilities. There is immediate access to six United States' highways and seven interstate highways. The Cleveland Innerbelt Modernization Plan is focused on improving safety, reducing congestion and traffic delays, and modernizing interstate travel along

I-71, I-77 and I-90 through downtown Cleveland. This investment by the State of Ohio will rehabilitate and reconstruct the Innerbelt Freeway system – including construction of two new bridges to carry I-90 traffic – and address operational, design, safety and access shortcomings that severely impact the ability of the Innerbelt Freeway system to meet the transportation needs of Northeast Ohio. The Ohio Department of Transportation ("ODOT") commenced construction of a new Innerbelt Bridge that was completed in the fall of 2013. Demolition of the old bridge has been completed. ODOT plans to build a new eastbound bridge in its place. Upon completion of the proposed second bridge, there will be a two bridge system to accommodate the more than 138,000 vehicles that cross the bridge each day. Total costs for the construction of the two bridges are estimated to be $560.4 million.

In the fall of 2014, ODOT commenced construction on its Opportunity Corridor Project, a three-mile, approximately $331 million road project that is designed to improve the transportation system and support planned economic development within the City in the areas between I-490/I-77 and University Circle. The Opportunity Corridor encompasses nearly 1,000 acres on the City's southeast side and is anchored by University Circle and the Cleveland Clinic. In addition to transportation benefits, it is anticipated that the Opportunity Corridor Project will bring new economic development and new jobs to the community. The Opportunity Corridor Project supports an economic development plan of the City and Greater Cleveland Partnership for the area through enhanced mobility, direct access to freeways and the University Circle area, new frontage for potential development, improved visibility and improved multi-modal access. The Opportunity Corridor Project is being funded in part by bonds issued by the Ohio Turnpike and Infrastructure Commission. The Opportunity Corridor Project is divided into three stages: the first stage commenced in the fall of 2014; the second stage is expected to commence in the spring of 2016; and the third stage is expected to commence in 2017.

Cleveland Hopkins International Airport ("Hopkins") is the primary commercial service airport for northeastern Ohio. Hopkins is situated approximately ten miles southwest of the downtown area and is accessible via highway from multiple directions. In addition, heavy rail rapid transit service to Hopkins is also provided from downtown by the Greater Cleveland Regional Transit Authority ("RTA"). Hopkins is owned by the City and operated by the City's Department of Port Control. As of November 2015, Hopkins was served by eight major and national carriers, 18 regional and commuter airlines, one foreign-flag airline, and three all-cargo airlines.

Hopkins' approximately 935,000 square foot terminal complex includes a multi-level terminal building and four concourse buildings which support 63 jet gates and 33 commuter aircraft parking positions. Due to the announcement by United Airlines ("UAL") in early February 2014 that UAL will significantly reduce service at Hopkins, Concourse D was closed on June 4, 2014 and all of UAL's operations were consolidated in Concourse C. There are three runways at Hopkins. On October 17, 2011 airport officials broke ground on a new FAA-funded Air Traffic Control Tower and Terminal Radar Approach Control Facility, which will replace the existing tower, originally built in 1988. The $69 million tower will be equipped with the latest technology—allowing the airport to seamlessly integrate with the FAA's move toward a Next-Generation (or "NextGen") Air Transportation System. The new tower was completed in 2015. The tower is one element in a coordinated program of airfield safety improvements.

Burke Lakefront Airport ("Burke"), which also is City-owned and operated, has two parallel runways on 480 acres adjacent to downtown Cleveland. Burke recorded 66,862 terminal operations in 2014, the majority by air taxi operators serving business activity in downtown Cleveland, and the remainder almost entirely by corporate and private general aviation aircraft operators.

The Port of Cleveland (the "Port") is an interlake and international shipping center located on the shores of Lake Erie and the Cuyahoga River. The Port primarily handles steel and bulk commodities and is a heavy lift port which is favorable for such items as automobile manufacturing equipment, presses and raw and finished steel and factory components. The Port brings more than $572 million into the Greater Cleveland economy annually through payroll for the approximately 17,800 jobs dependent upon maritime activities. Approximately $900 million in merchandise and material is shipped to and from the Port each year. The Port averages 13 million tons of cargo per year.

Norfolk Southern and CSX chose the City as their gateway to the Northeast and Midwest after the respective railroads restructured the rail systems following the acquisition of Conrail.

The City is also served by the RTA. The RTA owns and operates a public mass transit system, providing transportation to a 457 square mile service area which includes 59 municipalities, one of which is the City.

Financial Services

The City is a regional financial center and is the headquarters for the Fourth District Federal Reserve Bank, serving Ohio, the western portion of Pennsylvania and portions of Kentucky and West Virginia.

Education

Within the County are 13 public and private two-year and four-year colleges and universities, including, among others, Case Western Reserve University, John Carroll University, Cleveland State University, Cuyahoga Community College, Baldwin Wallace University, Notre Dame College, Ursuline College, the Cleveland Institute of Music, and the Cleveland Institute of Art.

Health Care

There are over 20 hospitals, including acute care and private psychiatric hospitals, in the County. Among these institutions are the Cleveland Clinic Foundation, University Hospitals Health System (affiliated with Case Western Reserve University School of Medicine), and the MetroHealth System, all headquartered in the City.

Recreation and Entertainment

The City is noted for its many cultural institutions, including the internationally acclaimed Cleveland Orchestra and the Cleveland Museum of Art, the latter of which completed a $350 million renovation and expansion project in late 2013. The project included refurbishing

historic galleries and adding 35,000 square feet of gallery space, which opened in stages. Theaters and entertainment centers include Playhouse Square (a complex of six theaters, currently with seating for over 10,000), Public Auditorium, Karamu House, and Severance Hall. Other cultural institutions include the Cleveland Play House, Great Lakes Theater, Cleveland Public Theater, Apollo's Fire (the Cleveland Baroque Orchestra), Verb Ballet, and Dance Cleveland.

The Rock and Roll Hall of Fame and Museum, a 150,000 square foot facility located at North Coast Harbor, opened in 1995 and has attracted more than 10 million visitors to date.

The Great Lakes Center for Science and Technology, located on North Coast Harbor next to the Rock and Roll Hall of Fame and Museum, opened in July 1996. With more than 400 hands-on exhibits and a six-story Omnimax theater, the Center gives visitors the chance to explore science, environment and technology and their relationships to the Great Lakes.

Other museums include Museum of Contemporary Art Cleveland, Cleveland Botanical Gardens, Cleveland Museum of Natural History, Dunham Tavern Museum, Cleveland Children's Museum and Western Reserve Historical Society. Recreational facilities in the County include the 18,800-acre Metropolitan Park System, Cleveland Metroparks Zoo, Wade Park, Rockefeller Park, Cultural Gardens, Lakefront State Park and, outside the City, the Cuyahoga Valley National Park. The Cleveland Metroparks Zoo opened its new $2 million carousel and play area in May of 2014 and opened Stillwater Place, its new $2.3 million 10,000 square foot event center, in May of 2015. The Crawford Auto Aviation Museum, part of the Western Reserve Historical Society, reopened in January 2013 after a $4 million renovation.

Professional sports are available to area residents at various facilities located in the City's downtown. FirstEnergy Stadium (previously known as Cleveland Browns Stadium), located on the lakefront in downtown Cleveland on the same site as the former Cleveland Municipal Stadium, was completed in August 1999 and is the home of the National Football League's Cleveland Browns. The facility consists of an open-air stadium with approximately 68,000 seats. In addition to NFL football, the facility is suitable for major league soccer and open-air concerts.

The Gateway Sports Complex, located in the central business district of the City, includes Progressive Field (home of the American League's Cleveland Indians), Quicken Loans Arena (home of the National Basketball Association's Cleveland Cavaliers, the American Hockey League's Lake Erie Monsters, and the Arena Football League's Cleveland Gladiators), and a parking garage.

Progressive Field, which opened in April 1994, is an open air, natural turf baseball stadium with a current seating capacity for approximately 36,000 people. Quicken Loans Arena, which opened in October 1994, is a multi-functional, indoor facility for sporting and entertainment events and seats approximately 20,000 people. The Gateway common areas consist of approximately 13 acres and include Gateway Square, an area for outdoor entertainment and activities.

The City hosted the Rock and Roll Hall of Fame inductions in April of 2015 and hosted the 2014 Gay Games which featured more than 35 sports, band and choral competitions and community and cultural events. More than 10,000 participants from more than 65 countries

attended. In addition, the City hosted the National Senior Games in July 2013. The Games attracted over 10,000 registered athletes and had a total attendance of approximately 65,000.

The Republican National Committee has selected the City to host the 2016 Republican National Convention in July of 2016. The Convention is expected to draw 50,000 attendees.

Downtown and other Economic Development

The City continues to focus its strategies on identified clusters of regional strength including health technology, paints and coatings, lighting and electrical, information technology, automotive and automotive related, aerospace, and banking and finance. Retention efforts include two visitation programs: the Cleveland Industrial Retention Initiative for all manufacturers, distribution and related supply chain businesses and a specific downtown stakeholders' visitation program through Downtown Cleveland Alliance. In 2015, Oatey Co., a long time Cleveland manufacturer of plumbing products with locations around the world, announced they would build a new 52,000 square foot headquarters building in Cleveland's Emerald Corporate Park.

One of the central focus areas is the Health Tech Corridor, a three-block wide transit-oriented development running from Cleveland State University to University Circle, encompassing 1,600 acres. The Health Tech Corridor includes three colleges and universities and three major hospitals. In 2014, RTA completed a new transit station in University Circle on Cedar Avenue and completed construction of a new transit station on Mayfield Road in August 2015. These stations complement RTA's Health Line transportation system by connecting it to the City's heavy rail system, with direct access to the airport, and providing better connectivity to the City's second "downtown."

Hemmingway Development has agreed to purchase a ten acre brownfield site that the City assembled and cleaned-up and plans to develop another Health Technology Campus to be anchored by a 40,000 square foot University Hospitals clinic and a 50,000 square foot speculative tech center, both expected to break ground in 2016. The Frost Building will become home to an 15,000 square foot co-working space. The space is expected to be home to over 400 member businesses within two years, creating a new location for entrepreneurs. The City and its partners received a grant of over $700,000 from the Economic Development Administration to create the first commercially available 100 Gigabit network in the United States in the Health Tech Corridor. The network is expected to be built out by the summer of 2016. The City of Cleveland also started clean-up activities on another 2.5 acre site to keep pace with demand in the Health Tech Corridor and is working with a potential buyer.

University Circle has seen a great deal of new investment. The Cleveland Clinic commenced construction of a $276 million 377,000 square foot cancer hospital. It will also complete a new $36 million, 3,000 space employee parking garage with over 12,000 square feet of retail in the area. First Interstate Properties announced a new 28-story residential tower called One University Circle that is expected to begin construction in 2016. Centric, a residential and office complex by Midwest Development Partners and the Coral Company, plans to start construction in 2016. The two projects represent a total investment of over $230 million dollars.

The proposed UC3 Development, a multi-block mixed use project with over 700 apartments and 150,000 square feet of retail, is expected to break ground in 2016.

In 2013, the Cleveland Clinic demolished a block of buildings across from its campus to make way for a new medical school in partnership with Case Western Reserve University. Originally planned as a 165,000 square foot medical education building, the project has now been expanded to become a Health Technology Campus of in excess of 200,000 square feet and a cost in excess of $200 million. The project broke ground on October 1, 2015.

In 2015, Integrated CC LLC, as the developer, commenced construction of a 276-room all service Holiday Inn Hotel located on the Cleveland Clinic Campus. The hotel will have a full service restaurant and lounge. The hotel is designed to serve the public and the families of overnight patients at the Cleveland Clinic. The $45 million project was financed with bonds issued by the Cleveland-Cuyahoga County Port Authority. Construction is expected to be completed and the hotel is expected to open for business in June of 2016.

The Uptown neighborhood has seen a great deal of investment including the new $27.2 million Museum of Contemporary Art Cleveland which opened in 2012; the 153-room, $27 million Courtyard by Marriot which opened in 2013; and the 158,000 square foot, $44.5 million Phase I of the Uptown project including restaurants, retail and 102 market rate apartments which opened in 2012. The $21 million Uptown Phase II, which includes an additional 43 apartments, dormitories for students of the nearby Cleveland Institute of Art, and additional retail space, opened in August 2014.

The Cleveland Institute of Art completed the $30 million Phase I of its eight-year expansion and renovation project that included the McCullough Center for the Visual Arts in 2013. In 2015, it completed construction on the $33.5 million Phase II. Phase II included a 79,000 square foot addition to the McCullough Center. The expansion is named the George Gund Building and includes a new 300-seat theater for the Cinematheque and a new art gallery.

The County's $465 million Convention Center and Global Center for Health Innovation project in downtown Cleveland was completed in 2013. The project included an integrated facility for (i) exhibition space and showrooms for medical devices and equipment and related functions (the Global Center for Health Innovation) and (ii) exhibition, tradeshow and conference facilities, meeting rooms and related functions. A $260 million Hilton convention center hotel, funded through bonds issued by the County, is on track to open in mid-2016 in time for the Republican National Convention. The 30-story, 650-room Hilton is being built on the site of the former County administration building, which was razed in early 2014. In preparation for the Republican National Convention, several other hotels were either recently completed or will be completed in time for the convention. They include the 481-room Westin, which opened in the spring of 2014; the 150-room Metropolitan, which opened in the fall of 2014; the 120-room Kimpton Schofield which opened in March of 2016, and the 180-room Drury Plaza Hotel, expected to open in the summer of 2016.

Geis Companies completed and opened the Metropolitan, 120 luxury apartments, as well as meeting and banquet spaces in the Ameritrust complex at Ninth Street and Euclid Avenue. As part of the $240 million renovation, Geis also renovated the Cleveland Trust Rotunda, built in

1906. The same complex is home to a second renovated office building, which includes a Heinen's grocery store, other retail and office space, and residential units. The grocery store opened in late February of 2015. A new 222,000 square foot office structure was constructed and leased to the County to serve as the new County Administration building.

Stark Enterprises purchased two buildings and a surface parking lot and has announced plans to create a 2.6 acre site that will become "nuCLEus", a $250 million development that will include 500 residential units, 200,000 square feet of office space, 130,000 square feet of retail space, two new parking garages and a new hotel. The planned hotel is designed to be a "bridge" across the two towers.

The $275 million Cleveland Flats East Development Project Phase I was completed in 2013. This project consisted of an approximately 476,000 square foot, 18-story office tower, an approximately 550-space parking garage, a 150-room Aloft hotel, and approximately 31,000 square feet of restaurant and retail space. The office tower is nearly 90% leased and restaurant and retail spaces are 100% leased. The $146 million Phase II project is underway and includes 243 apartments and 80,000 square feet of ground floor restaurant and retail as well as 48,000 square feet of entertainment space. A new 1,200 linear foot river walk has been completed and provides access to the riverfront. The Phase II grand opening was held in October 2015. The Metroparks announced plans to open a water taxi service that will connect both sides of the river and a lakefront beach at Wendy Park, furthering the tourist draw to this area.

Steelyard Commons, the $120 million retail redevelopment project on the site of a former steel mill, opened in early 2007. Steelyard Commons has a 97% occupancy rate, maintaining most of its original tenants including Home Depot, Marshalls, Target, Wal-Mart, Old Navy and restaurants Steak 'n Shake, Applebee's, I-Hop, Chipotle, and Burger King. Additional restaurants opened in 2012 followed by an Aldi grocery store in 2013. Construction on the $24.2 million, 90,000 square foot Phase II has been completed and is anchored by Burlington Coat Factory, which opened in 2014. Several outlot buildings have been constructed and occupied as well.

The Ohio City neighborhood continues to grow and be a destination for residents and tourists. The neighborhood includes the City-owned West Side Market, the oldest continually operating market in the country, which recently celebrated its 100th anniversary. The West 25th Street Lofts Project is under construction to create 83 loft style apartments and some rooftop penthouse units, as well as 9,100 square feet of commercial space. It is expected to be completed in late 2016. Total project cost is over $60 million. Abode Modern Lifestyle Developers has assembled 4 acres of land with hopes to construct a $40 to $50 million new mixed use development. The area has seen more than $15 million in new investment to redevelop over 198,000 square feet of vacant or underutilized mixed use space, including historic renovations and some new construction.

The Detroit Shoreway neighborhood adjacent to Ohio City is also enjoying an arts and entertainment resurgence. The Cleveland Public Theater and Capital Theater were joined in 2015 by a new performing arts venue, the Near West Theatre. The Templin Bradley, a 30-unit mixed income apartment building with both market rate and low-income units, opened in June of 2015.

The City's manufacturing sector continues to be a large employer in the region. Manitowoc's Commercial Oven Division purchased local company Cleveland Range and determined it would close its Fort Wayne, Indiana location, and expand the Cleveland site to be the Division headquarters.

In late 2008, the City established a new program called the Vacant Property Initiative, providing a combination of loans and forgivable loans to companies that redevelop a vacant building or lot to create and retain jobs in the City. The goal of the program is to keep existing companies in the City and attract new companies by leveling the playing field with suburban locations. The City funded 75 companies by providing $28.8 million in City funds that leveraged $650 million for projects totaling over $686 million in new investment. The program created 4,397 jobs and retained another 2,019 jobs. In addition to keeping and increasing payroll taxes, the program increases real estate taxes through the investment in these vacant properties.

Housing and Neighborhood Development

The City continues to implement strategies to position the City and its neighborhoods to capitalize on future opportunities. The City has identified areas (i) where significant needs must be addressed, (ii) where need and market potential overlap, (iii) where scattered site rehabilitation will be sustainable, and (iv) where the City can create new housing opportunities, particularly for very low income households. In these target areas, the City has focused market-building and stabilization efforts, specifically focused on demolition, housing renovation and land reutilization.

Housing

Cleveland's housing strategy embodies the core principles of creating healthy mixed-income neighborhoods of choice while preserving and expanding housing opportunities for low and moderate income households. The City has used its funds to develop homes and apartments for low income families, for seniors, and to continue the City's long term commitment to preserving the existing stock of HUD-assisted rental housing.

All housing development supported by City funding, including tax abatement, complies with the City's Green Building Standards, which were updated and re-issued in 2013. The standards further the City's commitment to both green building and the ongoing development of quality housing, particularly for low and moderate income families. The City's Green Building Standards incorporate National standards such as LEED and Enterprise Community Partners' Green Community Standards.

Overall, the downtown housing market has remained strong. The 97.8% occupancy rate has been maintained even with the addition of more units such as The 9 and The Residences at 1717. With the completion of new projects, the downtown population exceeds 13,000 residents. The Hanna Annex Building completed The Residences at Hanna in July 2013. In December 2013, the 102 apartments were fully occupied with a wait list in place. The total project cost was estimated to be $23 million. Another developer is working on several conversions into mixed use projects, which will add additional market rate rental units. The Langston's 318 market rate apartments in nine buildings in the Campus District area around Cleveland State University were

completed in August 2013. The $50 million Campus Village project will also have amenities including meeting rooms and a cyber café at the corner of East 23rd and Chester Avenue.

Large scale housing projects completed in 2014 include Kingsbury Tower, the renovation of a 126-unit apartment building for low income families; Magnolia On Detroit, the renovation of a 84-unit low income apartment complex; and Morningstar Tower, the renovation of a 200-unit senior apartment building. Construction started on several more projects, including Templin Bradley, Innova, a 177-unit, market rate apartment project, and New Community Place, a 147-unit apartment complex for low-income persons.

Another City focus is the renovation of single family housing. These efforts resulted in the completion and sale of 158 homes. Renovated homes in the City supported by City funding or tax abatement had a median sale price of $118,425. City programs also assisted people with maintaining or repairing their homes. In 2014, over 318 families received home repair assistance, including code or safety issues as well as energy efficiency improvements.

To the City's goals of energy efficient and healthy homes, the City and a consortium led by Environmental Health Watch performed a Warm & Healthy Home Initiative to combine healthy home medical interventions with energy efficient home improvements. The initiative, funded by a $1,206,000 grant from HUD and concluded in 2014, completed 187 projects. Under its Energy Savers Program, the City also performed energy audits, and provided low-cost loans for energy efficiency measures for 143 moderate and middle-income houses. The City has partnered with experienced non-profit housing developers and service providers to create permanent supportive housing units for chronically homeless persons with disabilities. Through the Housing First Initiative, over 600 units of such housing have been developed since 2006. Housing First's building management staff and supportive services staff work together to serve residents providing the support they need to remain housed and achieve other life goals. The most recent project supported by the City Emerald Alliance VIII, broke ground in 2014 for 66 permanent supportive housing apartments in the Detroit Shoreway neighborhood.

Habitat for Humanity initiated an expansion of its affordable home ownership model by commencing a rehabilitation project in two neighborhoods in 2013. By the end of 2014, ten homes have been renovated.

Other initiatives for housing renovation were started in 2013, including the Slavic Village recovery, a unique partnership between Safeguard, Inc., Forest City Enterprises and Slavic Village Development. This effort plans to perform moderate rehabilitation of 50 homes in a targeted part of the Broadway neighborhood. By the end of 2014, 22 homes had been completed.

In 2014, the City continued efforts to assist people in avoiding foreclosure and staying in their homes. A partnership with the Cleveland/Cuyahoga County Office of Homeless Services, Cleveland Tenants Organizations, and other partners, continued to assist residents threatened with the loss of housing.

Demolition

The City has continued an aggressive pursuit of demolition or renovation of distressed properties that have proliferated as a result of foreclosures. The City renewed its code

enforcement partnership with community based development corporations. In 2014, the City razed 271 condemned structures at the cost of $1.55 million. Between 2006 and 2014, the City razed 7,875 condemned structures at a cost in excess of $48 million.

These demolitions position the City to be able to eliminate blight and strategically assemble land and to begin site preparation in anticipation of a stronger economy. The local strategies for the re-use of vacant land have been broad, have involved many partners and have been innovative. Those ongoing efforts have been particularly focused on urban gardens and agricultural programs. The City uses Community Development Block Grant funds to contract with The Ohio State University Extension Program to administer the Summer Sprout Community Garden Program, which now includes more than 43 acres at 197 gardens and 3,675 gardeners in all wards of the City. The gardens have donated 29,990 pounds of product to the local food banks and hunger programs.

Land Reuse

The City has assembled land, and has begun site preparation, in anticipation of a rebound in the housing market. Several sites across the City are, or have been, assembled to be ready for development. One of those sites is under construction for a 72-unit housing project, Trailside at Morgana Run. The first ten units are completed and 4 of them have been sold.

Over the past few years, the City has made vacant land reuse a priority – particularly urban agriculture and urban farming. Repurposing vacant urban land into productive community gardens and urban farms makes the City's neighborhoods more vibrant and sustainable. Specifically, urban agriculture is a means of addressing food deserts, promoting healthy eating, supporting principles of sustainability, and encouraging community engagement and participation.

The Re-Imagining Cleveland Pilot Program encourages the redevelopment of land-bank property as green space, urban agriculture and the expansion of resident's side yards. This program piloted new policies and procedures for vacant land management, including long-term leases and subsidized access to the City's water system. It has invested nearly $1.4 million in NSP-1 and NSP-2 funding to fund 63 greening and urban agriculture sites and 73 side yard expansions. The 5 year pilot program is expiring, but most of the uses will continue, and some of the tenants are buying their sites as their leases expire. One such project is a bio-cellar to grow crops such as mushrooms year round in the foundation of a partially demolished house.

The most significant large-scale urban farming initiative is the 28-acre Urban Agriculture Innovation Zone. City-owned vacant land in this area is leased for several urban agriculture programs, including a market farm incubator operated by The OSU Extension.

Brownfield Redevelopment

The City is committed to the elimination of vacant and condemned houses and commercial structures and has identified seven priority areas for potential economic growth. Hundreds of structures have been demolished in these areas to clear and assemble sites for redevelopment. These priority areas were selected because of their proximity to highways and existing infrastructure and the opportunity for economic growth.

Since 2007, the City's Industrial-Commercial Land Bank facilitated the following reclamation efforts:

- The former Tops grocery store at Superior and Lakeview was taken into the Industrial Commercial Land Bank as a deed in lieu of foreclosure action. The property was marketed to a local developer who has created the Shops at Garrett Square attracting a Save-a-Lot and the first Ohio location for Forman Mills. The City and the developer share in the profits from the sale or lease of the outlots. An AutoZone is under construction on an outlot. The project has created over 85 jobs to date.

- Completed the environmental remediation of the 5.6-acre Trinity Building, securing a No Further Action Letter ("NFA") in 2012. The United States Environmental Protection Agency ("EPA") and the State of Ohio Clean Ohio Assistance Fund have provided funding to complete the project. The site is being redeveloped for the new 24,000 square foot City of Cleveland kennel, which will be double the size of the current facility. The City has budgeted over $5 million for the project.

- The former General Motors Fisher Auto Body facility on Coit Road underwent years of environmental remediation and remained vacant. The City worked with Forest City Enterprises to complete the environmental evaluation of the site. An NFA was issued in 2013. Quasar, a bio-digester facility, has purchased five acres in the northern section of the 25 acre site and has completed the City's first biodigester, creating a sustainable asset available to many businesses to turn waste into power.

- Worked with a local Community Improvement Corporation to use beneficial re-use strategies to redevelop a 60 acre parcel of land that was the former coke plant owned by ArcelorMittal. The site was certified as a Job Ready Site by the State of Ohio in early 2014. Approximately 15 acres of the site were sold to the Ohio Department of Transportation for construction of a maintenance facility. The remaining property is currently being marketed, with several interested parties.

- Using $10 million in HUD 108 and Brownfield Economic Development Initiative funds, financed the construction of a cooperatively-owned greenhouse on almost 10 acres in Cleveland's Central neighborhood. The greenhouse will employ neighborhood residents and produce fresh produce for market and local food banks. The greenhouse opened in early 2013 and currently has over 20 employees.

- Secured over $3 million in funding through Clean Ohio for the assessment and remediation of the Miceli Dairy Products Company expansion site. The remediation of the site will enable Miceli to expand production on almost 25 acres of brownfield property. Miceli has completed Phase I of its expansion. The $20 million Phase II, expected to be completed by 2016 will allow the addition of new product lines and will include a bio-digester to turn the company's waste stream into energy.

- Conducted asbestos abatement at the former Warner Swasey facility to prepare the site for redevelopment. The abatement of the facility, located at the prime corner of East 55th and Carnegie, was funded by over $1.1 million in USEPA Revolving Loan

Funds through the County and the State. Asbestos abatement was completed in 2013, and the City is currently marketing the site.

- Ohio EPA approved an Urban Setting Designation Study for the City of Cleveland. As a result, costs to remediate property in the City are now significantly decreased.

- The City undertook a feasibility analysis of dredge material and slag to cap an old landfill to support a solar field. Clean Ohio funds and a USEPA-NREL program were used to complete the environmental and solar feasibility studies of the site. NREL completed their analysis in 2013. The City has solicited proposals to redevelopment of the site as a solar farm.

- Secured $3 million in Clean Ohio funding to support the remediation of the Flats Phase II site, enabling the development of waterfront entertainment and residential district.

- Moved forward with land acquisition in multiple areas of the City in order to consolidate parcels and create future greenfield development sites, enabling the City to better compete for relocations and expansions.

- Completed a multifaceted remediation project in North Broadway that involved the demolition of two large long-vacant industrial buildings and remediation of several properties using both State funds and federal EPA funds. The parcels will be redeveloped as Trailside at Morgana Run, which will have 95 homes of different types, new infrastructure and green space. In 2013, construction began on the first units, with initial homeowners moving in in late 2013.

Pension Obligations

Present and retired employees of the City are covered under two statewide public retirement (including disability retirement) systems. The Ohio Police and Fire Pension Fund ("OP&FPF") is applicable to uniformed members of the police and fire departments. All other City employees are covered by the Ohio Public Employees Retirement System ("OPERS").

OPERS currently reports 348,235 active contributing accounts statewide. The number of active members included in the plan for the City as of February 2016 was approximately 5,279. Employees contribute at a statutory rate of 10.0% of earnable salary or compensation, and the City contributes 14.0% (actuarially established for OPERS) of the same base. The City's contribution to OPERS for the year ending December 31, 2015 was approximately $36,356,000.

OP&FPF covers approximately 2,270 City full-time police and fire department employees and currently reports over 27,000 active accounts statewide. Police and fire employees contributed at a statutory rate of 10.75% of gross earnings from July 2, 2013 through July 1, 2014. As of July 2, 2014, the active member contribution increased to 11.5%. A final increase to 12.25% was instituted in July of 2015. The City currently contributes 19.5% for police personnel and 24.0% for fire personnel. The City's contribution to OP&FPF for the year ending December 31, 2015 was approximately $35,179,000.

For further information on these pension plans, see the financial statements of the City located at www.auditor.state.oh.us. Financial and other information for OPERS and OP&FPF can also be found on the respective website for each retirement system including its Comprehensive Annual Financial Report.

The City's current employer contributions to OPERS and OP&FPF have been treated as current expenses and are included in the City's operating expenditures, except to the extent that they are paid from the proceeds of a Police and Fire Pension levy collected by the City.

OPERS and OP&FPF are two of five statewide public employee retirement systems created by and operating pursuant to Ohio law, all of which currently have unfunded actuarial accrued liabilities. The General Assembly has the power to amend (and in the past has amended) the format of those systems and could revise rates or methods of contributions to be made by the City and its employees to OPERS and/or OP&FPF and could also revise benefits or benefit levels. On September 12, 2012, the General Assembly passed five separate pension reform bills intended to assist each of the five retirement systems in addressing its unfunded actuarial accrued liabilities. The bills passed with respect to OPERS and OP&F provide for (i) no change in the City contribution rates with respect to its employees' earnable salaries, (ii) no change in OPERS employee contribution rate, and (iii) an increase in the OP&F employee contribution rate from 10% to 12.25% in annual increments of 0.75% beginning on July 2, 2013. With certain transition provisions applicable to certain current employees, the bills increase minimum age and service requirements for retirement and disability benefits, revise the calculation of an employee's final average salary on which pension benefits are based to include the five highest years (rather than the three highest years), provide for OPERS pension benefits to be calculated on a lower, fixed formula, change provisions with respect to future cost–of-living adjustments to limit those adjustments to the lesser of any increase in the Consumer Price Index or three percent, and make other changes. The OP&F bill also authorizes the OP&F board to further adjust member contribution rates or further adjust age and service requirements after November 1, 2017, if, after an actuarial investigation, the board determines that an adjustment is appropriate. Those bills became effective on January 7, 2013.

Federal law requires City employees hired after March 31, 1986 to participate in the federal Medicare program, which requires matching employer and employee contributions, each being 1.45% of the wage base. Otherwise, City employees who are covered by a State retirement system are not currently covered under the federal Social Security Act. OPERS and OP&FPF are not now subject to the funding and vesting requirements of the federal Employment Retirement Income Security Act of 1974.

Employees

As of March 21, 2016 the City had approximately 7,000 full-time employees and 730 part-time employees. Thirty-one bargaining units represent approximately 5,640 employees. The larger units, together with the approximate number of employees represented by those units, include the American Federation of State, County and Municipal Employees, Local 100 (1,123); Cleveland Police Patrolmen's Association (1,220); the International Association of Firefighters (759); Municipal Foreman and Laborers Union, Local 1099 (428); Teamsters Local 244 (375);

and the Fraternal Order of Police (274). There have been no significant labor disputes or work stoppages within the City within the last twenty-five years.

The City's three-year labor agreements with its labor unions will expire on March 31, 2016. Negotiations have begun with several unions on new three-year contracts. Until new contracts are approved, the previous contracts remain operational. Once new contracts are approved and entered into, any changes in pay under the new contracts are retroactive to the expiration of the prior contracts. Non-union employees receive the same wage increases as those negotiated with the various unions.

City Council by ordinance establishes schedules of salaries, wages and other economic benefits for City employees. Generally, the terms of these ordinances have been the product of negotiations with representatives of the employees or bargaining units, and increases in economic benefits have normally been provided on an annual basis.

Chapter 4117 of the Revised Code (the "Collective Bargaining Law"), establishes procedures for, and regulates, public employer-employee collective bargaining and labor relations for the City and other state and local governmental units in Ohio. The Collective Bargaining Law creates a three-member State Employment Relations Board (the "SERB"), which administers and enforces the Collective Bargaining Law. Among other things, the Collective Bargaining Law: (a) creates rights and obligations of public employers, public employees and public employee organizations with respect to labor relations; (b) defines the employees it covers; (c) establishes methods for (i) the recognition of employees and organizations as exclusive representatives for collective bargaining and (ii) the determination of bargaining units; (d) establishes matters for which collective bargaining is either required, prohibited, or optional; (e) establishes procedures for bargaining and the resolution of disputes, including (i) negotiation, (ii) mediation and (iii) fact finding; and (f) permits all covered employees to strike, except certain enumerated classes of employees, such as police and fire personnel.

Disputes with employees who are prohibited from striking are to be resolved by binding arbitration starting with best offer, on an issue-by-issue basis. In the event that a legal strike presents a clear and present danger to the public health or safety, the appropriate Ohio court of common pleas may issue a temporary restraining order against the strike for a period not to exceed 72 hours, and in such a case the employer may request authorization of the SERB to enjoin the strike beyond the period of the temporary restraining order. The SERB determines whether a clear and present danger to public health or safety exists, and if it so determines, the court of common pleas issuing the temporary restraining order has jurisdiction to enjoin the strike for a period of 60 days after the expiration of the temporary restraining order or when agreement is reached, whichever occurs first. Thereafter, no court has jurisdiction to issue injunctions or other orders, and a strike may be resumed at the end of the 60-day period. The law established certain employers, employee and employee organization unfair labor practices and remedies for unfair labor practices.

Police and Fire Overtime

The City compensates the members of the City's police and fire departments at a rate of one and one-half times the regular rate of pay for overtime and holidays worked. It has been the City's practice that, for each one hour unit of overtime or holiday time worked, those employees receive one hour of pay at their regular rate and, at the discretion of the employee, one-half hour of pay at the regular rate or a credit of one-half hour of compensatory time for which they will be paid, to the extent that time is not used as vacation time, at their then current rate of pay upon termination of employment because of death, retirement, resignation, layoff or dismissal. A collective bargaining agreement between the City and the Cleveland Police Patrolmen's Association provides that employees may request payment of all or part of their compensatory time in money, and that if the requests exceed the total budgeted amount, the requests will be paid on a pro rata basis. Similar agreements have been reached with the Association of Cleveland Firefighters and the Fraternal Order of Police. The total estimated amount of the City's liability for accumulated compensatory time as of December 31, 2014 was $40,198,000, and that amount is unfunded. That amount was calculated based on the employees' then existing regular rate of pay. The rate of pay used to determine any cash payments made upon retirement or other termination or upon the request of the employee will be the rate of pay in effect at the time of such determination. The City has budgeted $4,990,000 for payment in 2016 of accrued overtime and other separation benefits anticipated to be payable in connection with retirements. The City has budgeted $625,000 in 2016 for payments requested by employees.

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APPENDIX B

DEFINITIONS AND SUMMARY OF CERTAIN TERMS AND PROVISIONS OF THE TRUST AGREEMENT

When used in this Official Statement, the following terms have the meanings set forth below; provided that the definitions set forth below are qualified in their entirety by reference to the Trust Agreement, as defined below, which is available from the Trustee.

"Additional Bonds" means the Additional Bonds authorized to be issued pursuant to the Trust Agreement, irrespective of whether such Additional Bonds are designated bonds or notes.

"Additional Bonds Test" means the requirements for the issuance of Additional Bonds described below in paragraph (ii) under "Authorization of the Issuance of Bonds."

"Adjusted Net Revenues" means, for any period, Net Revenues Available for Debt Service adjusted to reflect increases or decreases resulting from the following:

(a) Rates in effect at the time of issuance of the proposed Series of Additional Bonds;

(b) Revenues anticipated to be generated based upon rates established (and to be placed in effect within two years from the date of issuance of the proposed Series of Additional Bonds) in a rate ordinance adopted by the Council prior to the issuance of the proposed Series of Additional Bonds;

(c) Additional Net Revenues anticipated to result from any addition, extension or improvement to the System that the City has constructed and that is in operation on the date of issuance of the proposed Series of Additional Bonds, but that was not in operation during the entire period;

(d) Additional Net Revenues anticipated to result from the operation of existing facilities or existing collection systems that the City will acquire or commence operating with the proceeds of the proposed Series of Additional Bonds; and

(e) Changes in Operating Expenses anticipated to result from any additions, extensions and improvements to the System to be financed with the proposed Series of Additional Bonds.

"Appreciated Principal Amount" means, as of any date of computation with respect to any Capital Appreciation Bond, an amount equal to the principal amount thereof at its initial offering plus an amount of interest that, based on semiannual compounding on each Compounding Date from the date of delivery on the basis of a 360-day year of twelve 30-day months, will produce a yield approximately equal to the yield to maturity for such Bond specified in or pursuant to the applicable Supplemental Bond Ordinance. A schedule setting forth the Appreciated Principal Amount of Capital Appreciation Bonds as of the Compounding Dates may be set forth in the form of such Capital Appreciation Bond. The Appreciated Principal Amount with respect to any date other than the Compounding Dates is the Appreciated Principal Amount on the next preceding Compounding Date or dated date as noted on the form

thereof, as the case may be, plus the fraction of the difference between the Appreciated Principal Amount on such next preceding Compounding Date or dated date and the next succeeding Compounding Date that equals the ratio of (a) the number of days from such next preceding Compounding Date or dated date to the date for which the determination is being calculated to (b) the total number of days from such next preceding Compounding Date or dated date to the next succeeding Compounding Date.

"Assumed Amortization Period" means the period of time specified in paragraph (a) or paragraph (b) below, as selected by the Director of Finance:

(a) Five years; or

(b) The period of time, exceeding five years, set forth in a written opinion delivered to the City, of a Financial Expert, as being not longer than the maximum period of time over which indebtedness having comparable terms and security issued or incurred by similar issuers of comparable credit standing would, if then being offered, be marketable on reasonable and customary terms.

"Assumed Interest Rate" means the rate per annum (determined as of the last day of the calendar month next preceding the month in which the determination of Assumed Interest Rate is being made) set forth in an opinion delivered to the City of a Financial Expert, as being not lower than the lowest rate of interest at which indebtedness having comparable terms, security and federal tax status amortized on a level debt service basis over a period of time equal to the Assumed Amortization Period, and issued or incurred by similar issuers of comparable credit standing would, if being offered as of such last day of the calendar month, be marketable on reasonable and customary terms; provided that such rate may not be higher than the highest rate, if any, permitted by law at which such Bonds could be sold on that day.

"Balance Subfund" means the Subfund of that name established pursuant to the Trust Agreement.

"Balloon Bonds" means any Bonds or Bond Anticipation Notes (a) 25% or more of the principal payments of which are due in a single year, excluding any such principal payments that are subject to mandatory sinking fund requirements in a prior year, or (b) 25% or more of the principal of which may, at the option of the holder or holders thereof, be redeemed at one time.

"Bond Anticipation Notes" means any notes issued in anticipation of the issuance of a Series of Additional Bonds.

"Bond Counsel" means any law firm appointed by the City with nationally recognized standing in the field of municipal law whose approving opinions are generally accepted by purchasers of municipal bonds.

"Bondholder" or "Holder" or "Owner" or words of similar import mean, when used with reference to any Bond, any Person in whose name the Bond is registered on the Register, or his duly authorized attorney in fact, representative or assigns.

"Bond Registrar" means the Trustee acting in the capacity of Bond Registrar under the Trust Agreement, unless and until a successor Bond Registrar becomes such pursuant to

applicable provisions of the Trust Agreement, each of which must be a transfer agent registered in accordance with Section 17(A)(C) of the Securities Exchange Act of 1934.

"Bond Service Charges" means the principal of and the interest and any premium on the Bonds.

"Bonds" means, collectively, the Series 2016 Bonds and any Additional Bonds.

"Bond Year" means, as to any Series of Bonds, each one-year period (or shorter period from the date of issue of the Bonds of that Series) that ends on the close of business on the last day of a compounding interval used in computing the yield on the Bonds of that Series for purposes of the Code.

"Book-Entry Form" or "Book-Entry System" means a form or system in or under which (i) the beneficial right to the Bonds of any Series and the Bond Service Charges on the Bonds of such Series may be transferred only through a book entry, and (ii) physical bond certificates in fully registered form are issued by the City only to a Depository or its nominee as registered owner, with the bonds "immobilized" in the custody of the Depository, and the book entry maintained by a person or persons other than the City or the Trustee being the record that identifies the owners of beneficial interests in the Bonds of such Series and the Bond Service Charges on the Bonds of such Series.

"Capital Appreciation Bonds" means Bonds, the interest on which, during their entire term or any portion thereof, is payable only as a component of the Appreciated Principal Amount at maturity or redemption pursuant to the applicable Supplemental Bond Ordinance, and includes Compound Interest and Income Bonds.

"Capital Costs" means (but in all cases only if and to the extent consistent with applicable law and with the tax-exempt status of any Bonds funding such Capital Costs) the costs of acquisition and construction of Waste Water Facilities of the System, the cost of acquisition of all land, rights-of-way, property rights, easements, franchise rights, and interests required by the City for such acquisition and construction, the cost of demolishing or removing any buildings or structures on land so acquired, including the cost of acquiring any lands to which such buildings or structures may be moved, the cost of diverting highways, interchange of highways, and access roads to private property, including the cost of land or easements therefor, the cost of all machinery, furnishings, and equipment, financing charges, interest prior to and during construction and after completion of acquisition or construction, engineering, expenses of research and development with respect to Waste Water Facilities, legal expenses, plans, specifications, surveys, estimates of cost and revenues, working capital, other expenses necessary or incident to determining the feasibility or practicability of acquiring or constructing any such project, administrative expense, and such other expense as may be necessary or incident to the acquisition or construction of the project, the financing of the acquisition or construction, including the amount authorized in the ordinance of the City providing for the issuance of Bonds to be paid into any special funds from the proceeds of those Bonds and the financing of the placing of any such project in operation.

"Certificate" or "Certification" means, as the case may be, either (i) a signed document attesting to or acknowledging the circumstances, representations or other matters therein stated

or set forth or (ii) a signed document setting forth matters to be determined by the Mayor, the Director of Finance, the Director of Public Utilities, or the Law Director of the City pursuant to the General Bond Ordinance, the Trust Agreement, any Supplemental Bond Ordinance or any Supplemental Trust Agreement.

"City" means the City of Cleveland, Ohio.

"Code" means the Internal Revenue Code of 1986, as amended, including, when appropriate, the statutory predecessor of the Code and all applicable Treasury regulations (including any proposed regulations in force) thereunder, including any regulations issued and proposed and in force and effect under the statutory predecessor of the Code.

"Compound Interest and Income Bonds" means any Bonds as to which accruing interest is not paid prior to the Interest Commencement Date specified in or pursuant to the Supplemental Bond Ordinance providing for issuance of such Bonds, and the Appreciated Principal Amount for such Bonds is compounded prior to the Interest Commencement Date for such Bonds, all as so designated by the Supplemental Bond Ordinance providing for the issuance of such Bonds.

"Compounding Date" means, as to a series of Capital Appreciation Bonds, each date on which interest thereon is compounded pursuant to the applicable Supplemental Bond Ordinance.

"Construction Fund" means the Fund by that name established pursuant to the Trust Agreement.

"Consulting Engineer" means any engineer, firm of engineers, or expert or firm of experts, independent of the City, experienced in the financing, construction and operation of plants and systems such as the System, designated by the City.

"Contingency Fund" means the Fund of that name established pursuant to the Trust Agreement.

"Contingency Fund Requirement" means $1,000,000 or such higher amount as the Director of Public Utilities may from time to time specify (in writing delivered to the Trustee) as the Contingency Fund Requirement. Subsequent to any such specification, the Director of Public Utilities may reduce the Contingency Fund Requirement from the specified amount, but not below $1,000,000.

"Costs of Issuance" means all items of expense, directly or indirectly payable or reimbursable and related to the authorization, sale and issuance of Bonds, including but not limited to printing costs, costs of preparation and reproduction of documents, electronic posting, filing and recording fees, initial fees and charges of any fiduciary, Bond Counsel fees and other legal fees and charges, fees and disbursements of Financial Advisors, Financial Experts, consultants and professionals, costs of credit ratings, fees and charges for preparation, execution, transportation and safekeeping of Bonds, costs of expenses of refunding, premiums for the insurance of the payment of the Bonds, initial fees for any letter of credit or other credit or liquidity facility securing payment of the Bonds or funding all or part of the Debt Service Reserve Requirement and any other cost, charge or fee in connection with the original issuance of Bonds.

"Council" means the Council of the City.

"Credit Obligations" means amounts payable by the City to Credit Providers with respect to credit enhancement or liquidity support other than as Bond Service Charges, including, without limitation, providers of any instrument funding all or part of the Debt Service Reserve Requirement.

"Credit Provider" means, with respect to a series of Bonds, the provider of a municipal bond insurance policy or other credit enhancement or liquidity support, if any, for such series of Bonds specified in the Trust Agreement or the applicable Supplemental Trust Agreement.

"Debt Service" for any period means, except as otherwise provided in the Trust Agreement, as of any date of calculation with respect to any Series, an amount equal to the sum of (i) interest due during such period on the Bonds of such Series, (ii) the principal amount, as of the beginning of such period, of Outstanding Bonds of such Series that mature on any future date in such period, reduced by the aggregate principal amount of such Bonds that would at or before that future date cease to be Outstanding by reason, but only by reason, of the application of Sinking Fund Installments at or before that future date, and (iii) the amount of any Sinking Fund Installment with respect to the Bonds of such Series required to be paid on any future date in such period (calculated on the assumption that Bonds Outstanding at the date of calculation will not cease to be Outstanding except by reason of payment on the due date thereof or redemption from Sinking Fund Installments); provided that for purposes of computing the Additional Bonds Test, the Rate Covenant, and any Debt Service Reserve Requirement described hereunder, the interest component of Debt Service with respect to a Series of Bonds will be reduced by any accrued interest or capitalized interest funded from the proceeds of Bonds, and, for the first annual period following the issuance of such Series of Bonds, may not include more than one year's interest.

"Debt Service Fund" means the Fund of that name established pursuant to the Trust Agreement.

"Debt Service Reserve Fund" means the Fund of that name established pursuant to the Trust Agreement.

"Debt Service Reserve Requirement" means, as of any date of calculation with respect to any Series of Bonds, the Debt Service Reserve Requirement for such Series specified in the Supplemental Trust Agreement applicable thereto, but subject to the requirements of the Trust Agreement.

"Defeasance Obligations" means non-callable (i) Government Obligations, (ii) Government Certificates, and (iii) any obligation covered by paragraph (b) or (c) in the definition of "Eligible Investments" that is, at the time of investment, rated in the highest rating category by at least two Rating Agencies.

"Depository" means any securities depository that is a clearing agency under federal law operating and maintaining, with its participants or otherwise, a book entry system to record beneficial ownership of Bonds or Bond Service Charges, and to effect transfers of Bonds, in book entry form, and includes and means initially The Depository Trust Company (a limited purpose trust company), New York, New York.

"Director of Finance" means the Director of Finance of the City and any Person succeeding to the powers, duties or functions of such office with respect to the General Bond Ordinance or the Bonds.

"Director of Law" means the Director of Law of the City and any Person succeeding to the powers, duties or functions of such office with respect to the General Bond Ordinance or the Bonds.

"Director of Public Utilities" means the Director of Public Utilities of the City and any Person succeeding to the powers, duties or functions of such office with respect to the General Bond Ordinance or the Bonds.

"Direct Payment" means a credit payment allowed pursuant to Section 54AA(g) of the Code with respect to Direct Payment BABs that is payable to the City by the U.S. Treasury, as provided in Section 6431 of the Code, or any other payment by the U.S. Treasury to the City to subsidize or reimburse the City for all or a portion of the interest cost that the City may incur on Bonds or Subordinated Indebtedness that qualifies for such payment under any successor or substantially similar program to Direct Payment BABs.

"Direct Payment BABs" means Bonds or Subordinated Indebtedness that are "Build America Bonds" within the meaning of Section 54AA(d) of the Code and that are qualified bonds within the meaning of Section 54AA(g), the interest on which is includible in gross income for federal income tax purposes and with respect to which the City makes an irrevocable election to receive one or more Direct Payments.

"Eligible Investments" means and include any of the following:

(a) Government Obligations and Government Certificates;

(b) Obligations issued or guaranteed by any of the following: (i) Federal Home Loan Bank System, (ii) Export-Import Bank of the United States, (iii) Federal Financing Bank, (iv) Government National Mortgage Association, (v) Federal Home Loan Mortgage Company, (vi) Federal Housing Administration, (vii) Federal National Mortgage Association, (viii) Federal Farm Credit Bank, (ix) Resolution Funding Corporation, and (x) Rural Economic Community Development Administration (formerly, Farmers Home Administration, or any indebtedness issued or guaranteed by any instrumentality or agency of the United States;

(c) Pre-refunded municipal obligations rated in the highest rating category by at least one Rating Agency and meeting the following conditions: (i) such obligations are not to be redeemed prior to maturity or the Trustee has been given irrevocable instructions concerning their call for redemption, and (B) the issuer of such obligations has covenanted not to redeem such obligations other than as set forth in such instructions; (ii) such obligations are secured by Government Obligations or Government Certificates that may be applied only to interest, principal, and premium payments on such obligations; (iii) the principal of and interest on such Government Obligations or Government Certificates (plus any cash in the escrow fund with respect to such pre-refunded obligations) are sufficient to meet the liabilities of the obligations; (iv) the Government Obligations or Government Certificates serving as security for the

obligations are held by an escrow agent or trustee; and (v) such Government Obligations or Government Certificates are not available to satisfy any other claims, including those against the trustee or escrow agent;

(d) Direct and general long-term obligations of any state of the United States of America or the District of Columbia (a "State"), to the payment of which the full faith and credit of such State is pledged and that are rated in either of the two highest rating categories by at least two Rating Agencies;

(e) Direct and general short-term obligations of any State, to the payment of which the full faith and credit of such State is pledged and that are rated in the highest rating category by at least two Rating Agencies;

(f) Interest-bearing demand or time deposits with, or interests in money market portfolios issued by, state banks or trust companies, national banking associations or savings and loan associations that are members of the Federal Deposit Insurance Corporation ("FDIC"), including the Trustee or any of its affiliates. Such deposits or interests must be (i) continuously and fully insured by FDIC or (ii) fully secured by Government Obligations, Government Certificates or obligations described in clause (b) of this definition ("clause (b) securities") or (iii) secured by surety company bonds held by the Trustee that, when executed, must be for an amount equal to the amount of such interest-bearing demand or time deposits that is not secured by (i) or (ii) above. Such Government Obligations, Government Certificates or clause (b) securities must have a market value at all times at least equal to the principal amount of the deposits or interests. The Government Obligations, Government Certificates or clause (b) securities must be held by a third party (who may not be the provider of the collateral), or by any Federal Reserve Bank or depositary, as custodian for the institution issuing the deposits or interests. Such third party must have a perfected first lien in the Government Obligations, Government Certificates or clause (b) securities serving as collateral, and such collateral is to be free from all other third party liens;

(g) Repurchase agreements, (i) the maturities of which are thirty (30) days or less or (ii) the maturities of which are longer than thirty (30) days, provided the securities subject to such agreements are marked to market at least weekly, and in either case are entered into with a Qualified Financial Institution. Each repurchase agreement must be in respect of Government Obligations and Government Certificates or clause (b) securities (collectively, "Repurchase Securities"), which must be required to be acquired at the market price and maintained in an amount equal to at least 102% of the total principal and accrued interest under the repurchase agreement. In addition, the provisions of the repurchase agreement must meet the following requirements: (A) a third party (the "Custodian"), which may not be the provider of the repurchase agreement and which must be acceptable to the Trustee (for Trustee-held funds) or to the City (for City-held funds) will be required to have possession of the Repurchase Securities; (B) failure by the repurchase agreement provider to cure any deficiency in the requisite levels of Repurchase Securities within two (2) business days (or such shorter period as may be specified in the repurchase agreement) will require the Custodian to liquidate the Repurchase Securities immediately; and (C) the Custodian will be required to have a perfected, first-priority security interest in the securities;

(h) Money market accounts of any state or federal bank, including the Trustee or any of its affiliates, or bank whose holding parent company is rated in the top two short-term or long-term rating categories by at least two Rating Agencies;

(i) Investment agreements the issuer of which is a Qualified Financial Institution;

(j) Any debt or fixed income security, the issuer of which is rated in the highest rating category by at least two Rating Agencies;

(k) Star Ohio, the Ohio Subdivision's Investment Fund created and administered by the Treasurer of the State pursuant to Section 135.45 of the Ohio Revised Code;

(l) demand deposits, including interest-bearing money market accounts, trust deposits, time deposits or bankers acceptances (in each case having maturities of not more than 360 days) of a domestic bank (including the Trustee and any bank affiliated with the Trustee), including a branch office of a foreign bank, which branch office is located in the United States, provided that such bank at the time of purchase, has a short-term bank deposit rating of "Prime-1" or better by Moody's and a rating of "A-1" or better by Standard & Poor's; and

(m) money market mutual funds, including, without limitation any mutual fund for which the Trustee or an affiliate of the Trustee serves as investment manager, administrator, shareholder servicing agent, and/or custodian or subcustodian, notwithstanding that (a) the Trustee or an affiliate of the Trustee receives fees from such funds for services rendered, (b) the Trustee charges and collects fees for services rendered pursuant to the Trust Agreement, which fees are separate from the fees received from such funds, and (c) services performed for such funds and pursuant to the Trust Agreement may at times duplicate those provided to such funds by the Trustee or its affiliates.

Any Supplemental Trust Agreement may provide that any obligations (other than Government Obligations or Government Certificates) otherwise constituting Eligible Investments will not constitute Eligible Investments for all or any portion of the moneys on deposit in any one or more Special Funds if the officer or officers of the City executing any Supplemental Trust Agreement on behalf of the City determine that such provision is necessary in order to enhance the perceived creditworthiness of the Bonds or any Series of Bonds (which determination will be conclusively evidenced by the execution by such official or officials of any Supplemental Trust Agreement containing such provision). Any investment that would, at the time of such investment, qualify as a lawful investment for interim moneys under Section 135.14 of the Ohio Revised Code (or any statute that supersedes that section) for any governmental body subject to its restrictions, will also qualify as an Eligible Investment.

"Event of Default" means any event of default under the Trust Agreement.

"Financial Advisor" means any financial advisory firm or firms retained by the Director of Finance of the City, from time to time, pursuant to the Trust Agreement, each of which must

be a municipal advisor registered with the Securities and Exchange Commission and the Municipal Securities Rulemaking Board under the Securities Exchange Act of 1934, as amended.

"Financial Expert" means (i) a Financial Advisor, or (ii) an investment banker selected by the City and experienced in underwriting indebtedness of the character of the Bonds.

"First Supplemental Trust Agreement" means the First Supplemental Trust Agreement, dated the same date as the Master Trust Agreement, between the City and the Trustee, pursuant to which the Series 2016 Bonds are issued and secured by the Trust Agreement.

"Fiscal Year" means a period of twelve consecutive months commencing on the first day of January of any year and ending on the last day of December of the same year, both inclusive; or such other consecutive twelve-month period as may hereafter be established as the fiscal year of the City for budgeting and accounting purposes to be evidenced, for purposes of the Trust Agreement, by a Certificate of an officer of the City filed with the Trustee.

"Funds" and "Subfunds" means the Funds and Subfunds established or provided for pursuant to the Trust Agreement.

"GAAP" means generally accepted accounting principles for entities such as the City as prescribed by (i) the pronouncements of the American Institute of Certified Public Accountants and (ii) the Financial Accounting Standards Board and the Governmental Accounting Standards Board, as applicable.

"General Bond Ordinance" means the Ordinance No. 29-16, passed by the Council on January 25, 2016, and constituting a part of the Master Trust Agreement.

"Government Certificates" means evidences of ownership of proportionate interest in future interest or principal payments of Government Obligations, including depository receipts thereof. Investments in such proportionate interest must be limited to circumstances wherein (i) a bank or trust company acts as custodian and holds the underlying Government Obligations; (ii) the owner of the investment is the real party in interest and has the right to proceed directly and individually against the obligor of the underlying Government Obligations; and (iii) the underlying Government Obligations are held in a special account, segregated from the custodian's general assets, and are not available to satisfy any claim of the custodian, any person claiming through the custodian, or any person to whom the custodian may be obligated.

"Government Obligations" means direct and general obligations of, or obligations the timely payment of principal and interest on which are unconditionally guaranteed by, the United States of America.

"Interest Commencement Date" means with respect to any Compound Interest and Income Bond, an Interest Payment Date prior to the scheduled maturity date for such Bond, as set forth in or pursuant to the Supplemental Bond Ordinance authorizing the issuance of such Bonds, after which interest accruing on such Bonds will be payable on each Interest Payment Date, with the first such payment beginning on the applicable Interest Payment Date immediately succeeding such Interest Commencement Date.

"Interest Payment Dates" means, with respect to any Series of Bonds, any date on which interest is due on such Bonds pursuant to the Supplemental Bond Ordinance authorizing the issuance of such Bonds.

"Interest Rate Hedge Agreement" means an interest rate swap, an interest rate cap or other such arrangement obtained by the City, in accordance with applicable law in effect at the time, with the goal of (i) lowering the effective interest rate to the City on Bonds, (ii) hedging the exposure of the City with respect to its obligations on Bonds against fluctuations in prevailing interest rates; (iii) otherwise reducing risk (including, without limitation, in addition to interest rate risk, tax risk or liquidity renewal risk) or otherwise managing exposure to changing markets (including, without limitation, in advance of issuances of Bonds); (iv) achieving a more appropriate matching of its assets and liabilities; and (v) otherwise pursuing an optimal capital structure.

"Mandatory Redemption Dates" means, with respect to any Series of Bonds, the dates, if any, on which such Bonds are required to be redeemed pursuant to the Supplemental Bond Ordinance authorizing the issuance of such Bonds.

"Master Trust Agreement" means the Trust Agreement authorized pursuant to the General Bond Ordinance, to be entered into by the City and the Trustee.

"Mayor" means the Mayor of the City and any Person succeeding to the powers, duties or functions of such office with respect to the General Bond Ordinance or the Bonds.

"Net Revenues" means the Revenues in any Fiscal Year or other specified period net of Operating Expenses for that period.

"Net Revenues Available for Debt Service" means, for purposes of satisfying the Additional Bonds Test and the Rate Covenant, Net Revenues for any period, excluding (i) proceeds from any insurance, other than business interruption insurance, or condemnation award, (ii) any profit or loss on the sale or other disposition, not in the ordinary course of business, of investments or fixed or capital assets, (iii) any gift, grant or contribution or other such extraordinary and non-recurring item, (iv) proceeds of Bonds or Subordinated Indebtedness; (v) revenues resulting from the early extinguishment of debt; and (vi) interest earnings on or credited to the Construction Fund.

"Operating Expenses" means the operating expenses of the City for the System in any Fiscal Year or other specified period but excluding any allowance for depreciation, principal amortization and interest (whether on capitalized leases or other debt), and accrued unfunded pension and other post-employment benefit liabilities, all as determined (except as otherwise specified in this definition) in accordance with GAAP for entities such as the City consistently applied.

"Operating Subfund" means the Subfund of that name established in the Revenue Fund pursuant to General Bond Ordinance.

"Original Purchaser(s)" means, with respect to any Series of Bonds, the original purchaser or purchasers named in the Bond Purchase Agreement entered into with respect to such Series.

"Outstanding Bonds" or "Bonds Outstanding" or "Outstanding" with respect to Bonds means all Bonds that have been authenticated and delivered or are then being delivered, by the Trustee, as of any date, under the Trust Agreement, except: (i) Bonds surrendered for exchange or registration of transfer and cancelled or cancelled because of payment or redemption on or prior to such date; (ii) Bonds (or the portions thereof) for the payment, redemption or purchase for cancellation of which sufficient moneys have been deposited on or prior to such date with the Trustee or any Paying Agents (whether upon or prior to the maturity or redemption date of any such Bonds), or which are deemed to have been paid and discharged pursuant to the provisions of the Trust Agreement; provided that if such Bonds are to be redeemed prior to the maturity thereof, notice of such redemption must have been given or arrangements satisfactory to the Trustee must have been made therefor, or waiver of such notice satisfactory in form to the Trustee must have been filed with the Trustee, and provided further that if such Bonds are to be purchased for cancellation, a firm offer for sale stating the price has been received and accepted; and (iii) lost, stolen, mutilated or destroyed Bonds in lieu of or in exchange for which others have been authenticated (or payment of which, when due, is made without replacement) under the Trust Agreement; provided, however, that in determining whether the Holders of the requisite principal amount of Outstanding Bonds have given any request, demand, authorization, direction, notice, consent or waiver under the Trust Agreement, Bonds owned by the City will be disregarded and deemed not to be Outstanding.

"Parity Swap Obligations" means the portion of the payment obligations of the City under any Interest Rate Hedge Agreement that (i) represents the equivalent of interest on the notional amount of the Interest Rate Hedge Agreement (but does not represent, among other things, any termination payment that may be payable by the City thereunder), and (ii) pursuant to the General Bond Ordinance, is secured by a pledge of Net Revenues on a parity with the pledge of Net Revenues that secures the Bonds.

"Paying Agent" means any bank or banks or trust company or trust companies designated as the paying agency or place of payment of any Series, and their successor or successors and any other bank or trust company that may at any time be substituted in their place pursuant to the Trust Agreement and will also mean the Trustee where the context permits.

"Person" or words importing persons means firms, associations, partnerships (including without limitation, general and limited partnerships), joint ventures, societies, estates, trusts, corporations, public or governmental bodies, other legal entities and natural persons.

"Predecessor Bond" of any particular Bond means every previous Bond evidencing all or a portion of the same debt as that evidenced by the particular Bond. For the purposes of this definition, any Bond authenticated and delivered under the Master Trust Agreement in lieu of a lost, stolen or destroyed Bond will, except as otherwise provided in the Master Trust Agreement, be deemed to evidence the same debt as the lost, stolen or destroyed Bond.

"Purchase Price" means, with respect to any Series of Bonds, the Purchase Price specified in or pursuant to the Supplemental Bond Ordinance related to such Additional Bonds.

"Qualified Financial Institution" means either (a) a bank, trust company, savings and loan association, national banking association, insurance company or other financial services company, including the Trustee or any of its affiliates, whose long-term debt obligations (or if

such financial institution has no long-term debt obligations, then whose deposits) or those of its parent corporation (provided such corporation unconditionally guarantees its subsidiary's obligations with respect to the Bonds or the City) are rated in one of the two highest rating categories by at least two Rating Services, or (b) any institution listed as a primary government securities dealer of the Federal Reserve Bank of New York.

"Rate Covenant" means the covenant of the City described in below in paragraph (iv) under "Covenants of the City."

"Rate Stabilization Fund" means the Fund of that name established pursuant to the Trust Agreement.

"Rating Agency" or "Rating Agencies" means Moody's Investors Service, Inc., New York, New York ("Moody's"), Fitch Investors Service, Inc., New York, New York ("Fitch"), and Standard & Poor's Corporation, New York, New York ("Standard & Poor's"), or any successors thereto, and any other nationally recognized credit rating agency then maintaining a rating on any Bonds at the request of the City.

"Rebate Fund" means the Fund of that name established pursuant to the Trust Agreement.

"Register" means the books kept and maintained by the Bond Registrar for registration and transfer of Bonds pursuant to the Trust Agreement.

"Reimbursement Agreement" means, with respect to a Series of Bonds, any agreement or agreements between one or more Credit Providers and the City under or pursuant to which credit enhancement or liquidity support for such Series of Bonds is issued or provided, which agreement sets forth the respective obligations of the City and of such one or more Credit Providers.

"Revenue Fund" means the Fund of that name established pursuant to the Trust Agreement.

"Revenues" means the revenues derived by the City from its ownership or operation of the System and income derived by the City from the investment of the Funds and Subfunds, all as determined (except as otherwise provided in this definition) in accordance with GAAP for entities such as the City consistently applied; provided, however, that Revenues do not include (i) amounts received by the City under agreements with any person or other political subdivision in connection with any Waste Water Facilities project, title to which vests, or may vest in regular and due course in accordance with any such agreement, in a person or political subdivision other than the City, or (ii) any gift, grant or contribution to the extent restricted by law or by the donor or grantor to a particular purpose inconsistent with its use for the payment of Bond Service Charges.

"Serial Bonds" means the Bonds of any Series maturing in annual installments.

"Serial Maturity Dates" means, with respect to any Series of Bonds, the dates specified in or pursuant to the Supplemental Bond Ordinance as the dates on which those Bonds that are Serial Bonds, if any, are to be retired in accordance with their stated terms.

"Series" means all of the Bonds authenticated and delivered on original issuance in a simultaneous transaction and any Bonds thereafter authenticated and delivered in lieu of or in substitution therefor pursuant to the General Bond Ordinance, the Trust Agreement, any Supplemental Bond Ordinance and any Supplemental Trust Agreement, regardless of variations in maturity, interest rate or other provisions.

"Sinking Fund Installment" means, as of any particular date of calculation, the amount required to be deposited by the City in the Debt Service Fund on a particular future date for the retirement of Outstanding Term Bonds of any Series that mature after that future date.

"Special Funds" means, collectively, the Construction Fund, the Debt Service Fund, the Debt Service Reserve Fund, the Subordinated Debt Service Fund, the Rate Stabilization Fund, the Contingency Fund, and the Balance Subfund.

"State" means the State of Ohio.

"Subordinated Debt Service Fund" means the Fund of that name established pursuant to the General Bond Ordinance.

"Subordinated Indebtedness" means any obligation or evidence of indebtedness incurred by the City in accordance with the Trust Agreement, the debt service charges on which are payable and secured on a basis subordinate to the Debt Service on the Bonds. Without limiting the generality of the previous sentence, "Subordinated Indebtedness" may include any obligations payable by the City under an Interest Rate Hedge Agreement that are not Parity Swap Obligations. An obligation of the City need not be evidenced by a Supplemental Trust Agreement or expressly secured by the Trust Agreement for such obligation to qualify as Subordinated Indebtedness or for the holders thereof to be entitled to payment of debt service charges thereon to be paid from the Subordinated Debt Service Fund.

"Supplemental Bond Ordinance" means any ordinance supplemental to or amendatory of the General Bond Ordinance, adopted by the City in accordance herewith.

"Supplemental Trust Agreement" means any trust agreement entered into pursuant to Article VII of the Master Trust Agreement, amending, supplementing or otherwise modifying the Master Trust Agreement as previously amended, supplemented or otherwise modified.

"System" means the wastewater collection and disposal system owned and operated by the City, including all Waste Water Facilities thereof, as extant or as that system may be hereafter modified, improved, supplemented or extended.

"Tax-Exempt Bonds" means any Bonds issued with an opinion of Bond Counsel that the interest thereon is excluded or excludible from gross income for purposes of federal income taxation.

"Term Bonds" means the Bonds of any Series maturing on one principal maturity date, the principal of which is payable from fixed amounts required to be deposited in each year in the applicable Debt Service Fund for the payment of such principal on or prior to maturity.

"Term Maturity Dates" means, with respect a Series of Bonds, the dates, if any, specified in or pursuant to the Supplemental Bond Ordinance authorizing such Bonds as the dates on which those Bonds that are Term Bonds, if any, are to be retired at their stated maturities.

"Trust Agreement" means the Master Trust Agreement, including the General Bond Ordinance as a part thereof, as supplemented by the First Supplemental Trust Agreement and as it may be amended and further supplemented by any other Supplemental Trust Agreement.

"Trustee" means the person acting in the capacity of trustee under or pursuant to the Trust Agreement, and its successors and any corporation resulting from or surviving any consolidation or merger to which it or its successors may be a party, and any successor Trustee at the time surviving as successor Trustee under or pursuant to the Trust Agreement.

"Variable Rate Bond" means any Bond that does not bear interest throughout its term at a fixed rate, but that does bear interest at a rate that, during part or the entirety of the term of such Bond, varies from time to time based upon a formula or other method of determination set forth in the applicable Supplemental Bond Ordinance or Supplemental Trust Agreement. If the rate of interest on any Bond that had been a Variable Rate Bond becomes fixed for the balance of the period until its maturity, that Bond will cease to be a Variable Rate Bond. A Bond will not be deemed a Variable Rate Bond solely because the rate of interest thereon may be adjusted if such interest becomes includable in gross income for purposes of federal income taxation.

"Waste Water Facilities" means facilities for the purpose of treating, neutralizing, disposing of, stabilizing, cooling, segregating, or holding wastewater, including, without limiting the generality of the foregoing, facilities for the treatment and disposal of sewage or industrial waste and the residue thereof, facilities for the temporary or permanent impoundment of wastewater, both surface and underground, and storm and sanitary sewers and other systems, whether on the surface or underground, designed to transport wastewater, together with the equipment and furnishings thereof and their appurtenances and systems, whether on the surface or underground, including force mains and pumping facilities therefor when necessary.

"Working Capital Reserve" means a reasonable reserve for working capital, as determined by the City, not to exceed one-sixth of the Operating Expenses incurred during the preceding Fiscal Year.

"Working Capital Reserve Subfund" means the Subfund of that name established in the Revenue Fund pursuant to the General Bond Ordinance.

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The following is a summary of certain of the terms and provisions of the Trust Agreement.

Pledge of and Lien on Net Revenues and Special Funds

The City pledges and grants a lien on all Net Revenues and on moneys and securities on deposit in the Special Funds to secure the payment of the Bond Service Charges.

Funds and Subfunds

The City establishes and creates the following special trust funds and subfunds.

Special Trust Funds and Subfunds	Held By
Construction Fund	Trustee
Revenue Fund	City
Operating Subfund	
Working Capital Subfund	
Balance Subfund	
Debt Service Fund	Trustee
Interest Subfund	
Principal Subfund	
Debt Service Reserve Fund	Trustee
Subordinated Debt Service Fund	Trustee or City
Contingency Fund	City
Rate Stabilization Fund	City
Rebate Fund	Trustee

The Trustee will establish (i) separate subaccounts in the Interest Subfund and Principal Subfund of the Debt Service Fund for each Series of Bonds, (ii) separate subaccounts in the Debt Service Reserve Fund for any and each Series of Bonds to which the Debt Service Reserve Fund applies, (iii) separate subaccounts in the Construction Fund for each Series of Bonds with proceeds required to be deposited in the Construction Fund.

The Revenue Fund and the Subfunds therein, the Contingency Fund and the Rate Stabilization Fund each will be held by the City and not commingled with other moneys of the City except for purposes of investment. The Debt Service Fund, Debt Service Reserve Fund, Construction Fund and Rebate Fund each will be held by the Trustee. The Subordinated Debt Service Fund will be held by the City or by the Trustee or such trustee or other party as may be specified in the trust agreement or other contract between the City and the holder or holders of the Subordinated Indebtedness. All moneys or securities deposited with the Trustee pursuant to the Trust Agreement will be held in trust and applied only in accordance with the provisions of the Trust Agreement and will be considered trust funds for the purposes of the Trust Agreement.

Construction Fund

There will be deposited from time to time in the Construction Fund the portion of the proceeds of the Bonds required to be deposited therein by any Supplemental Bond Ordinance and Supplemental Trust Agreement. Amounts in the Construction Fund will be expended only to pay Capital Costs and Costs of Issuance or to cure any deficiency in the Debt Service Fund or the Debt Service Reserve Fund following the deposits required, as described under "Revenue Fund," which follows.

Revenue Fund

The City is required to cause all cash receipts that constitute Revenues of the System to be deposited into the Revenue Fund, along with any amounts that the City elects to transfer to the Revenue Fund from the Rate Stabilization Fund. The amounts in the Revenue Fund will be deposited in the Operating Subfund and first applied to the payment of Operating Expenses; thereafter, the balance in the Revenue Fund up to the Working Capital Reserve will be deposited in the Working Capital Reserve Subfund and retained as a reserve for the payment of Operating Expenses and used for that purpose if the Operating Subfund will have been fully expended; and thereafter, the balance in the Revenue Fund will be paid at the times and in the amounts as follows:

(A) To the Trustee for deposit in the Debt Service Fund as follows:

1. Monthly, on or before the last day of each month, the City will pay to the Trustee, for deposit in each subaccount in the Interest Subfund in the Debt Service Fund, an amount equal to one-sixth of the amount of interest due on the next Interest Payment Date on the Outstanding Bonds of the Series to which such subaccount relates; provided, however, that if fewer than six months precede the next Interest Payment Date, then, on or before the last day of each such month an amount that, if deposited in each such month, would equal the amount of interest due on the next Interest Payment Date on the Outstanding Bonds of the Series to which such subaccount relates will be deposited therein; provided further, however, such monthly payments need be made only to the extent that the amount then on deposit in any such subaccount is less than the amount required to be on deposit therein in accordance herewith; and provided yet further, however, that the frequency and amount of any deposits required by this paragraph will be subject to adjustment pursuant to the requirements of the General Bond Ordinance to make the deposits required for any Series of Bonds or any Parity Swap Obligations no less frequent than those for any other Series.

2. Monthly, on or before the last day of each of the twelve months that precede a month in which a Serial Maturity Date occurs or a Sinking Fund Installment is due, the City will pay to the Trustee for deposit in each applicable subaccount in the Principal Subfund in the Debt Service Fund an amount equal to one-twelfth of the amount necessary to provide for the payment of the principal due on the next Serial Maturity Date or of the next Sinking Fund Installment; provided, however, that if fewer than twelve months precede the next month in which a Serial Maturity Date occurs or a Sinking Fund Installment is due, then, on or before the last day of each such month an amount that, if deposited in each such month, would equal the amount of principal due on the next Serial Maturity Date or of the next Sinking Fund Installment for the Outstanding Bonds of the Series to which such subaccount relates will be deposited

therein; provided further, however, such monthly payments need be made only to the extent that the amount then on deposit in any such subaccount is less than the amount required to be on deposit therein in accordance herewith; and provided yet further, however, that the frequency and amount of any deposits required by this paragraph will be subject to adjustment pursuant to the requirements of the General Bond Ordinance to make the deposits required for any Series of Bonds or any Parity Swap Obligations no less frequent than those for any other Series.

(B) To the Trustee for deposit in the Debt Service Reserve Fund, monthly on or before the last day of each month, commencing on the date set forth in the applicable Supplemental Bond Ordinance with respect to any Series of Bonds for which the Debt Service Reserve Requirement is to be deposited in the Debt Service Reserve Fund, the City must pay to the Trustee for deposit in the subaccount established in the Debt Service Reserve Fund for such Series of Additional Bonds, an amount such that, if that same amount were deposited in equal monthly installments, the aggregate of the amounts so deposited in such subaccount would be equal to the Debt Service Reserve Requirement with respect to such Series of Additional Bonds on the fifth (5th) anniversary of such commencement date, after giving effect to any amounts already on deposit therein and pledged to such Series of Additional Bonds. Thereafter, if the amount on deposit in such subaccount at any time is less than the Debt Service Reserve Requirement with respect to such Series of Additional Bonds, the City must pay to the Trustee for deposit in such subaccount, on or the before the last day of each month following the date of such deficiency, an amount equal to one-twelfth of such deficiency. The frequency and amount of any deposits required by this paragraph will be subject to adjustment pursuant to the General Bond Ordinance so that the deposits required by the provisions summarized by this paragraph for any Series of Bonds will be no less frequent than the deposits required by paragraph (A) above for Bond Service Charges or Parity Swap Obligations.

(C) To any custodians of the Subordinated Debt Service Fund for deposit therein, such amounts as are required to be deposited therein on such dates as are specified in the Supplemental Trust Agreement or other contracts between the City and the holder or holders of the Subordinated Indebtedness. If any Supplemental Trust Agreement or other contract between the City and the holder or holders of Subordinated Indebtedness requires the establishment and funding a reserve account related to such Subordinated Indebtedness, then any required deposits to such reserve account may be made after any otherwise required deposits described by this paragraph (C) but before any required deposits described under paragraphs (D) and (E), below.

(D) To the Contingency Fund, monthly on or before the last day of each month on which the balance in the Contingency Fund is less than the Contingency Fund Requirement, not less than 1/12 of the amount by which the Contingency Fund Requirement exceeds the balance.

(E) To the Rate Stabilization Fund, on any date on which deposits are required to be made by the provisions described in paragraphs (A) through (D), above, and after all such deposits are made, any amounts that the City elects to make to the Rate Stabilization Fund.

(F) On the first day of each Fiscal Year or on the first practicable date thereafter, the City will deposit any balance remaining in the Operating Subfund and the Working Capital Reserve Subfund in excess of the Working Capital Reserve at the end of the preceding Fiscal Year to the credit of the Balance Subfund; provided, that the City may at any time provide for semiannual or quarterly deposits of such Net Revenues to the Balance Subfund.

(G) Upon receipt of any Direct Payment, the City or the Trustee will cause it to be deposited (i) in the appropriate subaccount in the Debt Service Fund if such Direct Payment relates to Bonds, and (ii) in the appropriate account in the Subordinated Debt Service Fund if such Direct Payment relates to Subordinated Indebtedness, and will cause it to be applied solely to the purposes to which the Trust Agreement permits funds in such subaccount and account to be applied.

Debt Service Fund

The Trustee will pay out of the applicable subaccounts in the Interest Subfund and in the Principal Subfund in the applicable Debt Service Fund to each Paying Agent (i) on or before each Interest Payment Date for the Bonds of each Series (x) the amount required for the interest payable on all such Outstanding Bonds on such date and (y) the amount, if any, required for the payment of the principal maturing on all such Outstanding Bonds on such date, and (ii) on or before the day preceding the redemption date for any such Bonds to be redeemed at the option of the City, the amount required for the payment of such Bonds then to be redeemed. Such amounts will be so applied by each Paying Agent on and after the due date thereof.

The City will pay to the Trustee for deposit in the Debt Service Fund any moneys held in any other Funds or Subfunds in excess of the amounts required to be maintained therein that the City has elected to apply to the redemption prior to maturity of any Bonds, and such amounts will be so applied by the Trustee in the manner and at the times provided in the Trust Agreement.

Amounts in the applicable subaccount in the Principal Subfund for the payment of any Sinking Fund Installment for the Term Bonds of any Series then Outstanding (together with amounts in the applicable subaccount in the Interest Subfund with respect to the Term Bonds for which such Sinking Fund Installment was established) will be applied by the Trustee to the redemption of Term Bonds of the Series and maturity for which such Sinking Fund Installment was established in an amount not exceeding that necessary to effect the mandatory redemption of such Term Bonds on the Mandatory Redemption Dates. The Trustee will, by giving notice as provided in the Trust Agreement, proceed to call for redemption on the due date of any such Sinking Fund Installment the Term Bonds of the Series and maturity for which such Sinking Fund Installment was established (except in the case of Term Bonds maturing on a Sinking Fund Installment date), in such amount as will be necessary to effect the mandatory redemption of such Term Bonds on the Mandatory Redemption Date in an amount equal to the amount of such Sinking Fund Installment. The principal amount of Term Bonds of such Series and maturity theretofore delivered by the City to the Trustee to satisfy such Sinking Fund Installment, in the manner and to the extent as may be provided for in the General Bond Ordinance or in the Supplemental Bond Ordinance or Supplemental Trust Agreement authorizing such Series, will be credited against the amount of such Sinking Fund Installment. Such notice will be given regardless of whether or not moneys therefor will have been deposited in the Principal Subfund in the applicable Debt Service Fund and without any instructions of the City. The Trustee will pay out of the Principal Subfund in the applicable Debt Service Fund to the appropriate Paying Agent, on or before the day preceding such Mandatory Redemption Date, the amount required for the redemption of the Term Bonds so called for redemption, and such amount will be applied by such Paying Agent to such redemption. All expenses in connection with such redemption of Term Bonds will be paid by the City from the Operating Subfund.

To the extent that a Series of Bonds is secured by credit enhancement or liquidity support provided by a Credit Provider pursuant to a Reimbursement Agreement, the Supplemental Bond Ordinance and Supplemental Trust Agreement with respect to that Series may provide that the Trustee will apply amounts in the Debt Service Fund first, to the payment of Bond Service Charges and any Parity Swap Obligations then due and, second, to the payment of any Credit Obligations then due.

Debt Service Reserve Fund

If on any Interest Payment Date, the amount in the subaccount in the Interest Subfund or the Principal Subfund for any Series of Bonds is less than the amount required to be therein on deposit pursuant to the Trust Agreement, the Trustee will, after any required deposits to Subordinated Debt Service Fund, Contingency Fund, Rate Stabilization Fund and Balance Subfund are made, apply the amounts in the applicable Subfund, if any, in the Debt Service Reserve Fund to the extent necessary to make good any such deficiency (or the entire amount in the applicable Subfund if less than sufficient). The City may, so long as it will not result in a reduction of the rating assigned to any of the Bonds, substitute for funds on deposit in the Debt Service Reserve Fund a letter of credit, surety bond or other credit enhancement facility issued by a Qualified Financial Institution.

Subordinated Debt Service Fund

If on the third day before any Interest Payment Date, the amount in any of the Subfunds in the Debt Service Fund is less than the amount required to be on deposit therein pursuant to the General Bond Ordinance, after deposits, if any, are made to the Contingency Fund, Rate Stabilization Fund and Balance Subfund, the City will, upon the request of the Trustee, transfer or cause the transfer of moneys from the Subordinated Debt Service Fund to the Trustee to make good any such deficiency (or the entire amount in the Subordinated Debt Service Fund if less than sufficient).

If on any day next following the date of a payment and deposit required into a Debt Service Reserve Fund, the amount in any of the Subfunds in the Debt Service Reserve Fund is less than the amount required to be on deposit therein, the City will, upon the request of the Trustee, transfer moneys from the Subordinated Debt Service Fund to the Trustee for deposit in the appropriate Subfund in the Debt Service Reserve Fund an amount sufficient to cure any such deficiency (or the entire amount in the Subordinated Debt Service Fund if less than sufficient).

Subject to the provisions relating to the Rebate Fund, the amounts on deposit in the Subordinated Debt Service Fund will be applied to the payment of the principal of and interest and any premium on Subordinated Indebtedness as and when due or to the required funding of any special reserve account established for any Subordinated Indebtedness.

Contingency Fund

If on the third day before any Interest Payment Date, the amount in any of the Subfunds in the Debt Service Fund is less than the amount required to be on deposit therein pursuant to the General Bond Ordinance, after deposits, if any, are made to the Rate Stabilization Fund and the Balance Subfund, the City will, upon the request of the Trustee, transfer or cause the transfer of

moneys from the Contingency Fund to the Trustee to the extent necessary to make good any such deficiency (or the entire amount in the Contingency Fund if less than sufficient).

If on any day next following the date of a payment and deposit required into the Debt Service Reserve Fund, the amount in any of the Subfunds in the Debt Service Reserve Fund is less than the amount required to be on deposit therein pursuant to the General Bond Ordinance, the City will, upon the request of the Trustee, transfer moneys from the Contingency Fund to the Trustee for deposit in the appropriate Subfund in the Debt Service Reserve Fund an amount sufficient to cure any such deficiency (or the entire amount in the Contingency Fund if less than sufficient).

Subject to the provisions described in the two proceeding paragraphs and the requirements of the General Bond Ordinance pertaining to the Rebate Fund, the amounts on deposit in the Contingency Fund may be used only for the payment of Capital Costs or for the purpose of redeeming Bonds.

Rate Stabilization Fund

If on the third day before any Interest Payment Date, the amount in any of the Subfunds in the Debt Service Fund is less than the amount required to be on deposit therein pursuant to the General Bond Ordinance, after deposits, if any, are made from the Balance Subfund, the City will, upon the request of the Trustee, transfer or cause the transfer of moneys from the Rate Stabilization Fund to the Trustee to make good any such deficiency (or the entire amount in the Rate Stabilization Fund if less than sufficient).

If on any day next following the date of a payment and deposit required into the Debt Service Reserve Fund, the amount in any of the Subfunds in the Debt Service Reserve Fund is less than the amount required to be on deposit, the City will, upon the request of the Trustee, transfer moneys from the Rate Stabilization Fund to the Trustee for deposit in the appropriate Subfund in the Debt Service Reserve Fund an amount sufficient to cure any such deficiency (or the entire amount in the Rate Stabilization Fund if less than sufficient).

Subject to the provisions of described in the two preceding paragraphs, the City may from time to time transfer amounts on deposit in the Rate Stabilization Fund to the Revenue Fund and cause them to be applied as Revenues.

Balance Subfund

If on the third day before any Interest Payment Date, the amount in any of the Subfunds in the Debt Service Fund is less than the amount required to be on deposit in that fund, the City will, upon the request of the Trustee, transfer moneys from the Balance Subfund to the Trustee to make good any such deficiency (or the entire amount in the Subordinated Debt Service Fund if less than sufficient).

If on any day next following a payment date, the amount in any of the Subfunds in the Debt Service Reserve Fund is less than the amount required to be on deposit in that fund, the City will, upon the request of the Trustee, transfer moneys from the Balance Subfund to the Trustee for deposit in the appropriate Subfund in the Debt Service Reserve Fund an amount

sufficient to cure any such deficiency (or the entire amount in the Balance Subfund if less than sufficient).

Subject to the provisions described in the two immediately preceding paragraphs and the provisions described in the discussion of the Rebate Fund, below, the amounts on deposit in the Balance Subfund will be applied by the City to any lawful System purposes, including, but not limited to, transfers to the Subordinated Debt Service Fund for payment of the principal of and interest on any Subordinated Indebtedness and the funding or replenishment of any special reserve account established for any Subordinated Indebtedness, but amounts on deposit in the Balance Subfund may not be transferred or credited to the Rate Stabilization Fund.

Rebate Fund

The General Bond Ordinance establishes and orders to be maintained as a separate deposit account (except when invested) in the custody of the Trustee a Rebate Fund. Any provision of the General Bond Ordinance or of the Trust Agreement to the contrary notwithstanding, the Rebate Fund is not pledged to the payment of Bond Service Charges, is not one of or part of any of the Special Funds, and is free and clear of any lien of the Trust Agreement. Separate accounts will be created in the Rebate Fund for each Series of Bonds, commencing with the Series 2016 Bonds.

At least five days prior to the end of the fifth Bond Year and each fifth Bond Year thereafter for each Series of Bonds, including the Series 2016 Bonds, and within five days after the date on which all outstanding Bonds of such Series are paid or deemed paid in full, the City will calculate or will furnish information to and engage an independent certified public accounting firm that is not unacceptable to the Trustee, to calculate, the Rebate Amount as of the end of that Bond Year or the date of such final payment. The Trustee may rely upon the City for the calculation of the Rebate Amount, provided that the Trustee receives a verification of the accuracy of such computations by a firm of independent certified public accountants or an opinion of nationally recognized bond counsel that such computations are in compliance with the Code. Within 30 days after the end of the fifth Bond Year and every fifth Bond Year thereafter, the Trustee, acting on behalf of the City and at the written direction of the City, will pay to the United States in accordance with Section 148(f) of the Code from the moneys then on deposit in the applicable account in the Rebate Fund an amount equal to 90% (or such greater percentage not in excess of 100% as the City may direct the Trustee in writing to pay) of the Rebate Amount determined from the date of the original delivery of the Bonds of that Series (less the portion of the Rebate Amount, if any, previously paid to the United States). Within sixty (60) days after the payment in full of all outstanding Bonds of such Series and upon written direction of the City, the Trustee will pay to the United States in accordance with Section 148(f) of the Code from the moneys then on deposit in the applicable account in the Rebate Fund an amount equal to 100% of the Rebate Amount determined from the date of the original delivery of the Bonds of that Series to the date of such payment (but less that portion of the Rebate Amount, if any, previously paid to the United States) and any moneys remaining in the applicable account in the Rebate Fund following such payment will be paid to the City. Notwithstanding anything in the General Bond Ordinance to the contrary, the moneys needed for any payment required by the preceding two sentences may be derived from balances in any of the Special Funds other than the Debt Service Fund and the Debt Service Reserve Fund, and the City agrees to apply those balances to that purpose in the following Order: the Balance Subfund, the Rate Stabilization Fund, the

Contingency Fund, the Subordinated Indebtedness Fund and any related reserve fund or account, and the Construction Fund.

Investment of Funds and Subfunds

Moneys held in the Debt Service Reserve Fund or in any special reserve account established for any Series of Bonds will be invested and reinvested, in accordance with written instructions received from the City, by the Trustee, to the fullest extent practicable, in Eligible Investments maturing not later than the final maturity of the Series of Bonds for which the applicable subaccount was established or subject to withdrawal at any time required for purposes of the Debt Service Reserve Fund.

Moneys held in the Debt Service Fund will be invested and reinvested, in accordance with written instructions from the City, by the Trustee, to the fullest extent possible, in Eligible Investments maturing within one year from the date of purchase but in any event not later than such times as is necessary to provide moneys for required payments to be made from such Funds or subject to withdrawal at any such time.

Except as otherwise provided in the General Bond Ordinance with respect to the investment of moneys in the Debt Service Fund and the Debt Service Reserve Fund or any special reserve account established for any Series of Bonds, and only to the extent otherwise provided, moneys held in the Funds and Subfunds are invested and reinvested by the City and the Trustee, as the case may be, to the fullest extent practicable, in Eligible Investments that mature or are subject to withdrawal not later than such times as are necessary to provide moneys for payments to be made from such Funds and Subfunds, as required by the General Bond Ordinance and in the applicable Supplemental Bond Ordinance or Supplemental Trust Agreement; provided, however, that the Trustee may make such investments only in accordance with instructions received from the City.

An investment that can be redeemed at the option of the holder or that may be sold at a pre-established price to a "Qualified Purchase Agreement Provider" (as hereinafter defined) pursuant to a written agreement within a specified time period will be deemed to mature within that specified time period for purposes of limitations in the General Bond Ordinance on the maximum maturity of investments, but only for as long as the redemption or sale option remains exercisable within the specified period. A "Qualified Purchase Agreement Provider" means a financial institution, including, without limitation, a bank (including the Trustee or any of its affiliates) or insurance company) that is rated, or the obligations of which are rated, in one of the top two rating categories by at least two Rating Agencies.

In computing the amount in any Fund or Subfund established under the provisions of the General Bond Ordinance, obligations purchased as an investment of moneys therein will be valued at the lower of market value or the amortized value thereof. Amortized value, when used with respect to an obligation purchased at par, means the par amount thereof, and, when used with respect to an obligation purchased at a premium above or a discount below par, means the value as of any given time obtained by dividing the total premium or discount at which such obligation was purchased by the number of interest payments remaining on such obligation after such purchase and deducting the amount thus calculated for each interest payment date after such purchase from the purchase price in the case of an obligation purchased at a premium and adding

the amount thus calculated for each interest payment date after such purchase to the purchase price in the case of an obligation purchased at a discount.

Except as otherwise provided in the General Bond Ordinance, the Trustee will sell at the best price obtainable, using reasonable diligence to determine such best price, or present for redemption, any obligation so purchased as an investment whenever it is so requested in writing by the Director of Finance or whenever it is necessary in order to provide moneys to meet any payment or transfer from any Fund held by it.

Investments purchased as an investment of moneys in any Fund or Subfund will be deemed at all times to be a part of such Fund or Subfund, and any losses suffered due to the investment thereof will be charged to such Fund or Subfund. A deficiency in a Subfund in the Debt Service Reserve Fund will not be deemed to exist as of a date of transfer if the amount on deposit therein is less than the applicable Debt Service Reserve Requirement with respect to any Series of Bonds solely because as of such date of transfer one or more deposits to such Subfund in the Debt Service Reserve Fund is not required in connection with the original issuance of such Series until a future date or dates.

The valuation and sale of investments may be amended in the Master Trust Agreement or any Supplemental Trust Agreement upon the request of a Credit Provider.

Authorization of the Issuance of Bonds

The Trust Agreement authorizes the issuance from time to time of Bonds for the purpose only of paying Capital Costs, or refunding, including advance refunding, for any lawful purpose, any Outstanding Bonds or Subordinated Indebtedness, where such refunding will not adversely affect the exclusion of the interest on any Tax-Exempt Bonds from gross income for the purposes of federal income tax.

Upon the original issuance of Additional Bonds, the City must deliver to the Trustee certain certifications, including the following:

(i) A Certificate of the Director of Finance and the Director of Law, certifying that, to their knowledge, the City is not, on the date of issuance of the Bonds, in default in the performance of any of its covenants provided in the General Bond Ordinance, in the Trust Agreement or in the Bonds;

(ii) For any Series of Additional Bonds that the City issues for the payment of Capital Costs or for the purpose of refunding Subordinated Indebtedness or other obligations other than Outstanding Bonds, either (A) a Certificate of the Director of Finance certifying that the Net Revenues Available for Debt Service for the preceding Fiscal Year or any consecutive 12 calendar months during the 18-month period immediately preceding the date of original issuance of the proposed Series of Bonds (in each case, the "Test Period") are in an amount not less than the greater of (1) 120% of maximum annual Debt Service on all Bonds then Outstanding and on the proposed Series of Additional Bonds or (2) 100% of the sum of: (A) maximum annual Debt Service on any Bonds then Outstanding and on the proposed Series of Bonds, (B) all required deposits to the Debt Service Reserve Fund, (C) all required deposits to the Subordinated Debt Service Fund and to any reserve fund securing Subordinated Indebtedness, (D) all

required deposits to the Contingency Fund, and (E) debt service payable on any other obligations payable from the Balance Subfund and deposits to any reserve funds securing such other obligations, during that Test Period (the amounts described in clauses (A) through (E) being referred to collectively as the "Test Amount"), or (B) a Certificate of the Director of Finance, which may be, but is not required to be, accompanied by or based upon a Certificate of a Consulting Engineer, certifying that the Adjusted Net Revenues for the Test Period are in an amount not less than the greater of (1) 120% of maximum annual Debt Service on any Bonds then Outstanding and on the proposed Series of Bonds or (2) 100% of the Test Amount during that Test Period. No certification under the provision described in this paragraph is required for any Series of Bonds being issued for the purpose of refunding Outstanding Bonds, provided that the issuance of the refunding Bonds will not result in any increase in Debt Service in any Fiscal Year; otherwise, the same certification is required as for a Series of Bonds for the payment of Capital Costs; and

For purposes of determining the City's compliance with the Additional Bonds Test, the amount of any Direct Payment received or expected to be received by the City or the Trustee in any Fiscal Year will be credited against (i) Debt Service in such Fiscal Year if such Direct Payment relates to Bonds or (ii) debt service on Subordinated Indebtedness in such Fiscal Year if such Direct Payment relates to Subordinated Indebtedness.

For purposes of determining compliance with the Rate Covenant and the Additional Bonds Test, the interest component of Debt Service with respect to a Series of Bonds will be reduced by any accrued interest or capitalized interest funded from the proceeds of Bonds, and, for the first annual period following the issuance of such Series of Bonds, will not include more than one year's interest.

Bond Anticipation Notes. In anticipation of the issuance of any Series of Additional Bonds, the City may issue one or more Series of Bond Anticipation Notes. All requirements of the Trust Agreement applicable to Bonds also apply to Bond Anticipation Notes, including, without limitation the conditions precedent for the issuance of a Series of Bonds. For the purpose of (i) determining compliance with the Rate Covenant for any period prior to the date of calculation, (ii) determining whether Additional Bonds may be issued in compliance with the requirements with the Additional Bonds Test when any Bond Anticipation Notes are outstanding, (iii) determining whether Bond Anticipation Notes may be issued in compliance with the requirements of the Additional Bonds Test, and (iv) determining the amount of the Debt Service Reserve Requirement attributable to such Bond Anticipation Notes, the Debt Service payable on any Bond Anticipation Notes will be computed for purposes of the Trust Agreement in accordance with in accordance with the provisions of the Trust Agreement pertaining to Balloon Bonds and described in the subsection below entitled "Balloon Bonds."

Debt Service Reserve Requirement. In connection with any issuance of Bonds, the City may, in the applicable Supplemental Trust Agreement, provide that no deposit to the Debt Service Reserve Fund will be made for or with respect to such Additional Bonds, in which case: (i) the holders of such Additional Bonds will have no right or entitlement to have any portion of the Bond Service Charges on such Additional Bonds be paid from any amounts in the Debt Service Reserve Fund, (ii) the lien of and pledge on the Debt Service Reserve Fund will not

extend to or be for the benefit of the holders of such Bonds, and (iii) the Debt Service on such Bonds will not be taken into account in determining the Debt Service Reserve Requirement.

In the event that the Supplemental Trust Agreement applicable to an issue of Additional Bonds provides as described in the preceding paragraph, then such Supplemental Trust Agreement may also provide for the creation of a special reserve account solely for such Bonds, separate from the Debt Service Reserve Fund, and may provide for the deposit therein, at the time of issuance or in one or more subsequent deposits thereto, of an amount specified in the applicable Supplemental Trust Agreement as the required reserve for such Bonds, or may require the City to provide a municipal bond insurance policy, a bank letter or line of credit, or any other form of credit or liquidity facility issued by a Qualified Financial Institution to enhance the security for such Bonds in lieu of a funded reserve account. If the Supplemental Trust Agreement applicable to such a Series of Bonds so provides for such a special reserve account and further requires that deposits be made thereto from Net Revenues at any time, then such deposits may be made only from moneys remaining in the Revenue Fund on the last day of a month after any required deposits to the Debt Service Reserve Fund.

If the Supplemental Trust Agreement applicable to a Series of Bonds provides for no reserve account, or for a special reserve account as permitted by the preceding paragraph, then such Supplemental Trust Agreement must also provide that the amount of any defaulted principal of or interest on such Series of Bonds that would have been timely paid had the Debt Service Reserve Requirement for such Bonds been funded in the Debt Service Reserve Fund may be paid only from moneys remaining in the Revenue Fund on the last day of a month after any required deposits to the Debt Service Reserve Fund.

Computation of Debt Service With Respect to Bonds

In the event that all or any portion of any Series of Bonds have been issued as or are proposed to be issued as Variable Rate Bonds, then, except as otherwise provided in the Trust Agreement, in order to compute the Debt Service on such Bonds for the purposes of the Trust Agreement, the following rules will apply:

Variable Rate Bonds. Regarding Variable Rate Bonds:

(i) For the purpose of determining compliance with the Rate Covenant for any period prior to the date of calculation, the rate of interest borne by any Outstanding Variable Rate Bonds will be deemed to be the actual weighted average rate in effect thereon during such period.

(ii) For the purpose of determining whether Bonds may be issued in compliance with the Additional Bonds Test, regardless of whether such Bonds are to be Variable Rate Bonds, the rate of interest to be borne by any Outstanding Variable Rate Bonds will be deemed to be the highest rate of interest borne by such Variable Rate Bonds during the preceding 24 months, but if such Variable Rate Bonds have not been outstanding for 24 months preceding the date of the computation, then it is to be assumed that such Variable Rate Bonds will bear interest at a fixed interest rate equal to 120% of the highest rate applicable within the 24 months immediately preceding the date of the computation under an index generally accepted in the securities industry for securities

having ratings and maturity or tender dates comparable to those of the Outstanding Variable Rate Bonds.

(iii) For the purpose of determining whether Bonds that are to be Variable Rate Bonds may be issued in compliance with the Additional Bonds Test, and for the purpose of determining the amount of the Debt Service Reserve Requirement, if any, attributable to such Variable Rate Bonds, the rate of interest to be borne by such Variable Rate Bonds will be deemed to be a fixed interest rate equal to 120% of the highest rate applicable within the 24 months immediately preceding the date of the computation under an index generally accepted in the securities industry for securities having ratings and maturity or tender dates comparable to those of the proposed Variable Rate Bonds.

(iv) For the purpose of determining whether any Outstanding Variable Rate Bond is deemed paid and discharged for purposes of the Trust Agreement, such Variable Rate Bond will be deemed to bear interest at the actual rate of interest borne thereby for the remainder of the period that such rate remains in effect, and for any subsequent period prior to the date on which such Variable Rate Bond is actually to be paid and discharged, such Variable Rate Bond will be deemed to bear interest at the maximum rate of interest such Variable Rate Bond may bear pursuant to the Supplemental Trust Agreement applicable thereto.

Balloon Bonds. Regarding any Bonds that are or are proposed to be issued as Balloon Bonds, for the purpose of (i) determining compliance with the Rate Covenant for any period prior to the date of calculation, (ii) determining whether Additional Bonds, regardless of whether they are to be Balloon Bonds, may be issued in compliance with the Additional Bonds Test when any Balloon Bonds are outstanding, (iii) determining whether Bonds that are to be Balloon Bonds may be issued in compliance with the Additional Bonds Test, and (iv) determining the amount of the Debt Service Reserve Requirement, if any, attributable to such Balloon Bonds, the Debt Service on such Bonds will be determined as follows:

(A) if such Balloon Bonds are not Capital Appreciation Bonds, by assuming that such Balloon Bonds are to be amortized on the basis of level debt service over the Assumed Amortization Period and that such Bonds bear interest at the Assumed Interest Rate; and

(B) if such Balloon Bonds are Capital Appreciation Bonds, by assuming that the Appreciated Principal Amount of such Bonds at maturity is to be amortized on the basis of level principal payments over the Assumed Amortization Period.

Capital Appreciation Bonds. Regarding any Bonds that are or are proposed to be issued as Capital Appreciation Bonds, for the purpose of (i) determining compliance with the Rate Covenant for any period prior to the date of calculation, (ii) determining whether Bonds, regardless of whether they are to be Capital Appreciation Bonds, may be issued in compliance with the Additional Bonds Test when any Capital Appreciation Bonds are outstanding, (iii) determining whether Bonds that are to be Capital Appreciation Bonds may be issued in compliance with the Additional Bonds Test, and (iv) determining the amount of the Debt Service Reserve Requirement attributable to such Capital Appreciation Bonds, the Debt Service on such Bonds will include the Appreciated Principal Amounts at maturity.

Interest Rate Hedge Agreements. If the City enters into an Interest Rate Hedge Agreement to simulate a fixed rate of interest on Variable Rate Bonds, then the City may (1) secure the portion of the payment obligations of the City thereunder that represents the equivalent of interest on the notional amount of the Interest Rate Hedge Agreement with a pledge of the Net Revenues on a parity with the pledge thereof that secures the Bonds, and (2) cause the debt structure that is simulated through the combination of the Variable Rate Bonds with such Interest Rate Hedge Agreement to apply for purposes of calculating or projecting the Debt Service on such Variable Rate Bonds for any period of time during which such Interest Rate Hedge Agreement is to be effective, provided that (i) the debt structure that is simulated through the combination of the Variable Rate Bonds with such Interest Rate Hedge Agreement complies with the restrictions of the Trust Agreement on the terms of and security for the Bonds applied to that structure as though it consisted entirely of Bonds and as though the portion of the payment obligations of the City thereunder that represents the equivalent of interest on the notional amount of the Interest Rate Hedge Agreement payable to the counterparty to the Interest Rate Hedge Agreement constituted Bond Service Charges, (ii) the counterparty to any swap agreement and the provider of any interest rate cap is rated either of the two highest rating categories by at least two Rating Agencies, (iii) no such Interest Rate Hedge Agreement purports to entitle the counterparty to the Interest Rate Hedge Agreement to payment by the City from any source other than the Net Revenues, but such Interest Rate Hedge Agreement may provide for securing any portion of the payment obligations of the City thereunder that represents the equivalent of interest on the notional amount of the Interest Rate Hedge Agreement (but does not represent, among other things, any termination payment that may be payable by the City thereunder) with a pledge of the Net Revenues on a parity with the pledge thereof that secures the Bonds; and (iv) the cost of obtaining such Interest Rate Hedge Agreement has been determined by the Director of Finance (who may obtain the written advice of a Financial Advisor) to be justified by the corresponding benefit to the City and to be commercially reasonable based on then current market conditions. In the event the City enters into any such Interest Rate Hedge Agreement, it may not exercise any option to terminate such Interest Rate Hedge Agreement unless the Variable Rate Bonds to which such Interest Rate Hedge Agreement had related and that will remain outstanding after the termination thereof would fulfill the requirements of the Trust Agreement as they apply to Variable Rate Bonds, as though such Variable Rate Bonds were being issued on the date of the termination of such Interest Rate Hedge Agreement. If deposits to the Debt Service Fund for Parity Swap Obligations created pursuant this paragraph are required to be made more frequently than had been required for Bond Service Charges on any Outstanding Bonds, then, from and after the time such deposits for Parity Swap Obligations commence, deposits to the Debt Service Fund must be required for all Outstanding Bonds with the same frequency as they are required for such Parity Swap Obligations.

If the City enters into an Interest Rate Hedge Agreement to simulate a variable rate of interest on fixed rate Bonds, then the City may (1) secure the portion of the payment obligations of the City thereunder that represents the equivalent of interest on the notional amount of the Interest Rate Hedge Agreement with a pledge of the Net Revenues on a parity with the pledge thereof that secures the Bonds, and (2) cause the debt structure that is simulated through the combination of the fixed rate Bonds with such Interest Rate Hedge Agreement to apply for purposes of calculating or projecting the Debt Service on such fixed rate Bonds for any period of time during which such Interest Rate Hedge Agreement is to be effective, provided that (i) the

debt structure that is simulated through the combination of the fixed rate Bonds with such Interest Rate Hedge Agreement complies with the restrictions of the Trust Agreement on the terms of and security for the Bonds applied to that structure as though it consisted entirely of Variable Rate Bonds and as though the portion of the payment obligations of the City thereunder that represents the equivalent of interest on the notional amount of the Interest Rate Hedge Agreement payable to the counterparty to the Interest Rate Hedge Agreement constituted Bond Service Charges, (ii) the counterparty to any swap agreement and the provider of any interest rate cap is rated either of the two highest rating categories by at least two Rating Agencies, (iii) no such Interest Rate Hedge Agreement purports to entitle the counterparty to the Interest Rate Hedge Agreement to payment by the City from any source other than the Net Revenues, but such Interest Rate Hedge Agreement may provide for securing any portion of the payment obligations of the City thereunder that represents the equivalent of interest on the notional amount of the Interest Rate Hedge Agreement (but does not represent, among other things, any termination payment that may be payable by the City thereunder) with a pledge of the Net Revenues on a parity with the pledge thereof that secures the Bonds; and (iv) the cost of obtaining such Interest Rate Hedge Agreement has been determined by the Director of Finance (who may obtain the written advice of a financial advisor) to be justified by the corresponding benefit to the City and to be commercially reasonable based on then current market conditions. In the event the City enters into any such Interest Rate Hedge Agreement, it may not exercise any option to terminate such Interest Rate Hedge Agreement unless the fixed rate Bonds to which such Interest Rate Hedge Agreement had related and that will remain outstanding after the termination thereof would fulfill the requirements of the Trust Agreement as they apply to fixed rate Bonds, as though such fixed rate Bonds were being issued on the date of the termination of such Interest Rate Hedge Agreement. If deposits to the Debt Service Fund for Parity Swap Obligations created pursuant this paragraph are required to be made more frequently than had been required for Bond Service Charges on any Outstanding Bonds, then, from and after the time such deposits for Parity Swap Obligations commence, deposits to the Debt Service Fund must be required for all Outstanding Bonds with the same frequency as they are required for such Parity Swap Obligations.

Covenants of the City

The City covenants under the General Bond Ordinance with the Holders of the Bonds as follows:

(i) The City will maintain the System, or will cause the System to be maintained, in sound operating condition and free from any liens or encumbrances which would interfere with the full and continuous use and operation thereof by the City.

(ii) The City will keep proper books of record and account so as to show the complete financial results of the operation of the System, including Revenues and Operating Expenses and all capital expenditures therefor, and will invest and deposit cash receipts that constitute Revenues in accordance with applicable laws, the General Bond Ordinance and the Trust Agreement.

(iii) The City will at all times prescribe, charge and collect such rates, rentals and other charges for the use of the services of the System and must restrict the Operating Expenses of the System in such a way that Net Revenues Available for Debt Service

(adjusted for purposes of clause (i) as described in the following sentence) in each Fiscal Year may be not less than the greater of (i) 120% of Debt Service on all Bonds then Outstanding payable during that Fiscal Year or (ii) 100% of the sum of: (A) Debt Service on all Bonds then Outstanding payable during that Fiscal Year, (B) all required deposits to the Debt Service Reserve Fund during that Fiscal Year, (C) all required deposits to the Subordinated Debt Service Fund and to any reserve fund securing Subordinated Indebtedness during that Fiscal Year, (D) all required deposits to the Contingency Fund during that Fiscal Year, and (E) debt service payable on any other obligations payable from the Balance Subfund and deposits to any reserve funds securing such other obligations during that Fiscal Year. For purposes of clause (i) of the preceding sentence, but not for the purpose of clause (ii) of the preceding sentence, "Net Revenues Available for Debt Service" for any Fiscal Year will include any amounts transferred from the Rate Stabilization Fund to the Revenue Fund during such Fiscal Year, but will be reduced by any amounts transferred from the Revenue Fund to the Rate Stabilization Fund during that period. For purposes of determining the City's compliance with the Rate Covenant, the amount of any Direct Payments received by the City or the Trustee in any Fiscal Year will be credited against (i) Debt Service in such Fiscal Year if such Direct Payment related to Bonds or (ii) debt service on Subordinated Indebtedness in such Fiscal Year if such Direct Payment related to Subordinated Indebtedness.

(iv) If in any Fiscal Year the Net Revenues Available for Debt Service is less than the amount required under paragraph (iii), above, the City will, within 30 days following receipt by the City of its audited annual financial statement and the written request of the Trustee, employ a Consulting Engineer to make recommendations within 45 days as to a revision of the rates, fees and charges or Operating Expenses or methods of operation of the System, if any, that will result in producing the amount so required in the next succeeding Fiscal Year. The City covenants and agrees, promptly upon its receipt of such recommendations, to revise the rates, fees, and charges or Operating Expenses or methods of operation of the System and to take such other action as is in conformity with such recommendations to the extent the City feasibly may do so and to notify the Trustee of the actions it has taken. Upon the taking by the City of such actions, its failure to comply with the requirement described in paragraph (iii), above, will not constitute an Event of Default.

(vi) The City will permit the authorized representative of the Trustee, or of any Holder or Holders of 25% of the aggregate principal amount of the Bonds at the time Outstanding, to inspect the System and all records, accounts, and data relating to Revenues and the collection thereof, at all reasonable times.

(vii) The City will not create nor suffer to be created any pledge, lien, or charge, except to secure Additional Bonds issued in accordance herewith, that would constitute a pledge or lien prior to, or on a parity with, the pledge of and lien on Net Revenues created by the General Bond Ordinance and the Trust Agreement to secure the Bonds.

(viii) The City will maintain a risk management program that provides protection against risks of damage or destruction of the System and liabilities for the operations of the System. In furtherance thereof, the City may, among other things: self-

insure in whole or in part; adopt alternative or supplemental risk management programs that the Council of the City determines to be reasonable; organize either solely or in connection with other political subdivisions, organizations or captive insurance companies; participate in programs of captive insurance companies organized by others; establish a self-insurance trust fund; participate in mutual or other cooperative insurance or other risk management programs or pools with other political subdivisions or organizations; participate in or enter into agreements with local, State or federal governments in order to achieve such insurance; or participate in other alternative risk management programs. In case of any damage to or destruction of any part of the System amounting to a loss of $500,000 or more, the City promptly must give or cause to be given written notice thereof to the Trustee generally describing the nature and extent of such damage or destruction. Regardless of whether the net proceeds of insurance, if any, received on account of such damage or destruction are sufficient for such purpose, the City promptly will commence and complete, or cause to be commenced and completed, the repair or restoration of the System as nearly as practicable to the value, condition and character thereof existing immediately prior to such damage or destruction, with such changes or alterations, however, as the City may deem necessary for proper operation of the System and as will not impair or diminish, in the judgment of the City, the suitability of the System as a water pollution control system.

(ix) The City will, for so long as any Bonds are Outstanding, retain under the control of the City all facilities from time to time comprising the System, subject to disposition in the ordinary course for substitution or removal upon obsolescence, and will, in the exercise of such control, comply with all applicable requirements of law. For purposes of the covenant described by this paragraph, the term "control" includes either the ownership or the operation of such facilities by the City.

(x) The City will observe and perform all its agreements and obligations provided for by the Series 2016 Bonds, any Additional Bonds, the General Bond Ordinance or the Trust Agreement.

(xi) The City will, for each Fiscal Year, prepare and adopt, within the time period and in a manner prescribed by law, a detailed budget of the estimated Revenues and Operating Expenses of the System. The budget must provide for all operating Revenues, including interest income, and all Operating Expenses and other commitments required under the Trust Agreement.

(xii) The City will take whatever action may be reasonably required to enforce its rights against any Person collecting charges on behalf of the City for the use of the System and to protect the City's interest in any funds collected by such Person that are payable to the City.

Events of Default and Remedies

The following constitute "Events of Default" under the Trust Agreement:

(a) Default in the payment of any interest on any Bond when and as the same becomes due and payable;

(b) Default in the payment of the principal of (including any Sinking Fund Installments) or any redemption premium on any Bond when and as the same becomes due and payable, whether at the stated maturity or due date thereof, by acceleration or by call for redemption pursuant to any mandatory or optional redemption requirements;

(c) Failure by the City to perform or observe any other covenant, agreement or condition on the part of the City contained in the General Bond Ordinance, any Supplemental Bond Ordinance, the Master Trust Agreement or any Supplemental Trust Agreement or in the Bonds (except if and to the extent that the City's failure to perform or observe any such other covenant, agreement or condition is expressly stated not to constitute an Event of Default), which failure or default continues for a period of 60 days after written notice, by registered or certified mail, to the City specifying the failure or default and requiring the same to be remedied, which notice may be given by the Trustee in its discretion and which notice must be given by the Trustee at the written request of the Holders of not less than 25% in aggregate principal amount of Bonds then Outstanding; and

(d) Certain acts of insolvency involving the City.

Upon the occurrence of an Event of Default described in paragraphs (a), (b) or (d) above, the Trustee may declare the principal of all Bonds then Outstanding (if not then due and payable) and the interest accrued thereon to be due and payable immediately, whereupon that portion of the principal of the Series 2016 Bonds thereby coming due and the interest thereon accrued to the date of payment will, without further action, become and be immediately due and payable, anything in the Trust Agreement or in the Series 2016 Bonds to the contrary notwithstanding. If, after the occurrence of an Event of Default, the principal and interest has been thus declared due and payable because of such Event of Default, and prior to the entry of judgment in a court of law or equity for enforcement under the Trust Agreement or the appointment, and the confirmation thereof, of a receiver thereunder after an opportunity for hearing by the City, all sums payable under the Trust Agreement except the principal of, and interest accrued after the next preceding Interest Payment Date on, the Bonds which have not reached their stated maturity dates and which are due and payable solely by reason of that declaration, have been duly paid or provided for by deposit with the Trustee or Paying Agents, and all existing defaults have been made good, then and in every such case such payment or provision for payment will constitute a waiver of such Event of Default and its consequences and an automatic recession and annulment of such declaration, but no such waiver and rescission will extend to or affect any such subsequent Event of Default or impair any rights consequent thereon.

Upon the occurrence of any Event of Default under the Trust Agreement, the Trustee, among other remedies, will be entitled to the appointment of a receiver to administer and operate the System on behalf of the City, upon the filing of a suit to enforce the rights of the Trustee and the Holders of Bonds.

Application of Moneys Received Pursuant to Remedial Action

All moneys received by the Trustee or receiver pursuant to any right given or action taken under the provisions the Trust Agreement will first be applied to the payment of the cost and expenses of the proceedings resulting in the collection of such moneys and of the fees,

expenses, liabilities and advances incurred or made by the Trustee or receiver. The balance of such moneys will then be applied first to the payment, without priority or preference, of the interest on and principal or redemption premium of, if any, on the Bonds, in the manner set forth in the Trust Agreement.

Rights and Remedies of Holders

No Holder of any Bond will have any right to institute any suit, action or proceeding for the enforcement of the Trust Agreement or for the execution of any trust thereof or for the appointment of a receiver or any other remedy under the Trust Agreement unless an Event of Default has occurred and is continuing, such Holder has previously have given to the Trustee written notice of such Event of Default, and the Holders of not less than 25% in aggregate principal amount of Bonds then Outstanding have made written request to the Trustee and have afforded the Trustee reasonable opportunity either to proceed to exercise the powers granted or to institute such action, suit or proceeding in its own name and have also offered to the Trustee an indemnity as provided in the Trust Agreement, and the Trustee thereafter fails or refuses to exercise the powers granted under the Trust Agreement, or to institute such action, suit or proceeding in its own name.

Rights and Remedies Related to Other Secured Obligations

In connection with the City's entering into any Interest Rate Hedge Agreement or Reimbursement Agreement or incurring or issuing any Subordinated Indebtedness, a Supplemental Trust Agreement may amend or supplement the provisions of the Master Trust Agreement related to Events of Default and consequent rights and remedies to provide for the protection and enforcement of the rights of the other party or parties to such Interest Rate Hedge Agreement or Reimbursement Agreement or the holder or holders of such Subordinated Indebtedness and for the application of moneys to the payment of Parity Swap Obligations, Credit Obligations or debt service charges on Subordinated Indebtedness, provided that all such provisions must be consistent with the Master Trust Agreement and with the priorities of payment and security established hereby, and no such provision may diminish or derogate from the rights conferred on the Holders of Bonds and the priority of Bond Service Charges in a manner inconsistent with those priorities. Without limiting the generality of the provision in the preceding sentence, no party entitled to the payment of Credit Obligations may be given the right, so long as any Bonds are outstanding or any Parity Swap Obligations are payable, to direct the acceleration of the maturity of any Bonds, nor may the holders of any Subordinated Indebtedness be given the right, so long as any Bonds are outstanding or any Parity Swap Obligations or Credit Obligations are payable, to direct the acceleration of the maturity of any Bonds. The Trust Agreement provides that the provisions described in this paragraph will not be interpreted to limit the rights of Credit Providers to the enforcement of rights to payment of Bond Service Charges, nor may it be interpreted to limit the rights of parties entitled to the payment of Parity Swap Obligations to the enforcement of those rights.

Supplemental Trust Agreements

The Trust Agreement provides that the City and the Trustee may, without the consent of, or notice to, any of the Holders, enter into a Supplemental Trust Agreement in connection with the issuance of Additional Bonds. In addition, the City and the Trustee may, without the consent

of, or notice to, any of the Bondholders enter into Supplemental Trust Agreements as may not, in the opinion of the City and the Trustee, be inconsistent with the terms and provisions of the Trust Agreement for any of the following purposes: (i) to cure any ambiguity, inconsistency or formal defect or omission in the Master Trust Agreement or any Supplemental Trust Agreement; (ii) to grant to or confer upon the Trustee for the benefit of the Bondholders any additional rights, remedies, powers or authority that may lawfully be granted to or conferred upon the Bondholders or the Trustee; (iii) to subject additional revenues or receipts to the pledge of the Trust Agreement; (iv) to add to the covenants and agreements of the City contained in the Trust Agreement other covenants and agreements thereafter to be observed for the protection of the Bondholders, or to surrender or limit any right, power or authority reserved to or conferred upon the City in the Master Trust Agreement, including the limitation of rights of redemption so that in certain instances Bonds of different Series will be retained in some prescribed relation to one another; (v) to evidence any succession to the City and the assumption by such successors of the covenants and agreements of the City contained in the Master Trust Agreement and the Bonds; (vi) to permit the exchange of Bonds, at the option of the Holders thereof, for coupon Bonds of the same series payable to bearer, in an aggregate principal amount not exceeding the unmatured and unredeemed principal amount of the Predecessor Bonds, bearing interest at the same rate or rates and maturing on the same date or dates, with coupons attached representing all unpaid interest due or to become due thereon if, in the opinion of Bond Counsel, that exchange would not result in the interest on any of the Bonds Outstanding becoming subject to federal income taxation; (vii) to permit the use of a Book-Entry system to identify the owner of an interest in an obligation issued by the City under the Trust Agreement, whether that obligation was formerly, or could be, evidenced by a tangible security; (viii) to permit the Trustee to comply with any obligations imposed upon it by law; (ix) to specify further the duties and responsibilities of, and to define further the relationship among, the Trustee, the Registrar and any Paying Agents; (x) to achieve compliance of the Trust Agreement with any applicable federal securities or tax law; (xi) to make any change necessary to obtain or maintain a rating on any of the Bonds by a Rating Service; and (xii) to make any other change to insert any provision into or delete or amend any provision of the Trust Agreement, provided that such insertion, deletion or amendment will not adversely affect the rating (other than any rating based upon credit enhancement or liquidity support provided by any party other than the City, which may not be taken into account for purposes of the clause described by this paragraph) then assigned to any Bonds Outstanding by any Rating Agency.

All other Supplemental Trust Agreements must be approved by the Holders of not less than 51% in aggregate principal amount of the Bonds then outstanding (excluding Bonds then held or owned by the City); provided that a Supplemental Trust Agreement permitting (i) an extension of the maturity of the principal of or the interest on any Bond, or a reduction in the principal amount of any Bonds or the rate of interest or redemption premium thereon, or a reduction in the amount or extension of the time of any payment required by any mandatory sinking fund requirements, may not be entered into without the consent of the Holder of each Bond so affected or (ii) a reduction in the aggregate principal amount of the Bonds required for consent to such Supplemental Trust Agreements, may not be entered into without the consent of the Holders of all of the Bonds then outstanding.

Defeasance

If the City pays or causes to be paid all principal, redemption premium, if any, and interest due or to become due on the Bonds, and provision is made for paying all other sums payable by the City under the Trust Agreement, the Trustee will release and discharge the Trust Agreement. All Outstanding Bonds will be deemed to have been paid or caused to be paid within the meaning of the Trust Agreement if (i) the Trustee and any Paying Agents hold, in trust for and irrevocably committed thereto, sufficient moneys, or (ii) the Trustee will hold, in trust for and irrevocably committed thereto, Defeasance Obligations which are of such maturities and interest payment dates and bear such interest as will, without further investment or reinvestment of either the principal amount thereof or the interest earnings therefrom, be sufficient together with moneys referred to in clause (i) above, for the payment, at their maturity or redemption date, of all principal, redemption premium, if any, and interest thereof and thereon to the date of maturity or redemption, as the case may be, or if default in such payment has occurred on such date then to the date of the tender of such payment; provided, that if any Bonds are to be redeemed prior to the maturity thereof, notice of such redemption must have been duly given or provision satisfactory to the Trustee have been duly made for the giving of such notice.

The Trustee

The Trustee agrees in the Trust Agreement to perform the trusts imposed upon it in the Trust Agreement as an ordinarily prudent trustee under a trust agreement. The Trust Agreement provides that the Trustee may act upon the opinion or advice of any attorney and will not be responsible for any loss or damage resulting from any action taken or not taken in good faith reliance upon such opinion or advice. In addition, the Trust Agreement provides that the Trustee will be protected in acting upon certain documents believed to be genuine and correct and to have been signed or sent by the proper persons.

Extent of City's Covenants; No Personal Liability

All covenants, stipulations, obligations and agreements of the City contained in the Trust Agreement will be deemed to be covenants, stipulations, obligations and agreements of the City to the full extent authorized by the Ohio Revised Code and permitted by the Constitution of the State and the Charter of the City. No covenant, stipulation, obligation or agreement contained in the Trust Agreement will be deemed to be a covenant, stipulation, obligation or agreement of any present or future member of the Council of the City or agent or employee of the City in his or her individual capacity, and neither the members of the Council of the City nor any official executing the Bonds will be liable personally on the Bonds or be subject to any personal liability or accountability by reason of the issuance thereof.

APPENDIX C

FORM OF BOND COUNSEL OPINION

April 14, 2016

To: City of Cleveland, Ohio

Morgan Stanley & Co. LLC

Zions Bank, a Division of ZB, National Association, as Trustee

We have served as bond counsel to our client the City of Cleveland, Ohio (the "City"), and not as counsel to any other person in connection with the issuance by the City of its $32,390,000 Water Pollution Control Revenue Bonds, Series 2016 (the "Series 2016 Bonds"), dated the date of this letter. The Series 2016 Bonds are issued pursuant to the authority of the Constitution and laws of the State of Ohio, including Article XVIII of the Ohio Constitution, the Charter of the City, and Ordinances No. 29-16 (the "General Bond Ordinance") and No. 28-16 (the "Series 2016 Ordinance"), each passed by the Council of the City on January 25, 2016, and under the Master Trust Agreement, dated April 14, 2016, which includes the General Bond Ordinance (together, the "Master Trust Agreement"), as supplemented by the First Supplemental Trust Agreement, dated April 14, 2016, which includes the Series 2016 Ordinance (the "First Supplemental Trust Agreement" and together with the Master Trust Agreement, the "Trust Agreement"), each between the City and Zions Bank, a Division of ZB, National Association, as trustee (the "Trustee"). Capitalized terms not otherwise defined in this letter are used as defined in the Trust Agreement.

In our capacity as bond counsel, we have examined the transcript of proceedings relating to the issuance of the Series 2016 Bonds, a copy of the signed and authenticated Series 2016 Bond of the first maturity, the Trust Agreement and such other documents, matters and laws as we deem necessary to render the opinions set forth in this letter.

Based on that examination and subject to the limitations stated below, we are of the opinion that under existing law:

1. The Series 2016 Bonds and the Trust Agreement are valid and binding obligations of the City enforceable in accordance with their respective terms.

2. The Series 2016 Bonds constitute special obligations of the City, and the principal of and interest (collectively, "debt service") on the Series 2016 Bonds, together with debt service on Bonds to be issued and outstanding, from time to time, under the Master Trust Agreement (collectively, the "Bonds"), are payable from and secured solely by the Net Revenues and Special Funds established under the Master Trust Agreement, all as described in the Trust Agreement. The Bonds and the payment of debt service on the Bonds are not secured by an obligation or pledge of any money raised by taxation, and the Bonds do not represent or

constitute a general obligation debt of the City or a pledge of the faith and credit of the City, the State of Ohio or any of its political subdivisions.

3. Interest on the Series 2016 Bonds is excluded from gross income for federal income tax purposes under Section 103 of the Internal Revenue Code of 1986, as amended (the "Code"), and is not an item of tax preference for purposes of the federal alternative minimum tax imposed on individuals and corporations; however, portions of the interest on the Series 2016 Bonds earned by certain corporations may be subject to a corporate alternative minimum tax. Interest on, and any profit made on the sale, exchange or other disposition of, the Series 2016 Bonds are exempt from all Ohio state and local taxation, except the estate tax, the domestic insurance company tax, the dealers in intangibles tax, the tax levied on the basis of the total equity capital of financial institutions, and the net worth base of the corporate franchise tax. We express no opinion as to any other tax consequences regarding the Series 2016 Bonds.

The opinions stated above are based on an analysis of existing laws, regulations, rulings and court decisions and cover certain matters not directly addressed by such authorities. In rendering all such opinions we assume, without independent verification, and rely upon (i) the accuracy of the factual matters represented, warranted or certified in the proceedings and documents we have examined, (ii) the due and legal authorization, execution and delivery of those documents by, and the valid, binding and enforceable nature of those documents upon, any parties other than the City, and (iii) the due authorization, signing and delivery by, and the binding effect upon and enforceability against, the Trustee of the Trust Agreement.

In rendering those opinions with respect to the treatment of the interest on the Series 2016 Bonds under the federal tax laws, we further assume and rely upon compliance with the covenants in the proceedings and documents we have examined, including those of the City. Failure to comply with certain of those covenants subsequent to issuance of the Series 2016 Bonds may cause interest on the Series 2016 Bonds to be included in gross income for federal income tax purposes retroactively to their date of issuance.

The rights of the owners of the Series 2016 Bonds and the enforceability of the Series 2016 Bonds and the Trust Agreement are subject to bankruptcy, insolvency, arrangement, fraudulent conveyance or transfer, reorganization, moratorium and other laws relating to or affecting creditors' rights, to the application of equitable principles, to the exercise of judicial discretion, and to limitations on legal remedies against public entities.

The opinions rendered in this letter are stated only as of this date, and no other opinion shall be implied or inferred as a result of anything contained in or omitted from this letter. Our engagement as bond counsel with respect to the Series 2016 Bonds has concluded on this date.

Respectfully submitted,

APPENDIX D
FORM OF CONTINUING DISCLOSURE AGREEMENT

$32,390,000
City of Cleveland, Ohio
Water Pollution Control Revenue Bonds
Series 2016

CONTINUING DISCLOSURE AGREEMENT

THIS CONTINUING DISCLOSURE AGREEMENT, dated as of April 14, 2016 (this "**Agreement**"), is made, signed and delivered by and between the **CITY OF CLEVELAND, OHIO**, a municipal corporation and political subdivision duly organized and existing under its Charter and the Constitution and laws of the State of Ohio (the "**City**"), and **ZIONS BANK, A DIVISION OF ZB, NA**, as trustee (together with its lawful successors, the "**Trustee**"), for the benefit of the Holders and Beneficial Owners (as defined herein) from time to time of the City's $32,390,000 Water Pollution Control Revenue Bonds, Series 2016 (the "**Series 2016 Bonds**"), authorized by Ordinance No. 29-16 passed by the Council of the City on January 25, 2016 (the "**General Bond Ordinance**") and Ordinance 28-16 passed by the Council of the City on January 25, 2016 (the "**Bond Ordinance**").

RECITAL

The City, by passage of the Bond Ordinance, has determined to issue the Series 2016 Bonds to provide funds for City purposes, and Morgan Stanley & Co., LLC, as representative of the underwriters (collectively, the "**Participating Underwriter**") identified in the Bond Purchase Agreement dated March 31, 2016 for the Series 2016 Bonds, has agreed to provide those funds to the City by purchasing the Series 2016 Bonds. As a condition to the purchase of the Series 2016 Bonds from the City and the sale of the Series 2016 Bonds to the Holders and Beneficial Owners, the Participating Underwriter is required to reasonably determine that the City has undertaken, in a written agreement for the benefit of the Holders and Beneficial Owners of the Series 2016 Bonds, to provide certain information in accordance with the Rule (as defined herein).

NOW, THEREFORE, in consideration of the foregoing and the mutual representations and agreements hereinafter contained, the City and the Trustee agree, in accordance with the provisions of the Rule, for the benefit of the Holders and Beneficial Owners from time to time of the Series 2016 Bonds, as set forth in this Agreement.

Section 1. **Purpose of Continuing Disclosure Agreement**. This Agreement is being entered into, signed and delivered for the benefit of the Holders and Beneficial Owners of the Series 2016 Bonds and in order to assist the Participating Underwriter of the Series 2016 Bonds in complying with Rule 15c2-12(b)(5) (the "**Rule**") promulgated by the Securities and Exchange Commission ("**SEC**") pursuant to the Securities Exchange Act of 1934, as may be amended from time to time.

Section 2. Definitions. In addition to the definitions set forth above, the following capitalized terms shall have the following meanings in this Agreement, unless the context clearly otherwise requires. Reference to "Sections" shall mean sections of this Agreement.

"Annual Filing" means any annual information filing provided by the City pursuant to, and as described in, Sections 3 and 4.

"Audited Financial Statements" means the audited financial statements of the Water Pollution Control Division (the "Division") of the City, prepared in conformity with generally accepted accounting principles for governmental units, except as may be modified from time to time and described in such financial statements.

"Authorized Disclosure Representative" means the person or persons at the time designated to act on behalf of the City by written notice, substantially in the form of Exhibit 4 to this Agreement, furnished to the Trustee, containing the specimen signature of such person or persons and signed on behalf of the City by the Director of Finance. The written notice may designate an alternate or alternates, each of whom shall have the same authority, duties and powers as such Authorized Disclosure Representative.

"Beneficial Owner" means any person that (a) has the power, directly or indirectly, to vote or consent with respect to, or to dispose of ownership of, any Series 2016 Bonds (including persons holding Series 2016 Bonds through nominees, depositories or other intermediaries), or (b) is treated as the owner of any Series 2016 Bonds for federal income tax purposes.

"Business Day" means any day other than a Saturday, a Sunday or a day on which the offices of the City are closed.

"Filing Date" means the last day of the ninth month following the end of the Fiscal Year (or the next succeeding Business Day if such day is not a Business Day), commencing September 30, 2016.

"Fiscal Year" means the 12-month period beginning on January 1 of each year or such other 12-month period as the Division shall adopt as its fiscal year.

"Holder" means, with respect to the Series 2016 Bonds, the person in whose name a Bond is registered in accordance with the Trust Agreement.

"MSRB" means the Municipal Securities Rulemaking Board.

"Obligated Person" means, any person, including the issuer of municipal securities (such as the Series 2016 Bonds), who is generally committed by contract or other arrangement to support payment of all or part of the obligations on the municipal securities being sold in an offering document (such as the Official Statement); the City is the only Obligated Person for the Series 2016 Bonds.

"Official Statement" means the Official Statement for the Series 2016 Bonds dated March 31, 2016.

"Specified Events" means any of the events with respect to the Series 2016 Bonds as set forth in Section 5(a).

"State" means the State of Ohio.

"Trust Agreement" means the Master Trust Agreement, effective as of April 14, 2016, as supplemented by the First Supplemental Trust Agreement dated as of April 14, 2016, by and between the City and the Trustee.

Section 3. Provision of Annual Information.

(a) The City shall provide (or cause to be provided) not later than the Filing Date to the MSRB and to the Trustee an Annual Filing, which is consistent with the requirements of Section 4. The Annual Filing shall be submitted in an electronic format and contain such identifying information as is prescribed by the MSRB, and may be submitted as a single document or as separate documents comprising a package, and may cross-reference other information as provided in Section 4; provided that the Audited Financial Statements of the Division may be submitted separately from the balance of the Annual Filing and later than the Filing Date if they are not available by that date. If the City's Fiscal Year changes, it shall give notice of such change in the same manner as for a Specified Event under Section 5.

(b) If the City is unable to provide to the MSRB an Annual Filing by the Filing Date, the City shall, in a timely manner, send a notice to the MSRB in an electronic format as prescribed by the MSRB.

Section 4. Content of Annual Filing. The City's Annual Filing shall contain or include by reference the following:

(a) Financial information and operating data of the type included in the Official Statement under the following: (1) the tables under the captions "OUTSTANDING BONDS AND OTHER OBLIGATIONS" and "DEBT SERVICE SCHEDULE;" (2) the table under the subcaption "THE DIVISION OF WATER POLLUTION CONTROL – Customer Base;" (3) the table under the subcaption "THE DIVISION OF WATER POLLUTION CONTROL – Historical Customer Accounts and Consumption;" (4) the table under the subcaption "THE WATER POLLUTION CONTROL – Projected Customer Accounts and Consumption;" (5) the table under the subcaption "THE DIVISION OF WATER POLLUTION CONTROL – Largest Customers of the Division;" (6) the table under the subcaption "THE DIVISION OF WATER POLLUTION CONTROL – Capital Improvement Programs;" and (7) the table "Summary of Historical Revenues, Expenses and Debt Service" under the subcaption "REVENUES, EXPENSES AND DEBT SERVICE - Historical Revenues and Expenses."

(b) The Audited Financial Statements.

The foregoing shall not obligate the City to prepare or update projections of any financial information or operating data. If any amendment of this Agreement is made concerning the operating data or financial information contained in the Annual Filing, the City shall include the information required by Section 6 of this Agreement.

Any or all of the items listed above may be included by specific reference to other documents, including a comprehensive annual informational report of the City or official statements of debt issues of the City or related public entities, which have been submitted to the MSRB or the SEC. The City shall clearly identify each such other document so included by reference.

(c) If the Trustee has not received the Annual Filing for a Fiscal Year by its close of business on the fifteenth Business Day preceding the Filing Date for that Fiscal Year, the Trustee shall provide a notice to the Authorized Disclosure Representative, not later than its close of business on the next Business Day, substantially in the form of the attached Exhibit 1, by facsimile transmission (or other means similarly prompt). If the Trustee has not received that Annual Filing by its close of business on the Filing Date, the Trustee shall provide a notice to the Authorized Disclosure Representative, not later than its close of business on the next Business Day, substantially in the form of the attached Exhibit 2, by facsimile transmission (or other means similarly prompt). The City shall be entitled to provide to the Trustee written evidence of the submission of the Annual Filing in accordance with Section 3 above, including a certificate of the Authorized Disclosure Representative as to the relevant facts, and, if applicable, a written statement regarding any failure to comply with Section 3 above. The Trustee shall be entitled to rely conclusively upon any written evidence provided by the City regarding the provision of that information to the MSRB. If, in any instance, the required information was not timely filed or the City fails to provide evidence by 3:00 p.m., Cleveland, Ohio time, on the second Business Day following the Filing Date of its timely filing with the MSRB, the Trustee shall send or cause to be sent promptly after receipt of any such evidence or statement from the City, but in any event not later than its close of business on the third Business Day following the Filing Date, a notice substantially in the form of the attached Exhibit 3, modified to reflect the pertinent facts, to the MSRB in an electronic format as prescribed by the MSRB.

Section 5. Reporting Specified Events.

(a) The City shall provide (or cause to be provided) to the MSRB, in an electronic format and containing such identifying information as is prescribed by the MSRB and in a timely manner but not later than ten Business Days after the occurrence of the event, and to the Trustee, notice of any of the following events with respect to the Series 2016 Bonds, as specified by the Rule:

(1) Principal and interest payment delinquencies;

(2) Non-payment related defaults, if material;

(3) Unscheduled draws on debt service reserves reflecting financial difficulties;

(4) Unscheduled draws on credit enhancements reflecting financial difficulties;

(5) Substitution of credit or liquidity providers, or their failure to perform; (a)

(6) (Issuance of) Adverse tax opinions, the issuance by the Internal Revenue Service of proposed or final determinations of taxability, Notices of Proposed Issue (IRS Form 5701-TEB) or other material notices or determinations with respect to the tax status of the security (*i.e.*, the Series 2016 Bonds), or other material events affecting the tax status of the security;

(7) Modifications to rights of security holders, if material;

(8) Bond calls, if material, and tender offers; (b)

(9) Defeasances;

(10) Release, substitution, or sale of property securing repayment of the securities, if material; (c)

(11) Rating changes;

(12) Bankruptcy, insolvency, receivership or similar event of the Obligated Person; Note: For the purposes of the event identified in this subparagraph, the event is considered to occur when any of the following occur: the appointment of a receiver, fiscal agent or similar officer for an Obligated Person in a proceeding under the U.S. Bankruptcy Code or in any other proceeding under state or federal law in which a court or governmental authority has assumed jurisdiction over substantially all of the assets or business of the Obligated Person, or if such jurisdiction has been assumed by leaving the existing governmental body and officials or officers in possession but subject to the supervision and orders of a court or governmental authority, or the entry of an order confirming a plan of reorganization, arrangement or liquidation by a court or governmental authority having supervision or jurisdiction over substantially all of the assets or business of the Obligated Person.

(13) The consummation of a merger, consolidation, or acquisition involving an Obligated Person or the sale of all or substantially all of the assets of the Obligated Person, other than in the ordinary course of business, the entry into a definitive agreement to undertake such an action or the termination of a definitive agreement relating to any such actions, other than pursuant to its terms, if material; and

(14) Appointment of a successor or additional trustee or the change of name of a trustee, if material.

Note:

(a) The City has not obtained or provided, and does not expect to obtain or provide, any credit enhancements or credit or liquidity providers for the Series 2016 Bonds.

(b) Any scheduled redemption of Series 2016 Bonds pursuant to mandatory sinking fund redemption requirements does not constitute a specified event within the meaning of the Rule.

(c) Repayment of the Series 2016 Bonds is not secured by a lien on any property capable of release or sale or for which other property may be substituted.

For the Specified Events described in Section 5(a), subsections (2), (6, as applicable), (7), (8, as applicable), (10), (13) and (14), the City acknowledges that it must make a determination whether such Specified Event is material under applicable federal securities laws in order to determine whether a filing is required.

(b) The Trustee shall promptly notify the Authorized Disclosure Representative upon becoming aware of the occurrence of any Specified Event (other than a non-payment related default, the giving of a notice of optional redemption of any Series 2016 Bonds, or defeasance of the Trust Agreement as to the Series 2016 Bonds or any portion thereof).

Section 6. Amendments. The City reserves the right to amend this Agreement, and noncompliance with any provision of this Agreement may be waived, as may be necessary or appropriate to (a) achieve its compliance with any applicable federal securities law or rule, (b) cure any ambiguity, inconsistency or formal defect or omission and (c) address any change in circumstances arising from a change in legal requirements, change in law or change in the identity, nature or status of the City or type of business conducted by the City. Any such amendment or waiver shall not be effective unless the Agreement (as amended or taking into account such waiver) would have materially complied with the requirements of the Rule at the time of the primary offering of the Series 2016 Bonds, after taking into account any applicable amendments to or official interpretations of the Rule, as well as any change in circumstances, and until the City shall have received either (i) a written opinion of bond counsel or other qualified independent special counsel selected by the City that the amendment or waiver would not materially impair the interests of the Holders or Beneficial Owners or (ii) the written consent to the amendment or waiver of the Holders of at least a majority of the principal amount of the Series 2016 Bonds then outstanding. An Annual Filing containing any revised operating data or financial information shall explain, in narrative form, the reasons for any such amendment or waiver and the impact of the change on the type of operating data or financial information being provided. If the amendment relates to the accounting principles to be followed in preparing Audited Financial Statements, (A) the City shall provide notice of such change in the same manner as for a Specified Event under Section 5 and (B) the Annual Filing for the year in which the change is made should present a comparison (in narrative form and also, if feasible, in quantitative form) between the financial statements or information as prepared on the basis of the new accounting principles and those prepared on the basis of the former accounting principles.

Section 7. Additional Information. Nothing in this Agreement shall be deemed to prevent the City from disseminating any other information, using the means of dissemination set forth in this Agreement or providing any other means of communication, or including any other information in any Annual Filing or providing notice of the occurrence of an event, in addition to that which is required by this Agreement. If the City chooses to include any information in any document or notice of occurrence of an event in addition to that which is specifically required by

this Agreement, the City shall have no obligation under this Agreement to update such information or include it in any future Annual Filing or notice of occurrence of a Specified Event.

Section 8. Remedy for Breach. This Agreement shall be solely for the benefit of the Holders and Beneficial Owners from time to time of the Series 2016 Bonds. The exclusive remedy for any breach of the Agreement by the City shall be limited, to the extent permitted by law, to a right of the Holders and Beneficial Owners to institute and maintain, or to cause to be instituted and maintained, such proceedings as may be authorized at law or in equity to obtain the specific performance by the City of its obligations under this Agreement in a court in Cuyahoga County, Ohio. Any such proceedings shall be instituted and maintained only in accordance with Section 133.25(B)(4)(b) or (C)(1) of the Revised Code (or any like or comparable successor provisions); provided that any Holder or Beneficial Owner may exercise individually any such right to require the City to specifically perform its obligation to provide or cause to be provided a pertinent filing if such a filing is due and has not been made. The Trustee may exercise any such right if requested to do so by the Holders of at least 25% in aggregate principal amount of the Series 2016 Bonds then outstanding, subject to the same conditions, limitations and procedures that would apply under the Trust Agreement if the breach were an "Event of Default" under the Trust Agreement. Any Beneficial Owner seeking to require the City to comply with this Agreement shall first provide at least 30 days' prior written notice to the City of the City's failure, giving reasonable detail of such failure, following which notice the City shall have 30 days to comply. A default under this Agreement shall not be deemed an "Event of Default" under the Trust Agreement, and the sole remedy under this Agreement in the event of any failure of the City to comply with this Agreement shall be an action to compel performance. No person or entity shall be entitled to recover monetary damages under this Agreement.

Section 9. Performance by the Trustee; Compensation.

(a) Solely for the purpose of (i) defining the standards of care and performance applicable to the Trustee in the performance of its obligations under this Agreement, (ii) the manner of execution by the Trustee of those obligations, (iii) defining the manner in which, and the conditions under which, the Trustee may be required to take any action at the direction of the Holders, including the condition that indemnification be provided, and (iv) matters of removal, resignation and succession of the Trustee under this Agreement, the Trust Agreement is hereby made applicable to this Agreement as if this Agreement were contained in the Trust Agreement; provided that the Trustee shall have only such duties under this Agreement as are specifically set forth in this Agreement.

(b) Subject to Section 10 of this Agreement, the City agrees to pay to the Trustee from time to time reasonable compensation for services provided by the Trustee under this Agreement and to pay or reimburse to the Trustee upon request all reasonable out-of-pocket expenses, disbursements and advances incurred or made by the Trustee in accordance with this Agreement and with prior notice to the City, except to the extent that any such expense, disbursement or advance is due to the negligence or bad faith of the Trustee.

(c) The obligations of the City under this Section 9 shall survive resignation or removal of the Trustee and termination of other provisions of this Agreement pursuant to Section 11 below.

(d) The Trustee is a party to this Agreement for and on behalf of the Holders and Beneficial Owners of the Series 2016 Bonds and shall not be considered to be the agent of the City when performing any actions required to be taken by the Trustee under this Agreement.

(e) The Trustee shall not have any obligation under this Agreement to investigate or determine whether any filing made under this Agreement complies with federal securities laws or rules.

Section 10. Appropriation. The performance by the City of its obligations under this Agreement shall be subject to the availability of funds and their annual appropriation to meet costs that the City would be required to incur to perform those obligations. The City shall provide notice to the MSRB in the same manner as for a Specified Event under Section 5 of the failure to appropriate funds to meet costs to perform the obligations under this Agreement.

Section 11. Termination. The obligations of the City under this Agreement shall remain in effect only for such period that the Series 2016 Bonds are outstanding in accordance with their terms and the City remains an Obligated Person with respect to the Series 2016 Bonds within the meaning of the Rule. The obligation of the City to provide the information and notices of the events described above shall terminate, if and when the City no longer remains such an Obligated Person. If any person, other than the City, becomes an Obligated Person relating to the Series 2016 Bonds, the City shall use its best efforts to require such Obligated Person to comply with all provisions of the Rule applicable to such Obligated Person.

Section 12. Dissemination Agent. The City may, from time to time, appoint or engage a dissemination agent to assist it in carrying out its obligations under this Agreement, and may discharge any such agent, with or without appointing a successor dissemination agent.

Section 13. Beneficiaries. This Agreement shall inure solely to the benefit of the City, any dissemination agent, the Participating Underwriter and Holders and Beneficial Owners from time to time of the Series 2016 Bonds, and shall create no rights in any other person or entity.

Section 14. Recordkeeping. The City shall maintain records of all Annual Filings and notices of Specified Events and other events including the content of such disclosure, the names of the entities with whom such disclosures were filed and the date of filing such disclosure.

Section 15. Governing Law. This Agreement shall be governed by the laws of the State.

[Signature Page Follows]

IN WITNESS WHEREOF, the City has caused this Continuing Disclosure Agreement to be duly signed and delivered to the Participating Underwriter, as part of the Bond proceedings and in connection with the original delivery of the Series 2016 Bonds to the Participating Underwriter, on its behalf by its officials signing below, all as of the date set forth above, and the Holders and Beneficial Owners from time to time of the Series 2016 Bonds shall be deemed to have accepted this Agreement made in accordance with the Rule.

CITY OF CLEVELAND, OHIO

By: _____
Sharon Dumas, Director of Finance

By: _____
Robert L. Davis, Director of Public Utilities

The legal form is approved.

Barbara A. Langhenry
Director of Law

By:_____
Assistant Director of Law

CERTIFICATE – CONTINUING DISCLOSURE AGREEMENT

As fiscal officer of the City of Cleveland, Ohio, I certify that the money required to meet the obligations of the City under the foregoing Continuing Disclosure Agreement during Fiscal Year 2016 has been lawfully appropriated by the City for those purposes and is in the City treasury or in the process of collection to the credit of an appropriate fund, free from any previous encumbrances. This Certificate is given in compliance with Sections 5705.41 and 5705.44 of the Revised Code.

Dated: April 14, 2016

_____

Sharon Dumas, Director of Finance
City of Cleveland, Ohio

EXHIBIT 1

$32,390,000
City of Cleveland, Ohio
Water Pollution Control Revenue Bonds
Series 2016

NOTICE TO CITY REGARDING ANNUAL FILING

TO: Director of Finance
City of Cleveland, Ohio

The undersigned, as the Trustee under the Master Trust Agreement, effective as of April 14, 2016, as supplemented by the First Supplemental Trust Agreement dated as of April 14, 2016, by and between the City and the Trustee securing the captioned bonds (the "**Bonds**"), and serving in that capacity under the Continuing Disclosure Agreement (the "**Agreement**") relative to the Bonds, between the Trustee and the City of Cleveland, Ohio (the "**City**"), hereby notifies you that the City, as of the date of this notice, has not provided or caused to be provided to the undersigned the Annual Filing (as defined in the Agreement) that is required under Sections 3 and 4 of the Agreement to be so provided not later than _____. The City is required under the Agreement to provide, or to cause to be provided, the Annual Filing both to the undersigned and to the MSRB not later than that date.

Zions Bank, a Division of ZB, National Association,
Cleveland, Ohio
as Trustee

Dated: _____

By: _____
Title: _____

EXHIBIT 2

$32,390,000
City of Cleveland, Ohio
Water Pollution Control Revenue Bonds
Series 2016

NOTICE TO CITY OF FAILURE TO FILE ANNUAL FILING

TO: Director of Finance
City of Cleveland, Ohio

The undersigned, as the Trustee under the Master Trust Agreement, effective as of April 14, 2016, as supplemented by the First Supplemental Trust Agreement dated as of April 14, 2016, by and between the City and the Trustee securing the captioned bonds (the "**Bonds**"), and serving in that capacity under the Continuing Disclosure Agreement (the "**Agreement**") relating to the Bonds, between the Trustee and the City of Cleveland, Ohio (the "**City**"), hereby notifies you that the City, as of the date of this notice, has not provided or caused to be provided to the undersigned the Annual Filing (as defined in the Agreement) that was required under Sections 3 and 4 of the Agreement to be so provided not later than _____.

Please provide the required Annual Filing to the undersigned, together with written evidence as to whether that information was provided to the MSRB and, if so, when it was provided. If in any instance the Annual Filing was not timely provided to the MSRB in accordance with Section 3 of the Agreement, you may submit a written statement regarding the failure to comply that would be provided to the MSRB, with the notice that the Trustee must give of the failure to comply under Section 4(c) of the Agreement. Any such written evidence or statement must be received by the undersigned not later than 3:00 p.m., Cleveland, Ohio time, on _____. Failure to receive written evidence by that time will result in the filing by the Trustee of a notice promptly thereafter with the MSRB, substantially in the form attached as Exhibit 3 to the Agreement.

Zions Bank, a Division of ZB, National Association,
Cleveland, Ohio,
as Trustee

Dated: _____

By: _____
Title: _____

EXHIBIT 3

$32,390,000
City of Cleveland, Ohio
Water Pollution Control Revenue Bonds
Series 2016

NOTICE TO MSRB OF FAILURE TO TIMELY FILE ANNUAL FILING

The undersigned, as the Trustee under the Master Trust Agreement, effective as of April 14, 2016, as supplemented by the First Supplemental Trust Agreement dated as of April 14, 2016, by and between the City and the Trustee securing the captioned bonds (the "**Bonds**"), and serving in that capacity under the Continuing Disclosure Agreement (the "**Agreement**"), between the Trustee and the City of Cleveland, Ohio (the "**City**"), hereby notifies you that the City has not provided any written evidence to the Trustee concerning the timeliness of its filing of the annual financial information required under Rule 15c2-12(b)(5) (the "**Annual Filing**") with the MSRB.

[1. The City, as of the date of this notice, has not provided or caused to be provided to the Trustee the Annual Filing for its fiscal year that ended _____. That Annual Filing was required under the Agreement to be provided to the Trustee and the MSRB not later than _____; and]

[1. The City provided or caused to be provided to the MSRB on _____ the Annual Filing that was required to be provided to it not later than _____; and]

2. The City has provided that attached statement concerning its failure to provide or cause to be provided the Annual Filing in accordance with the Agreement. The Trustee does not assume any responsibility for the accuracy or completeness of that statement and has not and will not undertake any investigation to determine its accuracy or completeness.

Zions Bank, a Division of ZB, National Association,
Cleveland, Ohio,
as Trustee

Dated: _____

By: _____
Title: _____

cc: Authorized Disclosure Representative

EXHIBIT 4

$32,390,000
City of Cleveland, Ohio
Water Pollution Control Revenue Bonds
Series 2016

DESIGNATION OF AUTHORIZED DISCLOSURE REPRESENTATIVE AND ALTERNATE

Zions Bank, a Division of ZB, National Association
600 Superior Ave., Suite 1300
Cleveland, OH 44114

The undersigned hereby certifies that pursuant to the Continuing Disclosure Agreement between the City of Cleveland, Ohio and Zions Bank, a Division of ZB, National Association, Cleveland, Ohio, dated as of April 14, 2016, the individuals listed below have been designated as Authorized Disclosure Representative and Alternate, respectively, and that the signatures opposite their names are their true signatures.

Name	Signature
Elizabeth C. Hruby Authorized Disclosure Representative	_____
Catherine M. Troy Alternate	_____

Date: _____

_____
Sharon Dumas, Director of Finance

APPENDIX E

FEASIBILITY REPORT

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City of Cleveland, Ohio

SERIES 2016 WATER POLLUTION CONTROL BONDS FEASIBILITY REPORT

March 31, 2016

SERIES 2016 WATER POLLUTION CONTROL BONDS FEASIBILITY REPORT

City of Cleveland, Ohio

Prepared for:

Division of Water Pollution Control

City of Cleveland, Ohio

601 Lakeside Avenue

Cleveland, OH

Said Abou Abdallah, P.E.

Associate Vice President

Prepared by:
Arcadis U.S., Inc.

1111 Superior Avenue

Suite 1300

Cleveland

Ohio 44114

Tel 216 781 6177

Our Ref.:

CL001053.B001

Date:

March 31, 2016

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1 INTRODUCTION AND PURPOSE

1.1 Introduction and Purpose

Arcadis U.S., Inc. (Arcadis), has been retained by the City of Cleveland, Ohio (City) and its Division of Water Pollution Control (WPC) to provide Consulting Services to prepare a Bond Feasibility Report (Report) for the planned issuance of Series 2016 Water Pollution Control Revenue Bonds. The proceeds of the bonds will be used to fund capital improvements to WPC's wastewater collection system (System). WPC's System includes sanitary sewers, combined sanitary and storm sewers, separated storm sewers, wastewater pump stations and related appurtenances and equipment to collect and transfer wastewater and stormwater flow. The System does not include the regulator weirs (structures on sewer systems that permit the discharge of excess flow into a relief sewer and the design dry weather flow to continue up to its intended outlet) or wastewater treatment facilities. All wastewater collected by the System is conveyed to the Northeast Ohio Regional Sewer District (NEORSD) facilities for further transportation and treatment.

The Report includes:

- A description and condition summary of the System.
- An overview of WPC's organization and management.
- A summary of the status of current and proposed capital projects and improvements.
- A review of regulatory compliance.
- A summary of historical and projected customer statistics.
- A review of current and projected financial information.
- Conclusions regarding the technical, organizational and financial aspects of WPC.

The technical and financial data reviewed in preparing this Report include existing records and documents provided by WPC. The Report summarizes the work completed up to the date of the issuance of the Report. Changed conditions occurring or becoming known after such date could affect the material presented and the conclusions reached herein to the extent of such changes.

Arcadis has not independently verified the accuracy of the information provided by WPC and others; however, our Report is based upon the supposition that such sources are reliable and the information obtained therefrom is appropriate for the analysis undertaken and the conclusions reached herein. To the extent that the information provided to Arcadis by WPC and others is not accurate or not fulsome to all relevant details, the conclusions and recommendations contained in this Report may vary, and are subject to change. In addition, the scope of Arcadis' review did not include any pending or threatened litigation that may impact WPC. The City has stated that there is no significant litigation that it believes would have any material impact on WPC's operations, and as such, Arcadis has prepared its Report accordingly and under the presumption of the accuracy of such statement.

In completing this Report for WPC, Arcadis is not serving in the role of a “municipal advisor” under the regulations of the Securities and Exchange Commission. As such, Arcadis is not: (a) recommending any action regarding municipal financial products or the issuance of municipal securities; and (b) is not acting as a registered municipal advisor to the City, and does not owe a fiduciary duty to the City pursuant to Section 15B of the Securities Exchange Act of 1934, as amended by the Dodd-Frank Wall Street Reform and Consumer Protection Act, with respect to the information and material prepared in connection with this Report. The City should discuss the information and material prepared in connection with this Report with any and all internal and external financial and other advisors whom the City deems appropriate to provide such guidance before acting on this information and material.

Accordingly, since Arcadis is not a municipal advisor registered with the Securities and Exchange Commission, the City acknowledges that: (a) it has retained the services of an independent registered municipal advisor, who, during the past two (2) years, was not associated with Arcadis, and that (b) Arcadis has complied with the requirements set forth in the federal Securities Exchange Act, Municipal Advisor Rule (17 CFR 200, 240, 249), which requires that the engineering company: (i) receive from the municipal entity a representation in writing that it is represented by, and will rely on the advice of, an independent registered municipal advisor; (ii) provide written disclosure to the municipal entity that Arcadis is not serving as a municipal advisor and, with respect to the municipal entity, is not subject to the statutory fiduciary duty applicable to municipal advisors under the federal Securities Exchange Act, and (iii) provide a copy of such disclosure to the municipal entity’s independent registered municipal advisor. Arcadis has provided such disclosures to WPC and its registered municipal advisor. Arcadis does not provide opinions on, or advocate for using, a financial product (issuing debt) or the choice of financial products employed. As such, Arcadis submitted its work products to the City’s independent registered municipal advisors, Phoenix Capital Partners, LLP, and Government Capital Management, LLC for review.

1.2 Information Reviewed

In preparing this Report, Arcadis has reviewed information from the documents listed below, which are documents prepared by or on behalf of WPC.

- City’s Codified Ordinances, Chapter 541-Sewer Connections and Sewer Use Code (Charter reference § 163)
- WPC 2015 Asset Inventory
- NPDES Permit # OHQ000003 – Authorization for Small Municipal Separate Storm Sewer System to Discharge Storm Water Under the NPDES, expiration date September 10, 2019
- Community Discharge Permit Issued to Cleveland, Ohio by Northeast Ohio Regional Sewer District, dated June 26, 1986
- Community Combined Sewer Discharge Permit Issued to Cleveland, Ohio by Northeast Ohio Regional Sewer District, dated March 16, 1995
- East 185th St/Lakeshore Blvd Area Sewer System Evaluation Survey, dated February 2014

- United States Army Corp of Engineers (USACOE) – Euclid Creek Cleveland, Ohio Flood Risk Management Project – revised Maintenance Requirements for Shoal Removal within the Project's Channel Section, dated March, 27, 2015
- Memorandum of Understanding between the Doan Brook Water Shed Partners and the City of Cleveland, dated February 5, 2013
- Memorandum of Understanding between the Cuyahoga County Soil and Water Conservation District and the Euclid Creek Watershed Council, dated August 30, 2012
- Memorandum of Understanding between the Cuyahoga County Soil and Water Conservation District and the City of Cleveland, dated January 2008
- Bond Ordinances Nos. 28-16 and 29-16, passed by City Council on January 25, 2016
- 2015 Budget Book Containing 2012 – 2014 Actual Financial Reports and 2015 Budget for WPC
- 2016 WPC Budget
- Annual Financial Audit Reports for 2012 – 2014 for Department of Public Utilities, Division of Water Pollution Control, prepared by Clark Schaefer and Hackett
- 2016 – 2020 Capital Improvement Plan with Funding Directives, provided by Division of Fiscal Control
- 2012-2014 Annual Compliance Checklist Reports to Northeast Ohio Regional Sewer District

1.3 Arcadis

Arcadis is an international leader in the fields of infrastructure, environment, and water. Dedicated to enhancing mobility, sustainability, and quality of life, the firm's multidisciplinary staff provides consultancy, design, engineering, and management services to public and private clients across the U.S. and around the globe. For over 100 years, Arcadis has been designing the human habitat, setting industry standards with patented technologies and published works. Arcadis currently generates more than $3.0 billion in gross revenue through its worldwide projects. Arcadis has access to approximately 22,000 engineers, scientists, planners, management consultants, and support staff worldwide. Currently the combined organization is working on environmental and infrastructure projects in more than 70 countries.

Arcadis has maintained a local office in Cleveland since 1996, and is familiar with WPC's System; both through preparation of this Report and from having provided historical general engineering services with respect to WPC sewer system improvements. Arcadis has provided engineering and professional services at various capacities to WPC since 2001.

1.4 Conventions and Acronyms

WPC's fiscal year begins on January 1st and ends December 31st. Throughout the Report, a fiscal year is identified as "FY" which is the same as the calendar year in which the fiscal year ends, i.e., FY2016 is the fiscal year from January 1, 2016 through December 31, 2016.

Appendix A contains a listing of acronyms or abbreviations of terms used in the Report. These terms appear in multiple sections of this Report, and are thus defined in Appendix A for reference.

2 ORGANIZATION AND MANAGEMENT

2.1 Division of Water Pollution Control

WPC is a division of the City's Department of Public Utilities that provides retail wastewater and stormwater collection services to customers in the City. WPC was established by the City on June 21, 1965, by Ordinance 1599-65. WPC was created for the purpose of supplying sewer services to customers within the Cleveland metropolitan area. Beginning with a rudimentary system in the late 1800s, the Cleveland sewer system developed as the City itself expanded. Until the early 1970s, the City operated the entire system and managed all aspects of sewage treatment and disposal.

In 1972, in response to a lawsuit brought on by several communities in Cuyahoga County, a court order created the Northeast Ohio Regional Sewer District (NEORSD). The ownership and operation of all wastewater treatment plants, interceptors, sewer regulator systems and devices, weirs, retaining basins, and some storm water handling facilities were transferred to NEORSD in December 1973. The City, however, retained responsibility for the local sewer collection system within the City limits of Cleveland. Over the years, various clarifications and agreements have modified the ownership responsibility of the City and NEORSD with respect to the assets located within the City's boundaries. The WPC System in 2016 includes approximately 1,445 miles of sewers, 12[1] pump stations, and related appurtenances.

In accordance with its governing legislation, WPC operates, maintains and manages the System on behalf of the City, pursuant to the requirements of the City's Community Discharge Permits (both the Community Combined Discharge Permit and Community Discharge Permit). The System transfers collected wastewater from its point of collection to a NEORSD interceptor sewer or treatment plant for processing. WPC is also charged with municipal regulation of the City's local stormwater system and as such, also regulates matters related to the pollution of watercourses.

.

2.2 Organization Structure

An overview of the current organization structure of WPC is shown on Figure 2-1. WPC is one of five (5) Divisions within the Department of Public Utilities of the City, which also includes Water, Fiscal Control, Utilities Administration and Cleveland Public Power. WPC is managed by the Commissioner of WPC, pursuant to the Charter and ordinances of the City, and under the direction of the Director of Public Utilities.

The WPC Commissioner reports to the Director of Public Utilities, who is appointed by and reports to the Mayor. The Board of Control, consisting of the Mayor and directors of several departments, establishes the rate and charges for WPC, subject to City Council approval. City Council is the final governing body for approval of rates and charges and other matters related to WPC. Several other City divisions both within and outside of the Department of Public Utilities also provide support services to WPC, including,

[1] WPC was originally responsible for 15 pump stations. However, 3 will be transferred to NEORSD by May 15, 2016 as discussed in Section 2.6.

but not limited to, the Division of Fiscal Control for financial management and Division of Water for billing and collection services. Figure 2-1 shows WPC's organization chart for FY2016.

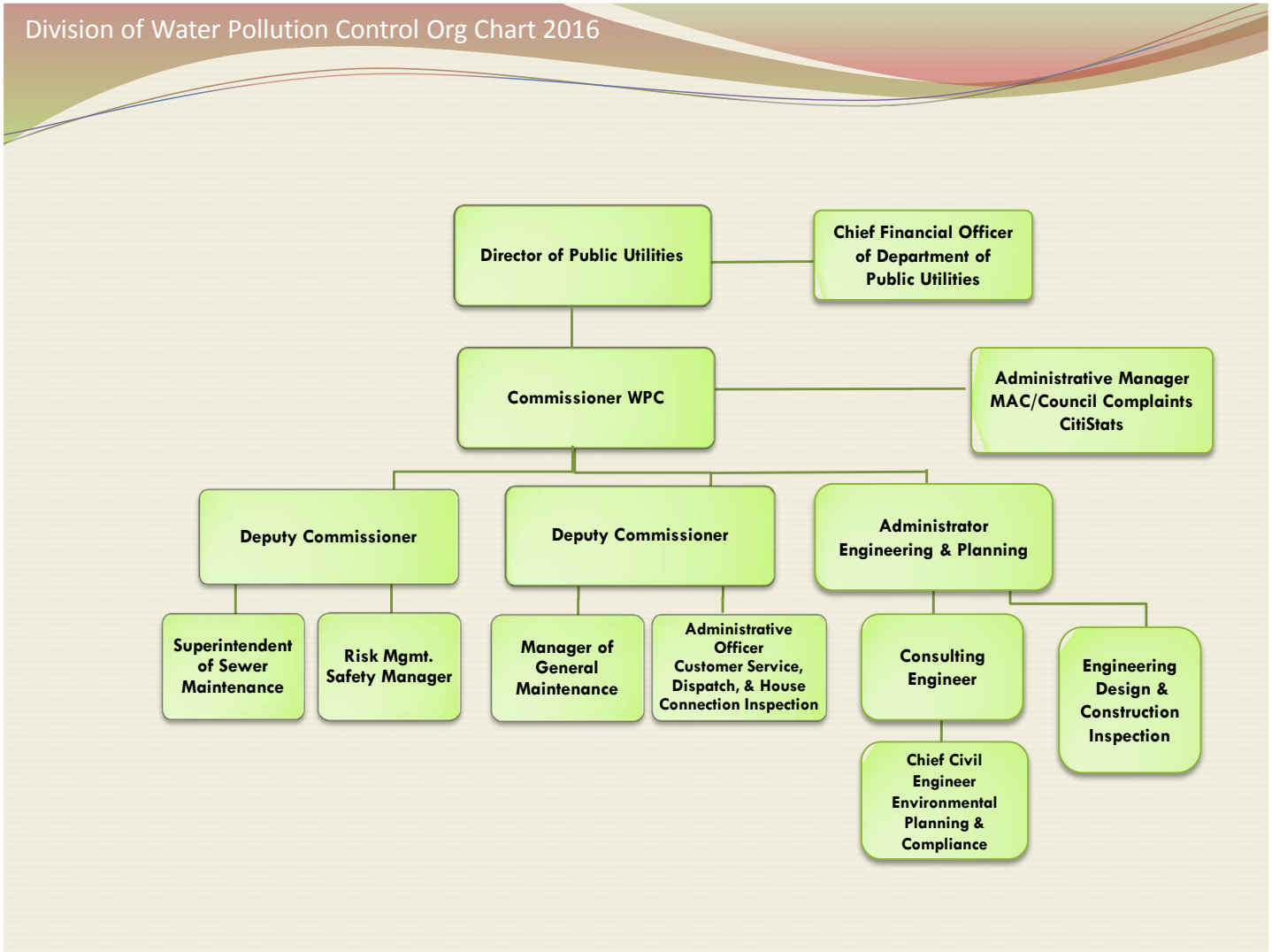

Figure 2-1. City of Cleveland Division of Water Pollution Control Organization Chart

Source: Cleveland Water Pollution Control. Organization Chart as of January 2016.

WPC's 2016 personnel budget includes 162 full-time and 4 part-time positions. Of these, 125 full time and 4 part-time positions are currently filled. The City recently reorganized and moved 16 financial, public affairs, geographic information systems (GIS), information technology (IT), and human resource type positions that were within WPC to other divisions within Public Utilities to consolidate responsibilities and enhance efficiency and coordination throughout the department. Although these positions were moved out of WPC's personnel budget into other Department of Public Utility's divisions, these staff continue to support WPC as required. In addition, the approved staffing level for WPC's FY2016 personnel budget remains the same as in FY2015 at 166, and WPC is in the process of hiring additional operation, maintenance and engineering staff to meet future needs.

All of the full-time and part-time employees of WPC are employees of the City, and approximately eighty-nine percent (89%) belong to one of several collective bargaining units represented within the City. There are currently 14 employees that do not belong to unions. They include primarily upper level professionals and administrators. All of the union contracts are effective from April 1, 2013, through March 31, 2016. The City is in the process of negotiating new three-year union contracts to replace the expiring contracts. At this time, WPC does not anticipate that the new contracts will have any material impact on WPC's operations or financial projections.

The organizational structure of WPC generally is consistent with that of wastewater utilities in many large cities. The proposed staffing plan adequately addresses all of the core functions of WPC, with certain key support services provided by other City divisions. WPC is in the process of expanding its in-house staff and capabilities to meet future needs.

WPC does have on-call professional services and other contracts in place to provide emergency and other required services as may be needed to supplement current staff availability and experience.

2.3 Management Staff

The WPC Commissioner is responsible for all technical and administrative operations of WPC, and the implementation of programs, policies and procedures at the direction of the Director of Public Utilities and the Mayor. Mr. Rachid Zoghaib, P.E., currently serves as the Commissioner of WPC, and has served in this position since April 2011. Prior to this, Mr. Zoghaib served as Deputy Commissioner for seven (7) years and Director of Engineering for almost ten (10) years. Mr. Robert L. Davis is the Director of Public Utilities, and began serving in this position in 2015. Mrs. Ramona Lowery is a new Deputy Commissioner for WPC, and reports to the Commissioner. Although Mrs. Lowery started in this position in 2015, she is a former employee of the City and NEORSD. WPC is in process of hiring a second deputy commissioner to assist in managing the anticipated increase in workload associated with WPC's plans, and anticipates that this position will be filled before the end of the year. Mr. Frank Badalamenti, Chief Financial Officer (CFO) of the Department of Public Utilities, is responsible for the financial management of WPC.

Biographies of key management and operational staff are provided below.

Robert L. Davis, Director of Public Utilities – Mr. Davis is the Director of Public Utilities. He is responsible for management and oversight of more than 1,700 employees, a budget of approximately $570 million, and all operations within WPC as well as Division of Cleveland Public Power, Division of Water, Utilities Fiscal Control, the City's Office of Radio Communications, and Cleveland's public access channel. Mr. Davis has over twenty-eight (28) years of experience as a utility professional. He graduated

with a Bachelor of Science from West Virginia University. He also has a Class IV Water License, and is a member of the American Water Works Association (AWWA) for more than twenty-five (25) years. He is the past Chair of the Ohio Section of AWWA, and served five (5) years on the Ohio Governing Board. Mr. Davis also served as Chair of the Membership Committee for the Ohio Section and works very closely with Young Professionals, Diversity, Strategic Planning, and Finance and Audit committees.

Rachid Zoghaib, P.E., Commissioner of WPC – Mr. Zoghaib has served as the Commissioner of WPC since 2011. He has more than twenty-five (25) years of wastewater, environmental and public management experience. Prior to this position, he served as the Deputy Commissioner from 2004 to 2011, and as the Director of Engineering from 1994 to 2004. As Commissioner, his responsibilities include managing and directing all administrative, operational and financial activities and programs of WPC. Mr. Zoghaib earned a Bachelor of Science Degree in Civil Engineering and a Master of Science Degree in Civil Engineering from Youngstown State University and a Leadership Certificate from Cleveland Management Academy at Cleveland State University. He is also a Licensed Professional Engineer in the State of Ohio.

Ramona Lowery, Deputy Commissioner of WPC – Mrs. Lowery has twenty (20) years of wastewater experience, and is responsible for managing the sanitary sewage and stormwater collection systems. Mrs. Lowery also oversees the implementation of pollution prevention programs through Public Involvement and Public Education (PIPE). She provides executive supervision in planning and managing WPC. She also assists the Commissioner of WPC in managing daily operations and coordinating reliable sewer maintenance operations. Prior to her current employment at WPC, she worked for NEORSD for seven (7) years as Manager of Operation and Maintenance. Mrs. Lowery has a Master of Business Administration and Master of Public Administration from the University of Phoenix, and a Bachelor of Science in Civil Engineering from Cleveland State University. She holds a Class II Wastewater Collection System License and a Class II Water Distribution System License. Mrs. Lowery is also affiliated with several professional organizations such as Water Environment Federation, Cleveland Engineering Society, and National Society of Black Engineers.

Frank Badalamenti, Chief Financial Officer of Department of Public Utilities – Mr. Badalamenti is responsible for all financial day-to-day responsibilities, including accounting, finance, forecasting, strategic planning, budgeting, rate setting, risk management, and procurement for Cleveland Public Power, Cleveland Water, and Water Pollution Control. Mr. Badalamenti served over twenty (20) years in the United States Air Force, retiring as a Master Sergeant. He holds a Master of Business Administration from Grand Canyon University and a Bachelor of Science in Finance from Wayland Baptist University. Prior to serving as CFO for the Department of Public Utilities, Mr. Badalamenti spent over twelve (12) years as the Manager of Internal Audit for the City of Cleveland. He has also authored and implemented numerous policy and procedure changes throughout every department within the City.

The management staff of WPC provide both the qualifications and institutional knowledge of the System to deliver the required services.

2.4 Interactions with the City

WPC is a division within the Department of Public Utilities of the City. The City also maintains several other departments and divisions that provide key administrative and support services to WPC. The costs

of these interdepartmental services are charged back to WPC, as WPC operations are financed primarily through direct charges to the users of its services. Several of these other divisions, such as the Division of Water, also function as separate utilities. WPC's allocable share of support services costs from these departments and divisions is based on the cost of the services provided. The interdepartmental services provided to WPC include but are not limited to the following:

- Billing and collection services;
- Procurement, accounts payable, and payroll services;
- Budget, capital improvement planning and bond issuance support;
- Technology Management Services (IT) and GIS support;
- Human Resources support for hiring, terminations, and retirements;
- Legal support for agreements by WPC and claims for and against WPC;
- Utilities administration;
- Motor vehicle maintenance; and
- Street construction maintenance and repair.

In FY2013 and FY2014, WPC paid the City approximately $3.5 million and $3.7 million, respectively, in interdepartmental charges, of which $2.5 million and $2.4 million were charges from Division of Water for billing and collection services. In FY2015, the interdepartmental charges were $5.2 million, of which $3.9 million was for billing and collection services. The large increase in Division of Water charges for FY2015 was primarily the result of a one-time payment adjustment by the Division of Water to WPC. Increases also were associated with implementation of a new automated meter (AMR) reading program, which should serve to increase revenues through elimination of estimated bills and improved reading accuracy. Although the Division of Water charges are anticipated to decrease to $3.15 million in FY2016, the WPC budget will maintain an estimated $5.1 million in interdepartmental charges for FY2016 to account for the additional charges associated with utilities administration and fiscal control. These additional charges result from WPC's prorated share of costs associated with transferred public affairs, GIS, IT, finance and related staff from WPC to other divisions within Public Utilities as discussed in Section 2.2.

Several City divisions retain independent stormwater responsibilities that are coordinated by WPC. For example, the City Department of Building and Housing has the responsibility for construction and post-construction stormwater runoff control and the City Department of Public Works is responsible for stormwater housekeeping.

2.5 Overview of Service Contracts and Agreements

WPC performs almost all of the management and operation services required for the System using in-house staff and resources either directly through WPC, or via City support services from other divisions. There are several Memorandums of Understanding (collectively, MOUs), on-call service contracts and interactions with other agencies that support the delivery of services. These include, but are not limited to:

- MOUs with the Cuyahoga County Soil and Water Conservation District and the Doan Brook Watershed Partners regarding provision of public education for stormwater control.
- On-call emergency services contracts with various private vendors to assist in conducting emergency repairs beyond the in-house resources of WPC.
- Professional services contracts to provide general engineering services to WPC. These services include wastewater system design and planning, sewer system design, regulatory support, bidding and construction services, surveying, field representation and operations support.

Although not governed by contract, WPC also coordinates the delivery of its services with the requirements of NEORSD; both as a condition of the City's discharge permits and as the host community for a significant portion of NEORSD's assets as discussed further below.

2.6 NEORSD Interactions

Prior to 1973, the City was responsible for both wastewater collection and wastewater treatment. Beginning in December 1973, the ownership and operation of all of the wastewater treatment plants, interceptors, sewer regulator systems and devices, weirs, retaining basins, and some stormwater facilities were transferred to NEORSD pursuant to a 1972 Court Order. Subsequently, further clarification of the maintenance responsibilities of NEORSD-related collection system assets located within the City limits, was confirmed by an Agreed Judgment Entry and Court Order (City of Cleveland v NEORSD, Cuyahoga County Court of Common Pleas, Case #455021) dated April 2005. This order specified that these assets should be maintained by NEORSD as part of its interceptor system and not maintained by the City. An MOU and Exhibits A-C that are attached to the Agreed Judgment Entry and Appendix D, Figure D-1 attached to this Report, provides maps and a description of the assets owned and maintained by each of the parties under the 2005 Agreed Judgment Entry and the Court Orders. The clarifications included confirmation, among other items, that the regulator weirs are the responsibility of NEORSD.

Updates to the allocation of responsibilities are conducted as needed by mutual agreement between WPC and NEORSD. Most recently, this included the transfer of the Mary Street, West 11th Street and Stones Levee pump stations from WPC to NEORSD by May 15, 2016. NEORSD and WPC are also in the process of finalizing a stormwater agreement in year 2016, which, under the authority of Title V of NEORSD's Stormwater Management Code, will transfer the operation and maintenance of the assets that are part of the Regional Stormwater System, including two of WPC's detention basins, from WPC to NEORSD. These transfers of responsibility are consistent with NEORSD's Project Clean Lake Long Term Control Plan (LTCP), as required by USEPA, and Regional Stormwater Management Program (RSMP) goals. These transfers are by prior mutual agreement between WPC and NEORSD because NEORSD has identified these assets as key components for NEORSD's systems. Upon the transfer, these assets will not be considered part of the System, and were not included in the condition assessment review conducted by Arcadis as part of the preparation of this Report.

WPC discharge of flows to NEORSD's interceptor system for transfer and treatment at NEORSD's wastewater treatment plants is governed by the Community Combined Discharge Permit with NEORSD (dated March 1995) for combined sewers (NEORSD Title IV code) and the Community Discharge Permit with NEORSD (dated June 1986) for sanitary sewer (NEORSD Title III code). Neither permit has an expiration date; however, the permits may be modified if necessary to reflect the changing needs and

circumstances of NEORSD and WPC. In addition, there are several levels of coordination between WPC and NEORSD as summarized below:

- Monthly NEORSD engineering coordination meetings with WPC engineering participation.
- Quarterly NEORSD Water Action Committee meetings with WPC representation and participation.
- NEORSD Project Clean Lake LTCP Implementation and Coordination with City on green infrastructure projects, problem area studies and flow monitoring.
- City representation as voting members on NEORSD governing board; of the 7 members of NEORSD governing board, 3 are appointed by the City.
- NEORSD inclusion in the approval process and requirements for any new development and redevelopment within the City.

WPC consistently complies with its Community Discharge Permit and Community Combined Discharge Permit (collectively referred to as the Community Discharge Permits), requirements that require implementation of a number of best management practices. Under the Community Discharge Permit for the sanitary sewer system, a peak flow limitation of 14.284 million gallons per day (MGDs) is imposed at the Grayton Road connector. WPC and NEORSD have not reported any exceedance of this limitation.

With respect to the Community Combined Discharge Permit, there are no peak flow limits or restrictions imposed at this time. NEORSD, however, has the right to impose future limitations, if required. No limitations are anticipated to be imposed within the forecast period as (a) the quantity of wastewater flow currently delivered to NEORSD is less today than it was when the discharge permits were issued, (b) the System is essentially fully developed, and (c) the City maintains a policy of managing the amount of wastewater flow discharged to its collection system to result in peak flow rates at, or less than, what the System was designed for. In addition, as part of its LTCP required by USEPA consistent with a consent decree, NEORSD is undertaking several improvements for handling wet weather flows, including designing a tunnel for capture and storage of peak wastewater flows.

In addition to wastewater treatment responsibility, NEORSD also supports the WPC stormwater management program. On September 15, 2015, the Ohio Supreme Court ruled that NEORSD was authorized to implement its RSMP and had authority to charge stormwater fees to fund it. Under the RSMP, NEORSD will collect stormwater management fees from its customers, using the City as its billing agent, and use this money as authorized by Ohio Revised Code (ORC) Chapter 6119 for the use, services or benefits of a water-resource project afforded to property owners in its defined regional stormwater area, including in the City. That includes NEORSD assuming responsibility for millions of dollars of necessary maintenance along streams across the region. It is unknown when NEORSD will begin assessing the RSMP charges, but collection is anticipated by the end of FY16. Pursuant to NEORSD's RSMP's Community Cost Share policy, member communities are eligible to receive funding for qualifying community-specific stormwater projects pursuant to an application process. A minimum of twenty-five percent (25%) of the total annual stormwater fee revenue collected in each member community is allocated to the Community Cost-Share Account for each member community. The City is in regular communication with NEORSD regarding the RSMP and cost share opportunities.

3 SYSTEM DESCRIPTION

3.1 Service Area

The City is located in northeastern part of the State of Ohio on the southern shore of Lake Erie; approximately sixty (60) miles west of the Pennsylvania border. The City is the county seat of Cuyahoga County, which is the most populous county in the State of Ohio. The City itself is the second largest city in the State of Ohio, covering a land surface area of 77.6 square miles. It is geographically located near the mouth of the Cuyahoga River, and extends along the southern shore of Lake Erie. Founded in 1796, the City is a diverse city of approximately 390,000 people.

WPC's service area includes all areas within the City limits and represents a significant portion of the entire metropolitan area tributary to NEORSD interceptor and treatment plants. The service area includes residential, commercial, industrial, government and non-profit customers located with the City limits. It also includes customers that may be physically outside the City limits, but have their water meters located within the City limits. A map of WPC's service area and the types of sewer service provided within the service area is shown in Appendix D.

3.2 Description of Major Assets

WPC is responsible for operating and maintaining its wastewater and stormwater collection systems. The System currently includes 12[2] pump stations, over 1,445 miles of both sanitary and stormwater collection systems, mostly in the form of a combined sewer collection system, 36,000 manholes, 43,500 catch basins, and related equipment as discussed further below.

3.2.1 WPC Sewer Collection System

WPC's System includes three (3) types of sewers. The most prevalent type are sewer pipes that collect and transfer combined sanitary and storm flows. The System also includes separate sanitary and storm sewers, and a third category of segmented block pipes wherein separate sanitary and storm sewers are encased within the same segmented block pipe. Appendix D, Figure D-2 provides a map that illustrates the areas within the City served by combined and separated (including segmented block) sewers. As illustrated in Appendix D, Figure D-2, the majority of the System is served by combined sewers.

Ninety percent (90%) of the sewers have an equivalent pipe diameter of less than sixty (60) inches; with the most common pipe size being twelve (12) inches in diameter. The most common type of construction (forty-nine percent (49%)) is vitrified clay pipe (VCP), but with significant quantities of both brick (twenty-eight percent (28%)) and reinforced concrete pipe (eighteen percent (18%)). The segmented block pipe accounts for approximately 42,000 linear feet or less than one percent (1%) of the total linear feet of pipe. The majority of the System was constructed in the 1920s, and is estimated to have an average age of

[2] WPC was originally responsible for 15 pump stations. However, 3 will be transferred to NEORSD by May 15, 2016 as discussed in Section 2.6.

approximately eighty (80) years. Portions of the System, however, date back to the 1870s. Figure 3-1 presents a graphical summary of WPC's collection system by age and type.

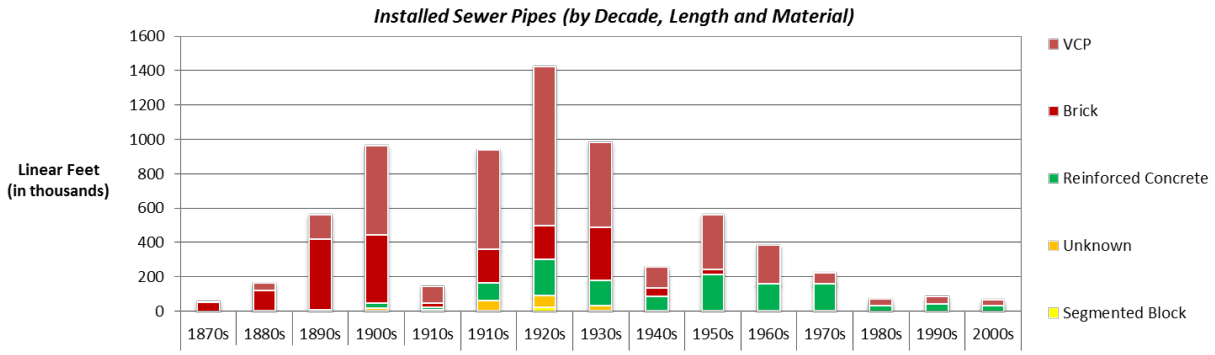

Figure 3-1. WPC Sewer Pipe Age and Type

The estimated useful life of a typical sewer is approximately one hundred (100) years, and approximately twenty-seven percent (27%) of WPC's collection system is over one hundred (100) years old. WPC has developed a Capacity, Management, Operation, and Maintenance (CMOM) Plan and a comprehensive capital program, which includes conducting a Sewer System Evaluation Survey (SSES) to address this aging infrastructure. The CMOM is discussed further in Section 4.3 of this Report.

WPC is responsible for the operation and maintenance of the System within the City limits, including up to one (1) foot from the curb line of the roadway. The laterals and connections from the building to one (1) foot from the curb line are the responsibility of the building owners. The interceptors and regulators that transfer flow from the WPC collection system to NEORSD treatment facilities are the responsibility of NEORSD, as described in Section 2, and are therefore not included as WPC assets.

3.2.2 Stormwater System

The City's stormwater system consists primarily of approximately forty-three thousand five hundred (43,500) catch basins, which lead primarily to the combined sewer collection system and approximately two hundred six (206) miles of separated storm sewers. A significant portion of these separated storm sewers are within a segmented block pipe installed in the 1920s. Separate storm and separate sanitary flows are channelled within their respective open channel pipe, both enclosed within the same casing. Any storm sewer flows in excess of the pipe's capacity will overflow into the sanitary system.

The City's stormwater system is operated and maintained pursuant to its Ohio Environmental Protection Agency, National Pollution Discharge Elimination System (NPDES) permit (NPDES No. OHQ000003) for Small Municipal Separate Storm Sewer Systems (Ohio EPA NPDES MS4 permit). The remainder of the City's stormwater is managed as part of its wastewater collection system as approximately ninety two percent (92%) of the WPC's System is a combined sanitary and storm sewer system.

The WPC Commissioner serves as the City's Stormwater Manager for implementing and coordinating the best management practices required as part of a stormwater management plan required under its Ohio EPA NPDES MS4 permit. A Stormwater Steering Committee consisting of representatives from WPC, Cuyahoga County Soil and Water Conservation District, the City's Departments of Public Works, Law, and Building and Housing, Cleveland City Planning Commission, Office of Sustainability, Division of Waste Collection, Office of Capital Projects, and City Council implement the stormwater management plan across multiple departments for compliance with the City's Ohio EPA NPDES MS4 permit.

WPC's responsibilities include overall implementation, management, enforcement and reporting on stormwater compliance including compliance with the Six Minimum Control Measures as specified in the City's Ohio EPA NPDES MS4 permit. Table 3-1 summarizes the allocation of responsibility for each of these Six Minimum Control Measures.

Table 3-1. Allocation of Responsibility for MS4 Stormwater Permit Six Minimum Control Measures

Number	Minimum Control Measure	Responsibility
1	Public Education and Outreach	WPC
2	Public Involvement and Participation	WPC
3	Illicit Discharge Detection and Elimination	WPC
4	Construction Site Stormwater Runoff Control	Department of Building and Housing
5	Post Construction Stormwater Management in New Development and Redevelopment Control	Department of Building and Housing
6	Pollution Prevention and Good Housekeeping	Department of Public Works and other affected departments

NEORSD's RSMP, described in Section 2.6, is anticipated to start shortly and will serve to further supplement the City's stormwater management program, including providing access to funding for qualifying stormwater projects.

3.2.3 Sewage Pump Stations

WPC operates and maintains twelve (12) sewage pump stations, which are utilized to transport sanitary wastes where gravity is not a viable transport option. This number of pump stations does not include the three (3) pump stations that are transferring from WPC to NEORSD by May 15, 2016 as discussed in

Section 2.6. The remaining sewage pump stations are relatively small, with less than seven hundred (700) gallons per minute capacity.

A listing of the wastewater pump stations is included in Section 5 of this Report. The condition of the pump stations is discussed in Section 5.

4 OPERATIONS AND MAINTENANCE

4.1 Sewer Legislation

Legal authority for WPC's System derives from the City's Charter and codes and ordinances, which are reviewed and revised as needed. All ordinances are available on the City's website. WPC's key sewer ordinances are: *City of Cleveland Code of Ordinances, Part 5 – Municipal Utilities and Services Code, Title VII – Sewers and Sewage Disposal, Chapter 541- Sewer Connections and Sewer Use Code*; *Charter reference § 163.*

This statutory framework regulates the use of public and private sewers and drains in the City Right-of-Way. It also requires, among other things, that individual property owners maintain their sewer service connections from one foot beyond the curb to the property to be drained and control unacceptable discharges to the collection system, including natural watercourses. If a blockage is in the main sewer or between the main sewer and a point one foot beyond the existing or proposed curb line of the roadway, repairs shall be made by WPC. If the blockage is between the building and one (1) foot beyond the existing or proposed curb line of the roadway, WPC advises the owner to engage a licensed and bonded sewer contractor to make the necessary repairs.

WPC also complies with its obligations under its Ohio EPA NPDES MS4 permit and NEORSD's Code of Regulations, Title III "Separate Sanitary Sewer Code" and Title IV "Combined Sewer Code" and associated Community Combined Sewer Discharge Permit and Community Discharge Permit.

4.2 Sewer Extensions and Connections

The System is fully developed. Should new connections to the existing collection system or redevelopment projects by developers be proposed, approval by WPC, Cleveland City Planning Commission and the Department of Building and Housing, as well as NEORSD will be required. In addition, pursuant to City policy, the connections must maintain compliance with WPC's policy of managing the amount of wastewater flow discharged to its collection system as previously discussed in Section 2.6.

4.3 Operation and Maintenance

WPC maintains a sewer system preventative maintenance program that includes implementation of best management practices. Historically, WPC's program focused first on addressing any complaints and known problem areas and second on conducting preventative maintenance. While these priorities remain, WPC is increasing both its preventative maintenance efforts and resources and rate of replacement of sewers which are past their useful life. WPC has developed a comprehensive draft CMOM plan to increase preventative maintenance investment and reduce potential for sanitary sewer overflows (SSOs) in a cost-effective manner. Key elements of the program include:

- Using a computerized maintenance and management system (CMMS) to track all activities including inspections, cleaning, and rehabilitation and replacement projects.

- Continual updating and maintaining of WPC sewer maps in GIS.
- Conducting condition assessments in order to budget and plan for upcoming rehabilitation and replacement programs.
- Maintaining an adequate inventory of equipment and spare parts to allow immediate response to emergencies and prevent sanitary sewer overflows.
- Developing Standard Operating Procedures (SOPs).
- Establishing performance measures and goals.
- Expanding safety and training for employees.
- Establishing operation and maintenance programs for cleaning and inspections.

Currently, the cleaning of the sewers is done by means of a priority schedule in which blocked main sewers are completed first. Known problem areas that do not present an immediate concern are then prioritized based on available information and scheduled for cleaning. As work load permits, preventative maintenance cleaning is then performed. The goal of WPC is to increase preventative maintenance cleaning so that all mains are cleaned, at least, once every five (5) years.

WPC performs catch basin cleaning to provide a clear pathway for water to enter the sewer system. WPC typically cleans more than ten thousand (10,000) catch basins per year. The schedule of catch basin cleaning is based on known problem areas, visual observation and by requests. WPC's goal is to increase catch basin cleaning so that each catch basin is cleaned, at least, once every three (3) years. The City Division of Streets contributes to WPC preventative maintenance program by performing all of the street cleaning in the City.

WPC performs routine inspections of its sewer lines through a staff of house connection inspectors who investigate any complaint and determine the process of resolution, including follow up. They also make visual inspections while on rounds to detect problems in the sewer lines including (among other items) observations of sunken or flooded areas, ground cracks along pipe routes, damage to special structures and manholes, manhole covers above or below pavement levels, and evidence of flooding or seepage from sewers. This information is recorded in the City's CMMS which is also used to log customer service requests and complaints and to generate and manage maintenance work orders and performance measures.

Based on the available information, WPC's maintenance crews are deployed to determine the problem and to perform necessary repairs. Cleaning and closed circuit television (CCTV) inspections are conducted and a permanent video log of the inspections is kept for further study. WPC immediately repairs problems with in-house staff. Maintenance crews are dispatched by a priority rating system with customer "no relief" status given the highest priority. For large scale emergencies, WPC employs on-call contractors to insure immediate repairs.

WPC received 4,526 sewer service requests in 2013 and 4,430 requests in 2014. In 2015, approximately 4,357 sewer service requests were recorded and investigated by WPC. Sewer flow issues, including sewer back up and flooding, accounted for approximately twenty-six percent (26%) of these calls. Of the complaints received, many were attributable to poor private lateral conditions. Typically, the private laterals account for the majority of the poor sewer conditions due to lack of maintenance on the

homeowners' part as well as the increased vulnerability of the laterals due to root intrusion. The remainder of the complaints were to report, or to request, an investigation related to blocked catch basins, missing or broken manhole covers, sewer cave-ins, or damaged catch basins.

Some of the calls or requests were to report an illicit discharge to the System, such as a chemical spill, septage discharge or the presence of chemical odors. Some were for illegal dumping into the System such as dumping of debris.

The classifications of calls and requests are generally typical from year to year. After the receipt of a call or a service request, WPC issues a work order for investigation of the request. The work order may be followed by several other work orders based on the need for the particular investigation. A service call may generate more than three work orders depending on the situation.

WPC reports on its maintenance performance to NEORSD on an annual basis. Table 4-1 provides a summary of WPC's key maintenance performance measures since 2011.

Table 4-1. Maintenance Performance Measures

Maintenance Performance Measure	2011	2012	2013	2014	2015
Catch Basins Cleaned	17,953	18,266	11,860	10,831	9,006
Linear Feet of Sewer Lines Cleaned	260,536	416,585	490,897	449,621	425,716
Catch Basin Brick Work	257	258	260	349	357
Catch Basin Connection Repairs	87	99	89	86	132
Linear Feet of Sewer Lines Televised	188,277	193,797	264,697	239,970	212,706
House Connection Repairs	199	212	193	353	384
Main Sewer Repairs	27	27	29	58	59
Manhole Repairs	No data	No data	9	19	21

Source: WPC records as of February 19, 2016

As illustrated above, the performance metric for any line item varies from year to year depending on the priority needs of the System which, in turn, are affected by customer service requests, weather conditions, emergencies and availability of equipment. For example, in 2012 there was an increased need to clean catch basins. Whereas in 2013, there was an increased need for cleaning and televising sewer lines. In 2014 and 2015, WPC was able to increase the number of catch basin brick work and house connection repairs.

WPC also maintains records relating to the a) average number of days to complete house connection repair work; b) average number of days to complete catch basin repair work; and c) average response time for customer service complaints. WPC uses this to benchmark and improve future performance. On average, WPC responds to customer complaints within seventy-five (75) minutes and completes repair work within two (2) weeks.

WPC also conducts regular inspection of its pump stations including performing preventative maintenance and repair and replacement of components as needed for proper operations and to maintain redundancy.

WPC's historical operation and maintenance efforts have served to fulfill customer requests and meet the immediate needs of the System. WPC has established a proactive approach to meet both the current and long-term needs of the System. To assist in implementing this approach, WPC received approval for the implementation of a new fixed charge component to the sewer rate beginning in FY2015. A discussion of the rates and charges is presented in Section 7 of this Report. The new fixed charge was established to provide for increased investment in infrastructure, which will facilitate replacement of aging sewers and related equipment. This will result in fewer emergency repairs and an increase in proactive cleaning and inspections.

4.4 Key Operating Permits and Regulatory Status

Regulations affecting the collection and discharge of wastewater from WPC are contained in the City's Community Discharge Permits with NEORSD and its Ohio EPA NPDES MS4 permit. NEORSD discharge permits regulate the sanitary and the combined sanitary and storm sewers. These permits do not contain any expiration date, but are subject to modifications as necessary. Pursuant to these discharge permits, WPC is required to submit an Annual Compliance Report Checklist to NEORSD. The compliance report checklist demonstrates WPC's compliance with the permit and NEORSD's best management practice (BMP) guidelines. NEORSD's BMP guidelines contain five (5) key BMPs:

- BMP 1: General Conditions
- BMP 2: Operation and Maintenance and Inspection Program
- BMP 3: Source Control
- BMP 4: Record System
- BMP 5: BMP Frequencies
    - Sewer System Cleaning
    - Sewer System Inspection
    - Catch Basin Cleaning
    - Street Sweeping
    - Storage Device Cleaning
    - Leaf Pickup
    - Pump Station Operations

Discharge of stormwater for WPC is regulated by the City's Ohio EPA NPDES MS4 permit. As discussed in Section 4, only a small portion of the System consists of separated storm sewers. The majority of the System is combined storm and sewer system that is regulated under the previously discussed discharge permits. A new Ohio EPA NPDES MS4 permit was issued on September 11, 2014. Key requirements of the permit include compliance with the Six Minimum Control Measures and submission of annual reports to the Ohio Environmental Protection Agency (OEPA) on compliance activities as previously discussed in Section 3.2.2.

4.4.1 Regulatory Compliance

WPC is not under any administrative orders, and to the best of our knowledge, there are no issued notices of violations. NEORSD, however, is under a Consent Order to eliminate combined sewer overflows and improve water quality in Lake Erie through the implementation of its LTCP.

WPC is implementing NEORSD best management practices (BMP) included in its permits and reporting on its compliance to NEORSD through submission of its Annual Compliance Report Checklist as required by its permits. WPC is also supporting NEORSD in implementing its LTCP. The BMPs included in WPC's discharge permits provide various BMP frequency guidelines, such as cleaning of sewers once every five (5) years. These guidelines represent goals for WPC, and are not a condition or requirement of the permits.

With respect to its Ohio EPA NPDES MS4 permit for stormwater, WPC documents its compliance activities in its annual reports to the Ohio EPA. Based on a review of these reports, WPC appears to be in compliance with the requirements.

At this time, there are no other known regulations forecasted to be promulgated in the near future that would have a significant impact on WPC's operations, or would require significant capital expenditures in addition to those already anticipated by WPC. As a condition of WPC's discharge permit, NEORSD has the ability to impose peak flow limits, which could require significant capital investment in excess of currently contemplated capital expenditures. At this time, there is no indication that NEORSD would impose any such limits. See Section 2 regarding WPC and coordination with NEORSD.

4.5 Discussion

WPC maintains a sewer system preventative maintenance program that includes implementation of BMPs and is in compliance with its regulatory permits. As part of these BMPs, the permits identify key goals and frequencies to measure BMP achievements. These goals and frequencies are not requirements, but guidelines. WPC has established System BMP goals that appear to appropriately balance costs and benefits and a sound basis for measurement of these goals. Historically, WPC's program focused first on addressing any complaints and known problem areas and second on conducting preventative maintenance. While these priorities remain, WPC is increasing its resources (both through establishment of the new fixed charge to fund capital, and through increased staffing for maintenance and repairs) to provide for increased preventative maintenance and a higher rate of replacement of sewers and equipment that are past their useful lives. WPC's continued investment in capital and staffing will allow WPC to substantially achieve the BMP goals within the forecast period.

5 CONDITION OF SYSTEM

5.1 Condition Assessment Approach

Arcadis conducted a condition assessment review of the System in January 2016 and February 2016 in connection with the proposed Series 2016 Bond issuance. The condition assessment review included site visits to a representative number of pump stations and storm facilities and a desk-top review of available information regarding the age and condition of buried infrastructure (linear assets) and equipment used to maintain the System. The condition assessment also included a review of existing reports provided by WPC, discussions and interviews with WPC staff, and confirmation of the visual observations made during the limited field visits of the facilities. The key purpose of this condition review was to confirm anticipated conditions, assess general maintenance and repair of the System, and assess the adequacy of WPC's proposed capital improvement plan.

The critical assets of WPC include twelve (12[1]) pump stations. The condition assessment focused primarily on a limited field inspection of these pump stations that occurred during the week of January 6, 2016. During these site visits, issues facing WPC were discussed, including the steps being taken to address these issues. The condition of the pump stations and ancillary equipment also was discussed. Key findings are discussed in subsequent sections.

For buried infrastructure (i.e., sewer interceptors, storm sewers, sanitary and combined sewers, laterals, force mains, manholes, catch basins, etc.) and the maintenance equipment, the condition assessment relied on existing information provided by WPC. This included information regarding the type, age and reported condition of maintenance equipment and preliminary GIS information regarding the age and materials of pipes to the extent that information is available, as well as reports made available regarding compliance activities. No field investigations were conducted on the buried infrastructure or maintenance equipment. Arcadis has no reason to question the soundness of the information provided by WPC based on historical knowledge of the WPC System, and as such, accepted the information as provided.

The condition assessment of the critical above-ground assets was based on numerical ratings within the following categories.

- Reliability
- Operations and maintenance (O&M) performance
- Capacity
- Regulatory Compliance

Using the above categories, an overall risk rating was assigned to each of the pump stations. The risk ratings for each of the above four categories are outlined in Table 5-1. The risk rating assigned to each pump station is presented in Section 5-2 following the description of the pump stations.

[1] WPC was originally responsible for 15 pump stations. However, 3 will be transferred to NEORSD by May 15, 2016 as discussed in Section 2.6.

Table 5-1. Summary of Rating System

Numerical Rating	Interpretation of Rating	Description
1	Little to no risk	Relatively new and in good physical and operating condition
2	Some risk	Good condition, no known capital requirements
3	Moderate risk	Aged or worn, but generally in good operating condition; may require capital investment within five-years
4	Significant risk	Operational, but nearing end of life and/or requires investment to bring to full operating condition
5	High risk	Should be on high priority for renewal and/or replacement

When considering the rating assigned to a critical asset, the criticality of that asset was also considered. For assets that are critical to the overall operation and regulatory compliance of the facility, the allowable risk is small. Thus, highly critical assets would tend to be given more stringent ratings.

5.2 Condition of Sewage Pumping Stations

Limited field visits to twelve (12) remote sewage pump stations were conducted beginning on January 6, 2016. In general, based on our limited visual inspection during these site visits, it appears that most of WPC's pumping stations are in fair to good physical condition, fairly reliable, and have adequate hydraulic capacity. Appendix D, Figure D-3 provides a map that shows the relative location of WPC's pump stations within System. The following provides a description of the pump stations and their risk review score (see Table 5-3).

WPC utilizes two distinct types of sewage pumping stations. All the stations are duplex stations, either of the centrifugal suction lift type, or the centrifugal flooded suction type. The suction lift stations are typically housed in an above-ground fiberglass enclosure over a precast concrete wet well. The pumping equipment, controls, valves and piping are all housed above ground in the enclosure. The centrifugal flooded suction type stations utilize a pre-fabricated "can" station, which includes a dry well containing the pumps, controls and valves and a separate wet well that collects the wastewater before pumping.

All of WPC's pump stations are relatively small in size (under 700 gallon per minute capacity), with pump inlets of either four (4) inch or six (6) inch suction pipe size. With the exception of the Burke Lakefront Airport Pump Station; most were constructed in the late 1980s and 1990s, and are within the expected design lives for the facilities. WPC has no ownership responsibilities of the Burke Lakefront Airport Pump Station. Another City department retains such ownership, but WPC is responsible for operations and maintenance of the station.

WPC pump stations are generally reliable and no major issues are reported. The Incinerator Pump Station is essentially new, having been replaced with all new components in 2015; however, the stations that are forty (40) or more years old, such as the pump stations located at Nottingham Road, Panna Lane, and Harsax Avenue, will require significant upgrades in the next few years.

WPC's pumping stations have two pumps; only one of which is required for normal peak conditions. During severe wet weather events, however, both pumps may need to operate to meet peak conditions due to excessive inflow and infiltration (I/I) into the System. WPC also has portable pumps that are transported and set up at pump stations to assist with pumping the flow during wet weather events. The vast majority of pumping stations have adequate, and even excess capacity, and thus run infrequently. Table 5-2 identifies the run times of the pump stations under average day dry weather conditions.

Table 5-2. Summary of Pumping Station Run Times

Pump Station		Capacity (GPM)	Run Time under Dry Weather Flow In Hours/Day
1.	Bradley Road Station	150	1.8
2.	Burke Lakefront Airport Station	300	0.4
3.	East 37 Street Station	700	2.8
4.	Edgewater Drive Station	250	2.0
5.	Inner Harbor Station	180	2.4
6.	Incinerator Station	600	3.2
7.	Nottingham Road Station	100	2.6
8.	Panna Lane Station	350	2.4
9.	Harsax Avenue Station	300	2.2
10.	Service Yard Station	350	3.0
11.	Spring Hill Station	195	3.0
12.	Stevedore Station	425	1.5

WPC utilizes a special design of its duplex pumping systems, which consists of one of the two duplex pumps being equipped with both an electric motor, as well as an internal combustion engine operating on natural gas. This engine driven pump, when operated during a power loss event, would produce adequate power to operate the system controls including telemetry and operate a single pump to maintain system operations. Because the pump stations are equipped with these engine driven pumps, back-up generators are not required. WPC also can provide portable pumps during power outages to provide redundancy to the affected pump stations.

WPC has two maintenance contracts in place to provide service and repair on all of its pump stations. These service contracts appear to be effective at maintaining the pump stations and associated telemetry and control systems since each are in good working condition.

In order to remotely monitor the pumping stations, all of the stations are equipped with radio telemetry to transmit critical signals and alarms to WPC's Supervisory Control and Data Acquisition (SCADA) system. Should this radio system fail, WPC dispatches crews to monitor the pump stations on a daily basis.

While many of the pumping stations are in residential neighborhoods, WPC does not report any significant issue with odors. Table 5-3 represents a summary of the pumping station condition assessment.

Table 5-3. Summary of Pumping Station Condition Assessments

Pump Station		Capacity (GPM)	Original Installation Date	# of Pumps	Type of Pump Station	Rating[1]
1.	Bradley Road Station	150	1997	2	Centrifugal Suction Lift	2
2.	Burke Lakefront Airport Station	300	1957	2	Centrifugal Flooded Suction	5
3.	East 37 Street Station	700	1989	2	Centrifugal Flooded Suction	4
4.	Edgewater Drive Station	250	1997	2	Centrifugal Suction Lift	2
5.	Inner Harbor Station	180	1987	2	Centrifugal Flooded Suction	4
6.	Incinerator Station	600	1989 (upgraded 2015)	2	Centrifugal Flooded Suction	1
7.	Nottingham Road Station	100	1965		Centrifugal Flooded Suction	5
8.	Panna Lane Station	350	1971	2	Centrifugal Flooded Suction	5
9.	Harsax Avenue Station	300	n/a	2	Centrifugal Flooded Suction	4
10.	Service Yard Station	350	1998	2	Centrifugal Suction Lift	2
11.	Spring Hill Station	195	1992	2	Centrifugal Suction Lift	4
12.	Stevedore Station	425	1999	2	Centrifugal Suction Lift	3

1) See Table 5-1 for a description of the ratings.

As noted, the above ratings reflect the general condition of the pump stations. For the most part, the pump stations appear to be reliable; however a few present potential risks and require additional engineering investigations to develop capital improvement requirements and recommendations. Such improvement requirements could require up to $2 million of capital investment over the next five (5) years.

At this time, WPC does not have any specifically identified capital projects to address pump station upgrades. WPC, however, does have undesignated funding set aside in its five-year capital plan to address future capital improvement requirements. Such funding is available for improvements to upgrade the condition of WPC's pump stations. Details of the capital plan are provided in Section 6.

5.3 Condition of the Collection System

As presented in Section 3 of this Report, the estimated average age of the WPC collection system is eighty (80) years, and it is estimated that approximately twenty-seven percent (27%) of the system is older than one hundred (100) years.

Over the past five (5) years, WPC has invested approximately $5 million per year on a cash basis for its capital program, which includes renewal and replacement of its aging infrastructure. Recognizing that an increased level of investment is needed to preserve the long-term reliability of its assets, it is embarking on an accelerated program to increase targeted investment in renewal and replacement. Beginning in FY2016, WPC's operating budget includes an appropriation of $1 million each year for a five (5) year period to conduct sewer system evaluation surveys (SSESs) to identify sewer pipes in need of repair and replacement. The SSES includes infrastructure inspections, condition assessment, flow monitoring, smoke and dye testing and a public information program. A hydraulic sewer computer model also will be developed for sewer pipes ten (10) inches in diameter and greater, along with associated rehabilitation recommendations, including criticality evaluation of the risks and challenges. In addition, WPC's capital budget, which has been increased to an average of $12.8 million per year for the next five (5) years, includes the hiring of a Program Manager (PMT project) to assist WPC in the implementation of its targeted capital plan. This capital plan represents more than a doubling of historical investment into the System. Section 6 presents additional details regarding WPC's capital plan.

WPC is serving its customers and maintaining compliance with its permit requirements. A review of customer service requests indicates that approximately twenty percent (20%) of the requests are potentially associated with collapses and breaks due to deteriorated pipe conditions. WPC conducts CCTV inspection of pipes to examine sewers/connections as these service requests are called in by residents. While the number of service requests has declined over the past three (3) years, the number of repairs has increased over the past two (2) years. The recent increase in repairs is due in part to WPC proactively addressing some deferred maintenance, but also may be due to increased requirements associated with the aging infrastructure.

WPC also undertakes ongoing condition assessments, as well as contracting for professional engineering services to conduct comprehensive SSES of known problem areas. The most recent of such studies is a Technical Report dated February 2014 for the East 185th Street/Lakeshore Blvd Area SSES. Moving forward, WPC has allocated in its operating budget an annual expenditure of $1 million per year to conduct SSESs and develop a targeted capital improvement plan.

The East 185th St./Lakeshore Blvd. study area is one of the few areas in the City with separated sewers. The local sanitary sewer system in the area conveys flow to the Euclid Creek Pump Station through the Lakeshore Boulevard Interceptor Sewer. In 2015, NEORSD increased the capacity of the system by upgrading the Euclid Creek Pump Station and constructing a new relief sewer.

Although NEORSD has increased the capacity of the Euclid Creek Pump Station and constructed the new relief sewer, the study identified that additional improvements are needed within the WPC collection system to reduce wet weather flows. Several alternatives were presented with the most cost effective alternative being the construction of a conveyance/relief sewer and rehabilitation of the existing sewers at an estimated cost of $2.7 million. WPC is finalizing its plan for the East 185th Street/Lakeshore Boulevard study area, and anticipates undertaking the major capital requirements of the project in FY2018 and FY2019. In the meantime, WPC is undertaking a number of improvements along Euclid Creek as part of its ongoing repair program and capital projects budget for catch basins and manholes.

5.4 Condition of Collection System Equipment

WPC maintains a fleet of more than eighty (80) vehicles ranging from pickup trucks used by supervisors and staff to conduct inspections and respond to complaints; to heavy equipment used by staff to respond to emergencies. Approximately half of the vehicles are used daily to maintain the System. Figure 5-1 provides a summary status of the key equipment needed to maintain the System, including number of allotted vehicles (includes all planned vehicles regardless of operational status), number in fair-to-good operational condition, and average number deployed on a daily basis. In most instances sufficient equipment in good operational condition were available to meet the System demands. This was not the case for the vac-all and jet-vac vehicles, where equipment deemed to be in poor or unreliable condition needed to be deployed to meet the System needs.

The average age of the vehicles listed in Figure 5-1 is 8 years. WPC typically establishes a ten (10) year life for vehicles. The vehicles that are in poor condition are reportedly being scheduled for replacement as budget permits. In the meantime, they serve as backup vehicles. WPC purchased twelve (12) new vehicles in 2014, and has budgeted $1.9 million in its FY2016 capital budget for the purchase and/or lease of five (5) new vacuum-trucks. WPC's capital budget for the next five (5) years includes a total of $6.5 million for collection system equipment to be funded with cash to meet its vehicle allocation plan.

WPC's ability to address sewer system investigations and repairs in the past has been limited by the availability of good operating vehicles and maintenance equipment. The acquisition of the twelve (12) new vehicles in 2014 resulted in an increase in completed repairs. The equipment allocation established should serve to further increase preventative maintenance activity, television inspections and completion of repairs.

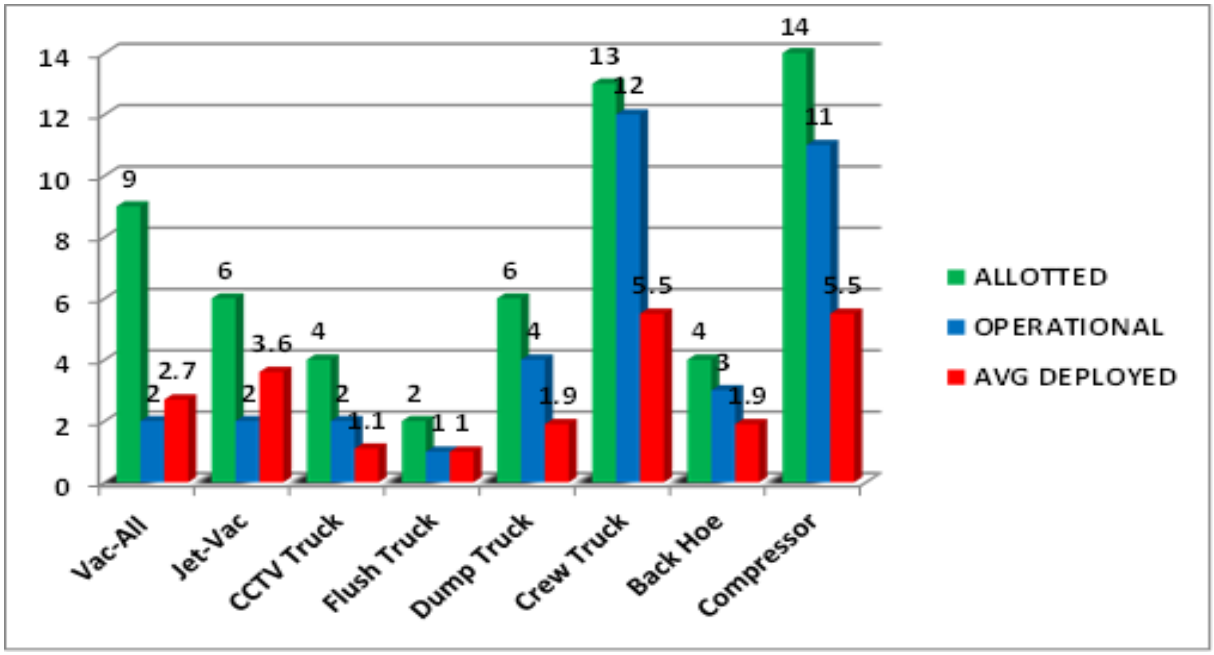

Figure 5-1. Summary Status of Key Equipment Needed to Maintain the System

5.5 Discussion

Based on January 2016 visual inspections, Arcadis has made some observations of the System that, if addressed, could further improve facility performance and operations. Many of these already are addressed in WPC's five-year capital plan. They include:

- The wastewater and stormwater collection system has an average age of approximately eighty (80) years with parts of the system exceeding the typical useful life. Conducting SSESs and detailed investigation of customer service requests will help to identify priorities for renewal and replacement. The use of the CMMS and GIS will facilitate development of a comprehensive asset management program, and the PMT Project should result in the cost effective and efficient implementation of WPC's capital improvement plan. WPC should further target increasing its capital expenditures over the long term, with a goal of renewal and replacement of one percent (1%) of its system per year, subject to results of ongoing sewer inspections. This likely will require a continued annual capital investment at least equal to that of the five-year plan beyond FY2020.
- WPC conducts regular inspection and preventative maintenance of the pump stations to ensure operability. Several pump stations (Burke Lakefront Airport Station, Nottingham Road Station and Panna Lane Station), however, represent a high priority for renewal and replacement investment to minimize potential risks. The Nottingham Road Station and Lakefront Airport Station likely will require replacement within the next five (5) years. Capital improvements for the Burke Lakefront Airport Station, however, remain the Department of Port Control's responsibility. The E37th Street Station, Harsax Avenue Station and Spring Hill Station are not that old, but will require some refurbishment. The cost of these improvements could be up to $2 million. WPC will

evaluate the refurbishment needs of these pump stations and implement the needed improvements over the next five (5) years. It is anticipated that funding for the improvements will come from WPC's undesignated project allocation contained in its five-year plan.

- The allocation of capital funds established for maintenance equipment appears sufficient to meet the needs of the System. Much of the existing equipment will require replacement. The capital plan addresses these requirements.

Overall, the System appears to have sufficient capacity to meet current and projected future wastewater needs of the service area. Based on the limited visual observations and review of information, the System appears to be functioning satisfactorily. Portions of the System are past their useful life and therefore poses additional risk for potential sewer line failures. The WPC capital plan has been developed with a focus on minimizing any such risks.

Available data indicates that WPC is operating and maintaining the System in general compliance with its permit requirements with NEORSD and the conditions of its MS4 stormwater permit. Based on our observations and discussions with WPC staff, operation and maintenance of the System continues to be enhanced through implementation of technology and improved information management systems, increases in staff for operations, and increased capital investment.

With respect to the pump stations, WPC's pump stations appear to be reliable and have sufficient capacity. Several pump stations present some potential risk and require investment; however, all of the pump stations are relatively small and run only a few hours per day.

While it is difficult to assess the condition of the collection system, WPC understands the risks of its older system, and is therefore undertaking SSESs to investigate its condition and develop cost effective renewal and replacement plans.

6 CAPITAL IMPROVEMENT PROGRAM

Arcadis was provided with the five-year capital plan for WPC, which covers the fiscal years 2016 through 2020. The fiscal year 2015 plan was also provided for reference. This plan is provided as Appendix B and summarized below.

6.1 Past Capital Improvements

WPC historically employed a pay-as-you-go capital improvement plan, supplemented with grants and low cost loans from the Ohio Public Works Commission (OPWC) and Ohio Water Development Authority (OWDA) for eligible projects. Past improvement efforts focused primarily on addressing both street and basement flooding, acquiring equipment needed to maintain the System and implementing information technology systems to better manage its assets. For example, WPC implemented Cityworks as its CMMS and integrated the City's GIS to view storm and wastewater assets and analyzing preventive maintenance issues without being on site. WPC prioritized its capital improvement program based on known problem areas, results of sewer investigations, customer surveys, service calls and complaints and data available from its information management systems. Past CIPs included construction of relief main sewers, replacement of collapsed and deteriorated sewers, drainage improvements to prevent flooding and lining of storm and sanitary sewers. Table 6-1 presents a summary of the historical annual level of capital expenditure on the System:

Table 6-1. Historical Capital Improvement Funding

Capital Investment	FY2011	FY2012	FY2013	FY2014	FY2015	Five-year Total
Sewer Replacements	$3,775,818	$2,829,022	$2,897,448	$5,715,378	$7,666,006	$22,883,672
Other Improvements[1]	$1,151,240	$486,254	$2,903,956	$731,764	$1,613,671	$6,886,885
Total Annual Expenditure[2]	$4,927,058	$3,315,276	$5,801,404	$6,447,142	$9,279,677	$29,770,557

Notes:
1) Consists primarily of equipment purchases and investment in technology.
2) All capital expenditures were cash funded.

WPC is embarking on an expanded renewal and replacement program to reduce the age of its infrastructure, reduce potential for overflows, and preserve the long-term use and reliability of the assets.

6.2 Projects to be Funded with the Series 2016 Bonds

WPC plans to use $35 million of proceeds from the Series 2016 Bonds to fund the majority of its capital improvement plan for the next three (3) years. This revenue bond funding will be supplemented with the

cash funding of vehicles and equipment and additional technology projects. Table 6-2 presents a summary of the projects to be funded with the Series 2016 Bonds. A more detailed description of each of these projects is presented in Appendix B.

Table 6-2. Projects Funded with Series 2016 Bonds

Type	Project Name	FY2016	FY2017	FY2018	Description
Stormwater					
	Euclid Creek	$2,500,000			Stream channel restoration
Sewer Replacement					
	Sewer Replacement	$1,000,000	$1,000,000	$1,000,000	Renewal and replacement of sewers at various locations
	Sewer Lining	$800,000	$800,000	$800,000	Lining of sewers at various locations
	Sewer Connections	$1,000,000	$1,000,000	$1,000,000	Rehabilitating and replacing sewer service connections at various locations
	Major Sewer Replacement Projects	$8,750,000	$3,150,000	$2,750,000	Installation and replacement of sanitary and storm sewers. Refer to Appendix B for a detailed listing and description of specific projects.
Building and Misc. Improvements					
	PMT	$1,700,000	$1,400,000	$1,300,000	Securing a professional firm to provide capital improvements program management for a five-year period to meet WPC goals and objectives including specific levels of service for sewer reliability, functionality and sustainable operations.
	Professional Services	$500,000	$500,000	$500,000	The Division issues RFPs every two years for professional consultants to provide capital improvement engineering services to WPC as required including wastewater system design and planning, sewer system design, regulatory support, bidding and construction services, surveying, and field representation.
Undesignated					
	Undesignated		$1,550,000	$2,000,000	Capital investment in pump station upgrades and other improvements.
	Total	**$16,250,000**	**$9,400,000**	**$9,350,000**	**$35 million in proceeds over the three-year period.**

As illustrated in Table 6-2, it is anticipated that $35 million of proceeds from the Series 2016 Bonds will be used to primarily fund sewer rehabilitation and replacement projects for FY2016 through FY2018, with approximately $16.25 million expended the first year, and $9.40 and $9.35 million expended in each of the next two (2) years. The remainder of WPC's capital improvement program is anticipated to be funded with cash or future bond proceeds as discussed in the next section.

6.3 Future Capital Improvements

In addition to the projects anticipated to be funded by the Series 2016 Bonds, WPC anticipates cash funding of equipment purchases and technology investments and issuing additional revenue bonds as needed to assist in funding WPC's long term capital plan. Table 6-3 provides a summary of the anticipated sources of funding for WPC's long-term capital plan.

Table 6-3. Summary of Capital Plan Sources of Funding

Source of Funding	Annual Capital Cash Flow Forecast					Total	% of Total
	FY2016	FY2017	FY2018	FY2019	FY2020		
Cash/Reserves	$2,940,080	$2,071,600	$2,215,800	$2,382,500	$3,250,000	**$12,859,980**	20%
2016 Bonds	$16,250,000	$9,400,000	$9,350,000	$0	$0	**$35,000,000**	55%
Future Bonds	$0	$0	$0	$8,042,000	$8,091,000	**$16,133,000**	25%
Total	**$19,190,080**	**$11,471,600**	**$11,565,800**	**$10,424,500**	**$11,341,000**	**$63,992,980**	**100%**

WPC's level of investment into the System will average approximately $12.8 million per year over the next five (5) years, with more than sixty percent (60%) or an average of $8.1 million per year (including $5.7 million for identified sewer replacement projects and $2.4 million for SSES recommended improvements) dedicated to sewer rehabilitation and replacement. This level of investment is significantly higher than historical investment, and is more than the current annual depreciation expense associated with all of WPC's assets of approximately $5.7 million per year. WPC's objective is to increase capital spending to reduce the average age of the collection system.

A summary breakdown of the five-year capital plan is provided in Table 6-4 below. The full plan including project descriptions is provided in Appendix B.

Table 6-4. Summary of Five Year Capital Plan by Category ($ 000's)

	Annual Capital Cash Flow Forecast (FY, $000's)					Total (M)	% of Total
Category	**FY2016**	**FY2017**	**FY2018**	**FY2019**	**FY2020**		
Stormwater	$2,950	$450	$450	$450	$450	$4,750	7.4%
Sewer Replacement	$11,550	$5,950	$5,550	$2,800	$2,800	$28,650	44.8%
Sewer Work on Roadway Projects	$500	$500	$500	$500	$500	$2,500	3.9%
Building and Misc. Improvements	$2,200	$1,900	$1,800	$1,700	$1,700	$9,300	14.5%
Equipment	$1,900	$1,000	$1,200	$1,200	$1,200	$6,500	10.2%
Technology Projects	$90	$120	$70	$30	$100	$410	0.6%
Undesignated	--	$1,550	$2,000	$3,740	$4,590	$11,880	18.6%
Total	$19,190	$11,470	$11,570	$10,420	$11,340	$63,990	100%

*Totals may not add due to rounding.

The five-year capital plan represents WPC's anticipated plan based on the information available as of January 2016. WPC is currently procuring a professional capital improvements manager (PMT project) and conducting SSESs to refine the proposed plan. Therefore, the five-year plan may be subject to change as new information becomes available. WPC has established the "undesignated" category to account for unknowns and provide a funding allocation to address SSES recommendations and pump station improvement requirements.

6.3.1 Stormwater System – Planned Improvements

A total of $4.75 million is budgeted for stormwater system upgrades over the next five (5) years. Of this amount, $2.5 million is budgeted to remove shoaling deposits and in stream vegetation; and provide stream bank stabilization along Euclid Creek pursuant to the agreement reached with the USACOE dated March 27, 2015. The remainder is for annual catch basin and manhole rehabilitation and reconstruction throughout the System. The five-year capital plan also includes $2.5 million in sewer work on roadway projects. These projects include approximately $0.5 million per year of sewer and storm drainage improvements at various locations within the City that are coordinated with the City's Public Works Department to enhance efficiency of operations. A more detailed description of these projects is provided in Appendix B.

As described in Section 2.6, NEORSD will implement its stormwater management user charge to recover NEORSD expenses related to management of combined stormwater interceptors and overflows and implementation of green infrastructure and other initiatives in compliance with its NPDES Permit and its approved LTCP. The City and NEORSD are in discussions regarding NEORSD potential assistance in funding WPC stormwater projects through a portion of revenues collected by NEORSD through its stormwater user charge. Options for such funding assistance may include NEORSD providing funding for qualifying stormwater projects undertaken by the City. At this time, however, no such funding has been included in the CIP. Once this funding becomes available, it is anticipated that additional stormwater projects will be added to the CIP as appropriate.

6.3.2 Wastewater Collection System – Planned Improvements

The majority of WPC's five-year capital budget is related to sewer replacement projects. Of the $28.7 million five-year expenditure identified under the category Sewer Replacement, approximately $14 million is budgeted for ongoing annual sewer replacements, sewer lining, and service connection replacements at various locations throughout the City. The remainder consists of specific projects related to rehabilitation and replacement of aging sewers that are anticipated to be funded from the proceeds of the Series 2016 Bonds or future bonds.

In addition to the above, the plan also includes $11.9 million under the Undesignated Projects category. WPC, through its operating budget, will hire professional engineers to conduct SSESs of the System to assess the condition of the collection system and identify targeted improvements. The recommended capital improvements from these SSESs will be funded through the undesignated projects budget. Capital improvements identified to upgrade WPC pump stations and address other improvements identified as a result of condition assessments would also be funded from this category.

6.3.3 Building and Other Planned Improvements

The five-year capital plan includes $16.2 million in capital expenditures for building and other improvements. Of this amount, $0.4 million represents WPC's continued investment in technology, $6.5 million in equipment to maintain the System (both to be funded with cash), $6.8 million for program management to ensure implementation of a cost effective capital improvement program, and $2.5 million in professional services for sewer design and related services (to be funded with revenue bonds).

6.4 Discussion

Our review of the proposed capital improvements projects resulted in the following findings:

1. Contracts for Sewer Replacement, Sewer Lining, and Sewer Connections represent a continuation of efforts by WPC from the last two (2) years. We are of the opinion that that these services provided good value to WPC and that it is reasonable for WPC to appropriate additional funds to continue receiving these valued services.
2. The remaining capital projects proposed for 2016, 2017 and 2018 appear to have been prepared with reasonable judgment to the scope of the proposed improvements and to the value of the anticipated construction costs. We are of the opinion that the proposed costs are equivalent to the proposed scope of these projects.

It is anticipated that WPC will continue its practice of updating its five-year capital plan as soon as the results of ongoing studies, such as the SSESs, are completed and in no event less than once per year. WPC is also expected to make the necessary changes to the capital plan and not delay acquisition of funding to support the required capital projects.

7 CUSTOMER STATISTICS AND SEWER USE RATES

7.1 Customer Base

The City has a population of approximately 390,000 (based on 2010 U.S. Census Bureau statistics) and is the [4]8th largest city in the US, and the second largest in Ohio. WPC provides services to approximately 126,000 active metered water connections that serve all residential, commercial, and industrial, government and non-profit customers. Table 7-1 illustrates the current number of active water metered accounts broken down by meter size. Fewer than ten (10) customers in the southwest side of the City are served by septic systems.

Table 7-1. Number of Accounts by Water Meter Size

Water Meter Size	# of Accounts
1/2 IN	1
5/8 IN	107,807
Homestead (1)	10,993
3/4 IN	420
1 IN	2,266
1½ IN	1,181
2 IN	1,717
3 IN	716
4 IN	419
6 IN	218
8 IN	66
12 IN	7
Total	**125,811**

(1) Homestead customers are provided a discounted rate regardless of meter size.

Ninety-six percent (96%) of WPC's customers are residential, while the remaining four percent (4%) of the customers are primarily commercial, governmental and industrial accounts.

Table 7-2 provides the historical trend in number of billed customers and consumption over the past five (5) years. As illustrated in this table, the number of customers and consumption have varied from year to year, but overall billable consumption has declined over the past five (5) years. Overall, WPC has seen an average annual decline in consumption of less than two percent (2%) over the past few years. The

rate of decline in the future, however, is expected to be less over time as the City has diversified its economic base and as economic conditions in and around the City continue to improve. A continued decline in billed consumption per account, however, is expected as customers continue to upgrade to more water-efficient fixtures and increase water conservation efforts.

Table 7-2. Historical Customer Accounts and Consumption

Description	FY2012	FY2013	FY2014	FY2015
Water Meter Connections	125,577	126,555	125,768	125,811[1]
Annual Customers Change	--	0.78%	(0.62%)	0.03%
Consumption (MCF)[1,2]	1,826,888	1,822,980	1,823,983	1,790,291
Annual Consumption Change	--	(0.21%)	0.06%	(1.85%)

Notes:
1) Source: WPC Sales Service Data provided by Fiscal Office. Data for FY2011 not available.
2) MCF represents thousand cubic feet (equivalent to 7,480 gallons).
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In 2014, WPC experienced a slight increase in billable consumption. This increase was due in part to the Division of Water implementing an automated meter reading (AMR) system, which served to reduce the number of estimated bills and improve meter reading accuracy. This is anticipated to be a one-time increase in response to the improved AMR system. Over the long-term however, consumption per account is anticipated to decrease in response to water conservation efforts. This reduction is offset somewhat by WPC's minimum bill policy as described in Section 7.2.

WPC is projecting continued decreases in consumption in FY2016 through FY2020. A two and two-tenths percent (2.2%) annual decrease in consumption billing revenue is projected throughout the planning period to account for potential decreases in customer usage and other unknowns. This projection is consistent with the Division of Water's projections for water consumption. With respect to number of billing customers, the total number of customers is projected to remain static during the forecast period. The number of customers is based on buildings with metered water service that are connected to the System. As long as the buildings are connected to the System they will continue to be billed a fixed charge regardless of the amount of water consumed. WPC does not anticipate any net disconnection of buildings or establishments from its System during the forecast period. A summary of the projected fixed charge customer accounts and billable consumption for the forecast period is shown in Table 7-3.

Table 7-3. Projected Customer Accounts and Consumption

Description	FY2016	FY2017	FY2018	FY2019	FY2020
Fixed Charge Customer Accounts	125,811	125,811	125,811	125,811	125,811
Annual Change in Customers	(0%)	(0%)	(0%)	(0%)	(0%)
Consumption Forecast (MCF)	1,750,905	1,712,385	1,674,712	1,637,869	1,601,835
Annual Consumption Change	(2.2%)	(2.2%)	(2.2%)	(2.2%)	(2.2%)

WPC's Top Ten Customers by billed consumption are provided in Table 7-4. The Top Ten Customers comprise approximately twelve and six-tenths percent (12.6%) of the billable consumption.

Table 7-4. Overall Top Ten Customers in 2015

Group Description	Total Accounts	Billable Consumption (MCF)	% of Total Consumption
Cuyahoga Metropolitan Housing Authority	700	72,396.20	4.0%
Cleveland Clinic	37	43,453.40	2.4%
Case Western Reserve University	62	22,631.80	1.3%
Cuyahoga County	22	17,009.60	1.0%
University Hospitals Health System	15	16,543.10	0.9%
Metro Health Medical Centers	11	15,465.10	0.9%
Cleveland Thermal	5	12,292.30	0.7%
Cleveland Board of Education	80	9,357.80	0.5%
Cleveland State University	27	9,197.80	0.5%
National Aeronautics and Space Administration	3	7,429.00	0.4%
Subtotal % of Total Consumption			**12.6%**

Although the City has seen significant population decline for more than a decade, in large part due to declines in large industry and manufacturing jobs, that decline is stabilizing as the City has become more diversified. The City's Top Ten Customers represent primarily service companies.

The City's unemployment rate, as of December 2015, was slightly higher than the State of Ohio and national averages (at approximately six and nine-tenths percent (6.9%)), but remains lower than unemployment rates in 2010, which were at ten and eight-tenths percent (10.8%). As a result, we believe that WPC currently has a stable and diverse customer base, and the projected customer declines are

likely overstated. The median household income of $26,179 per household in the City (2014 census data), however, remains significantly lower than the State of Ohio average of $48,849 (2014 census data). Maintaining affordable rates remains a priority for the City. It is for this reason that WPC provides a homestead rate as discussed further in the next sections.

7.2 User Rates and Charges

WPC is responsible for establishing sewer rates and other fees and charges for its customers, subject to approval by the Board of Control and City Council. Pursuant to Section 112 Fixation of Utility Rates of the City Charter: "Rates for the service or products of all non-tax supported public utilities, owned and operated by the City which are managed and supervised by the Director of Public Utilities, shall be fixed by the Board of Control, subject, however to approval by Council." Furthermore, in Section 80 Board of Control of the City Charter: "The Mayor and the directors of the several departments, whether established by this Charter or by ordinance, shall constitute a Board of Control. The Mayor shall be ex-officio President of the Board. The vote of a majority of all members of the Board shall be necessary to the adoption of any question, motion or order. All votes shall be by "yeas" and "nays" and entered on the record. All meetings of the Board shall be public, a record of its proceedings shall be kept, and an abstract of its proceedings shall be printed in the *City Record*.

The rates must be at sufficient levels to pay for O&M costs of the System, and debt service on outstanding debt, and to provide for appropriate contingency, coverage, and reserves in accordance with the requirements of the Trust Agreement and General Bond Ordinance. User charges are set after accounting for other revenue, collection rate assumptions, and requirements for debt service coverage and contingency reserves.

Table 7-5 shows the historical user rates for WPC for FY2011 through FY2015. The rates include two key charge components:

- **User Consumption Charges**: Historically, WPC relied solely on user charges that were based on each customer's billable metered water consumption. Under this consumption charge component, customers are charged the same rate per MCF of metered water consumption. The only exception is that certain customers pay a discounted Homestead rate if they are eligible and have been approved based on established economic and social criteria. The Homestead rate provides customers with a discounted rate which is approximately forty percent (40%) lower than the regular rate to maintain affordability of such rates. The consumption charge also includes a minimum bill charge based on a minimum consumption of one MCF per quarter.
- **Fixed Meter Charges**: Beginning in FY2015, WPC instituted a new fixed charge component that varies based on the size of the customer's incoming water meter service. Billing of customers for this new fixed charge began during the second quarter of FY2015. See Table 7-6 for a breakdown of charges by meter size. The fixed charge is in addition to the consumption charge and was instituted to address the costs of maintaining the infrastructure of the System. The fixed charge is assessed even if the customer had zero water consumption. Similar to the consumption charge, a forty percent (40%) discounted Homestead fixed charge rate is available for qualifying customers to maintain affordability of the rates.

As shown in Table 7-5, unit consumption rates from FY2011 through FY2014 were held flat at $12.53 per MCF for regular customers and $7.43 per MCF for homestead customers. In 2015, the fixed charge component was implemented, while the unit consumption rate remained the same as in previous years.

Table 7-5. Historical Sewer Rates and Rate Structure

Description	FY2011	FY2012	FY2013	FY2014	FY2015
Consumption Rate per MCF					
• Regular Customers	$12.53	$12.53	$12.53	$12.53	$12.53
• Homestead Customers	$7.43	$7.43	$7.43	$7.43	$7.43
Fixed Charge Based on 5/8 Inch Meter Size					
• Regular Customers	$0.00	$0.00	$0.00	$0.00	$12.00
• Homestead Customers	$0.00	$0.00	$0.00	$0.00	$7.20
Minimum Bill Per Quarter					
• Regular Customers	$12.53	$12.53	$12.53	$12.53	$24.53
• Homestead Customers	$7.43	$7.43	$7.43	$7.43	$14.63
Effective Increase in WPC Rate	-	0%	0%	0%	30%[3]

Notes:
1) Bills are issued quarterly.
2) Source: Cleveland Water Pollution Control Approved Rates.
3) Although the consumption rate did not change in FY2015, a new fixed charge was implemented depending on customer water meter size. The above table provides the fixed charge rate for a 5/8 inch meter. Table 7-6 below provides the detailed breakdown by meter size. The new charge went into effect April 1, 2015. The net effect of this new charge is to increase quarterly billing rates in FY2015 by almost 30% over FY2014 quarterly billing rates.

At the same time as approving the new fixed charge component based on meter charge for FY2015, the Board of Control established and City Council approved a rate increase package through 2019. The approved rates will have periodic increases for five years from FY2015 through FY2019.Table 7-6 provides the approved fixed rate and consumption rates for FY2015 through FY2019.

Table 7-6. Approved Quarterly Fixed Charge and Consumption Rates

Fixed Charge Meter Size	FY2015	FY2016	FY2017	FY2018	FY2019
½ to 1 inch	$12.00	$12.00	$15.00	$15.00	$18.00
1½ to 2 inch	30.00	30.00	40.00	40.00	55.00
3 to 4 inch	99.00	99.00	115.00	115.00	140.00
6 inch	198.00	198.00	225.00	225.00	260.00
8 Inch	300.00	300.00	325.00	325.00	370.00
12 inch	450.00	450.00	500.00	500.00	550.00
Homestead	7.20	7.20	9.00	9.00	10.80
Minimum Consumption Charge	**FY2015**	**FY2016**	**FY2017**	**FY2018**	**FY2019**
Regular (per MCF)	$12.53	$12.53	$12.53	$12.53	$12.53
Homestead (per MCF)	7.43	7.43	7.43	7.43	7.43

Notes:
1) Source: City Codified Ordinance 543.02.
2) In FY2015 the new rates went into effect on April 1, 2015. All other rate changes go into effect on January 1st at the beginning of the fiscal year.

As illustrated in Table 7-6, fixed charges will increase in FY2017 and FY2019, but unit consumption rates are anticipated to remain unchanged through FY2019.

7.3 Billing and Collection

In accordance with WPC's billing policies, customers are billed quarterly based on their meter size and metered water consumption. Bills are issued by the City's Division of Water on behalf of WPC. The Division of Water charges WPC its allocable share of the cost of billing services provided based on the Division of Water's cost and number of bills issued. Billing, collections and customer service are included in the services provided by the Division of Water. Consistent with historical collection rates, WPC anticipates an average annual collection rate of ninety percent (90%).

The Division of Water, as billing agent for WPC, anticipates changing from quarterly billing to monthly billing in FY2017 in response to customer requests. While the overall charges from the Division of Water are anticipated to increase commensurate with increased costs associated with more frequent printing and postage costs, the charges per bill will decrease due to frequency. Furthermore, because no increase in staffing is anticipated to be required and the increased postage and printing costs will be split between WPC, Water and Waste Collection, the net increase in WPC's share of the cost of billing services due to changing to monthly billing is anticipated to be minimal. A small increase in billing costs is projected to be incurred in FY2017 and has been included in WPC's forecast.

8 FINANCIAL FEASIBILITY

8.1 Overview

As part of this Report, a financial plan and forecast were prepared for WPC for FY2016 through FY2020. The purpose of this section is to present the financial plan and forecast, the projection of revenues and expenses, debt service coverage, and to document the assumptions used in the preparation of the forecast in order to provide an engineer's opinion as to the financial feasibility of the Series 2016 Bonds.

The financial forecast represents an estimate of the probable results of operations and reflects WPC's judgment as to the most likely set of conditions and course of action based upon present circumstances. Arcadis worked closely with WPC to obtain the information necessary to prepare the financial forecast, and various financial and system information was provided by WPC to support its development.

8.2 Historical Revenues and Expenses

Table 8-1 presents historical financial results, including historical revenues, for FY2011 through FY2015. As illustrated in Table 8-1, WPC revenues exceeded expenses, excluding depreciation expense. WPC has been able to cash fund capital improvements and has not issued any debt during the period. As shown in Table 8-1, a significant increase in sewer user charge revenue was realized in FY2015 as a result of the implementation of the Fixed Charge component of the rate structure to fund infrastructure costs.

Table 8-1. Historical Financial Performance

	FY2011	FY2012	FY2013	FY2014	FY2015
Operating Revenues					
Charges for Services	$22,199,000	$22,876,000	$22,549,000	$22,547,000	$28,744,000
Total Operating Revenues	$22,199,000	$22,876,000	$22,549,000	$22,547,000	$28,744,000
Operating Expenses					
Operations	$9,315,000	$10,194,000	$9,232,000	$10,670,000	$10,818,000
Maintenance	$8,835,000	$9,075,000	$8,091,000	$8,459,000	$8,170,000
Total Operating Expenses	$18,150,000	$19,269,000	$17,323,000	$19,129,000	$18,988,000
Operating Income (Loss) Before Depreciation	$4,049,000	$3,607,000	$5,226,000	$3,418,000	$9,756,000
Depreciation	$4,734,000	$5,108,000	$5,221,000	$5,579,000	$5,700,000
Operating Income (Loss) After Depreciation	($685,000)	($1,501,000)	$5,000	($2,161,000)	$4,056,000
Non-Operating Revenue (Expense)					
Investment Income	$53,000	$59,000	$58,000	$54,000	$51,000
Interest Expense	($121,000)	($102,000)	($82,000)	($61,000)	($40,000)
Other	$53,000	$21,000	$99,000	$86,000	$80,000
Total Non-Operating Revenue (Expense), Net	($15,000)	($22,000)	$75,000	$79,000	$91,000
Income (Loss) Before Other Contributions	($700,000)	($1,523,000)	$80,000	($2,082,000)	$4,147,000

Notes:

1) Information from the FY2011 though FY2014 annual audited financial statements, based on accrual basis, exclusive of depreciation Audited by independent auditors Clark Schaefer and Hackett
2) FY2015 based on WPC estimates provided by Fiscal Control on March 9, 2016.
3) WPC's outstanding debt consists of low cost loans from the Ohio Public Works Commission and Ohio Water Development Authority.

8.2.1 Existing Debt Service

WPC's existing debt includes loans from the OPWC and OWDA. Table 8-2 summarizes the principal amount of OPWC and OWDA obligations outstanding as of January 1, 2016.

Table 8-2. Existing Obligations as of January 1, 2016

Loans	Maturity Date	Outstanding Principal Balance
OPWC Loan – Kerruish Stormwater Management Facility	FY2023	$78,404
OPWC Loan – Hamlet Ave and Adolpha Rd Area Storm Sewer	FY2021	64,295
OWDA Loan – Sewer Construction	FY2017	88,529
OWDA Loan – Spring Rd Sewers	FY2017	16,536
OWDA Loan – Ivermere Ave Sewers	FY2017	56,024
OWDA Loan – Cooley Ave Repl Sewers	FY2017	449,332
Total Amount of Principal Outstanding as of 1-1-16		**$753,120**

Notes:
1) Source: WPC 2014 Financial Statement.

Table 8-3 provides a summary of the schedule of payments associated with the existing outstanding debt obligations. The outstanding debt service information was provided by WPC. These obligations are subordinate to the Series 2016 Bonds.

Table 8-3. Existing Debt Service Schedule (Principal and Interest Payments)

Period Ending	Existing OPWC (1)		Existing OWDA		Subtotal Existing Debt
	Principal	Interest	Principal	Interest	
FY 2016	$24,060	-	$457,578	$10,352	$491,990
FY 2017	$24,060	-	$152,841	$2,294	$179,195
FY 2018	$24,060	-	-	-	$24,060
FY 2019	$24,060	-	-	-	$24,060
FY 2020	$24,060	-	-	-	$24,060
FY 2021	$11,201	-	-	-	$11,201
FY 2022	$11,201	-	-	-	$11,201
TOTAL	**$142,702**		**$610,419**	**$12,646**	**$765,767**

Notes:
1) Source: WPC 2014 Financial Statements.

8.3 Future Revenue and Expenses

8.3.1 Projected Revenues

Historical billed consumption, customer accounts, and sewer revenues from FY2011 through FY2015 were reviewed, along with budget projections prepared and provided by WPC for FY2016, to develop

projections of revenues for the period FY2016 to FY2020. The System revenues include revenues derived from user consumption charges, fixed meter charges and other miscellaneous revenue sources. The revenue projection includes the impacts of the City Council approved wastewater rates and rate adjustments for both the consumption and fixed rate components, for the period FY2016 through and including FY2019 as discussed in Section 7. The projected sewer rate revenue over the forecast period is provided at the end of this section in Table 8-6.

8.3.2 Projected O&M Expenses

The O&M expenses of the System include personnel (salaries, fringe benefits, and overtime), materials and supplies (general offices supplies, sewer and stormwater material supplies, minor equipment, etc.), utilities, contracted professional services and City interdepartmental charges. The interdepartmental charges include billing services provided by the Division of Water, financial services provided by Division of Fiscal Control, vehicle maintenance provided by the Division of Motor Vehicle Maintenance and other administrative services (human resources, information technology and public relations support) provided by various other City divisions.

The projection of O&M expenses is based on historical expense levels and changes thereto. The O&M expense projection assumptions are summarized in Table 8-4. These projections were developed based on discussions with WPC regarding System operational changes, historical cost trends, and anticipated future billed consumption trends.

Table 8-4. O&M Expense Projection Assumptions

Expense Category	Annual Percent Increase	Basis / Rationale
Personnel	2.0%	WPC estimate.
Benefits	5.0%	Projection based on historical trends.
Inflation	3.0%	WPC estimate.
Utilities	3.0%	WPC estimate.

8.3.3 Capital Improvement Funding Plan

WPC plans to fund the capital plan described in Section 6 with a mix of cash and debt, as summarized in Table 6-3. Approximately $35 million or eighty-five percent (85%) of sewer and stormwater conveyance projects over the next three (3) years (FY2016-FY2018) will be funded with the proceeds of the Series 2016 Bonds. The remaining amount of the CIP over these three (3) years is anticipated to be funded with cash. In FY2019 and FY2020, WPC anticipates funding its CIP with a combination of cash and additional revenue bonds. To the extent that WPC can secure grants and loans from other sources, this will serve to expedite the capital plan and offset the need to issue additional revenue bonds. The projections, however, currently do not include these potential low cost funding opportunities. WPC also plans on

continuing to fund a number of renewal and replacement projects on a pay-as-you-go basis with available cash in excess of its operating reserve goals.

8.3.4 Projected Non-Operating Expenses

Table 8-5 provides the debt service schedule of payments associated with the Series 2016 Revenue Bonds, which was provided by WPC's financial advisor on March 31, 2016 based on final pricing for a thirty (30) year term.

Table 8-5. Series 2016 Revenue Bond Debt Service

Year	Principal Payment	Interest Payments	Subtotal Annual Debt Service
FY2016	-	928,546.53	928,546.53
FY2017	550,000.00	1,584,250.00	2,134,250.00
FY2018	570,000.00	1,562,250.00	2,132,250.00
FY2019	590,000.00	1,545,150.00	2,135,150.00
FY2020	610,000.00	1,521,550.00	2,131,550.00
FY2021	635,000.00	1,497,150.00	2,132,150.00
FY2022	660,000.00	1,471,750.00	2,131,750.00
FY2023	695,000.00	1,438,750.00	2,133,750.00
FY2024	730,000.00	1,404,000.00	2,134,000.00
FY2025	765,000.00	1,367,500.00	2,132,500.00
FY2026	805,000.00	1,329,250.00	2,134,250.00
FY2027	845,000.00	1,289,000.00	2,134,000.00
FY2028	885,000.00	1,246,750.00	2,131,750.00
FY2029	930,000.00	1,202,500.00	2,132,500.00
FY2030	980,000.00	1,156,000.00	2,136,000.00
FY2031	1,025,000.00	1,107,000.00	2,132,000.00
FY2032	1,080,000.00	1,055,750.00	2,135,750.00
FY2033	1,130,000.00	1,001,750.00	2,131,750.00
FY2034	1,190,000.00	945,250.00	2,135,250.00
FY2035	1,245,000.00	885,750.00	2,130,750.00
FY2036	1,310,000.00	823,500.00	2,133,500.00
FY2037	1,375,000.00	758,000.00	2,133,000.00
FY2038	1,445,000.00	689,250.00	2,134,250.00
FY2039	1,515,000.00	617,000.00	2,132,000.00
FY2040	1,590,000.00	541,250.00	2,131,250.00
FY2041	1,670,000.00	461,750.00	2,131,750.00
FY2042	1,755,000.00	378,250.00	2,133,250.00

Year	Principal Payment	Interest Payments	Subtotal Annual Debt Service
FY2043	1,845,000.00	290,500.00	2,135,500.00
FY2044	1,935,000.00	198,250.00	2,133,250.00
FY2045	2,030,000.00	101,500.00	2,131,500.00
Total	**32,390,000.00**	**30,399,156.53**	**62,789,146.53**

The projections of other future non-operating expenses were developed based on a review of existing debt obligations, discussions with WPC regarding anticipated capital outlay requirements, and historical cost trends. These expense projections are shown at the end of this section in Table 8-6. The key assumptions associated with these projections include:

- Repayment of existing OPWC and OWDA debt obligations per the existing established debt service schedules (see Section 8.2.1 for debt service schedule for existing debt).
- Issuance and repayment of the Series 2016 Bonds pursuant to the anticipated debt service presented in this Report.
- Cash funding of capital improvements pursuant to the capital improvement funding plan discussed in the subsequent section.
- Issuance of additional revenue bonds as planned by WPC to fund the long-term capital improvement plan.

In addition to the above, the forecast includes maintenance of a $1 million dollar contingency fund, six (6) months of cash reserves, and a debt service reserve fund for all revenue bond debt. The debt service reserve for the Series 2016 Bonds was assumed to be capitalized as part of the issuance of the Series 2016 Bonds.

For the purposes of projecting estimated new debt service anticipated to be required beginning in FY2019, an interest rate of five percent (5%) and an amortization period of thirty (30) years was assumed. In lieu of additional revenue bond issues, however, WPC intends to pursue grants and low interest loans for projects as they become available. This will include grants and loans anticipated to be available from NEORSD for stormwater projects in the near future.

8.3.5 Debt Service Coverage Requirements

The Trust Agreement and General Bond Ordinance contain specific debt service coverage requirements that must be met by WPC. Specifically, the rate covenant requires that the City set its rates so as to cause its Net Revenues to meet the greater of a) one hundred twenty percent (120%) of the annual debt service on WPC's outstanding Bonds or b) one hundred percent (100%) of the sum of the outstanding debt including any required deposits to reserves. Projected debt service coverage results are presented in Table 8-6, and are anticipated to be sufficient to satisfy the requirements of Section 20 (c) and (d).

8.3.6 WPC Management Policies

WPC anticipates funding the majority of its capital plan with revenue bonds and low cost loans or grants as they become available, and using cash for equipment and technology investments and repairs as needed.

WPC has maintained operating reserves to address potential liquidity concerns and improve the ability of the utility to absorb potential short-term operating deficits in months when billed consumption may decline and where fixed costs are high. The City has also covenanted in the Trust Agreement to maintain a Contingency Fund to further improve its ability to respond to emergencies. For the purpose of projections included in Table 8-6, a minimum operating reserve target equal to six (6) months (180 days) of annual O&M expenses was assumed. These reserves are in addition to the Contingency Fund Requirement, which has been established in the General Bond Ordinance at a minimum of $1,000,000.

In addition to the rate covenant requirements described in this Report, WPC has established an internal management target for debt service coverage of 150 percent.

8.4 Cash Flow Projection

Presented in Table 8-6, are financial projections for WPC for the period of FY2016 through FY2020. Notes for each of the line items in Table 8-6 are presented in Appendix C. Based on the projections, it is anticipated that WPC will achieve its debt service coverage requirements in FY2016 through FY2019 with the sewer user rate increases that have been adopted. It is also anticipated that the City will need to approve an additional sewer rate increase in FY2020 to fund its proposed capital improvement plan and meet its debt service coverage requirements for FY2020. On Table 8-6 a rate increase of 4.5 percent is assumed for FY2020 and based on the revenue and expense projections results in a debt service coverage ratio of 164 percent. This targeted rate increase has not yet been formally proposed to, and will require approval by, City Council. A comprehensive rate study will be undertaken in 2019 to establish the recommended rates for FY2020 through FY2025.

These projections reflect WPC's views and assumptions with respect to future events as of the date of this Report, and are subject to future economic conditions and other risks and uncertainties. Arcadis believes that these projections are reasonable based upon the information provided to it; however, actual and future results and trends could differ materially from those presented herein. Accordingly, Arcadis makes no warranty or representation that the projected values or results contained in this document will actually be achieved. In the event that the anticipated costs and funding levels differ from the projections presented herein, WPC will adjust the rates accordingly.

Table 8-6. Cash Flow Projection (FY2016 through FY2020)

Line No.	Description	Fiscal Year Ending December 31st FY2016	FY2017	FY2018	FY2019	FY2020
	Customer Information					
1	Number of Customer Meter Accounts	125,811	125,811	125,811	125,811	125,811
2	Regular Consumption User Charge ($/MCF)	12.53	12.53	12.53	12.53	13.09
3	**Annual Rate Increase**	0.0%	1.6%	0.0%	2.0%	4.5%
	Revenues					
4	Consumption User Charge Revenue	$ 20,374,361	$ 19,926,125	$ 19,487,750	$ 19,059,020	$ 19,478,509
5	Fixed Charge Revenue	6,013,969	7,476,678	7,476,678	9,039,355	9,446,126
6	Miscellaneous Revenue	865,000	865,000	865,000	865,000	865,000
7	**Total Revenues**	$ 27,253,330	$ 28,267,803	$ 27,829,428	$ 28,963,375	$ 29,789,635
	Operating Expenses					
8	Personnel Expenses	$ 11,146,302	$ 11,476,837	$ 11,819,363	$ 12,174,390	$ 12,542,448
9	Other Training and Professional Dues	25,500	26,265	27,053	27,865	28,700
10	Utilities	389,513	401,198	413,234	425,631	438,400
11	Contractual Services	1,679,910	1,730,307	1,782,217	1,835,683	1,890,754
12	Materials and Supplies	447,000	460,410	474,222	488,449	503,102
13	Maintenance	1,904,000	1,961,120	2,019,954	2,080,552	2,142,969
14	Claims	983,780	1,013,293	1,043,692	1,075,003	1,107,253
15	Interdepartmental Charges	5,125,544	5,279,310	5,437,690	5,600,820	5,768,845
16	**Total Operating Expenses**	$ 21,701,549	$ 22,348,741	$ 23,017,425	$ 23,708,393	$ 24,422,472
17	**Net Revenues Available for Debt Service**	$ 5,551,781	$ 5,919,061	$ 4,812,003	$ 5,254,982	$ 5,367,163
	Debt Service					
18	Existing Debt Service	491,990	179,195	24,060	24,060	24,060
19	Series 2016 Revenue Bond	928,547	2,134,250	2,132,250	2,135,150	2,131,550
20	Future Revenue Bond	-	-	-	567,936	1,139,332
21	**Total Debt Service**	$ 1,420,536	$ 2,313,445	$ 2,156,310	$ 2,727,145	$ 3,294,942
	Debt Service Coverage					
22	DSC (Total Revenue Bonds,120% Min Target)	598%	277%	226%	194%	164%
23	DSC (All-Debt)	391%	256%	223%	193%	163%
	Capital Project Funding					
24	Cash from Operating Fund	2,940,080	2,071,600	2,215,800	2,382,500	3,250,000
25	Series 2016 Revenue Bond	16,250,000	9,400,000	9,350,000	-	-
26	New Debt	-	-	-	8,042,000	8,091,000
27	**Total Capital Project Funding**	$ 19,190,080	$ 11,471,600	$ 11,565,800	$ 10,424,500	$ 11,341,000
28	**Minor Capital Expense**	$ 200,000	$ 206,000	$ 212,180	$ 218,545	$ 225,102
29	**Total Revenue Requirements**	$ 23,322,085	$ 24,868,186	$ 25,385,914	$ 26,654,084	$ 27,942,515
30	**Revenues Over (Under) Expenses**	$ 3,931,245	$ 3,399,617	$ 2,443,514	$ 2,309,291	$ 1,847,120
	WPC Cash Fund Balance					
31	**Beginning Balance**	$ 16,600,000	$ 17,591,165	$ 18,919,182	$ 19,146,896	$ 19,073,687
32	**Revenues Over (Under) Expenses**	3,931,245	3,399,617	2,443,514	2,309,291	1,847,120
33	Transfers to Capital Projects Fund	(2,940,080)	(2,071,600)	(2,215,800)	(2,382,500)	(3,250,000)
34	**Ending Balance**	$ 17,591,165	$ 18,919,182	$ 19,146,896	$ 19,073,687	$ 17,670,807
35	Minimum Operating Reserve (6 Months)	$ 10,850,775	$ 11,174,371	$ 11,508,712	$ 11,854,196	$ 12,211,236

Notes:

1) The revenue and coverage projections for FY2020 reflect a targeted rate increase that has not yet been proposed to or approved by City Council.

9 CONCLUSION

9.1 Considerations and Assumptions

The principal considerations and assumptions are provided throughout this Report, some of which are as follows:

- Arcadis has made no determination as to the validity and enforceability of any contracts, agreement, existing law, rule, or regulation applicable to WPC and its operations; however, for purposes of this Report, we have assumed that all such contracts, agreements, laws, rules and regulations will be fully enforceable and complied with in accordance with their terms.
- WPC will continue its policies of employing qualified and competent personnel; will properly operate and maintain the System in accordance with generally accepted engineering practices and all recommended operations and maintenance protocols; and operate the System in a prudent and sound manner.
- NEORSD will not modify WPC's operating permit to include peak flow restrictions that would result in the need for WPC to implement unplanned capital improvements, or to establish BMP frequency requirements (versus goals) for items such as cleaning of sewers or catch basins that are substantially in excess of current WPC BMP frequencies.
- The proposed CIP reflects the anticipated requirements of the System, and the CIP is expected to be largely implemented by WPC as planned.
- If any unforeseen expenses occur, which may include items such as O&M expenses and capital expenditures to address a change in law, increased priority or uninsured catastrophic event, WPC would undertake internally driven actions such as reducing non-essential programs and implementing cost reduction measures to cover costs associated with these expenses, or adjust wastewater user charges as needed to satisfy the rate covenants, as may be needed.
- WPC will update its capital plan at least annually based on the findings and recommendations of the SSES studies and PMT requirements.
- WPC will continue to control costs and implement rate increases as required to meet its obligations.

In completing this Report for WPC, Arcadis is not serving in the role of a "municipal advisor" under the regulations of the Securities and Exchange Commission. As such, Arcadis is not: (a) recommending any action regarding municipal financial products or the issuance of municipal securities; and (b) is not acting as a registered municipal advisor to WPC, and does not owe a fiduciary duty to WPC pursuant to Section 15B of the Securities Exchange Act of 1934, as amended by the Dodd-Frank Wall Street Reform and Consumer Protection Act, with respect to the information and material prepared in connection with this Report. WPC should discuss any information and material prepared in connection with this Report with any and all internal and external financial and other advisors that it may deem appropriate before acting on this information and material.

Accordingly, since Arcadis is not a municipal advisor registered with the Securities and Exchange Commission, WPC acknowledges that: (a) it has retained the services of an independent registered municipal advisor, who, during the past two years, was not associated with Arcadis, and that (b) Arcadis has complied with the requirements set forth in the Securities Exchange Act, Municipal Advisor Rule (17 CFR 200, 240, 249), which requires that the engineering company: (i) receive from the municipal entity a representation in writing that it is represented by, and will rely on the advice of, an independent registered municipal advisor; (ii) provide written disclosure to the municipal entity that Arcadis is not serving as a municipal advisor and, with respect to the municipal entity, is not subject to the statutory fiduciary duty applicable to municipal advisors under the Securities Exchange Act, and (iii) provide a copy of such disclosure to the municipal entity's independent registered municipal advisor. Arcadis provided such disclosures to WPC and its registered municipal advisor. Arcadis does not provide opinions on or advocate for using a financial product (issuing debt) or the choice of financial products employed. As such, Arcadis submitted its work products to WPC's independent registered municipal advisor for review and approval.

9.2 Conclusions

Set forth below are the principal opinions that Arcadis has reached regarding our review of the System and WPC-prepared budget forecasts. For a complete understanding of these opinions, the Report should be read in its entirety.

1. The System has been maintained and operated in accordance with generally accepted utility standards, and notwithstanding its age, appears overall to be in good general repair and operating condition. WPC has taken a number of steps and is working diligently to improve the condition of the collection system through targeted rehabilitation and replacement of aging sewers.
2. The estimated costs of the CIP developed by WPC have been prepared using sound estimating practices and methods. The identified capital projects are in general alignment with our observed requirements for the System.
3. The financial forecast is based upon revenue and expense projections that appear to be reasonable.
4. WPC has historically funded its capital program with cash and therefore has no existing debt, other than approximately $753,000 in principal associated with some low and zero rate interest loans from OWDA and OPWC.
5. With the rate increases that have been adopted for FY2016 through FY2019, and the projected rate increase needed in FY2020, the one hundred eighty (180) day cash reserve target and the debt service coverage ratios are expected to be achieved over the Forecast Period that meet or exceed those required by the General Bond Ordinance, i.e., the One Hundred Twenty Percent (120%) Debt Service Coverage Ratio Requirement.
6. Based on the considerations, assumptions, and conclusions described above, the financial forecast presented in this Report is reasonable, and the City's plans to meet the obligations of the Series 2016 Water Pollution Control Bonds are financially feasible.

In preparation of this Report and opinions contained herein, Arcadis has relied upon financial, engineering and operational data and assumptions prepared by and/or provided to it by WPC. In addition, other entities working on behalf of WPC have provided Arcadis with information and projections, which Arcadis used in assembling this Report. We believe such sources are reliable, and the information obtained to be appropriate for the review undertaken and the conclusions reached in this Report. To the best of our knowledge, information and belief, the information does not omit material facts necessary to make the statements herein not misleading. Notwithstanding the above, Arcadis has not independently verified the accuracy, scope or breadth of the information provided by the City, WPC and any others. In addition, the scope of the Arcadis review did not include any pending or threatened litigation against WPC. To the extent that the information provided to Arcadis is not accurate, the findings and conclusions contained in this Report, and in particular the results shown on Table 8-6, may vary and are subject to change.

Arcadis devoted effort in making such opinions consistent with (i) that degree of care and skill ordinarily exercised by members of the same profession currently practicing under same or similar circumstances and (ii) the time and budget available for its work in its efforts to endeavor to provide such opinions. The opinions are based on information provided by, and consultations with, WPC. No responsibility was assumed for inaccuracies in reporting by WPC or any third party data source used in preparing such opinions. Arcadis' opinions represent its professional judgment. Neither Arcadis, nor its parent corporation, or their respective subsidiaries and affiliates, makes any representation or warranty, expressed or implied, with respect to such opinions.

APPENDIX A

List of Acronyms and Abbreviations

APPENDIX A
LIST OF ACRONYMS AND ABBREVIATIONS

AFSCME	American Federation of State, County and Municipal Employees
AMR	Automated Meter Reading
Arcadis	Arcadis U.S., Inc.
AWWA	American Water Works Association
BMPs	Best Management Practices
CCTV	Closed Circuit Television
CFO	Chief Financial Officer
CFS	Cubic Feet Per Second
CIP	Capital Improvement Plan
City	City of Cleveland, Ohio
CWD	City of Cleveland Division of Water
CMMS	Computerized Maintenance and Management System
CMOM	Capacity, Management, Operations and Maintenance
EPA	Environmental Protection Agency
FY	Fiscal Year
GIS	Geographic Information System
I/I	Inflow and Infiltration
IT	Information Technology Management Services
LTCP	Long Term Control Plan
M	Million
MCF	Thousand Cubic Feet
MGD	Million gallons per day
MOU	Memorandums of Understanding
MS4	Small Municipal Separate Storm Sewer System
NEORSD	Northeast Ohio Regional Sewer District
NPDES	National Pollution Discharge Elimination System
OEPA	Ohio Environmental Protection Agency

O&M	Operations and Maintenance
OPWC	Ohio Public Works Commission
ORC	Ohio Revised Code
OWDA	Ohio Water Development Authority
PIPE	Public Involvement and Public Education
PMT	Program Management Team
RSMP	Regional Stormwater Management Plan
SCADA	Supervisory Control and Data Acquisition
SEIU	Service Employees International Union
SEME	Federal Credit Union
Series 2016 Bonds	Series 2016 Water Pollution Control Revenue Bonds
SOP	Standard Operating Procedures
SSES	Sewer System Evaluation Survey
SSO	Sanitary Sewer Overflow
USACOE	United States Army Corps of Engineers
USEPA	United States Environmental Protection Agency
VCP	Vitrified Clay Pipe
WPC	Division of Water Pollution Control
WWTP	Wastewater Treatment Plant or Facility

APPENDIX B

WPC Capital Improvement Program FY2016-FY2020

APPENDIX B
WPC CAPITAL IMPROVEMENT PROGRAM FY2016-FY2020

		2016	2017	2018	2019	2020
	Projects Related to Stormwater					
1	Euclid Creek	2,500,000				
2	Catch Basin/Manholes	450,000	450,000	450,000	450,000	450,000
	Sewer Replacements					
3	Sewer Replacement	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000
4	Sewer Lining	800,000	800,000	800,000	800,000	800,000
5	Sewer Connections	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000
6	Rocky River	1,500,000				
7	East 75	2,000,000				
8	East 85	2,700,000				
9	Bryant Ave	800,000				
10	E 150	600,000				
11	W 48	700,000				
12	W 134	450,000				
13	Dartmouth/W 165		1,000,000			
14	E 186		600,000			
15	E 89		350,000			
16	Rockport Ave		900,000			
17	W 83		300,000			
18	APL Property			300,000		
19	Canterbury Avenue			450,000		
20	Cornell Avenue			500,000		
21	Dalton Avenue			1,000,000		
22	Wakefield Avenue			500,000		
	Sewer Work on Roadway Projects					
23	Roadway drainage	500,000	500,000	500,000	500,000	500,000
	Building and Miscellaneous Improvements					
24	Equipment	1,900,000	1,000,000	1,200,000	1,200,000	1,200,000
25	PMT	1,700,000	1,400,000	1,300,000	1,200,000	1,200,000
26	Professional Services	500,000	500,000	500,000	500,000	500,000
27	***Technology***	90,080	121,600	65,800	32,500	100,000
28	***Undesignated***	-	1,550,000	2,000,000	3,742,000	4,591,000
	Total	19,190,080	11,471,600	11,565,800	10,424,500	11,341,000
	Cash Funded	2,940,080	2,071,600	2,215,800	2,382,500	3,250,000
	Debt Funded	16,250,000	9,400,000	9,350,000	8,042,000	8,091,000
		19,190,080	**11,471,600**	**11,565,800**	**10,424,500**	**11,341,000**

Project Descriptions

1. Euclid Creek Restoration Project (2016)

Description: The Euclid Creek project is located on the east side of the City of Cleveland in Cuyahoga County. The area of concern for this project is Euclid Creek between Lake Erie and Villaview Road.

The original flood control project was constructed by the Works Projects Administration (WPA) in the late 1930's. In 1984, the portion of the WPA project from the Lake Shore Boulevard bridge to approximately 2,000 feet upstream, also known as the Euclid Creek Flood Control project, was redeveloped by the Buffalo District Corps of Engineers (COE). The initial design of the Euclid Creek Flood Control project was to provide a non-damaging channel with a capacity of 10,400 cubic feet per second (cfs) throughout the project limits. The upstream portion of the COE project ties into the original WPA project. Flood waters due to heavy rains that occurred on May 25 and May 26, 1989 severely damaged the original WPA project. This initial damage was further aggravated by additional rainfall during the period of May 31 through June 9, 1989. The Emergency Streambank Rehabilitation project was designed and constructed under the supervision of the Buffalo District COE and was completed on May 29, 1992. The City of Cleveland was the local sponsor and superintendent. The City inspected the completed work and acknowledged satisfaction with the project. The City accepted the project along with the required maintenance of the creek and its structures from the U. S. Federal Government on January 29, 1993. Based on annual joint inspections between the U.S. Army Corps of Engineers and the City of Cleveland, it has been determined that the current condition of the local flood protection project on Euclid Creek has major deficiencies that will adversely impact the flood control project. In addition, the original WPA project has deficiencies also that need to be addressed (See photos, Attachment C). It is increasingly possible that the Euclid Creek Flood Control project will not withstand a major flood event and provide the necessary level of protection for which the project was designed. This study resulted in a comprehensive understanding of a multitude of problems, some of which are:

- Woody growth and shoaling have reduced the channel capacity therefore drastically reducing the level of protection.
- Declining integrity of grouted fabric slope protection from large sediment bars with vegetation growth.
- Eventual failure of the levee due to encroachment of trees and bushes.
- Missing rip-rap and heavy vegetation and wood growth within the rip-rap.
- Holes and undermining in the concrete slopes downstream of Villaview Road.

Project components include removal of shoaling deposits within thickness tolerances (between 12" and 30" above concrete liner) and in stream vegetation; stream bank stabilization; project limits from beginning of concrete lined channel to approx. 1,600 feet downstream of Lake Shore Blvd bridge(total length approx. 3,600 lf).

2. Constructing and Rebuilding Catch Basins and Manholes

This project consists of Constructing and Rebuilding Catch Basins and Manholes at Various Locations throughout the City of Cleveland.

3. Constructing and Installing Replacement Sewers and Rehabilitating Existing Sewers

This project consists of constructing and installing replacement sewers and rehabilitating existing sewers throughout the City of Cleveland. Work is scheduled as needed, and mostly performed in emergency situations.

4. Rehabilitating and Lining Sewers

This project consists of rehabilitating and lining sewers at various locations in the City of Cleveland.

5. Rehabilitating and Replacing Sewer Connections

The work consists of rehabilitating and replacing sewer service connections at various locations throughout the City of Cleveland.

6. Rocky River (2016)

Location: Rocky River Drive between Puritas Avenue and Melgrave Avenue

Description: The project consists of installing approximately 1,200 feet of sanitary and 1,200 feet of storm sewers on Rocky River Drive, between Puritas Avenue and Laverne Avenue; and approximately 250 feet of sanitary and 250 feet of storm sewers on Rocky River Drive, between Sedalia Avenue and Melgrave Avenue. The existing sewers were built in 1926. This causes property flooding on Rocky River Drive.

7. East 75 Street Sewer Pump Station Project (2016)

Location: East 75 Street between Dell Avenue and Grand Avenue

Description: The existing combined sewer on East 75 Street was built in 1903. When the Norfolk Southern railroad tracks and East 75 Street Bridge were constructed, the combined sewage on East 75 Street was rerouted and connected directly to the existing Kingsbury storm culvert located under the railroad tracks. This is considered an illicit discharge. The project will be undertaken to correct the problem. This project consists of installing approximately 300 LF of combined sewer on East 75 Street, under existing bridge, and approximately 120 LF of sewer on East 75 Street between Woodland Avenue and Grand Avenue in the City of Cleveland. The construction methods for this project include open cut excavation and boring/jacking.

8. East 85 Street Sewer Project (2016)

Location: Area bordered by East 85 Street, Crawford Road, Hough Avenue, East 105 Street and Chester Avenue

Description: The project consists of relocating the exiting Spring Brook Culvert.

The project is located in Ward 5 in the area bordered by East 85 Street, Crawford Road, Hough Avenue, East 105 Street and Chester Avenue, which is a residential area. The existing culverted storm sewer (Spring Brook Culvert) runs under properties in this area and discharges into Doan Brook. The sewer was built in the early 1900's. The sewer is structurally deteriorated.

9. Bryant Avenue Sewer Project (2016)

Location: Bryant Avenue between East 105 Street and Parkwood Drive

Description: The project consists of replacing approximately 3,000 feet of 8” and 10” sanitary sewer and 18” storm sewer on Bryant Avenue between East 105 Street and Parkwood Drive. The existing sewers were built in 1908. The sewers are structurally deteriorated. This causes basement and street flooding.

10. East 150 Street Sewer Project (2016)

Location: East 150 Street between St Clair Avenue and Aspinwall Avenue

Description: The project consists of replacing approximately 1,640 feet of 9” sanitary with a 15 inch sanitary sewer on East 150 Street between St Clair Avenue and Aspinwall Avenue. The existing sewers were built in 1906. The sewers are structurally deteriorated, which causes basement and street flooding.

11. West 48 Street Sewer Project (2016)

Location: Between Park Dr. and Memphis Ave

Description: The project consists of installing approximately 1,700 feet of storm relief sewer and replacing approximately 300 feet of deteriorated sewers on West 48 Street between Park Drive and Memphis Avenue. The existing combined sewer was built in 1916. The sewer is undersized and structurally deteriorated, which causes basement flooding on West 48 Street.

12. West 134 Street Sewer Project (2016)

Location: Between San Diego Avenue and West Avenue

Description: The project consists of installing approximately 900 feet of storm relief sewer on West 134 Street, between San Diego Avenue and West Avenue. The existing combined sewer was built in 1926. The sewer is undersized, which causes basement flooding on West 134 Street.

13. Dartmouth Avenue and West 165 Street Sewer Project (2017)

Location: Dartmouth between Rocky River Drive and West 165 Street; West 165 Street between Lucille Avenue and Ferndale Avenue

Description: The project consists of replacing approximately 3,900 feet of 12” sanitary and 12”-20” storm main sewers on Dartmouth between Rocky River Drive and West 165 Street and West 165 Street between Lucille Avenue and Ferndale Avenue. The existing sewers were built in 1924. The sewers are structurally deteriorated, which causes basement and street flooding.

14. East 186 Street Sewer Project (2017)

Location: Between Invermere Avenue and Harvard Avenue

Description: The project consists of replacing approximately 2,000 feet of 8” sanitary and 12”, 15” 21”, 24” and 27” storm main sewers on East 186 Street between Invermere Avenue and Harvard Road. The existing sewers were built in 1950. The sewers are structurally deteriorated, which causes basement and street flooding.

15. East 89 Street Sewer Project (2017)

Location: Between Woodland Avenue and Buckeye Avenue

Description: The project consists of replacing approximately 1,000 feet of No. 2 brick combined main sewer on East 89 Street between Woodland Avenue and Buckeye Avenue. The existing sewer is built in 1892. The sewers are structurally deteriorated, which causes basement and street flooding.

16. Rockport Avenue Sewer Project (2017)

Location: Between Lydian Avenue and Triskett Road

Description: The project consists of replacing approximately 2,700 feet of 12" sanitary and 20"-24" storm main sewers on Rockport between Lydian Avenue and Triskett Road. The existing sewers were built in 1926. The sewers are structurally deteriorated, which causes basement and street flooding.

17. West 83 Street Sewer Project (2017)

Location: Between Detroit Avenue and Lake Avenue

Description: The project consists of replacing approximately 800 feet of No. 2 brick combined main sewer on West 83 Street between Detroit Avenue and Lake Avenue. The existing sewer is built in 1906. The sewers are structurally deteriorated, which causes basement and street flooding.

18. APL Property (2018)

Location: 1729 Willey Avenue; within Animal Protective League (APL) parking lot

Description: The project consists of replacing approximately 200 feet of 36" brick combined main sewer located within the APL parking lot and adjacent to railroad tracks. The existing sewer was built in 1932. The sewer is structurally deteriorated.

19. Canterbury Road Sewer Project (2018)

Location: between East 185 Street and Lakeshore Boulevard

Description: The project consists of replacing approximately 1,350 feet of 8" sanitary and 12" storm sewers on Canterbury Road between East 185 Street and Lakeshore Boulevard. The existing sewer was built in 1911. The sewers are structurally deteriorated. This causes basement and street flooding.

20. Cornell Road Sewer Project (2018)

Location: Between Euclid Ave and Random Road

Description: The project consists of replacing/lining approximately 1,600 feet of 18", No. 2 and No. 6 brick sewers on Cornell Road between Euclid Ave and Random Road. The existing sewers were built in 1887. The sewer is structurally deteriorated, which causes basement and street flooding.

21. Dalton Avenue Sewer Improvements Project (2018)

Location: Between East 49 Street and East 52 Place

Description: The project consists of replacing approximately 700 feet of 12" and 15" combined main sewer and installing a sewer pump station on Dalton Avenue between East 49 Street and East 52 Place. The existing sewer was built in 1903. The sewer is structurally deteriorated.

22. Wakefield Avenue Sewer Project (2018)

Location: Between West 65 Street and West 74 Street

Description: The project consists of replacing approximately 1,300 feet of No. 3 and No. 2 brick sewer on Wakefield Avenue between West 65 Street and West 74 Street. The existing sewers were built in 1898. The sewers are structurally deteriorated, which causes basement and street flooding.

23. Roadway Drainage (2016 – 2020)

WPC share of Department of Transportation (DOT) project costs for drainage construction.

24. Equipment (2016 – 2020)

For FY2016 the equipment includes the acquisition of five new vacuum trucks. In future years, equipment expenditures required to continue to replace vehicles and equipment that is past its useful service life.

25. PMT (2016 – 2020)

In January 2016, WPC issued a request for proposals for Capital Improvements program Management Services. The program goals and objectives included establishment of specific levels of service for sewer reliability, functionality and sustainable operations. This also included program, management services such as quality management, Management Information System, risk management, asset management coordination, CIP program procedures and controls including design standards, budget assistance and operation management. The duration of this program is expected to last five (5) years starting in mid-2016 with an estimated cost of $6.8 million dollars.

26. Professional Services (2016 – 2020)

The Division issues RFPs every two (2) years for professional consultants to provide general engineering services to WPC. These services include wastewater system design and planning, sewer system design, regulatory support, bidding and construction services, surveying, field representation and operations support. At times sewer system investigations would include sewer line cleaning and jetting and short circuit television inspections. Services related to stormwater systems are typically included such as NPDES permitting and MS4 compliance assistance, BMP evaluations, eco-restoration, stream modeling and public education services.

27. Technology (2016 – 2020)

WPC direct cost share from IT for WPC technology investment projects.

28. Undesignated (2017 – 2020)

Includes projects to address recommendations of the WPC's five year SSES program and condition assessments.

APPENDIX C

Significant Assumptions

APPENDIX C

SIGNIFICANT ASSUMPTIONS

Summary of Significant Forecast Assumptions and Notes to WPC Financial Forecast

The accompanying Forecast, consisting of Table 8-6, "Cash Flow Projection (FY2016 through FY2020)", presents WPC's calculations of the Debt Service Coverage Ratio Requirement and the financial results of operations on a cash basis for FY2016 and the subsequent four fiscal years ending December 31, 2020 (collectively referred to as the Forecast Period). It should be noted that projected revenues and expenses for FY2016 were based on amounts included in WPC's Proposed Operating Budget for FY2016, dated January 8, 2016.

This summary of significant assumptions provides information regarding the basis of the Forecast and support for the underlying assumptions. Arcadis has reviewed these underlying assumptions for purposes of providing an opinion and making certain conclusions regarding the reasonableness of the Forecast, as shown in Section 8 of this Report. The Forecast has been prepared based on assumptions concerning future events and circumstances and WPC's most likely courses of action and best estimate of the financial results of operations during the Forecast Period.

Significant Assumptions WPC:

Customer Information:

1. **Number of Customer Meter Accounts**

   Number of metered customer accounts used by WPC to calculate fixed fee revenues. Meters vary from ½-inch to 12-inch diameter and includes homestead customers. For the purposes of these projections, the net number of customer metered accounts was projected to remain flat over the Forecast Period.

2. **Consumption User Charge**

   This is the regular volumetric charge of per MCF of billable metered water consumption. The volumetric user charges have been approved by City Council through FY2019. The FY2020 charge is a projected based on the proposed financial plan, and is subject to Board of Control and City Council approval. The WPC offers certain qualifying customers a discounted (60% off regular rate) "homestead" rate to qualifying customers. The homestead rate customers represent a very small number, approximately three percent (3%) of the annual revenues.

Billing Revenues:

3. **Consumption User Charge Revenues**

   Based on the FY2016 adopted budget and projected over the forecast period. These revenues were based on a 2.2 percent decline in billable metered water consumption and required rate revenue increases beginning in FY2020, subject to Board of Control recommendation and City Council authorization. Also assumes that billing collection rates will remain consistent with historic averages, which have averaged approximately ninety percent (90%) per year.

4. Fixed Charge Revenue

Beginning in FY2015, WPC charged customers a fixed fee based on meter size. WPC's authorized rates and projections were used for FY2016 through FY2019, which includes a rate increase in FY2017 and FY2019. A fixed charge rate increase is also projected for FY2020, subject to Board of Control recommendation and City Council authorization.

5. Rate Revenue from Increase

Required rate revenue increases from WPC's adopted schedule for FY2016 – FY2019 and as projected by the financial plan beginning in FY2020. FY2020 rate increases subject to Board of Control and City Council authorization.

6. Miscellaneous Revenue

Includes revenue from City Division & Cost Plus Jobs, Licenses & Permits, Interest Income, and any other offsets. These revenues were projected to be constant to match projections from WPC.

7. Total Revenues

Sum of Line Items 4, 5 and 6.

Operating Expenses

8. through 15.

Operating and maintenance expenses were grouped as personnel expenses, other training and professional services, utilities, contractual services, materials and supplies, maintenance, claims, interdepartmental charges, and minor capital expenses. These costs were based on WPC's Adopted Budget for FY2016. Future costs were projected using an escalation rate associated with each specific group. Escalation rates for the various expense groups was based on discussions with WPC regarding system operational changes, historical cost trends, and anticipated future wastewater flows and billed consumption trends.

16. Total Operating Expense

Sum of Line Items 8 through 15.

17. Net Revenues Available for Debt Service

Line Item 16 Total Operating Expenses, less Line Item 7 Total Revenues.

18. Existing Debt Service

Consists of the following debt service schedule: OPWC Loan - Kerruish Storm Water Management Facility, OPWC Loan - Hamlet Ave. & Adolpha Road Area Storm Sewer, OWDA Loan - Sewer Construction, OWDA Loan - Spring Road Sewers, OWDA Loan - Ivermere Avenue Vacinity Sewers, and OWDA Loan - Cooley Avenue Replacement Sewers.

19. Series 2016 Revenue Bonds

A debt service schedule was provided by the City's financial advisors and is based on final pricing dated March 31, 2016. Proceeds are used to fund $35 million of project disbursements, a fully funded debt service reserve fund, and costs of issuance.

20. New Debt Service for Capital Improvements

New Debt Service represents anticipated debt service payments required to fund planned capital project costs not funded by cash or remaining proceeds from the 2016 Revenue Bonds. New revenue bonds were assumed to be issued over a term of thirty (30) years at an interest rate of five percent (5%) per year. Proceeds are used to fund approximately $16.1 million of project disbursements, a fully funded debt service reserve fund and costs of issuance.

21. Total Debt Service

Sum of Line Items 18, 19, and 20.

Debt Service Coverage:

22. DSC (Revenue Bond Debt, 120% minimum)

Calculated by dividing Net Revenues Available for Debt Service Coverage (Line Item 17) by the Total Revenue Bond Debt (Sum of Line Items 19 and 20).

23. DSC (All Debt)

Calculated by dividing the Net Revenues Available for Debt Service Coverage (Line Item 17) by the Total Debt Service (Line Item 21).

24 Minor Capital Expense

Includes minor capital expenses such as computer hardware which are excluded from Net Revenues for purposes of calculating Debt Service Coverage.

25 Net Revenues before Cash Funding of CIP

Total Revenues (Line Item 7) less the sum of the Total Operating Expenses (Line Item 16), Total Debt Service (Line Item 21) and Minor Capital Expense (Line Item 24).

WPC Cash Fund Balance Summary:

26. Beginning Balance

The beginning cash balance of the enterprise fund was based on the ending unrestricted cash balance within the fund based on WPC's FY2015 unaudited financial statements.

27. Net Revenues before Cash Funding of CIP

Equal to Line Item 25.

28 Transfers to Capital Projects Fund

Transfer to Capital Projects Fund in an amount pursuant to the propose Capital Plan to cash fund certain capital improvements.

29. Ending Balance

Sum of Items 26, 27 and 28.

30. Minimum Operating Reserve (6 Months)

The minimum operating reserve target is at least six (6) months Operations and Maintenance Expense in FY2016 and remains at this level over the course of the Forecast Period.

Capital Project Funding"

31. through 33.

Summary of the sources of funding for WPC's Capital Plan.

34 Total Capital Project Funding

Sum of Line Items 31, 32, and 33.

APPENDIX D

Maps

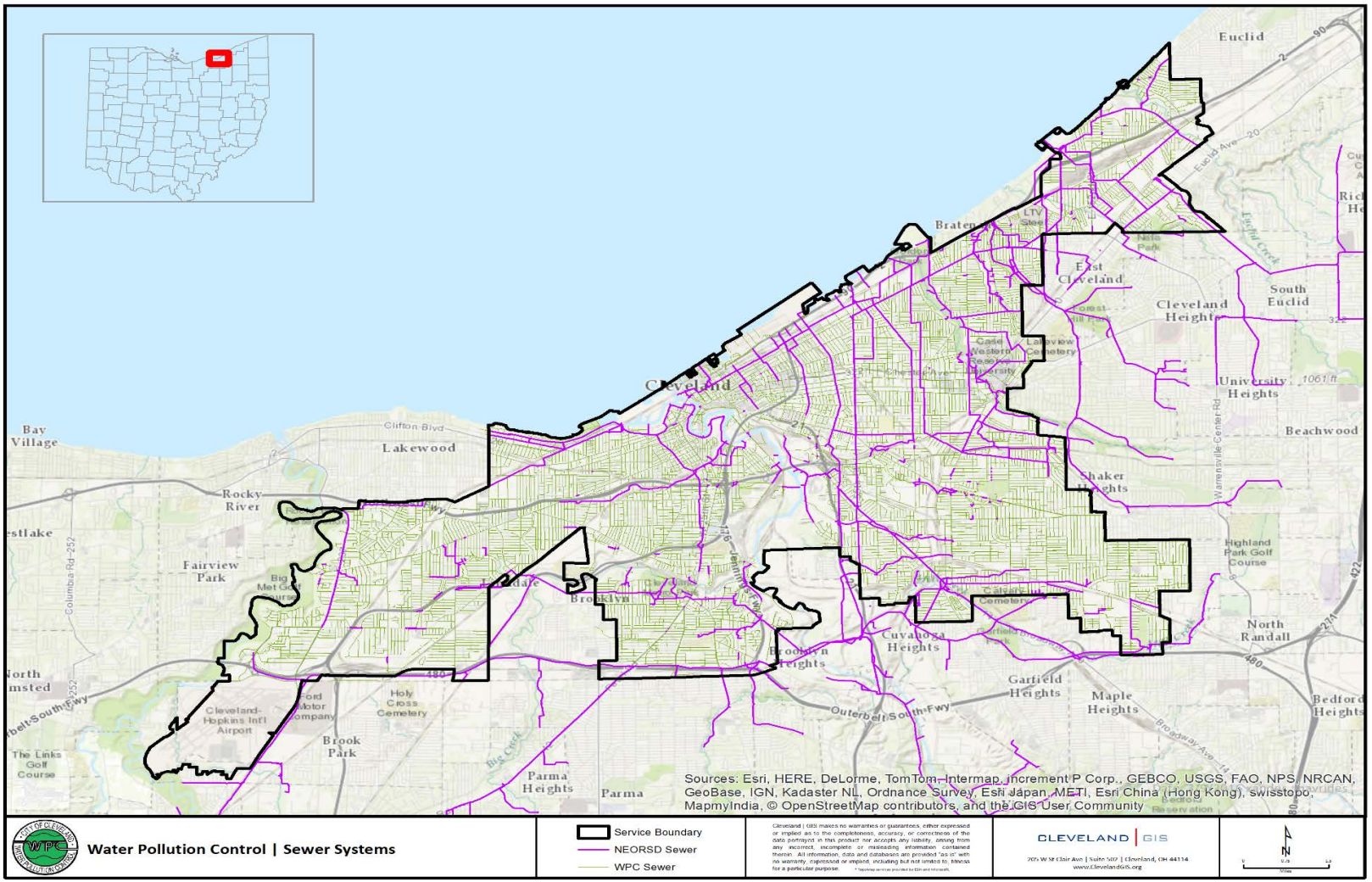

Figure D-1. Clarification of NEORSD and WPC Interceptor Responsibility

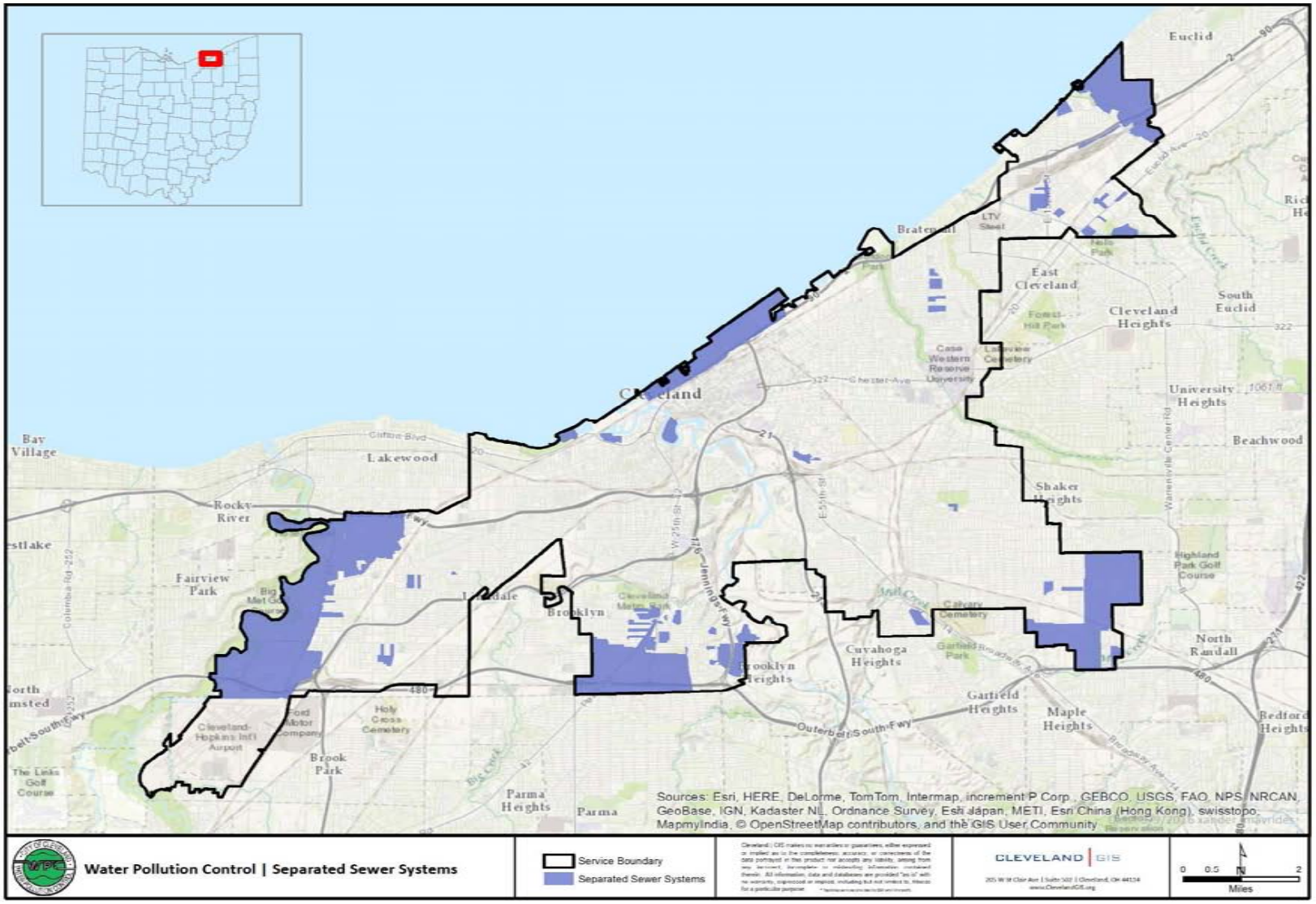

Figure D-2. Service Area Map by Type

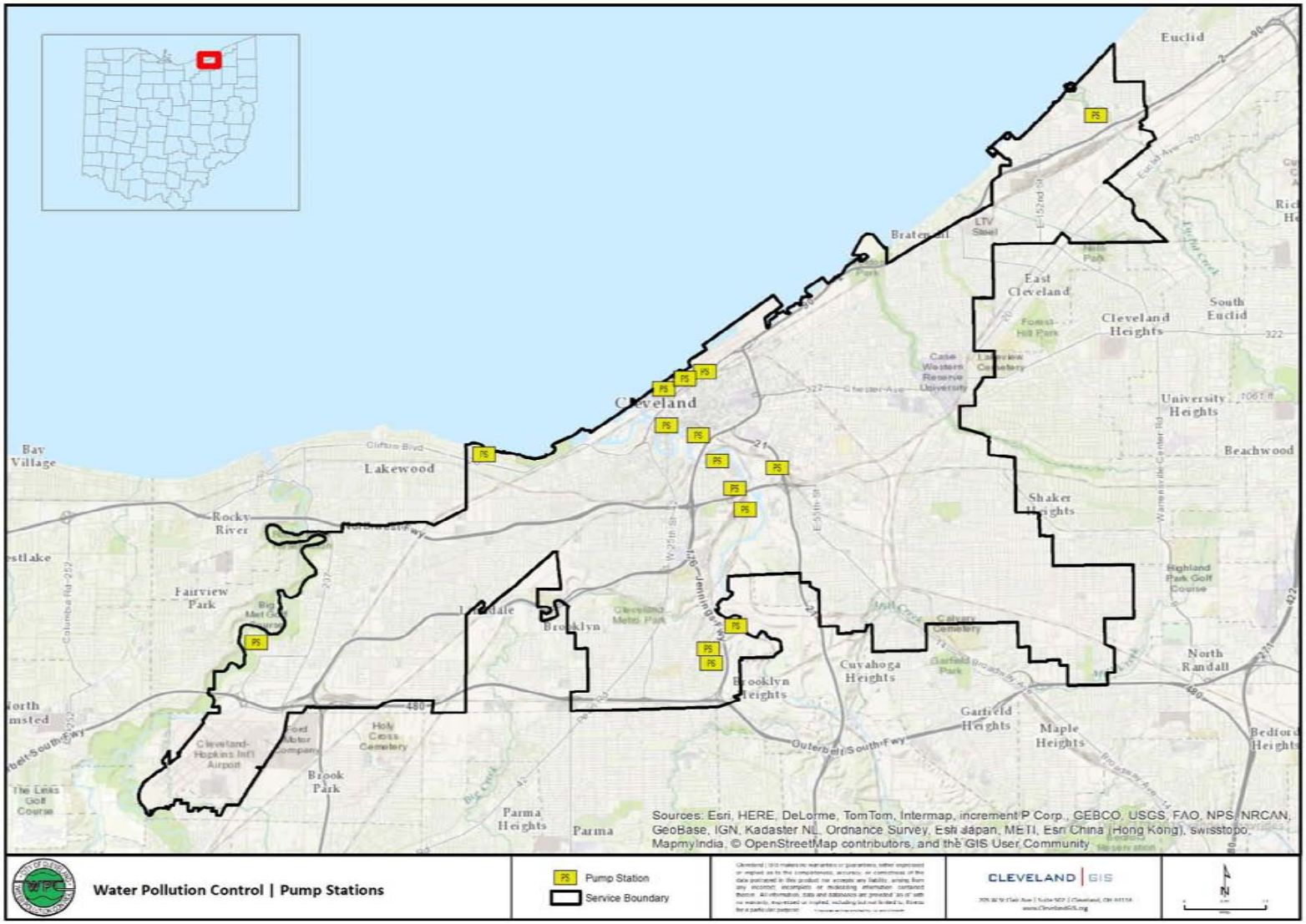

Figure D-3. Relative Location of Pump Stations within WPC System

Arcadis U.S., Inc.

1111 Superior Avenue
Suite 1300
Cleveland
Ohio 44114
Tel 216 781 6177

www.arcadis.com

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APPENDIX F

FORM OF UPDATES REGARDING GREEN PROJECTS AND GREEN BONDS

Amount of proceeds of the Series 2016 Bonds at the time of issuance: $32,390,000

Project Name	Project Description	Amount of Proceeds of Series 2016 Bonds That Have Been Spent as of December 31, ___ for Fiscal Year ___	Percentage of Proceeds of Series 2016 Bonds That Have Been Spent as of December 31, ___ for Fiscal Year ___

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APPENDIX G

BOOK-ENTRY ONLY SYSTEM

The Depository Trust Company ("DTC"), New York, NY, will act as securities depository for the securities (the "Series 2016 Bonds"). The Series 2016 Bonds will be issued as fully-registered securities registered in the name of Cede & Co. (DTC's partnership nominee) or such other name as may be requested by an authorized representative of DTC. One fully-registered Bond certificate will be issued for each stated maturity of the Series 2016 Bonds, each in the aggregate principal amount of such issue, and will be deposited with DTC. **So long as Cede & Co. is the registered owner of the Series 2016 Bonds, as partnership nominee for DTC, references herein to Bondholders, holders or owners of the Series 2016 Bonds (other than under the captions "Tax Matters" and "Continuing Disclosure Agreement" herein) shall mean Cede & Co. and shall not mean the Beneficial Owners of the Series 2016 Bonds.**

DTC, the world's largest securities depository, is a limited-purpose trust company organized under the New York Banking Law, a "banking organization" within the meaning of the New York Banking Law, a member of the Federal Reserve System, a "clearing corporation" within the meaning of the New York Uniform Commercial Code, and a "clearing agency" registered pursuant to the provisions of Section 17A of the Securities Exchange Act of 1934. DTC holds and provides asset servicing for over 3.5 million issues of U.S. and non-U.S. equity issues, corporate and municipal debt issues, and money market instruments (from over 100 countries) that DTC's participants ("Direct Participants") deposit with DTC. DTC also facilitates the post-trade settlement among Direct Participants of sales and other securities transactions in deposited securities, through electronic computerized book-entry transfers and pledges between Direct Participants' accounts. This eliminates the need for physical movement of securities certificates. Direct Participants include both U.S. and non-U.S. securities brokers and dealers, banks, trust companies, clearing corporations, and certain other organizations. DTC is a wholly-owned subsidiary of The Depository Trust & Clearing Corporation ("DTCC"). DTCC is the holding company for DTC, National Securities Clearing Corporation and Fixed Income Clearing Corporation, all of which are registered clearing agencies. DTCC is owned by the users of its regulated subsidiaries. Access to the DTC system is also available to others such as both U.S. and non-U.S. securities brokers and dealers, banks, trust companies, and clearing corporations that clear through or maintain a custodial relationship with a Direct Participant, either directly or indirectly ("Indirect Participants"). DTC has a Standard & Poor's rating of AA+. The DTC Rules applicable to its Participants are on file with the Securities and Exchange Commission. More information about DTC can be found at www.dtcc.com.

Purchases of Series 2016 Bonds under the DTC system must be made by or through Direct Participants, which will receive a credit for the Series 2016 Bonds on DTC's records. The ownership interest of each actual purchaser of each Security ("Beneficial Owner") is in turn to be recorded on the Direct and Indirect Participants' records. Beneficial Owners will not receive written confirmation from DTC of their purchase. Beneficial Owners are, however, expected to receive written confirmations providing details of the transaction, as well as periodic statements of their holdings, from the Direct or Indirect Participant through which the Beneficial Owner entered into the transaction. Transfers of ownership interests in the Series 2016 Bonds are to be accomplished by entries made on the books of Direct and Indirect Participants acting on behalf

of Beneficial Owners. Beneficial Owners will not receive certificates representing their ownership interests in Series 2016 Bonds, except in the event that use of the book-entry system for the Series 2016 Bonds is discontinued.

To facilitate subsequent transfers, all Series 2016 Bonds deposited by Direct Participants with DTC are registered in the name of DTC's partnership nominee, Cede & Co., or such other name as may be requested by an authorized representative of DTC. The deposit of Series 2016 Bonds with DTC and their registration in the name of Cede & Co. or such other DTC nominee do not effect any change in beneficial ownership. DTC has no knowledge of the actual Beneficial Owners of the Series 2016 Bonds. DTC's records reflect only the identity of the Direct Participants to whose accounts such Series 2016 Bonds are credited, which may or may not be the Beneficial Owners. The Direct and Indirect Participants will remain responsible for keeping account of their holdings on behalf of their customers.

Conveyance of notices and other communications by DTC to Direct Participants, by Direct Participants to Indirect Participants, and by Direct Participants and Indirect Participants to Beneficial Owners will be governed by arrangements among them, subject to any statutory or regulatory requirements as may be in effect from time to time. Beneficial Owners of the Series 2016 Bonds may wish to take certain steps to augment the transmission to them of notices of significant events with respect to the Series 2016 Bonds, such as redemptions, tenders, defaults, and proposed amendments to the offering documents. For example, Beneficial Owners of the Series 2016 Bonds may wish to ascertain that the nominee holding the Series 2016 Bonds for their benefit has agreed to obtain and transmit notices to Beneficial Owners. In the alternative, Beneficial Owners may wish to provide their names and addresses to the registrar and request that copies of notices be provided directly to them.

Redemption notices shall be sent to DTC. If less than all of a series of the Series 2016 Bonds within an issue are being redeemed, DTC's practice is to determine by lot the amount of the interest of each Direct Participant in such issue to be redeemed.

Neither DTC nor Cede & Co. (nor any other DTC nominee) will consent or vote with respect to Series 2016 Bonds unless authorized by a Direct Participant in accordance with DTC's MMI Procedures. Under its usual procedures, DTC mails an Omnibus Proxy to the City as soon as possible after the record date. The Omnibus Proxy assigns Cede & Co.'s consenting or voting rights to those Direct Participants to whose accounts Series 2016 Bonds are credited on the record date (identified in a listing attached to the Omnibus Proxy).

Redemption proceeds, distributions, and dividend payments on the Series 2016 Bonds will be made to Cede & Co., or such other nominee as may be requested by an authorized representative of DTC. DTC's practice is to credit Direct Participants' accounts upon DTC's receipt of funds and corresponding detail information from the City or the Trustee, on payable date in accordance with their respective holdings shown on DTC's records. Payments by Participants to Beneficial Owners will be governed by standing instructions and customary practices, as is the case with securities held for the accounts of customers in bearer form or registered in "street name," and will be the responsibility of such Participant and not of DTC, the Trustee, or the City, subject to any statutory or regulatory requirements as may be in effect from time to time. Payment of redemption proceeds, distributions, and dividend payments to Cede &

Co. (or such other nominee as may be requested by an authorized representative of DTC) is the responsibility of the Trustee, disbursement of such payments to Direct Participants will be the responsibility of DTC, and disbursement of such payments to the Beneficial Owners will be the responsibility of Direct and Indirect Participants.

A Beneficial Owner shall give notice to elect to have its Series 2016 Bonds purchased or tendered, through its Participant, to the Paying Agent, and shall effect delivery of such Series 2016 Bonds by causing the Direct Participant to transfer the Participant's interest in the Series 2016 Bonds, on DTC's records, to the Paying Agent. The requirement for physical delivery of Series 2016 Bonds in connection with an optional tender or a mandatory purchase will be deemed satisfied when the ownership rights in the Series 2016 Bonds are transferred by Direct Participants on DTC's records and followed by a book-entry credit of tendered Series 2016 Bonds to the Paying Agent's DTC account.

DTC may discontinue providing its services as depository with respect to the Series 2016 Bonds at any time by giving reasonable notice to the City or the Trustee. Under such circumstances, in the event that a successor depository is not obtained, Bond certificates are required to be printed and delivered.

The City may decide to discontinue use of the system of book entry only transfers through DTC (or a successor securities depository). In that event, Bond certificates will be printed and delivered to DTC.

The information in this section concerning DTC and DTC's book-entry system has been obtained from sources that the City believes to be reliable, but the City takes no responsibility for the accuracy thereof.

Direct Participants and Indirect Participants may impose service charges on book entry interest owners in certain cases. Purchasers of book entry interests should discuss that possibility with their brokers.

The City and the Trustee have no role in the purchases, transfers or sales of book entry interests. The rights of Beneficial Owners (i.e., book-entry interest owners) to transfer or pledge their interests, and the manner of transferring or pledging those interests, may be subject to applicable state law. Beneficial Owners may want to discuss with their legal advisers the manner of transferring or pledging their book entry interests.

The City and the Trustee have no responsibility or liability for any aspects of the records or notices relating to, or payments made on account of, book entry interest ownership, or for maintaining, supervising or reviewing any records relating to that ownership.

The City cannot and does not give any assurances that DTC, Direct Participants, Indirect Participants or others will distribute to the Beneficial Owners payments of debt charges on the Series 2016 Bonds made to DTC as the registered owner, or any redemption or other notices, or that the City will do so on a timely basis, or that DTC will serve and act in a manner described in this Official Statement.

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CITY OF CLEVELAND, OHIO • WATER POLLUTION CONTROL REVENUE BONDS, SERIES 2016 (GREEN BONDS)

Mixed Sources
Product group from well managed forests, controlled sources and recycled wood or fiber.

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